Space Exploration & Aerospace Engineering

Part I

Foundations of Space Engineering

Every space system — a weather balloon, a GPS satellite or a Mars rover — obeys the same physics and is built with the same engineering discipline. Part I explains the ecosystem, the environment a spacecraft must survive, how orbits work, how missions are engineered, the subsystems inside every spacecraft, and how data travels between space and the ground.



Chapters 1–7 · pages 1–124 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

Contents of Part I

PART I · FOUNDATIONS OF SPACE ENGINEERING

1 The Space Ecosystem: An Overview 2

1.1 Where Does Space Begin? 2
1.2 The Segments of a Space System 4
1.3 A Map of the Space Ecosystem 6
1.4 A Short History of Spaceflight 8
1.5 Space Activity in Numbers 9
1.6 Why Space Is Hard 14
1.7 Governance: The Rules of Space 16
1.8 The Engineering Disciplines of the Space Ecosystem 17
1.9 How This Manual Is Organised 18

2 The Space Environment 19

2.1 Why the Environment Drives Design 19
2.2 The Neutral Atmosphere 20
2.3 Vacuum 24
2.4 The Thermal Environment 24
2.5 Atomic Oxygen 26
2.6 Plasma and Spacecraft Charging 27
2.7 Ionising Radiation 28
2.8 Micrometeoroids and Orbital Debris 31
2.9 Mechanical Environments: Launch and Microgravity 33
2.10 Environments Beyond Earth Orbit 34
2.11 Designing for the Environment: Standards and Tools 35

3 Orbital Mechanics and Astrodynamics 38

3.1 From Newton to Kepler 38
3.2 Describing an Orbit: Orbital Elements and Reference Frames 42
3.3 Where Is the Satellite? Kepler's Equation 44
3.4 Orbit Perturbations 45
3.5 Orbital Manoeuvres 48
3.6 Rendezvous and Relative Motion 50
3.7 Orbit Determination 51
3.8 Ground Tracks and Coverage 52
3.9 Interplanetary Trajectories 53
3.10 Three-Body Dynamics and the Lagrange Points 55
3.11 The Δv Budget 57

4 Spacecraft Systems Engineering 59

4.1 What Systems Engineering Does 59
4.2 The Project Life Cycle 60
4.3 Requirements Engineering 62
4.4 Architecture and Trade Studies 63
4.5 Technical Budgets and Margins 64
4.6 Technology Readiness 65
4.7 Reliability and Redundancy 66
4.8 Risk Management 68
4.9 Model-Based Systems Engineering 69
4.10 Verification and Validation 70
4.11 Product Assurance and Safety 74
4.12 Cost and Schedule 75

4.13 Standards and Bodies of Knowledge 75

5 Spacecraft Subsystems I: Structures, Thermal Control, Power and Mechanisms 77

5.1 The Spacecraft Bus 77
5.2 Structures 78
5.3 Thermal Control 81
5.4 Electrical Power 84
5.5 Mechanisms 91
5.6 Integrated Design: Example Bus 92

6 Spacecraft Subsystems II: Attitude Control, Propulsion, Data Handling and Flight Software 94

6.1 Attitude Determination and Control 94
6.2 Spacecraft Propulsion 100
6.3 Command and Data Handling 104
6.4 Flight Software 107

7 Space Communications and the Ground Segment 110

7.1 The Role of Communications 110
7.2 Frequency Bands and Spectrum 111
7.3 Antennas 113
7.4 The Link Budget 114
7.5 Modulation and Coding 116
7.6 Optical (Laser) Communications 117
7.7 The Ground Segment 119
7.8 Space Data Standards: CCSDS 123

This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.

PART I

Foundations of Space Engineering

Every space system — a weather balloon, a GPS satellite or a Mars rover — obeys the same physics and is built with the same engineering discipline. Part I explains the ecosystem, the environment a spacecraft must survive, how orbits work, how missions are engineered, the subsystems inside every spacecraft, and how data travels between space and the ground.

1 The Space Ecosystem: An Overview
2 The Space Environment
3 Orbital Mechanics and Astrodynamics
4 Spacecraft Systems Engineering
5 Spacecraft Subsystems I: Structures, Thermal Control, Power and Mechanisms
6 Spacecraft Subsystems II: Attitude Control, Propulsion, Data Handling and Flight Software
7 Space Communications and the Ground Segment

Image: NASA/JSC

CHAPTER 1

The Space Ecosystem: An Overview

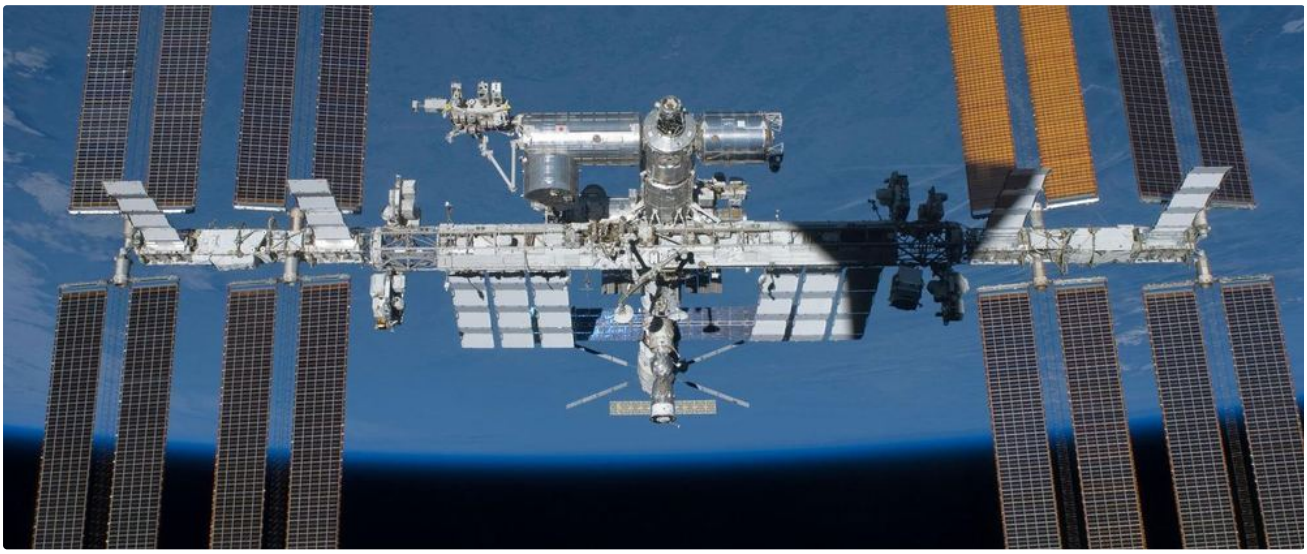


Image: NASA/JSC

> *Space is not one place or one industry. It is a layered environment that begins a few tens of kilometres above our heads and stretches to interstellar space, and an interconnected ecosystem of vehicles, ground systems, users, companies, governments and rules. This chapter maps that ecosystem, defines the regimes in which space systems operate, describes the segments of every space system, and summarises the history, economics, statistics and governance that shape it today.*

IN THIS CHAPTER

1.1 Where Does Space Begin?
1.2 The Segments of a Space System
1.3 A Map of the Space Ecosystem
1.4 A Short History of Spaceflight
1.5 Space Activity in Numbers
1.6 Why Space Is Hard
1.7 Governance: The Rules of Space
1.8 The Engineering Disciplines of the Space Ecosystem
1.9 How This Manual Is Organised

1.1 Where Does Space Begin?

There is no physical surface where the atmosphere stops and space starts. Air simply becomes thinner with height: at 5.5 km half of the atmosphere's mass is below you, at 16 km about 90 %, and at 50 km more than 99.9 %. Engineers therefore use *functional* boundaries — heights at which the physics that governs a vehicle changes.

The best-known boundary is the **Kármán line** at **100 km**. It is named after Theodore von Kármán. He noticed that an aircraft trying to fly at about this height would need to travel very fast to generate enough lift. In fact, it would already be close to orbital speed. Above it, wings are useless; a vehicle must either follow a ballistic path or be in orbit. The Fédération Aéronautique

Internationale (FAI) uses 100 km for astronautics records. The United States Air Force, NASA and the US Federal Aviation Administration award astronaut wings above **50 miles (80.5 km)**. Some scientists argue that 80 km is physically better justified. This is because satellites with very low perigees can briefly dip to about 80–90 km without immediately re-entering.

Neither number is a legal border. International law has never defined where national airspace ends and outer space begins. In practice, no satellite can stay in orbit for long below about 160 km, because atmospheric drag removes its energy within days.

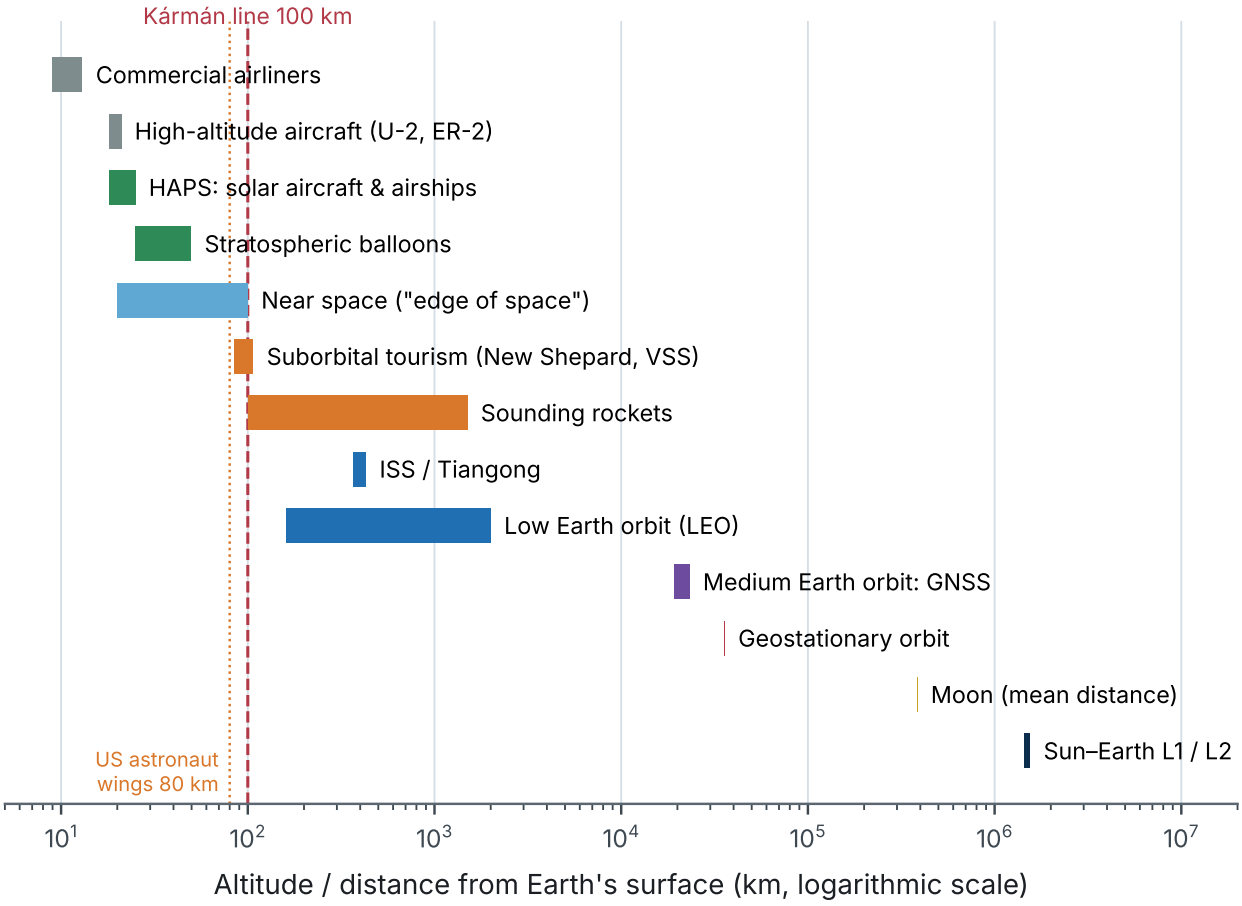


FIGURE 1.1 The regimes of near-Earth and cislunar space shown on a logarithmic distance scale. Each order of magnitude in altitude corresponds to a different class of vehicle and a different engineering problem: aerodynamics and buoyancy below 50 km, ballistic flight up to a few hundred kilometres, orbital mechanics beyond, and interplanetary trajectories past the Moon.

Figure 1.1 and Table 1.1 summarise the regimes used throughout this manual. Moving outward, each regime adds new challenges. Lower air density first removes lift and cooling, then oxygen for air-breathing engines, then protection from radiation and debris. Finally, distance itself becomes the problem, delaying radio signals by seconds, minutes or hours.

TABLE 1.1 Altitude regimes and the vehicles that operate in them.

Regime	Typical altitude or distance	Main vehicles	Dominant physics / engineering problem
Troposphere	0–12 km (0–18 km at the equator)	Aircraft, drones, weather balloons	Weather, lift, air-breathing propulsion
Stratosphere	12–50 km	HAPS (solar aircraft, airships), stratospheric balloons, U-2/ER-2	Low density, extreme cold (−56 °C), ozone, UV, long endurance

Regime	Typical altitude or distance	Main vehicles	Dominant physics / engineering problem
Mesosphere / lower thermosphere (“near space” / “ignorosphere”)	50–100 km	Sounding rockets, suborbital vehicles, re-entering spacecraft	Too thin for aircraft, too dense for satellites; meteors burn up here
Suborbital	apogee 80–1 500 km	New Shepard, sounding rockets, ballistic missiles, test vehicles	Ballistic flight, minutes of microgravity, re-entry heating
Low Earth orbit (LEO)	160–2 000 km	ISS, Tiangong, Starlink, Earth observation	Drag, atomic oxygen, debris, short ground contacts
Medium Earth orbit (MEO)	2 000–35 786 km	GPS, Galileo, GLONASS, BeiDou, O3b mPOWER	Van Allen radiation belts, longer links
Geosynchronous orbit (GEO)	35 786 km altitude (42 164 km radius)	Communications, weather, missile warning	Station-keeping, 0.24 s round-trip delay, crowded arc
Highly elliptical orbit (HEO)	perigee ~500 km, apogee ~40 000 km	Molniya/Tundra comms, early warning	Repeated belt crossings, high-latitude coverage
Cislunar space	to 384 400 km (Moon) and Earth–Moon L1/L2	Orion, lunar orbiters, landers, CAPSTONE	Three-body dynamics, deep-space communications
Sun–Earth Lagrange points	1.5 million km	SOHO, DSCOVR, JWST, Euclid, Roman	Halo orbits, thermal stability, station-keeping
Interplanetary / deep space	0.4–100+ AU	Planetary probes, orbiters, landers, Voyager	Light-time delay, power far from the Sun, autonomy

KEY CONCEPT — SPACE IS ABOUT SPEED, NOT HEIGHT

Reaching 100 km altitude needs only about 1.4 km/s of velocity change if you go straight up and come straight down. Staying in space at 400 km needs a horizontal speed of about **7.7 km/s** — roughly 28 000 km/h. Getting to orbit is therefore mostly a problem of going *sideways fast enough* that the ground curves away beneath you as quickly as you fall towards it. This is why an orbital rocket is roughly ten to twenty times larger than a suborbital rocket carrying the same payload.

1.2 The Segments of a Space System

Every space mission, whether a CubeSat built by students or a constellation of thousands of broadband satellites, can be described as a set of cooperating **segments** (Figure 1.2).

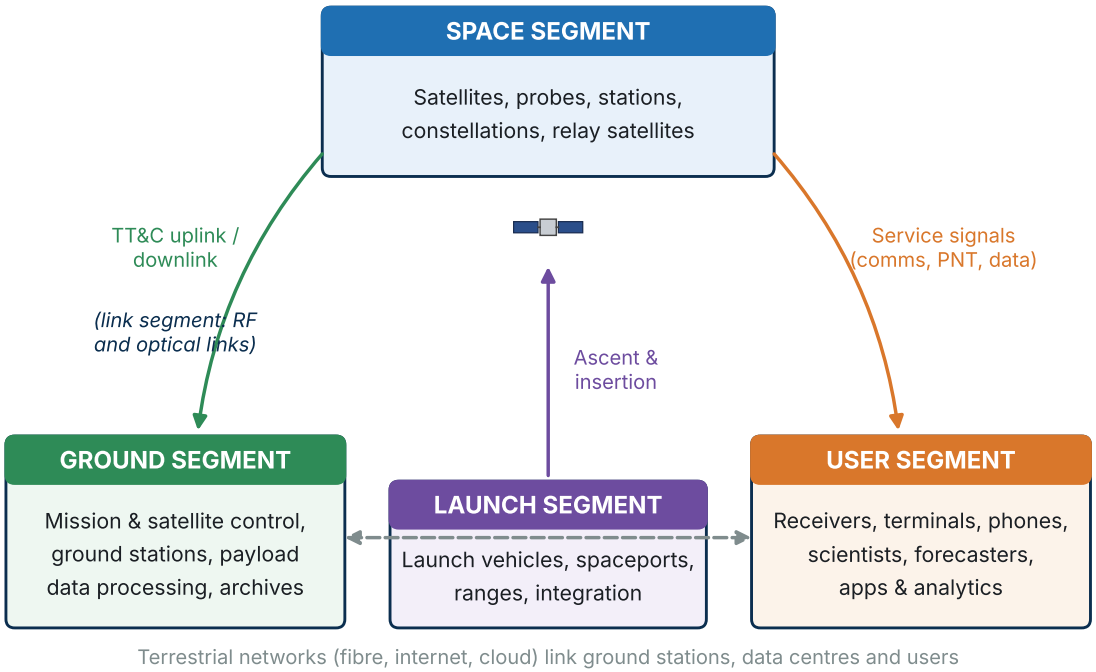


FIGURE 1.2 The four segments of a space system and the links that join them. The space segment is usually the most visible, but the ground and user segments often cost as much over a mission's life.

Space segment. The spacecraft themselves: satellites, probes, stations and their payloads. A spacecraft is normally divided into two parts. The **payload** is the part that performs the mission — a camera, a transponder, a telescope. The **bus** or **platform** is the part that keeps the payload alive — power, thermal control, attitude control, propulsion, computing and communications. Part I explains the bus in detail; the application Parts explain the payloads.

Ground segment. Everything on the ground that controls the spacecraft and handles its data: mission control centres, ground stations with antennas, networks, flight dynamics software, payload data processing, archives and distribution systems. Large ground networks — NASA's Deep Space Network, ESA's ESTRACK, commercial "ground-station-as-a-service" networks such as KSAT, AWS Ground Station and Leaf Space — are shared by many missions.

Launch segment. The launch vehicle, the spaceport, the range that tracks the rocket and protects the public, and the facilities that integrate the payload into the fairing. For a single satellite the launch is a one-time event, but its cost and risk shape the entire mission design.

User segment. The people and equipment that use the service. Examples are a GPS receiver in a phone, a satellite TV dish and a farmer's crop-monitoring app. Others are a forecaster using satellite data in a weather model, or a scientist analysing telescope images. For commercial systems the user segment is where revenue is earned; for navigation and broadband systems the user equipment is produced in hundreds of millions of units.

The **link segment** — radio-frequency (RF) and, increasingly, optical (laser) links — joins all of them. Links include telemetry, tracking and command (TT&C) for spacecraft control, high-rate payload data downlinks, user service links, and inter-satellite links that relay data through space. Chapter 7 explains how links are designed.

ENGINEERING NOTE — THINK IN SEGMENTS EARLY

A frequent cause of mission failure or cost overrun is a space segment designed in isolation. Examples include satellites that produce more data than the ground stations can download, spacecraft whose radio frequencies were never coordinated with regulators, and payloads whose data products nobody was funded to process. Systems engineering (Chapter 4) exists largely to prevent these interface failures.

1.3 A Map of the Space Ecosystem

The word *ecosystem* is used deliberately: space activities depend on each other like species in an environment. Cheap launch enables mega-constellations. Mega-constellations create demand for ground terminals, spectrum and collision avoidance, and collision avoidance needs space situational awareness. All of them depend on scientific understanding of space weather. Figure 1.3 maps the major domains covered in this manual.

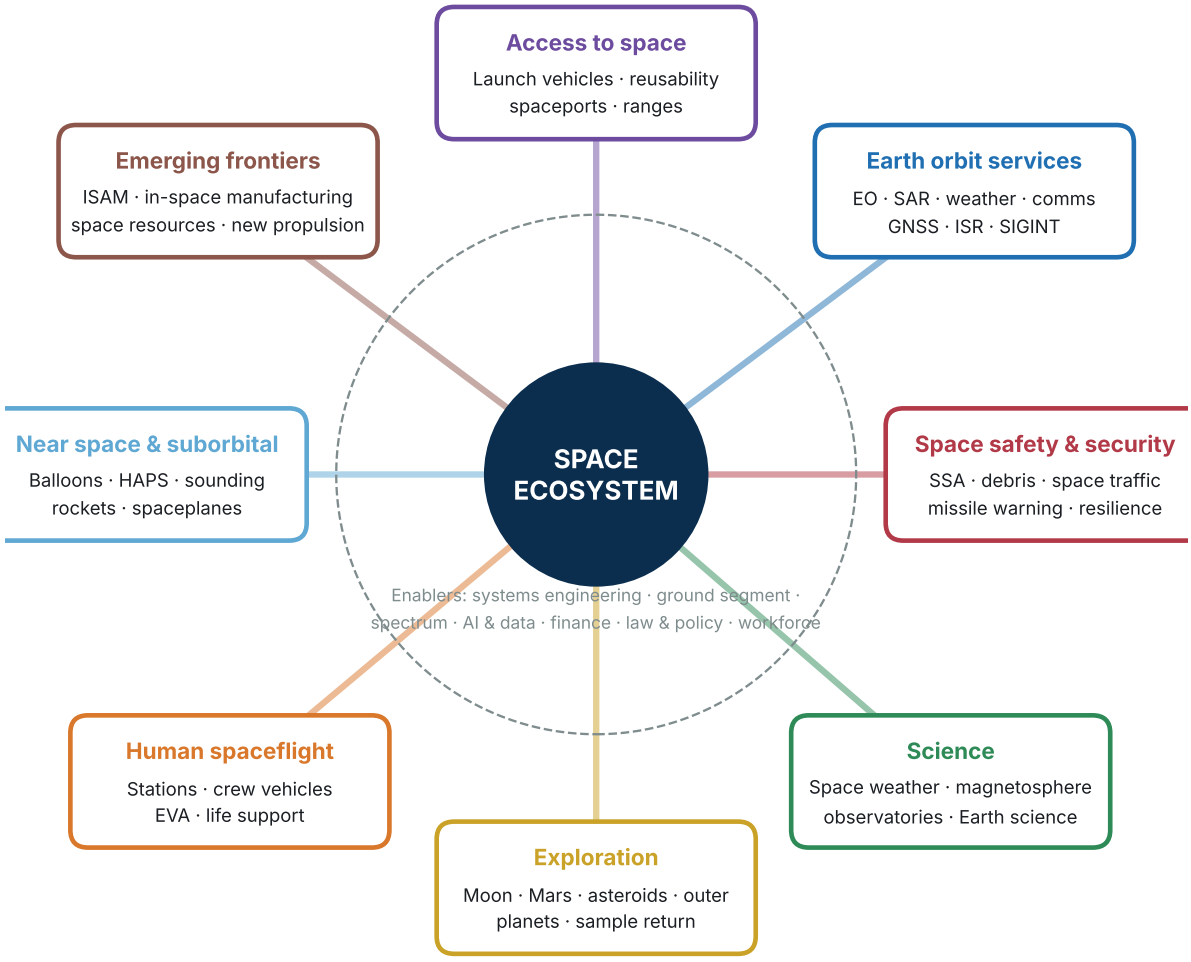


FIGURE 1.3 The space ecosystem as eight interlocking domains, each covered by one or more Parts of this manual, supported by common enablers such as systems engineering, ground infrastructure, spectrum management, artificial intelligence and law.

1.3.1 Classification by Mission Purpose

Spacecraft are commonly grouped by what they do. The classification below follows the structure of this manual.

- **Near-space and suborbital platforms** — stratospheric balloons, high-altitude platform stations (HAPS), sounding rockets, suborbital crewed vehicles and spaceplanes (Part II).
- **Earth observation (EO) and remote sensing** — optical, infrared, radar, atmospheric and oceanographic satellites (Part IV).
- **Communications** — fixed and mobile satellite services, broadcasting, broadband, data relay, direct-to-device (Part V).
- **Positioning, navigation and timing (PNT)** — GPS, GLONASS, Galileo, BeiDou and regional systems (Part V).
- **National security** — intelligence, surveillance and reconnaissance (ISR), signals intelligence, missile warning, military communications (Part VI).
- **Space safety** — space situational awareness, debris tracking and removal (Part VI).
- **Space science** — heliophysics, magnetospheric physics, astronomy and astrophysics (Parts VII and XI).
- **Exploration** — robotic orbiters, probes, landers, rovers and sample-return missions; human spaceflight (Parts VIII and IX).
- **Technology and in-space economy** — servicing, assembly, manufacturing, resource extraction, propulsion demonstrations (Part XII).

1.3.2 Classification by Size

Satellites range from a few grams to more than 400 tonnes (the International Space Station). The industry uses an informal mass scale (Table 1.2). The boundaries vary slightly between organisations, but the categories are widely understood.

TABLE 1.2 Informal spacecraft mass classes and examples.

Class	Wet mass	Typical examples	Typical cost range (spacecraft only)
Femto/chip satellite	< 100 g	KickSat "Sprites", ChipSats	< US$1 000 each
Picosatellite	0.1–1 kg	PocketQube, 1U CubeSat (≈1.3 kg is borderline)	US$10k–100k
Nanosatellite	1–10 kg	1U–6U CubeSats, Planet "Dove" (≈5 kg)	US$50k–2M
Microsatellite	10–100 kg	12U/16U CubeSats, ICEYE SAR (≈90 kg), Spire	US$1M–20M
Minisatellite / small satellite	100–500 kg	OneWeb (≈150 kg), Starlink V1.5 (≈300 kg), NovaSAR-1 (≈450 kg)	US$5M–100M
Medium satellite	500–2 500 kg	Starlink V2 Mini (≈800 kg), Iridium NEXT (≈860 kg), Sentinel-2 (≈1.1 t)	US$50M–500M
Large satellite	> 2 500 kg	Landsat 9 (≈2.7 t), GPS III (≈3.9 t), GEO communications satellites (3–7 t), JWST (6.2 t)	US$150M–10B

Class	Wet mass	Typical examples	Typical cost range (spacecraft only)
Space station	> 50 t	Tiangong (≈100 t), ISS (≈450 t)	US$10B–150B

The CubeSat standard (Chapter 11) deserves special mention: by defining a standard 10 cm cube "unit" (1U) and a standard deployer, it made space accessible to universities and start-ups. More than 2 500 CubeSats have been launched since 2003.

1.4 A Short History of Spaceflight

The modern space age began on 4 October 1957, when the Soviet Union launched **Sputnik 1**, an 84 kg sphere with four antennas that transmitted a simple beep. Within twelve years humans had orbited the Earth (Yuri Gagarin, 12 April 1961) and walked in space (Alexei Leonov, 1965). They had also walked on the Moon (Neil Armstrong and Buzz Aldrin, 20 July 1969). Figure 1.4 marks major milestones.

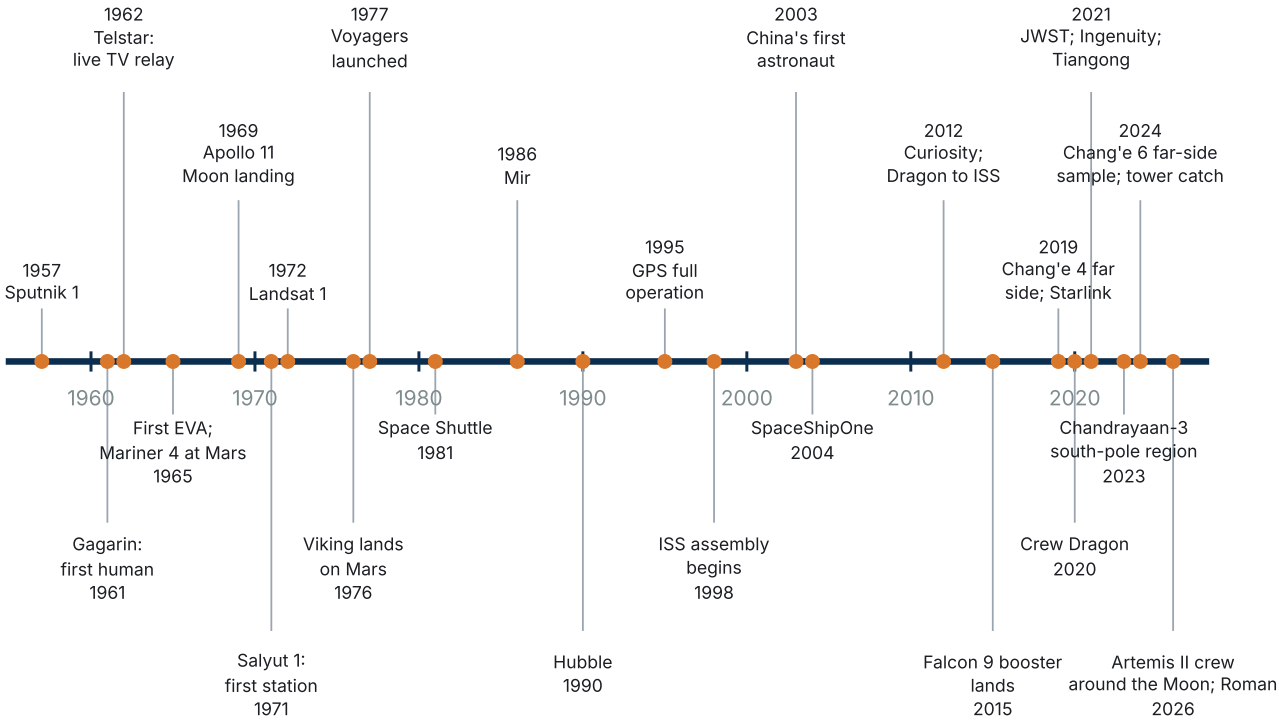


FIGURE 1.4 Selected milestones of the space age, 1957–2026. The pace of "firsts" was highest in the 1960s; since 2015 a second wave has been driven by reusable rockets, small satellites and new national programmes.

The history can be divided into five broad eras:

1. **The Space Race (1957–1975).** A competition between the USA and the USSR produced the first satellites, human flights, planetary probes, weather and communications satellites, reconnaissance satellites (CORONA, Zenit), and the Apollo lunar landings. Launch rates peaked at about 140 orbital attempts in 1967.

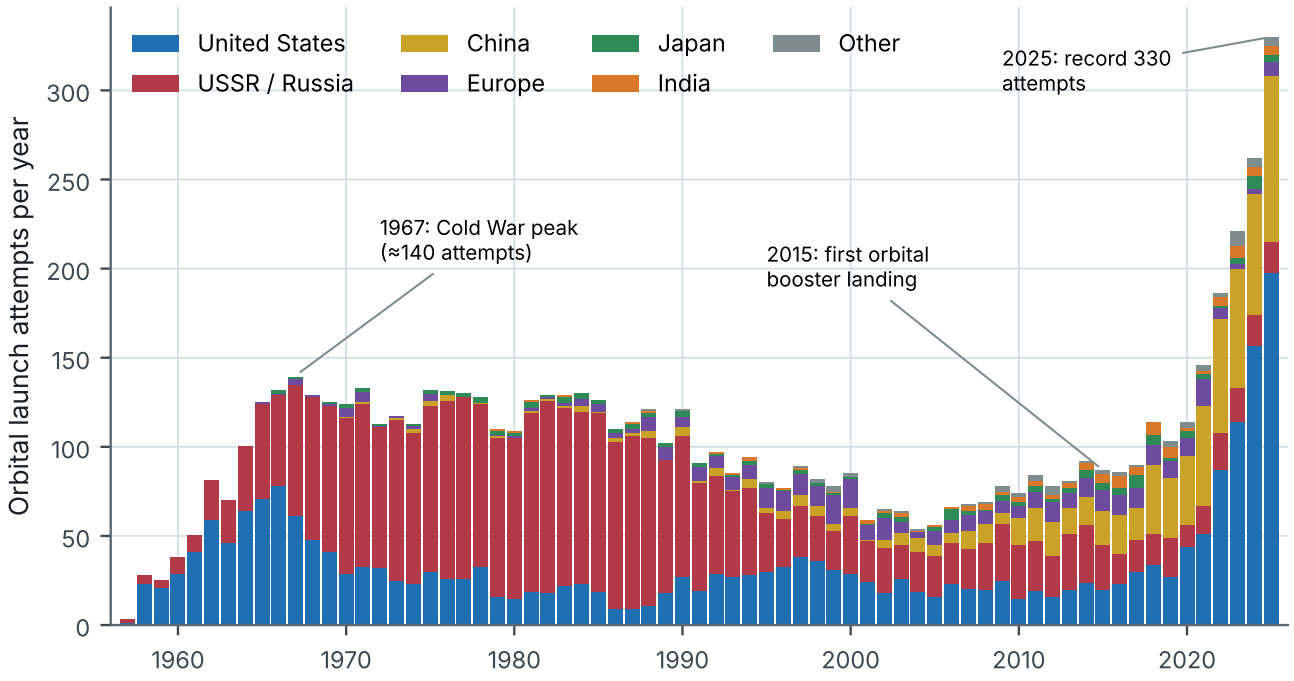
2. **Stations and shuttles (1971–2000).** The USSR operated Salyut stations and Mir; the USA flew Skylab and the reusable Space Shuttle. Satellite applications matured: Landsat (1972), GPS (first launch 1978, fully operational 1995), INTELSAT global communications, and the great observatories (Hubble 1990, Compton 1991, Chandra 1999).
3. **International cooperation and globalisation (1998–2010).** Assembly of the International Space Station began in 1998, permanently crewed since November 2000. China flew its first astronaut (Yang Liwei, 2003). Europe, Japan and India developed independent launch and exploration capabilities. Mars exploration intensified with orbiters and rovers.
4. **"New Space" (2010–2019).** Private capital entered the industry. SpaceX's Falcon 9 began flying in 2010, delivered cargo to the ISS in 2012 and landed an orbital-class booster in December 2015. CubeSats and small-satellite constellations (Planet, Spire) appeared. China landed on the far side of the Moon (Chang'e 4, 2019) and the first Starlink satellites launched in 2019.
5. **The industrial space age (2020–present).** This era brought commercial crew transport (Crew Dragon, 2020), mega-constellations with thousands of satellites, routine booster reuse and China's Tiangong station (2021–2022). It also saw the James Webb Space Telescope (2021), the first far-side sample return (Chang'e 6, 2024) and the first returning booster caught by its launch tower (Starship, October 2024). Then came the first crewed flight around the Moon since 1972 (Artemis II, April 2026) and the launch of the Nancy Grace Roman Space Telescope (August 2026).

> **HISTORY — THE SECOND SPACE RACE IS COMMERCIAL AND MULTIPOLAR**
>
> In 1967, two nations conducted nearly all space launches. In 2025, rockets from more than ten countries reached orbit, and private companies launched most of the payloads. The busiest single launch provider (SpaceX) flew more often than the whole world had in any year before 2021.

1.5 Space Activity in Numbers

1.5.1 Launch Rate

Figure 1.5 shows every orbital launch attempt since 1957, grouped by the nationality of the launching organisation. After the Cold War peak, launch rates fell for two decades as satellites became longer-lived and the Soviet programme contracted. Since 2016 the rate has risen steeply, driven first by reusable Falcon 9 boosters and Chinese commercial and government launches.



Data: J. McDowell, GCAT (planet4589.org), updated 13 Oct 2026. Includes failures.

FIGURE 1.5 Orbital launch attempts per year, 1957–2025, by country of the launch provider. The 2025 total of 330 attempts (counts vary slightly between sources, from 323 to 330, depending on how test flights are treated) is the highest in history. "United States" includes Rocket Lab launches from New Zealand.

TABLE 1.3 Orbital launch attempts by country of launch provider, 2025 and 2026 to mid-October (GCAT).

Country / region	2025 attempts	Share 2025	2026 (to 13 Oct)	Main vehicles
United States	198	60 %	146	Falcon 9, Falcon Heavy, Electron, Starship, Atlas V, Vulcan, New Glenn
China	93	28 %	71	Long March family, Zhuque, Kinetica-1, Ceres-1, Kuaizhou, Jielong, Gravity-1
Russia	17	5 %	13	Soyuz-2, Angara, Proton
Europe	8	2 %	7	Ariane 6, Vega-C
India	5	2 %	3	PSLV, GSLV, LVM3
Japan	4	1 %	3	H3, H-IIA (retired 2025)
Others (Iran, North Korea, South Korea, …)	5	2 %	1	Simorgh, Qaem-100, Nuri
World	**330**	100 %	**244**	

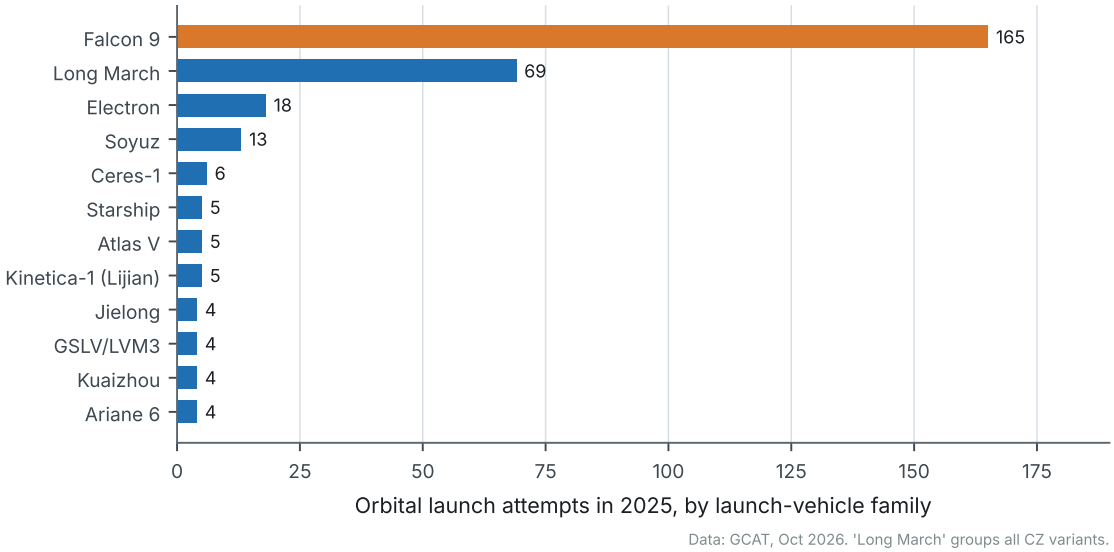


FIGURE 1.6 Orbital launch attempts in 2025 by launch-vehicle family. A single vehicle family, Falcon 9, accounted for half of all orbital launches worldwide.

1.5.2 Satellites in Orbit

The number of satellites has grown even faster than the number of launches, because one launch can now deploy dozens or hundreds of satellites. In 2025 alone about **4 500 payloads** were placed in orbit — more than in the first fifty years of spaceflight combined (Figure 1.7). Most of the growth comes from broadband constellations, especially SpaceX's Starlink, which had more than 10 600 satellites in orbit by October 2026. Next come Eutelsat OneWeb (≈650), China's Guowang and Qianfan constellations, and Amazon Leo (formerly Project Kuiper, about 400 satellites by mid-2026).

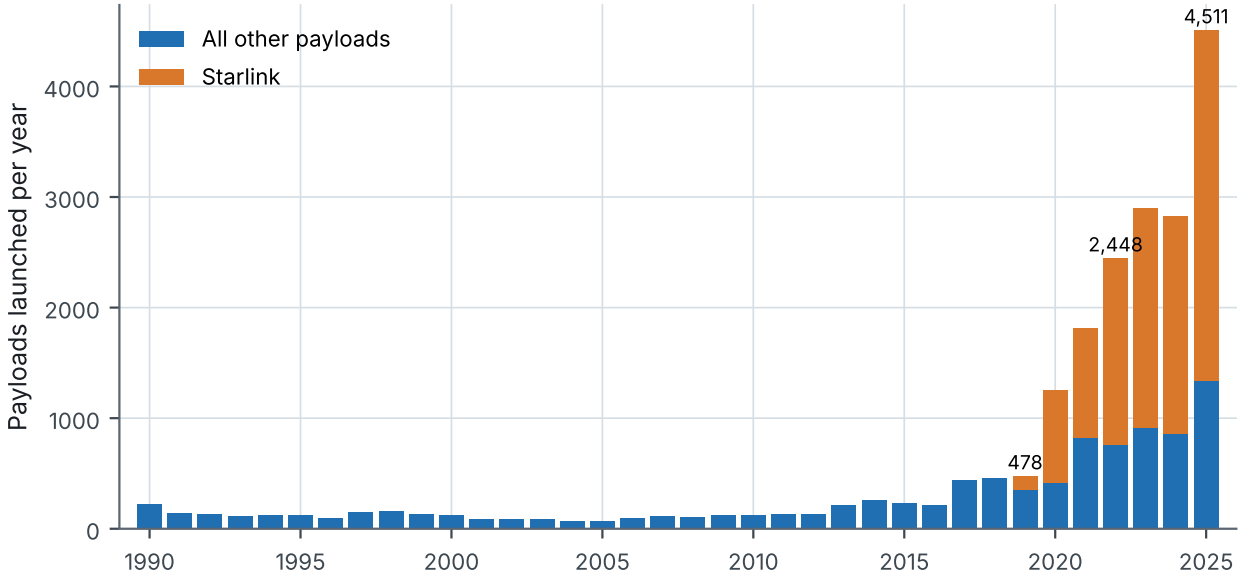
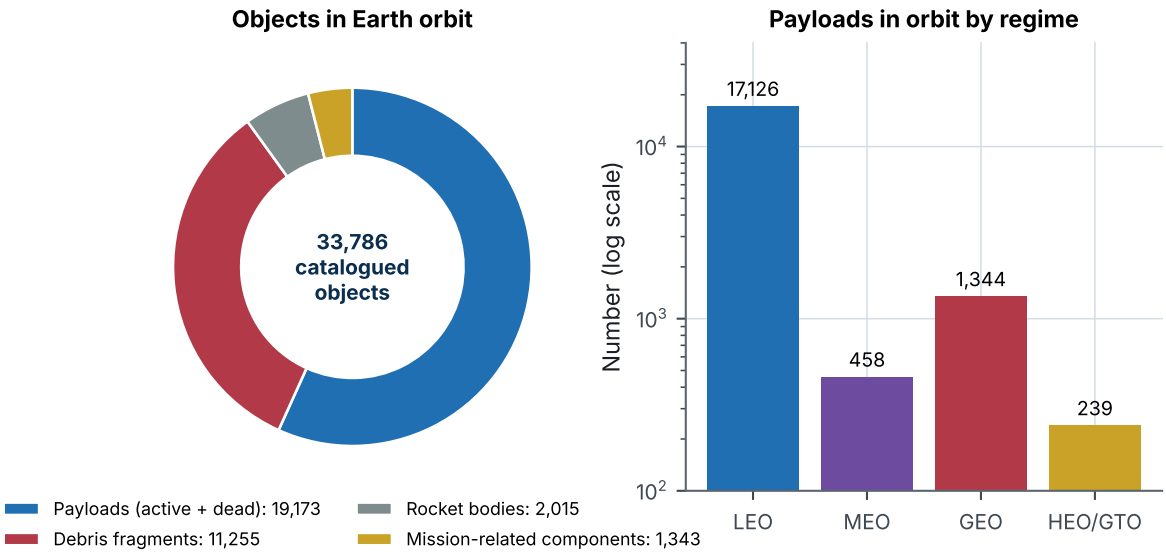


FIGURE 1.7 Payloads launched per year, 1990–2025, separating Starlink from all other payloads. The step change after 2019 is the arrival of mega-constellations.



Data: GCAT, 13 Oct 2026. Tracked objects ≥10 cm; ~1 million objects 1–10 cm are untracked.

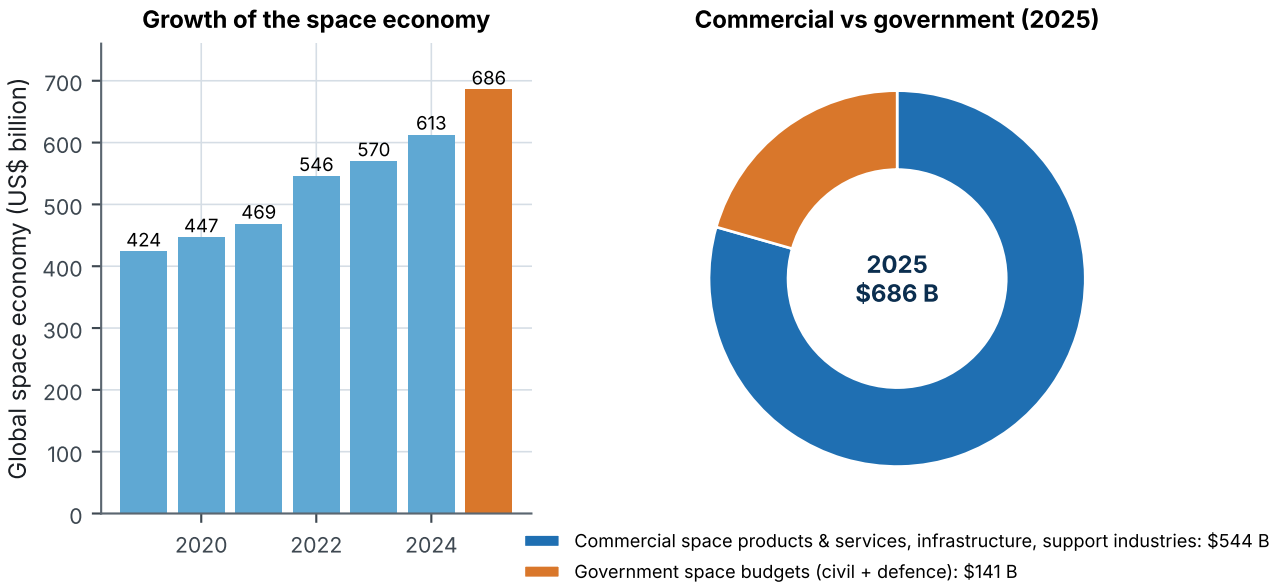
FIGURE 1.8 Left: the ~33 800 catalogued objects in Earth orbit by type. Right: payloads in orbit by orbital regime. Low Earth orbit dominates in numbers; GEO dominates in value per satellite.

DATA BOX — EARTH ORBIT POPULATION, OCTOBER 2026 (GCAT)

- Catalogued objects in orbit: **≈33 800**, of which payloads ≈19 200, debris fragments ≈11 300, rocket bodies ≈2 000, other mission objects ≈1 300.
- Payloads believed to be operating: **≈16 000–17 000** (GCAT estimate ≈16 000; CelesTrak lists ≈17 200 as operational; definitions of "active" differ).
- Starlink satellites in orbit: **≈10 650**.
- Payloads by regime: LEO ≈17 100, GEO ≈1 340, MEO ≈460, highly elliptical ≈240.
- Estimated objects too small to track (1–10 cm): **about one million**; objects 1 mm–1 cm: over 100 million (ESA debris model).

1.5.3 The Space Economy

The **space economy** is the total of all revenues and budgets generated by space activities. Estimates differ in what they include (for example, whether satellite TV subscriptions or GNSS-enabled phones count), but the Space Foundation's series is widely quoted: the global space economy reached about **US$686 billion in 2025**, up 12 % from US$613 billion in 2024 (Figure 1.9). About 79 % was commercial; government space budgets totalled about US$141 billion, of which the United States accounted for 57 %.



Source: Space Foundation, The Space Report (2019–2025 editions; 2025 data released July 2026). Values in nominal US$.

FIGURE 1.9 Left: growth of the global space economy, 2019–2025. Right: split between commercial activity and government budgets in 2025.

Three observations help interpret these numbers:

1. **Most money is made on the ground.** Satellite communications services, user terminals, GNSS chipsets and downstream data products earn far more than rockets or satellites themselves. The "upstream" manufacturing and launch sector is only roughly a tenth of the total.
2. **Governments remain the anchor customers.** Even in commercial markets, government contracts (for example NASA's Commercial Crew and Cargo programmes, the US Space Development Agency's constellation, or ESA's Galileo) fund the first systems and reduce investor risk.
3. **Defence spending is rising fastest.** Since 2022, military and intelligence space budgets have grown more quickly than civil budgets in the United States, Europe, Japan and elsewhere, reflecting the role of satellites in modern conflicts.

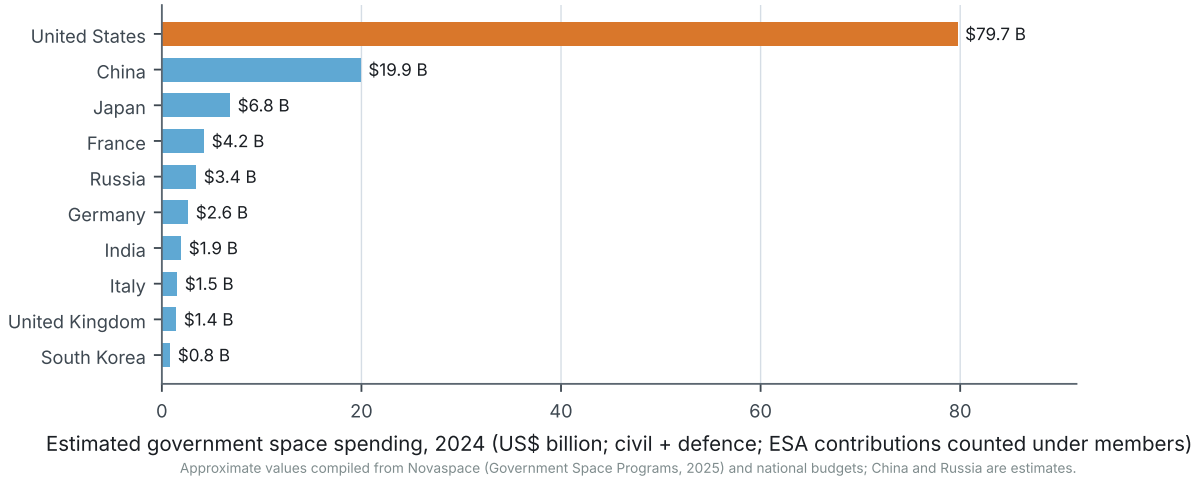


FIGURE 1.10 Estimated government space spending by country in 2024 (civil and defence combined). The United States spends roughly four times more than China and more than all other governments together.

TABLE 1.4 Major space agencies and organisations (selected).

Organisation	Country / members	Founded	Notable capabilities (2026)
NASA	USA	1958	Artemis (SLS/Orion), ISS, deep-space science, JWST, Roman, Deep Space Network
US Space Force / NRO / SDA	USA	2019 / 1961 / 2019	GPS, missile warning, military comms, reconnaissance, proliferated LEO architecture
NOAA (NESDIS)	USA	1970	GOES, JPSS weather satellites, space weather (SWFO-L1), commercial remote-sensing licensing
Roscosmos	Russia	1992 (successor to Soviet programme)	Soyuz crew and cargo, GLONASS, Angara, ISS Russian segment
CNSA / CMSA	China	1993 / 1992	Tiangong station, Chang'e lunar, Tianwen planetary, BeiDou, Long March
ESA	23 member states	1975	Ariane 6, Vega-C, Copernicus (with EU), Galileo (with EU), science missions
EUSPA / European Commission	EU-27	2021	Galileo, EGNOS, Copernicus programme management, IRIS² secure connectivity
JAXA	Japan	2003 (merger)	H3, Hayabusa missions, SLIM lunar lander, MMX, HTV-X cargo
ISRO	India	1969	PSLV, LVM3, Chandrayaan-3, Aditya-L1, NavIC, Gaganyaan
CNES / DLR / ASI / UKSA	France / Germany / Italy / UK	1961 / 1969 (as DFVLR) / 1988 / 2010	National programmes and major ESA contributors
CSA	Canada	1989	Canadarm robotics, RADARSAT, Artemis II astronaut
KASA	South Korea	2024	Nuri launcher, Danuri lunar orbiter
UAE Space Agency / MBRSC	United Arab Emirates	2014 / 2006	Hope Mars orbiter, astronaut programme, Gateway airlock (planned)

1.6 Why Space Is Hard

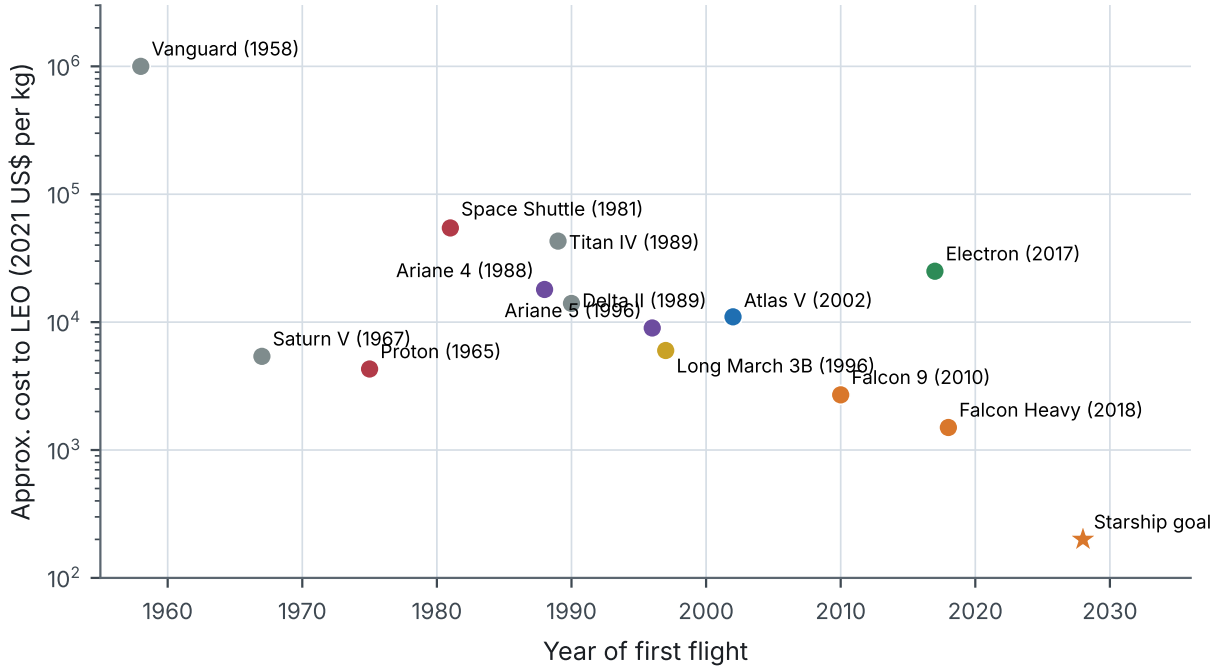
Space engineering combines three difficulties that are rarely found together in other industries.

1. The energy problem. To reach low Earth orbit a vehicle must gain about 9.3–9.6 km/s of velocity once gravity and drag losses are included. The rocket equation (Chapter 34) shows that this requires a vehicle that is typically 85–95 % propellant at lift-off; only 2–4 % of the lift-off mass reaches orbit as payload. Every kilogram of spacecraft therefore costs many kilograms of rocket.

2. The environment problem. In orbit, a spacecraft must survive vacuum, temperature swings of more than 200 °C and radiation that damages electronics and solar cells. It must also survive atomic oxygen, micrometeoroids and debris travelling at 10 km/s, and launch vibrations and shocks of tens of g (Chapter 2).

3. The no-repair problem. With very few exceptions (Hubble servicing, the ISS, the new generation of servicing vehicles), a spacecraft cannot be repaired after launch. It must work first time and keep working for 5–15 years. This drives redundancy, extensive testing, conservative design margins and long development schedules.

These difficulties explain why spacecraft traditionally cost tens of thousands of dollars per kilogram. Two trends are now lowering the cost. The first is reusable launch vehicles. The second is mass-produced small satellites, which accept higher individual failure rates because the constellation as a whole is resilient. Figure 1.11 shows how much the price of reaching orbit has fallen.



Approximate: list price ÷ maximum LEO payload (reusable vehicles at commercial price). After NASA/CSIS compilations; real costs vary by mission.

FIGURE 1.11 Approximate cost per kilogram to low Earth orbit for representative launch vehicles, in inflation-adjusted dollars. Reusable boosters reduced commercial launch prices by about an order of magnitude; fully reusable vehicles aim for another order of magnitude.

WORKED EXAMPLE — HOW MUCH DOES IT COST TO PUT A 6U CUBESAT IN ORBIT?

A 6U CubeSat has a mass of about 10 kg. On a rideshare mission (for example SpaceX's Transporter series), the list price in 2025 was about US$6 500 per kilogram. The minimum was about US$325 000 for 50 kg to sun-synchronous orbit. So a single 6U satellite would typically be launched through an integrator for roughly US$150 000–300 000, including the deployer. A dedicated small launcher (e.g., Rocket Lab Electron, about US$8.5 million per launch) is far more expensive per kilogram. However, it lets the customer choose the orbit and the launch date. The choice between *rideshare* and *dedicated* launch is one of the first trade-offs in a small-satellite mission (Chapter 11).

1.7 Governance: The Rules of Space

Space activities are regulated by a combination of international treaties, UN guidelines, international technical bodies and national laws.

TABLE 1.5 The core instruments of international space law.

Instrument	Year	Key provisions
Outer Space Treaty	1967	Space free for exploration by all states; no national appropriation of celestial bodies; no weapons of mass destruction in orbit; states responsible for national activities, including private ones ("authorisation and continuing supervision"); liability for damage
Rescue Agreement	1968	Assistance to astronauts in distress; return of space objects
Liability Convention	1972	Absolute liability for damage on Earth's surface or to aircraft; fault-based liability in space
Registration Convention	1975	States register launched objects with the UN (UNOOSA register)
Moon Agreement	1979	Moon resources as "common heritage of mankind" — ratified by few states, not by major space powers
ITU Constitution and Radio Regulations	updated at each World Radiocommunication Conference (latest WRC-23; next WRC-27)	Allocation of radio frequencies; coordination and registration of satellite networks and orbital positions
UN COPUOS Space Debris Mitigation Guidelines	2007	Limit debris release, avoid break-ups, post-mission disposal
UN Long-Term Sustainability Guidelines	2019	21 voluntary guidelines on safety, SSA, debris, regulation
Artemis Accords	2020	Non-binding principles for civil lunar exploration: transparency, interoperability, emergency assistance, registration, release of scientific data, preserving heritage sites, deconfliction "safety zones", space-resource use consistent with the OST. 76 signatories by late September 2026

National regulators turn these principles into licences. In the United States, for example, the FAA licenses launches and re-entries. The Federal Communications Commission (FCC) licenses spectrum and imposes debris rules, including a five-year post-mission disposal rule for LEO satellites adopted in 2022. NOAA licenses commercial remote-sensing systems. Other countries, including the UK (Space Industry Act 2018), France (French Space Operations Act 2008), Japan (Space Activities Act 2016), New Zealand, Luxembourg and the UAE, have comparable laws. China's International Lunar Research Station (ILRS) programme forms a second international lunar partnership alongside the Artemis Accords.

LESSON — DEBRIS IS A SHARED RISK

In 2007 China destroyed its Fengyun-1C weather satellite with a ground-launched missile, creating more than 3 500 trackable fragments, many still in orbit. In 2009 the active Iridium 33 and the defunct Kosmos 2251 collided at 11.7 km/s, creating about 2 300 more. In November 2021 a Russian anti-satellite test destroyed Kosmos 1408 and forced the ISS crew to shelter. Because debris does not respect national boundaries, debris mitigation and space traffic coordination (Chapter 21) have become central policy issues.

1.8 The Engineering Disciplines of the Space Ecosystem

A space programme employs almost every branch of engineering and science. Table 1.6 lists the main disciplines and where they appear in this manual.

TABLE 1.6 Engineering and science disciplines in the space ecosystem.

Discipline	What it does in space programmes	Main chapters
Systems engineering	Requirements, architecture, interfaces, budgets, verification, risk	4, 11, 24
Astrodynamics / flight dynamics	Orbit design, manoeuvres, navigation, orbit determination	3, 11, 21, 24
Structural and mechanical engineering	Primary structure, mechanisms, deployables, launch loads	5, 34
Thermal engineering	Radiators, heaters, insulation, heat pipes, cryogenics	5, 38
Electrical power engineering	Solar arrays, batteries, power conditioning, RTGs	5, 24
Guidance, navigation and control (GNC)	Attitude determination and control, autonomous navigation, landing	6, 27, 45
Propulsion engineering	Rocket engines, electric thrusters, propellant systems	6, 35, 42
Avionics and embedded software	Flight computers, data handling, fault protection, flight software	6, 45
RF and optical communications	Antennas, transponders, link budgets, lasercom	7, 17
Payload and instrument engineering	Optics, detectors, radars, spectrometers, particle instruments	12–16, 38–39
Aerothermodynamics	Re-entry heating, heat shields, atmospheric entry	10, 27
Human factors and life sciences	Life support, radiation protection, ergonomics, medicine	30–33
Ground systems and operations	Mission control, scheduling, data processing	7, 45
Data science and artificial intelligence	Onboard processing, analytics, autonomy, forecasting	43–53
Product assurance and safety	Reliability, quality, parts, materials, safety analysis	4
Law, policy and economics	Licensing, spectrum, insurance, finance, export control	1, 21, 41

1.9 How This Manual Is Organised

The remaining chapters follow a logical sequence. Part I builds the foundations: the environment (Chapter 2), orbits (Chapter 3), systems engineering (Chapter 4), spacecraft subsystems (Chapters 5 and 6) and communications (Chapter 7). Each later Part applies these foundations to a family of systems. Part XIII then shows how artificial intelligence is changing every one of these domains, from onboard image classification to AI weather models and autonomous rovers. Part XIV collects the reference data an engineer needs at hand.

KEY TAKEAWAYS

- Space has no physical boundary; 100 km (Kármán line) and 80 km are conventional limits. Orbit requires horizontal speed (≈7.8 km/s), not just height.
- Every mission consists of space, ground, launch and user segments joined by links; interfaces between segments are where many problems occur.
- In 2025 the world made a record 330 orbital launch attempts and placed about 4 500 payloads in orbit; about 16 000 satellites were operating in late 2026, two-thirds of them Starlink.
- The space economy was about US$686 billion in 2025, roughly four-fifths commercial; government budgets (≈US$141 billion) remain the anchor of the industry.
- Space is hard because of the energy needed to reach orbit, the hostile environment, and the near-impossibility of repair; reusability and mass production are changing the economics.
- International treaties set principles, while national licences, ITU spectrum rules and debris guidelines govern day-to-day activities.

REVIEW QUESTIONS

1. Why is reaching an altitude of 100 km not enough to stay in orbit? What horizontal speed is needed in low Earth orbit?
2. Name the four segments of a space system and describe one interface between two of them that could cause a mission to fail.
3. Classify each of the following by mission purpose and by size class: a 3U CubeSat that images crops, a 6-tonne geostationary communication satellite, and a Mars rover.
4. Describe how the number of orbital launch attempts per year has changed since 2015, and identify the main reasons for the change.
5. Roughly what fraction of the US$686 billion space economy in 2025 was commercial? Give two examples each of commercial and government space activities.
6. Give three reasons why space is “hard”, and for each describe an engineering approach used to overcome it.
7. What principles does the Outer Space Treaty establish? Which other rules and bodies govern spectrum use and debris mitigation in practice?
8. Choose one engineering discipline described in this chapter and explain how it contributes to a satellite mission from design to operations.

2 CHAPTER The Space Environment



Image: NASA/JSC

> *A spacecraft must survive two very different environments - the short, violent environment of launch and the long, quiet but hostile environment of space. This chapter explains each hazard in turn - the upper atmosphere and drag, vacuum, the thermal environment, atomic oxygen, plasma and charging, ionising radiation, micrometeoroids and debris, and the mechanical loads of launch - shows how each one varies with orbit, and gives the design rules engineers use to cope with them.*

IN THIS CHAPTER

2.1 Why the Environment Drives Design
2.2 The Neutral Atmosphere
2.3 Vacuum
2.4 The Thermal Environment
2.5 Atomic Oxygen
2.6 Plasma and Spacecraft Charging
2.7 Ionising Radiation
2.8 Micrometeoroids and Orbital Debris
2.9 Mechanical Environments: Launch and Microgravity
2.10 Environments Beyond Earth Orbit
2.11 Designing for the Environment: Standards and Tools

2.1 Why the Environment Drives Design

A spacecraft engineer's first question about any mission is *where* the spacecraft will fly, because the environment determines most design decisions. It sets the thickness of shielding, the type of electronic parts, the thermal coatings and the solar array degradation margin. It also sets the propellant needed to fight drag, and even the shape of the vehicle. Many famous failures were environmental failures. An early example is the Telstar 1 satellite, killed in 1962 by radiation from a high-altitude nuclear test. A recent one is the loss of 38 newly launched Starlink satellites to a minor geomagnetic storm in February 2022.

The space environment can be divided into the following components, each discussed in this chapter:

1. **The neutral atmosphere** — drag, atomic oxygen, and their dependence on solar activity.
2. **Vacuum** — outgassing, contamination, cold welding, discharges.
3. **The thermal environment** — sunlight, Earth albedo and infrared, eclipses, deep-space cold.
4. **Plasma** — charged particles that charge spacecraft surfaces and cause discharges.
5. **Ionising radiation** — trapped belts, solar particle events and galactic cosmic rays.

6. **Micrometeoroids and orbital debris (MMOD)** — hypervelocity impacts.
7. **Mechanical environments** — launch loads, vibration, acoustics, shock, and microgravity.

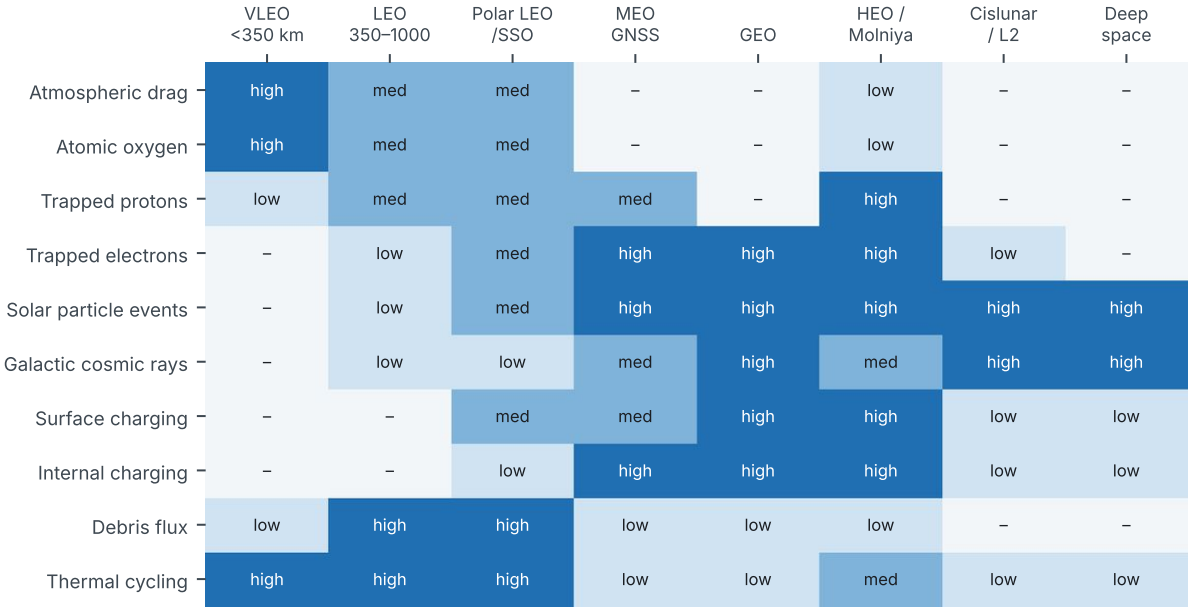


FIGURE 2.1 Relative severity of the main environmental hazards in different orbital regimes. No orbit is benign: low orbits suffer drag, atomic oxygen and debris; medium and high orbits suffer radiation and charging; deep space adds cosmic rays and solar particle events with no magnetospheric protection.

2.2 The Neutral Atmosphere

2.2.1 Structure of the Atmosphere

The atmosphere is divided into layers by the way temperature changes with height (Figure 2.2). In the **troposphere** temperature falls about 6.5 K per kilometre. In the **stratosphere** it rises again because ozone absorbs ultraviolet sunlight. It falls through the **mesosphere** to the coldest point on Earth, the **mesopause** at 85–90 km (about −90 °C), where noctilucent clouds form and meteors burn up. Above that, in the **thermosphere**, the very thin gas absorbs extreme-ultraviolet (EUV) radiation from the Sun and is heated to 700–1 400 K or more. Above about 500 km lies the **exosphere**, where atoms are so sparse that they follow ballistic trajectories and can escape to space.

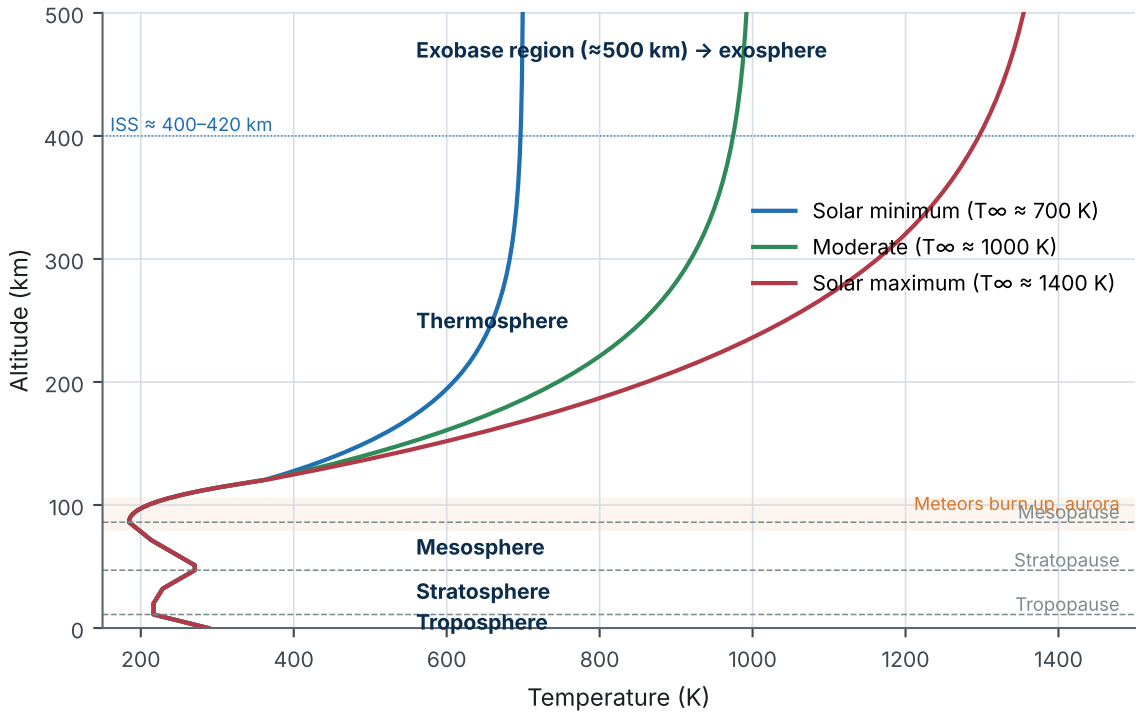


FIGURE 2.2 Temperature profile of the atmosphere. Below 86 km the US Standard Atmosphere 1976 is shown; above, the thermosphere approaches an "exospheric temperature" T_∞ that depends strongly on solar EUV output.

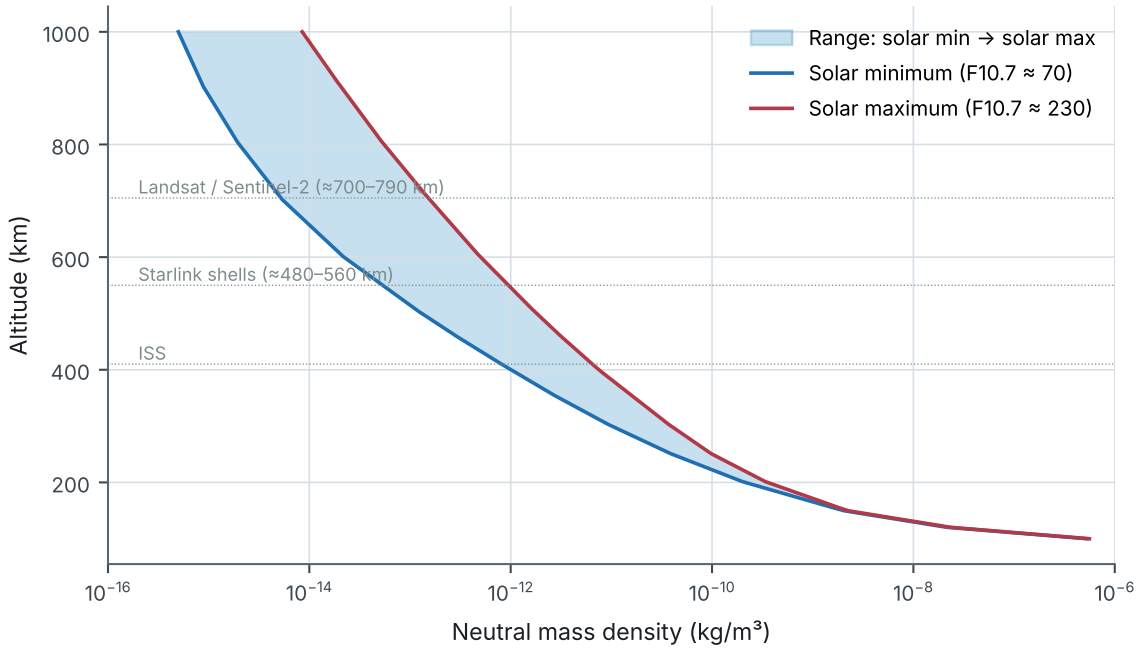
The composition also changes. Below about 100 km the atmosphere is well mixed (78 % N_2, 21 % O_2). Above, gases separate by molecular weight, and solar ultraviolet splits O_2 into **atomic oxygen (O)**, which becomes the dominant species between about 200 and 600 km. Higher still, helium and then hydrogen dominate.

2.2.2 Density and the Scale Height

In a column of gas in hydrostatic equilibrium, the pressure falls exponentially with altitude:

$$\rho(h) = \rho_0 \exp\left(-\frac{h - h_0}{H}\right), \qquad H = \frac{k_B T}{m g}$$

where H is the **scale height** — the height over which density falls by a factor of $e \approx 2.72$. Near the ground $H \approx 8.5$ km; at 400 km, where the gas is hot and light, H is about 50–60 km. The value of H depends on temperature, and thermospheric temperature depends on the Sun. So **density at a given LEO altitude can change by a factor of ten or more** between solar minimum and solar maximum. It can also change by a factor of two to five within hours during a geomagnetic storm (Figure 2.3).



Representative values from empirical models (NRLMSIS-type); density at 400 km varies by ~10× over a solar cycle.

FIGURE 2.3 Neutral density of the upper atmosphere for low and high solar activity. At typical LEO altitudes, the density — and therefore the drag on a satellite — varies by about an order of magnitude over the 11-year solar cycle.

Solar activity is tracked by two indices that drive the atmospheric models (NRLMSIS 2.0, JB2008, DTM-2020). The **F10.7 index** is the solar radio flux at 10.7 cm wavelength (a proxy for EUV output), typically 70 at solar minimum and 150–250 at maximum. The geomagnetic **Ap / Kp indices** measure the heating of the upper atmosphere by the solar wind (Chapter 22).

2.2.3 Atmospheric Drag

A satellite moving at velocity v through gas of density ρ experiences a drag force

$$F_D = \tfrac{1}{2}\,\rho\, v^2\, C_D\, A$$

where A is the cross-sectional area and C_D is the drag coefficient (typically 2.0–2.4 for satellites in free-molecular flow). The ratio $B = m/(C_D A)$ is the **ballistic coefficient**: a dense, compact satellite (large B) decays slowly; a satellite with large solar arrays (small B) decays quickly. For a circular orbit of radius a, drag reduces the orbit radius at the rate

$$\frac{da}{dt} = -\frac{\rho\sqrt{\mu\, a}}{B}$$

Counter-intuitively, drag makes a satellite *speed up*: as it loses energy it sinks to a lower orbit, where the orbital speed is higher. The decay accelerates as the satellite descends into denser air. So most of a satellite's orbital life is spent at the higher altitudes, and the final descent from 200 km takes only days. Figure 2.4 shows typical orbital lifetimes.

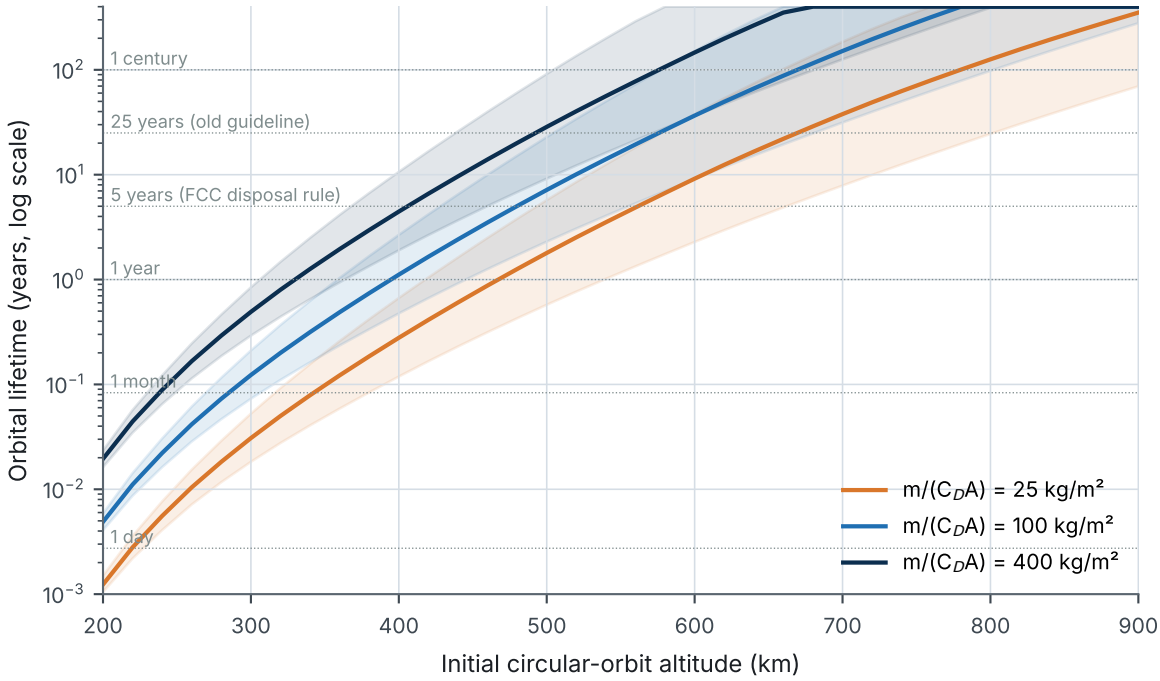


FIGURE 2.4 Approximate orbital lifetime versus initial altitude for three ballistic coefficients. A satellite at 400 km with m/(C_D A) = 100 kg/m² lasts about one to three years; at 600 km it lasts decades; at 800 km it lasts more than a century. The 5-year line marks the US FCC post-mission disposal rule (2022) for LEO satellites.

WORKED EXAMPLE — HOW FAST DOES THE ISS LOSE HEIGHT?

Take $a = 6\,778$ km (400 km altitude), $\rho = 3 \times 10^{-12}$ kg/m³ (moderate activity), mass 420 t, effective $C_D A \approx 2.2 \times 1\,900$ m² ≈ 4 200 m², so $B \approx 100$ kg/m².

$$\frac{da}{dt} = -\frac{3 \times 10^{-12} \times \sqrt{3.986 \times 10^{14} \times 6.778 \times 10^{6}}}{100} \approx -1.6 \times 10^{-3}\ \text{m/s}$$

That is about 140 m per day, or 4 km per month. The real ISS loses 1–5 km per month depending on solar activity and attitude, and is periodically re-boosted by the engines of Progress or Cygnus cargo vehicles. Without re-boosts it would re-enter within one to two years.

FAILURE LESSON — THE FEBRUARY 2022 STARLINK LOSS

On 3 February 2022, 49 Starlink satellites were released into a low 210 × 340 km parking orbit, a deliberately low orbit that ensures failed satellites re-enter quickly. A moderate geomagnetic storm the next day increased atmospheric density at that altitude by up to 50 %. The satellites, placed in a low-drag "knife-edge" attitude, could not climb out against the extra drag, and 38 re-entered. The lesson: in very low orbits, space-weather forecasts are as important as weather forecasts are for aircraft.

2.2.4 Very Low Earth Orbit (VLEO)

Orbits below about 350 km are attracting interest for high-resolution imaging (resolution improves in proportion to distance), lower latency and natural debris removal. Satellites such as ESA's GOCE, Japan's SLATS/Tsubame and a new generation of commercial VLEO imagers use streamlined shapes, electric propulsion and atmosphere-resistant materials. GOCE operated at 255 km from 2009 to 2013, using an ion engine to cancel drag. SLATS/Tsubame flew as low as 167 km (2017–2019). Some concepts even scoop up atmospheric gas to feed an electric thruster ("air-breathing electric propulsion").

2.3 Vacuum

At 400 km the pressure is about 10^{-6} Pa, and in GEO about 10^{-11} Pa — far better vacuum than most laboratory chambers. Vacuum causes several problems:

- **Outgassing.** Polymers, adhesives, paints and composites release trapped water and volatile compounds. The vapour can condense on cold optics, star trackers and solar cells, reducing their performance. Materials are screened with the ASTM E595 test: total mass loss (TML) below 1.0 % and collected volatile condensable material (CVCM) below 0.1 %. Sensitive instruments use "bake-out" before launch, contamination budgets and heated covers.
- **Cold welding and friction.** Clean metal surfaces in contact can bond (cold weld) in vacuum because there is no oxide layer or adsorbed gas to separate them. Mechanisms use dissimilar materials, dry lubricants (MoS_2, lead), or space-qualified oils with very low vapour pressure (perfluoropolyethers, multiply-alkylated cyclopentanes).
- **Sublimation.** Some materials (cadmium, zinc, uncoated magnesium) sublimate at elevated temperatures; tin whiskers grow on pure-tin finishes. Space-grade parts forbid pure tin, cadmium and zinc plating.
- **Electrical discharge.** Between about 1 and 1 000 Pa — the pressure inside a spacecraft just after launch — the breakdown voltage of a gap reaches a minimum (Paschen's law). High-voltage equipment must be vented or switched off during ascent. In RF equipment, high-power signals can cause **multipaction**, a resonant electron avalanche between surfaces in vacuum.
- **No convection.** Heat moves only by conduction and radiation, so electronics cannot be cooled by fans and air (except inside pressurised modules).

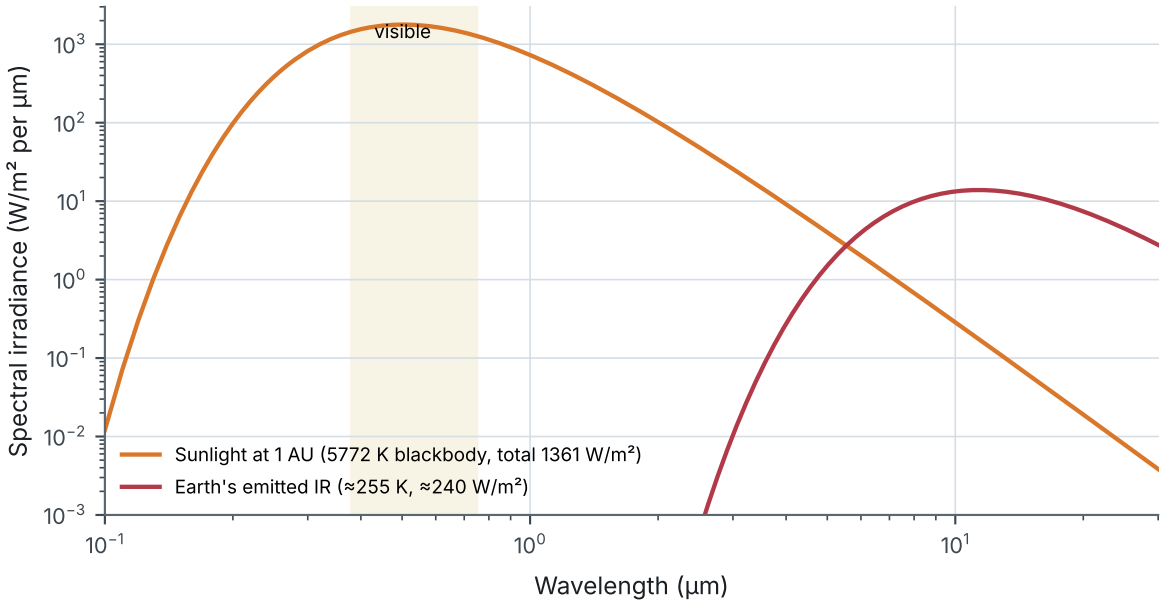
2.4 The Thermal Environment

2.4.1 Heat Sources

A spacecraft in Earth orbit receives heat from three external sources and loses heat only by radiation (Figure 2.5):

1. **Direct sunlight.** The **solar constant** at 1 AU is about $S = 1\,361$ W/m² (it varies ±3.4 % between January perihelion and July aphelion, and only ±0.1 % over the solar cycle). The spectrum peaks in the visible.
2. **Albedo.** The Earth reflects on average about 30 % of incident sunlight (0.25–0.40 depending on clouds, ice and ocean).
3. **Earth infrared.** The Earth radiates about 240 W/m² at its effective temperature of 255 K, with a peak wavelength near 10 µm.

In addition, internal electronics dissipate heat. Deep space appears as a sink at 2.7 K.



Spacecraft thermal design separates the 'solar' band (0.2–3 µm, absorptance α) from the 'thermal IR' band (3–50 µm, emittance ε).

FIGURE 2.5 Spectra of sunlight and of Earth's thermal emission. Because the two hardly overlap, a surface can be designed to absorb little sunlight (low α) while radiating strongly in the infrared (high ε) — the principle behind white paints and mirrors used on radiators.

2.4.2 Radiative Equilibrium

The temperature of a surface is set by the balance between absorbed and emitted power. For a flat plate facing the Sun with an insulated back:

$$\alpha S A = \varepsilon \sigma A T^4 \quad \Rightarrow \quad T = \left(\frac{\alpha}{\varepsilon} \frac{S}{\sigma} \right)^{1/4}$$

where α is the **solar absorptance**, ε the **infrared emittance** and $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} the Stefan–Boltzmann constant. The ratio α/ε controls temperature. For an isolated sphere (absorbing on its cross-section πr^2, radiating from its surface $4\pi r^2$) the result is $T = (\alpha S / 4\varepsilon\sigma)^{1/4}$. Figure 2.6 shows how surface coatings span a range of more than 300 °C in equilibrium temperature.

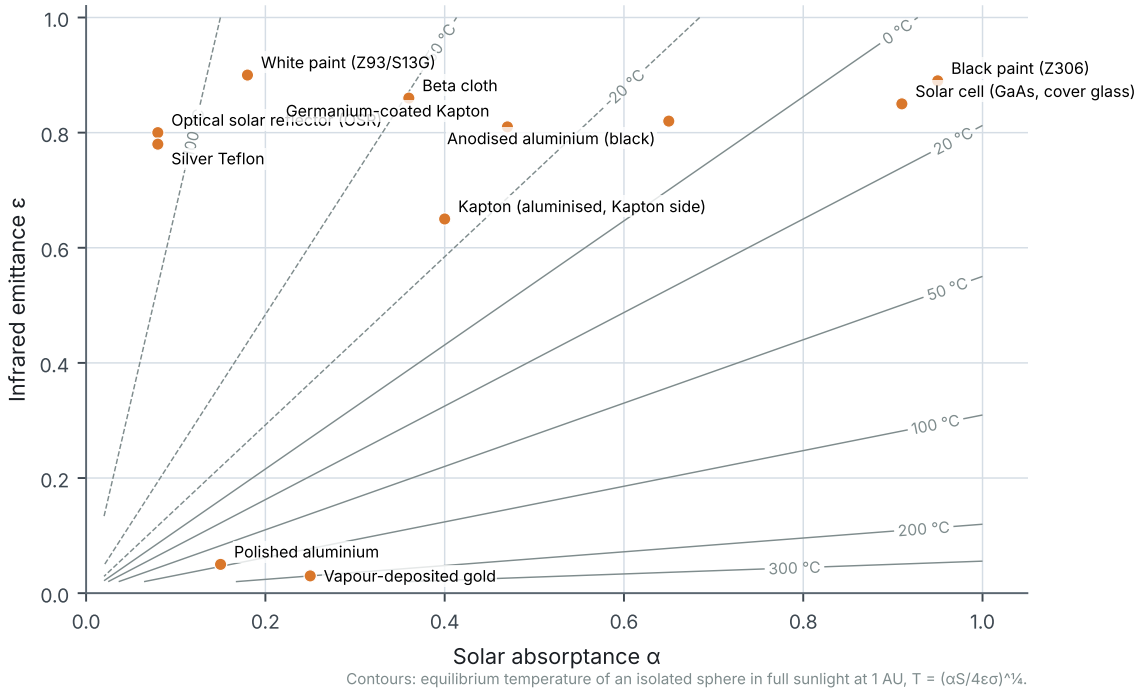


FIGURE 2.6 Absorptance–emittance map of common spacecraft surface finishes with lines of equal equilibrium temperature. Radiators use low-α/high-ε finishes (white paint, optical solar reflectors); blankets use low-ε outer layers to isolate; black paint maximises heat exchange inside instruments.

WORKED EXAMPLE — TEMPERATURE OF A CUBESAT IN SUNLIGHT AND ECLIPSE

A 1U CubeSat (10 cm cube) is covered in solar cells (α = 0.91, ε = 0.85). In a 500 km orbit it absorbs sunlight on (on average) about 1.5 faces, plus albedo and Earth IR on the Earth-facing side. A simple estimate of absorbed power in sunlight is

$$P_{in} \approx \alpha S A_{sun} + \alpha a S F A_{earth} + \varepsilon q_{IR} F A_{earth} + P_{int}$$
$$\approx 0.91 \times 1361 \times 0.015 + 0.91 \times 0.3 \times 1361 \times 0.5 \times 0.01 + 0.85 \times 240 \times 0.5 \times 0.01 + 1 \approx 18.6 + 1.9 + 1.0 + 1 \approx 22.5$$ W

The satellite radiates from 0.06 m², so $T = \left(22.5/(0.85 \times 5.67 \times 10^{-8} \times 0.06)\right)^{1/4} \approx 290$ K (17 °C). In eclipse (35 minutes of a 95-minute orbit) only Earth IR and internal power remain (≈2 W), and the equilibrium would be about 150 K. Thermal mass slows the cooling, so the real minimum is typically −10 to −30 °C. Batteries (operating range typically 0–40 °C for charging) usually need heaters or careful placement.

2.4.3 Eclipses and Thermal Cycling

A LEO satellite passes through Earth's shadow on most orbits — up to about 36 minutes of a 92-minute orbit at 400 km. It therefore experiences about 15–16 thermal cycles per day, or more than 50 000 cycles in ten years. These cycles fatigue solder joints, composite bonds and solar-array interconnects. Sun-synchronous dawn–dusk orbits avoid most eclipses. GEO satellites experience eclipses only during two **eclipse seasons** of about 45 days around the equinoxes, with a maximum eclipse of 72 minutes per day.

The thermal control techniques — insulation blankets, radiators, heat pipes, louvres and heaters — are described in Chapter 5.

2.5 Atomic Oxygen

Between about 200 and 700 km, atomic oxygen (AO) is the most abundant gas. A satellite moving at 7.7 km/s hits AO atoms with an energy of about 5 eV, enough to break chemical bonds. AO **erodes** organic materials such as Kapton, Mylar and carbon-fibre epoxy, oxidises silver, and degrades some paints. Erosion is characterised by the **erosion yield** — the volume of material removed per incident atom:

$$\Delta x = E_y \, \Phi$$

where Φ is the fluence (atoms/cm^2). Kapton H has $E_y \approx 3.0 \times 10^{-24}$ cm^3/atom. At 400 km during solar maximum, the ram-facing fluence can exceed 10^{22} atoms/cm^2 per year, removing about 0.3 mm of Kapton per year — enough to destroy an unprotected 25 µm film in weeks. Protective measures include silicon-dioxide or aluminium-oxide coatings, AO-resistant polymers (e.g., polysiloxane-based coatings, "CP1"), silicone-based paints and metal foils. NASA's Materials International Space Station Experiment (MISSE) series has exposed thousands of material samples on the ISS exterior since 2001 to measure these effects.

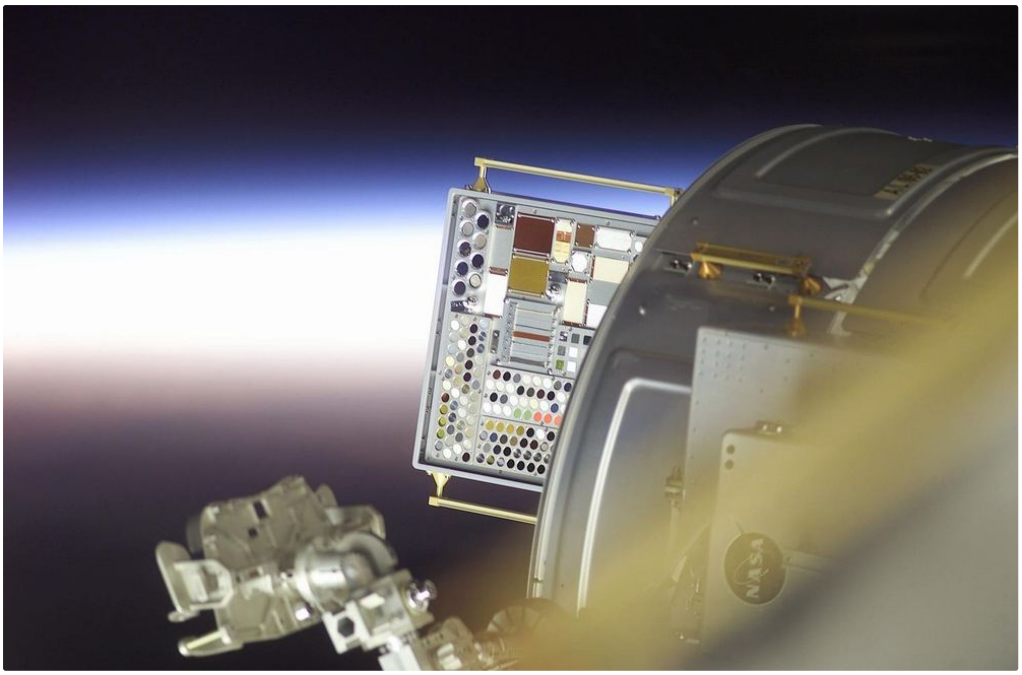


FIGURE 2.7 The first Materials International Space Station Experiment (MISSE-1) mounted on the outside of the ISS airlock in 2001, backdropped by an orbital sunrise. Each suitcase-like carrier exposed hundreds of material samples to atomic oxygen, ultraviolet light, radiation and thermal cycling for about 18 months before being returned to Earth for analysis. Image: NASA/MSFC.

AO also produces a faint orange **"spacecraft glow"** on ram-facing surfaces and can affect the calibration of sensitive optical instruments.

2.6 Plasma and Spacecraft Charging

Space is filled with **plasma** — ionised gas containing free electrons and ions. In LEO the plasma is cold (≈0.1 eV) and dense (10^{10}–10^{12} m^{-3}) — this is the ionosphere (Chapter 23). In GEO it is hot (keV energies) and tenuous. A spacecraft floating in plasma collects charge until the currents of electrons and ions to its surface balance, which sets its **floating potential**.

Surface charging is most severe in GEO and in high-inclination LEO orbits during geomagnetic substorms, when hot electrons (1–50 keV) are injected from the magnetotail. A sunlit surface emits photoelectrons and stays near zero potential, but shadowed dielectric surfaces may charge to thousands of volts negative. If neighbouring surfaces reach different potentials, an **electrostatic discharge (ESD)** can occur, injecting current pulses into cables, causing anomalies ("phantom commands"), or damaging solar arrays. Mitigations include conductive coatings (indium-tin oxide, Germanium-coated Kapton, carbon-loaded black Kapton), grounding all conductive parts, and limiting the voltage between adjacent solar cells.

Internal (deep-dielectric) charging occurs when energetic electrons (above about 0.5 MeV) penetrate shielding and accumulate inside cable insulation and circuit boards, eventually discharging. It is the dominant anomaly mechanism in GEO and MEO during periods of high outer-belt electron flux — for example after high-speed solar-wind streams. Guidelines (NASA-HDBK-4002B, ECSS-E-ST-20-06C) limit floating metal and require bleed paths for charge.

High-voltage solar arrays in LEO (above about 100 V) can arc into the surrounding plasma. The ISS uses 160 V arrays and a **plasma contactor unit** that emits xenon plasma to keep the structure near plasma potential, protecting astronauts during spacewalks.

2.7 Ionising Radiation

2.7.1 Sources

Three populations of energetic particles reach spacecraft:

1. **Trapped radiation (Van Allen belts).** Earth's magnetic field traps electrons and protons in two toroidal belts (Figure 2.8). The **inner belt** (1.2–2.5 Earth radii, R_E) contains protons of tens to hundreds of MeV, produced mainly by cosmic-ray neutron decay; it is very stable. The **outer belt** (3–7 R_E) contains electrons of 0.1–10 MeV; it varies dramatically with solar-wind conditions. Between them lies the **slot region**. GNSS satellites (≈4.2 R_E) fly in the heart of the outer belt; GEO satellites (6.6 R_E) on its outer edge.
2. **Solar particle events (SPEs).** Large solar flares and coronal mass ejections accelerate protons (and heavier ions) to tens or hundreds of MeV. Events last hours to days and can increase radiation doses by orders of magnitude. They are the dominant hazard for astronauts beyond LEO (Chapter 30). The largest events of the space age include August 1972 (between Apollo 16 and 17) and October 1989.
3. **Galactic cosmic rays (GCRs).** Fully ionised nuclei (87 % protons, 12 % helium, 1 % heavier ions up to iron and beyond) arriving from outside the Solar System with energies up to 10^{20} eV. Their flux is low but they are nearly impossible to shield, and heavy ions cause single-event effects in electronics. GCR flux is about 2–3 times higher at solar minimum, when the weaker solar wind lets more of them in.

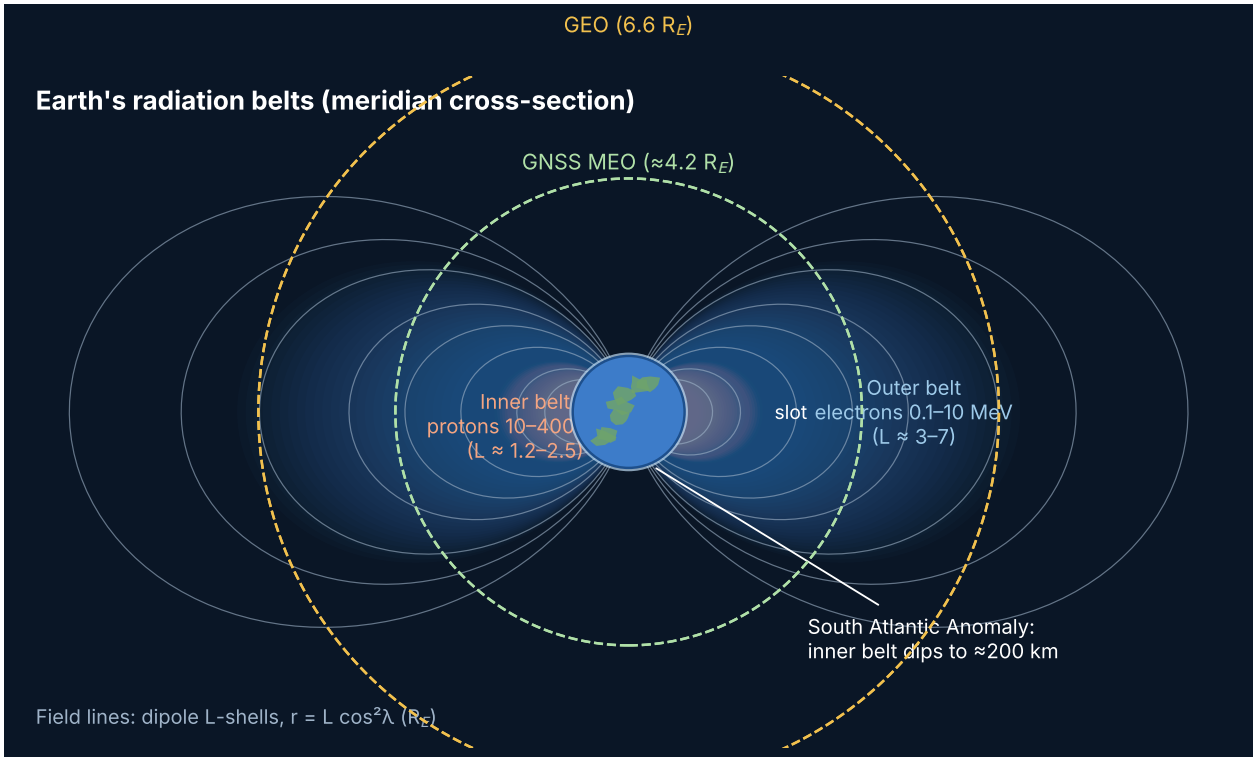


FIGURE 2.8 Cross-section of Earth's magnetic field and radiation belts. Low Earth orbit lies below the belts except over the South Atlantic Anomaly; navigation satellites fly through the outer electron belt; geostationary satellites sit at its outer edge.

Because Earth's magnetic axis is offset and tilted relative to its rotation axis, the inner belt comes closest to the surface over South America and the South Atlantic — the **South Atlantic Anomaly (SAA)**. LEO satellites, including the ISS and Hubble, receive most of their trapped-proton dose during SAA passes, and many

instruments switch off or flag their data there (Figure 2.9). In polar orbits, satellites also cross the "horns" of the outer belt at high latitudes and are exposed to solar particles, which reach low altitudes near the magnetic poles.

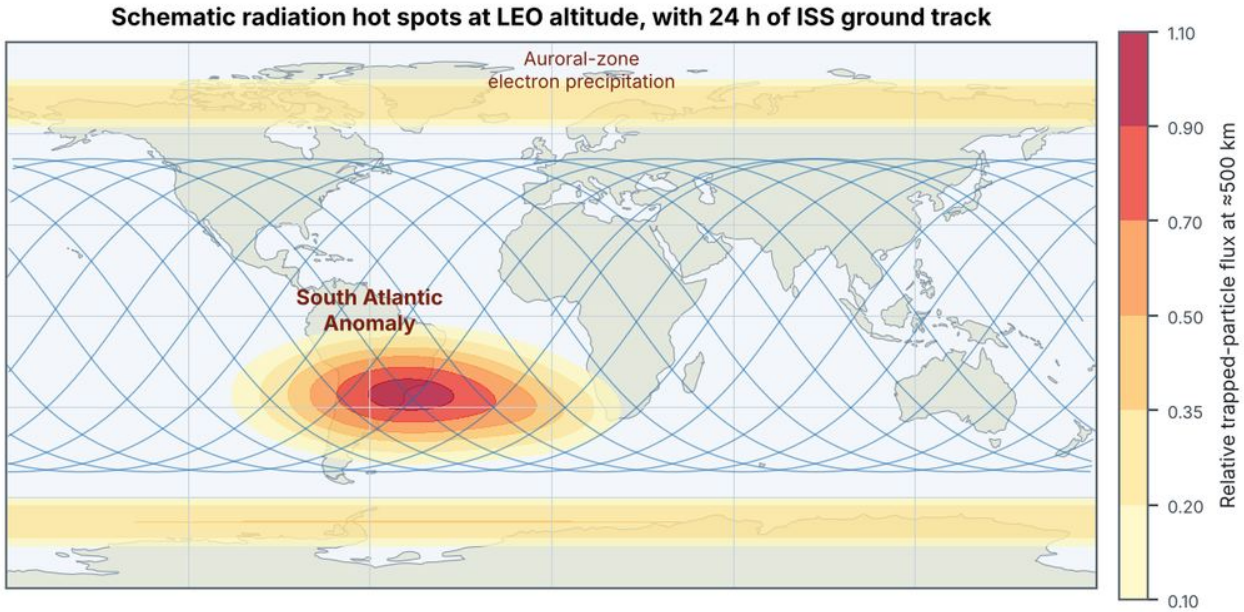


FIGURE 2.9 Schematic map of the regions where LEO satellites receive most of their radiation: the South Atlantic Anomaly and the high-latitude auroral zones, with one day of International Space Station ground track overlaid.

2.7.2 Radiation Effects on Electronics and Materials

Radiation affects spacecraft in three main ways:

- **Total ionising dose (TID).** Energy deposited by ionisation accumulates over the mission, trapping charge in oxides of transistors and shifting their threshold voltages and leakage currents until circuits fail. Dose is measured in gray (Gy = 1 J/kg) or, traditionally, **rad** (1 rad = 0.01 Gy), quoted for silicon as krad(Si).
- **Displacement damage dose (DDD).** Protons and neutrons knock atoms out of the crystal lattice, degrading solar cells, optocouplers, CCD and CMOS image sensors (more "hot pixels", worse charge-transfer efficiency) and bipolar transistors.
- **Single-event effects (SEEs).** A single heavy ion or energetic proton deposits a dense track of charge. This can flip a memory bit (**single-event upset, SEU**) or disturb an analogue output (**single-event transient, SET**). It can also latch a CMOS circuit into a high-current state that can burn it out (**single-event latch-up, SEL**), or destroy power MOSFETs (**single-event burnout, SEB; single-event gate rupture, SEGR**).

Table 2.1 summarises effects and mitigations.

TABLE 2.1 Radiation effects, typical causes and mitigations.

Effect	Main cause	Typical consequence	Mitigations
Total ionising dose (TID)	Trapped electrons and protons, SPE protons	Parameter drift, increased power, eventual failure	Shielding, rad-hard parts, derating, spot shielding, part screening (lot acceptance testing)
Displacement damage (DDD)	Protons, neutrons	Solar array power loss, detector dark current, CCD charge transfer loss	Cover glass, annealing, operating detectors cold, design margins (typically 15–30 % end-of-life array margin)
Single-event upset (SEU)	Heavy ions (GCR, SPE), protons in SAA	Bit flips in memories and registers	Error detection and correction (EDAC), triple modular redundancy (TMR), memory scrubbing, watchdogs

Effect	Main cause	Typical consequence	Mitigations
Single-event latch-up (SEL)	Heavy ions, protons	High current; destructive if not cleared	Latch-up-immune technologies (SOI), current limiters with automatic power cycling
Single-event burnout / gate rupture	Heavy ions	Permanent destruction of power devices	Voltage derating (e.g., operate at < 50–75 % of rated V_DS), rad-hard MOSFETs
Single-event functional interrupt (SEFI)	Heavy ions	Processor or FPGA hangs	Watchdog timers, reset logic, configuration scrubbing

Electronic parts are classified by their tolerance. Commercial off-the-shelf (COTS) parts often survive 5–30 krad(Si) and can be susceptible to latch-up. By contrast, **radiation-tolerant** parts are characterised to 30–100 krad, while **radiation-hardened** parts are designed for 100 krad to more than 1 Mrad and are latch-up immune. Rad-hard parts are expensive and lag commercial technology by one to two decades in performance — a major reason why AI processors in space are a challenge (Chapter 45).

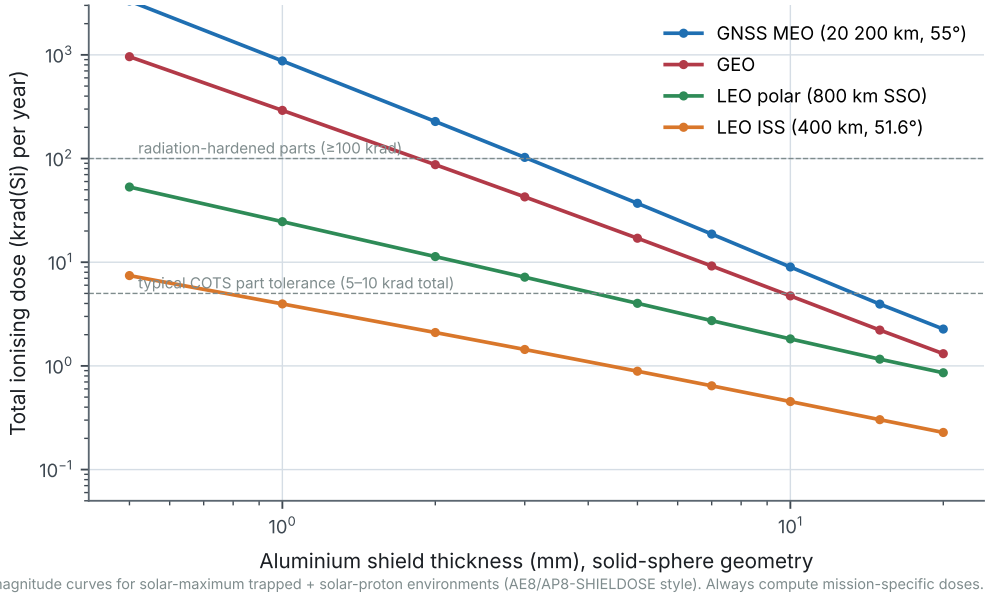


FIGURE 2.10 Illustrative annual total ionising dose behind spherical aluminium shielding for four orbits. MEO is the harshest common orbit; LEO at ISS altitude is comparatively benign. Shielding is effective against electrons (MEO, GEO) but less effective against high-energy protons.

2.7.3 Estimating Upset Rates

The susceptibility of a device to single-event effects is measured in particle accelerators. The result is a curve of **cross-section** σ (the effective sensitive area per bit or per device) versus **linear energy transfer (LET)**, the energy deposited per unit path length, usually given in MeV·cm²/mg (Figure 2.11). A device with a threshold LET above about 37 MeV·cm²/mg is considered effectively immune to heavy-ion upsets in most environments, because heavier, more ionising particles are very rare. The upset rate is found by folding the cross-section curve with the LET spectrum of the orbit (tools: CREME96, OMERE, SPENVIS).

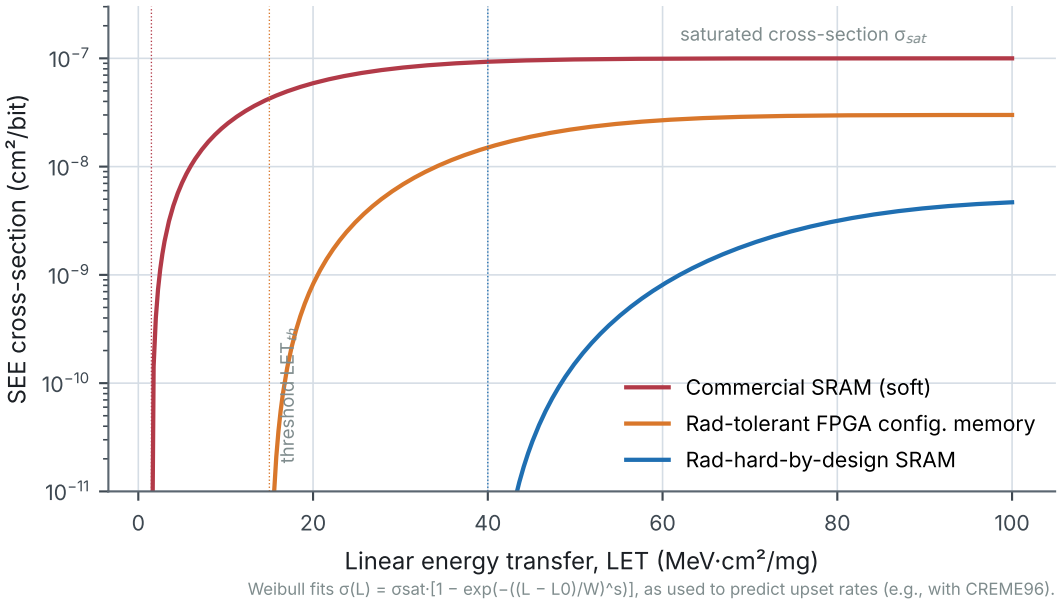


FIGURE 2.11 Single-event upset cross-section versus LET for three classes of memory, fitted with Weibull functions. Hardening raises the threshold LET and lowers the saturated cross-section, reducing upset rates by orders of magnitude.

ENGINEERING NOTE — RADIATION DESIGN MARGIN

Agencies require a **radiation design margin (RDM)** — the ratio of a part's tested capability to the predicted mission dose at its location. NASA and ESA practice is typically RDM ≥ 2 for TID (with lot-specific test data), rising for parts with large lot-to-lot variability. Dose predictions use standard models: AE8/AP8 or the newer **AE9/AP9-IRENE** for trapped particles, ESP/PSYCHIC or SAPPHIRE for solar protons, and ISO 15390 or Badhwar–O'Neill for cosmic rays. These are combined with ray-tracing through a 3-D model of the spacecraft (sector analysis) to find the shielding at each part.

2.8 Micrometeoroids and Orbital Debris

2.8.1 The Particle Population

Two kinds of small particles strike spacecraft at hypervelocity:

- **Micrometeoroids** — natural dust from comets and asteroids. They arrive from all directions at 11–72 km/s (average about 20 km/s), with annual showers such as the Perseids and Leonids adding short peaks. About 30–40 tonnes of meteoroid material enter Earth's atmosphere every day.
- **Orbital debris** — artificial objects: fragments from explosions and collisions, paint flakes, solid-rocket-motor slag and aluminium-oxide dust, and dead satellites. In LEO, debris impacts occur at an average of about 10 km/s, mainly from directions near the velocity vector.

For particles smaller than about 0.1 mm, meteoroids dominate; for larger particles in LEO, debris dominates. Objects larger than about 10 cm in LEO (1 m in GEO) are tracked and catalogued (Chapter 21). The roughly one million objects between 1 and 10 cm are not tracked and cannot be avoided. Yet they carry enough energy to destroy a satellite. A 1 cm aluminium sphere at 10 km/s has a kinetic energy of about 70 kJ — similar to a small car at 50 km/h.

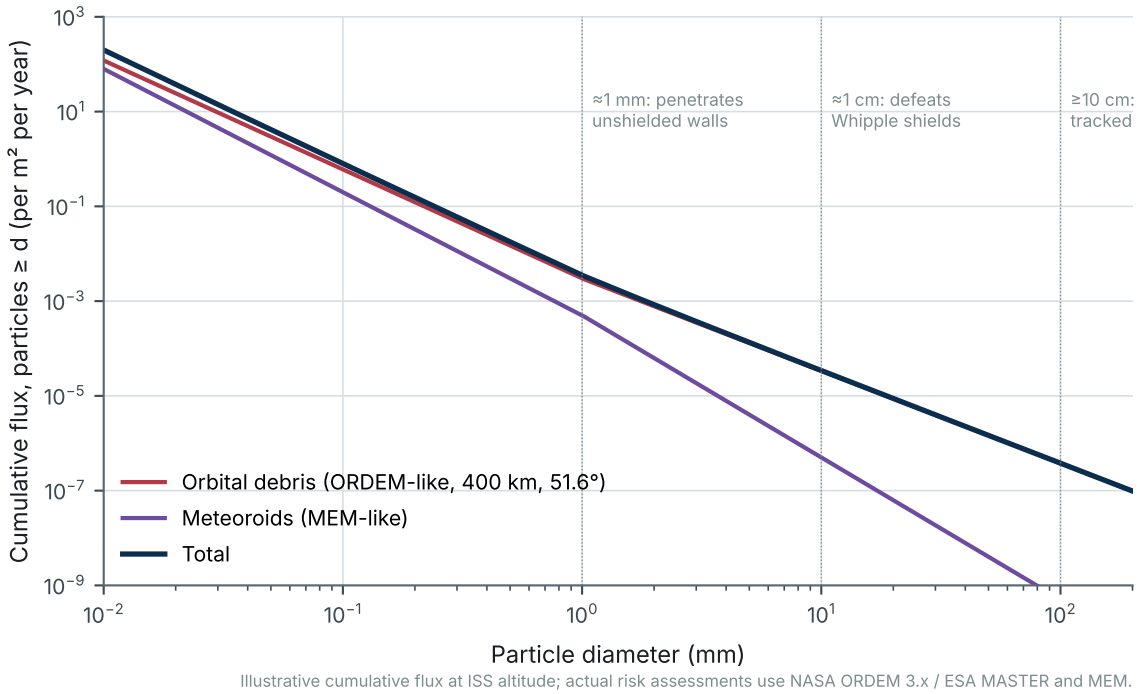


FIGURE 2.12 Illustrative cumulative flux of meteoroids and orbital debris at ISS altitude. The steep size dependence means that small impacts are frequent but survivable, while the rare large impacts are catastrophic.

2.8.2 Shielding

The standard protection is the **Whipple shield**, invented by astronomer Fred Whipple in 1947 (Figure 2.13). A thin sacrificial **bumper** placed some distance in front of the main wall shatters the incoming particle into a cloud of fragments and vapour, which spreads out and is stopped by the rear wall. A Whipple shield is far more mass-efficient than a single thick wall. Variants include the **stuffed Whipple shield** (with layers of Nextel ceramic fabric and Kevlar between bumper and wall, used on the ISS) and **multi-shock shields**. The ISS has more than 100 types of shield and is the most heavily protected spacecraft ever flown; the Bigelow-built BEAM and future inflatable habitats use multiple layers of fabric.

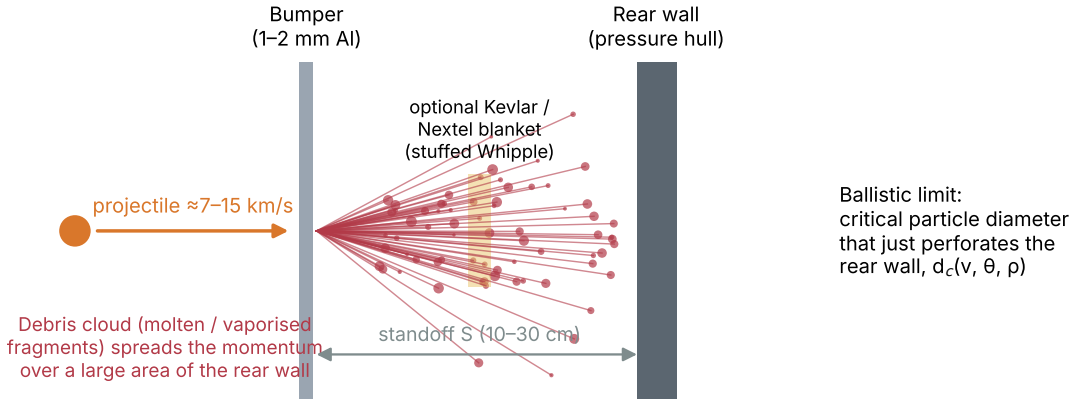


FIGURE 2.13 Principle of the Whipple shield. The bumper breaks up and partly vaporises the projectile; the resulting cloud spreads its momentum over a large area of the rear wall.



FIGURE 2.14 The Long Duration Exposure Facility (LDEF), a 9.7-tonne bus-sized cylinder carrying 57 experiments, being retrieved by Space Shuttle Columbia in January 1990 after almost six years in orbit. LDEF's surfaces recorded more than 30 000 impact craters visible to the naked eye and remain a cornerstone data set for meteoroid, debris and atomic-oxygen models. Image: NASA/JSC.

Shield performance is described by **ballistic limit equations** (BLEs). These are empirical fits to hypervelocity gun tests (light-gas guns reach about 7 km/s) and hydrocode simulations. They give the critical particle diameter d_c that just perforates the rear wall, as a function of velocity, impact angle, particle density and shield geometry. Combining the BLE with the flux model gives the **probability of no penetration (PNP)** over the mission, typically required to be above 0.95–0.99 for critical elements. Tools include NASA's BUMPER and ESA's ESABASE2/Debris.

> **DATA BOX — MMOD ON THE ISS AND HUBBLE**
>
> - The ISS has been struck by many thousands of small particles; impact pits are regularly found on windows (the Cupola windows have external shutters for protection), radiators and handrails.
> - In 2021 an untracked object punched a 5 mm hole through the thermal blanket of the Canadarm2 boom — the arm continued to work.
> - Solar arrays returned from Hubble servicing missions (1993, 2002) showed thousands of impact features, used to calibrate debris models.
> - The ISS performs a debris-avoidance manoeuvre when a tracked object's collision probability exceeds about 1 in 10 000 (or about 1 in 100 000 if the manoeuvre is easy). It has manoeuvred more than 40 times since 1999.

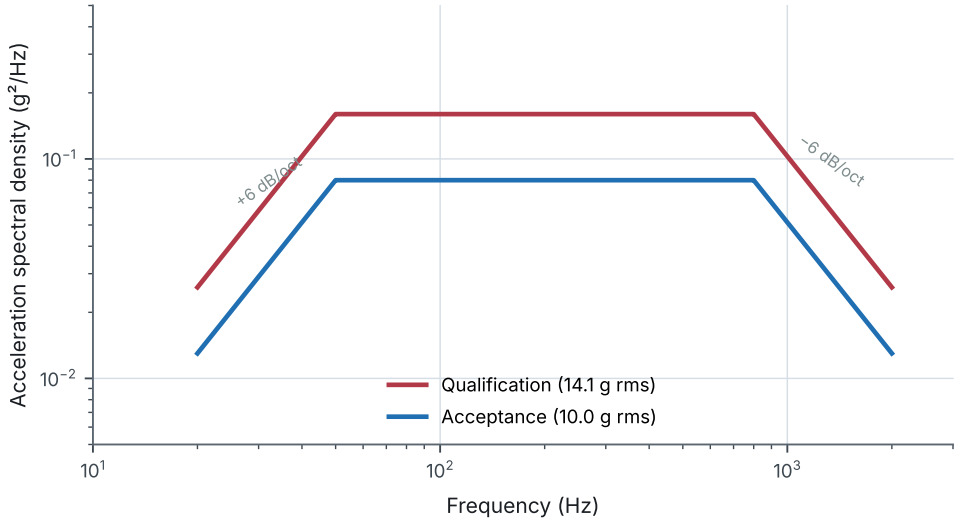
2.9 Mechanical Environments: Launch and Microgravity

2.9.1 Launch Loads

The most mechanically severe minutes of a spacecraft's life are during launch. Loads arise from:

- **Quasi-static acceleration.** Steady thrust produces axial accelerations of 3–6 g at burnout (Falcon 9 ≈ 4–5 g, Ariane 6 ≈ 4.5 g, Soyuz ≈ 4.3 g); low-frequency dynamic transients add lateral accelerations of 1–2 g.
- **Sine vibration.** Low-frequency (5–100 Hz) oscillations from engine ignition, pogo, wind gusts and staging. Spacecraft must have their primary structural natural frequencies above values specified by the launcher (typically > 20–25 Hz axial and > 10–15 Hz lateral) to avoid dynamic coupling.

- **Random vibration.** Broadband vibration (20–2 000 Hz) transmitted through the structure from engines and aerodynamic turbulence. Components are tested to spectra such as NASA's General Environmental Verification Standard (GEVS) shown in Figure 2.15.
- **Acoustic noise.** Sound pressure levels inside the fairing reach 135–145 dB overall at lift-off and in the transonic region, exciting light, large-area structures such as solar panels and antennas.
- **Shock.** Pyrotechnic separation events (fairing jettison, stage and spacecraft separation) produce short, high-frequency accelerations of thousands of g, characterised by a **shock response spectrum (SRS)**. Shock attenuates rapidly with distance, so sensitive units are placed away from separation interfaces.



Generalized random-vibration test levels for components ≤ 22.7 kg (NASA GSFC-STD-7000, GEVS). Duration 2 min/axis (qual), 1 min/axis (acceptance).

FIGURE 2.15 Random-vibration qualification and acceptance spectra from NASA's GEVS for small components. The plateau from 50 to 800 Hz at 0.16 g²/Hz corresponds to an overall level of 14.1 g rms.

2.9.2 Microgravity

Once in orbit, the spacecraft and everything in it are in free fall. Residual accelerations of 10^{-6} to 10^{-3} g arise from drag, gravity gradient, rotation, crew motion and machinery. Microgravity matters for fluid management, because propellant does not settle at the bottom of a tank (see Chapter 6). It matters for heat transfer: there is no buoyant convection, so boiling and combustion behave differently. It also matters for deployable structures, which must be tested on the ground with gravity-offloading rigs, and for the human body (Chapter 30). It is also a resource: in-space manufacturing (Chapter 40) exploits it to make materials impossible to make on Earth.

2.10 Environments Beyond Earth Orbit

Deep-space missions face additional environments. Table 2.2 and Figure 2.16 summarise some of the most demanding.

TABLE 2.2 Representative environmental challenges for planetary missions.

Destination	Key environmental challenges	Engineering responses (examples)
Moon	Surface −170 °C to +120 °C at the equator; 14-day nights; permanently shadowed craters near 25–40 K; abrasive, electrostatically charged dust; no magnetic field or atmosphere	Radioisotope heater units, lunar-night survival modes, dust-tolerant seals and connectors, radiation shielding for crews
Mars	Thin CO_2 atmosphere (≈0.6 % of Earth's pressure); dust storms reducing sunlight; −125 °C to +20 °C; perchlorate-laden dust; no global magnetic field	Entry heat shields and parachutes (Chapter 27), RTG or dust-tolerant solar power, heaters, dust-clearing strategies
Venus	Surface 465 °C and 92 bar, sulfuric-acid clouds; super-rotating winds	Pressure vessels and insulation (Venera landers survived up to 127 minutes); silicon-carbide electronics research

Destination	Key environmental challenges	Engineering responses (examples)
Mercury / near-Sun	Solar flux up to 11 times Earth's (Mercury) or ≈500 times (Parker Solar Probe at closest approach); intense heating from planet's surface	Heat shields (Parker's 11.4 cm carbon-composite shield), high-temperature solar arrays, sunshades, special orbit design
Jupiter system	The most intense radiation belts in the Solar System — Europa Clipper receives ≈2.8 Mrad behind 0.1 inch of aluminium	Radiation vaults (Juno's 1 cm titanium vault, Clipper's aluminium-zinc vault), orbits that minimise belt passages
Outer Solar System	Sunlight 1/100 to 1/1000 of Earth's; temperatures 40–100 K; light-hour communication delays	RTGs, low-temperature materials, high-autonomy operations
Interplanetary cruise	Solar particle events, GCRs, micrometeoroid streams near comets	Shielding, fault protection, safe modes, trajectory selection

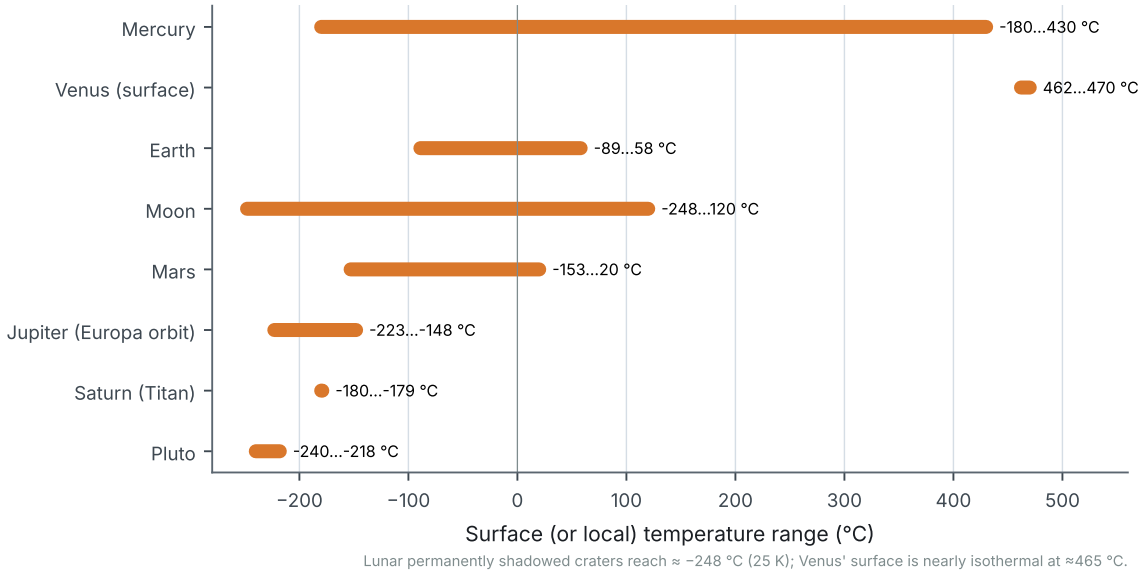


FIGURE 2.16 Temperature ranges at the surfaces (or local environments) of selected Solar System bodies. Thermal design must cover both extremes — and, for landers, the transition from cruise vacuum to surface conditions.

2.11 Designing for the Environment: Standards and Tools

Engineers capture the environment in a project **environmental specification**. It is derived from standards such as **ECSS-E-ST-10-04C** (space environment), **NASA-STD-7009/NASA-HDBK-1001** (terrestrial and natural space environments) and **ISO 15390** (cosmic rays). Other sources include the **AE9/AP9** models, **ORDEM 3.2** and **MASTER-8** (debris), **MEM 3** (meteoroids) and **NRLMSIS 2.0** (atmosphere). ESA's online **SPENVIS** system and NASA's **Space Environments and Effects (SEE)** program provide integrated tools.

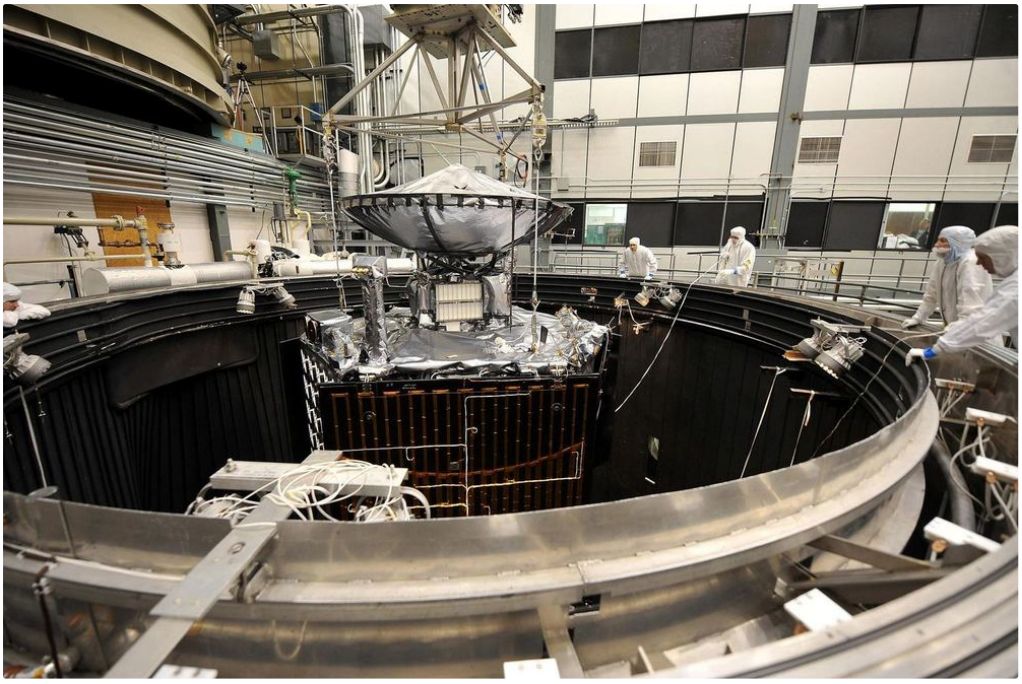


FIGURE 2.17 NASA's Juno spacecraft emerging from thermal-vacuum testing at Lockheed Martin in 2011. In the chamber the spacecraft was cycled between the hot and cold extremes predicted for its mission while operating, verifying both the thermal design and the electronics. Image: NASA/JPL (NASA/JPL-Caltech/Lockheed Martin).

The environmental specification feeds the verification programme: thermal-vacuum and thermal-balance tests, vibration, acoustic and shock tests, EMC tests, radiation tests of parts, and material screening (Chapter 4). The general principle is **"test as you fly, fly as you test"**: every environment the spacecraft will meet in flight should, as far as practical, be reproduced or bounded on the ground.

KEY TAKEAWAYS

- Atmospheric density at a given LEO altitude varies by ~10× over the solar cycle; drag sets satellite lifetime (about 1–3 years at 400 km, decades at 600 km, centuries at 800 km).
- Vacuum causes outgassing, cold welding and discharges; the thermal environment is governed by radiation balance and the α/ε ratio of surfaces.
- Atomic oxygen erodes polymers in LEO; plasma causes surface and internal charging, most severely in GEO and MEO.
- Radiation causes total dose, displacement damage and single-event effects; MEO is the harshest common orbit, and Jupiter the harshest planetary environment.
- Micrometeoroids and debris strike at 10–20 km/s; Whipple shields defeat particles up to about 1 cm, and larger objects must be tracked and avoided.
- Launch imposes quasi-static, sine, random, acoustic and shock loads that often size the structure; environmental testing reproduces these on the ground.

REVIEW QUESTIONS

1. Why does atmospheric density at 400 km change by about a factor of ten over a solar cycle, and what does this mean for satellite lifetime?
2. With a scale height of 60 km, by what factor does density fall between 400 km and 450 km altitude?
3. A plate at 1 au faces the Sun with $\alpha = 0.2$ and $\varepsilon = 0.8$ on its sunlit face and loses heat only from that face. Estimate its equilibrium temperature.
4. What is atomic oxygen, at which altitudes does it matter most, and how are exposed materials protected?
5. Distinguish surface charging from internal (deep dielectric) charging. In which orbits is each most severe?
6. Explain the differences between total ionising dose, displacement damage and single-event effects, and give one mitigation for each.
7. How does a Whipple shield work, and why must objects larger than about 1 cm be tracked and avoided instead?

8. List the main mechanical loads during launch and explain why they, rather than the space environment, often size a spacecraft's structure.

CHAPTER

3 Orbital Mechanics and Astrodynamics

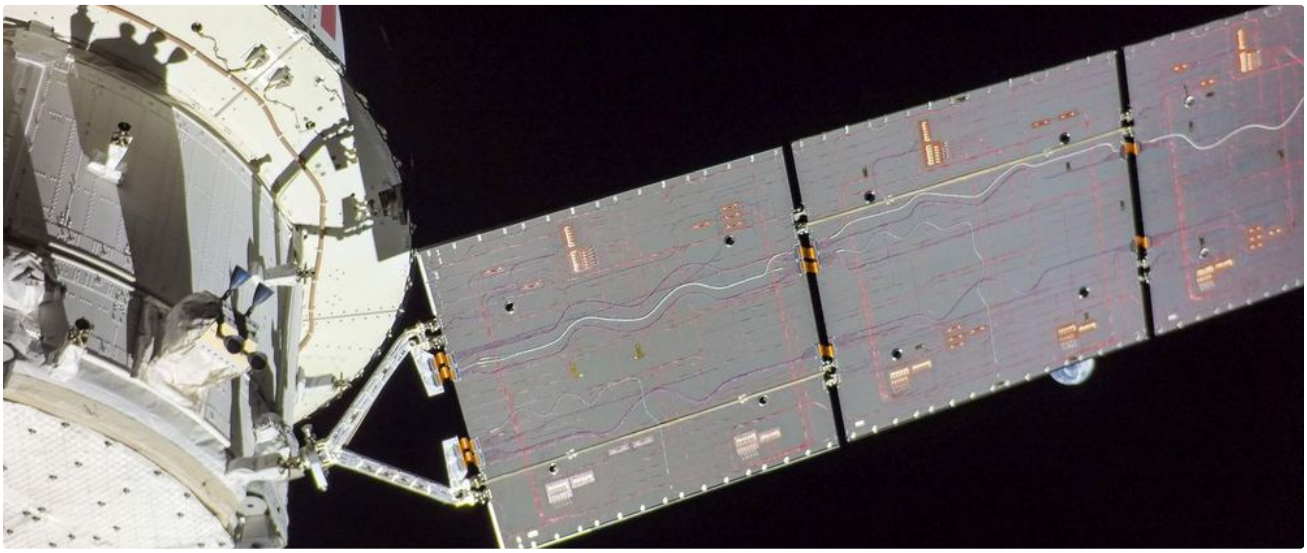


Image: NASA/JSC

> *Orbital mechanics explains how spacecraft move under gravity. It tells us why satellites stay up, how long an orbit takes, how much velocity change a mission needs, how orbits drift because of Earth's shape and the Sun and Moon, how to rendezvous with another spacecraft, and how to travel between planets. This chapter builds the subject from Newton's and Kepler's laws to the practical tools of mission design - Hohmann transfers, plane changes, ground tracks, Lambert's problem, gravity assists and the Lagrange points - with worked examples throughout.*

IN THIS CHAPTER

3.1 From Newton to Kepler
3.2 Describing an Orbit: Orbital Elements and Reference Frames
3.3 Where Is the Satellite? Kepler's Equation
3.4 Orbit Perturbations
3.5 Orbital Manoeuvres
3.6 Rendezvous and Relative Motion
3.7 Orbit Determination
3.8 Ground Tracks and Coverage
3.9 Interplanetary Trajectories
3.10 Three-Body Dynamics and the Lagrange Points
3.11 The Δv Budget

3.1 From Newton to Kepler

3.1.1 Gravity and the Two-Body Problem

Isaac Newton's law of universal gravitation states that two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between them:

$$F = \frac{GMm}{r^2}, \qquad G = 6.674 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$$

For a satellite of negligible mass orbiting a planet of mass M, it is convenient to use the **gravitational parameter** $\mu = GM$, which is known far more precisely than G or M separately. For Earth, $\mu_\oplus = 398\,600.4418$ km³/s²; for the Sun, $\mu_\odot = 1.327\,124 \times 10^{11}$ km³/s². The equation of motion of the satellite relative to the planet's centre is

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\,\mathbf{r}$$

This is the **two-body problem**. It has an exact solution: the satellite moves on a **conic section** — a circle, ellipse, parabola or hyperbola — with the planet at one focus (Figure 3.1). All the complexity of real orbits comes from small departures from this ideal, called **perturbations**.

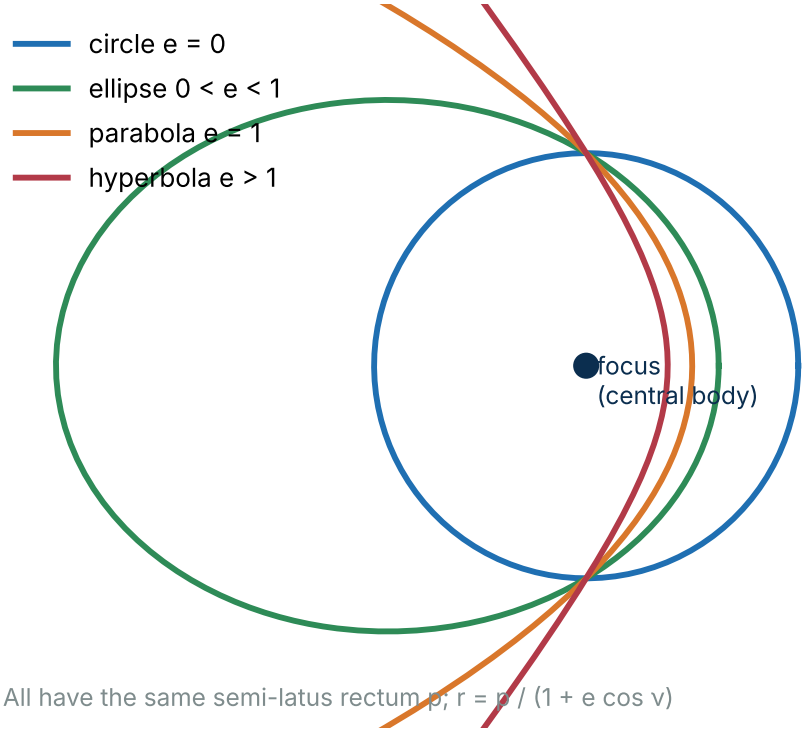


FIGURE 3.1 The four conic sections. Bound orbits are circles and ellipses; parabolas are the boundary case of escape; hyperbolas describe flybys and escape trajectories. All share the polar equation r = p/(1 + e cos ν) with the attracting body at a focus.

3.1.2 Kepler's Three Laws

Johannes Kepler deduced three laws from Tycho Brahe's observations of Mars, decades before Newton explained them:

1. **The orbit of a planet is an ellipse with the Sun at one focus.**
2. **A line joining the planet and the Sun sweeps out equal areas in equal times** (Figure 3.2). Physically, this is conservation of angular momentum: the satellite moves fastest at its closest point (**periapsis**) and slowest at its farthest point (**apoapsis**).
3. **The square of the orbital period is proportional to the cube of the semi-major axis:**

$$T = 2\pi\sqrt{\frac{a^3}{\mu}}$$

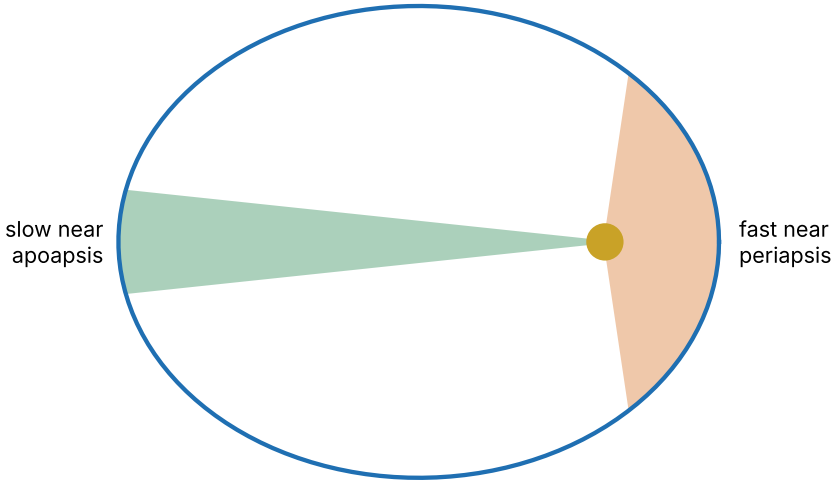


FIGURE 3.2 Kepler's second law: the two shaded sectors have equal area and are swept in equal times, so the satellite moves much faster near periapsis.

For Earth orbits, the words perigee and apogee are used; around the Sun, perihelion and aphelion; around the Moon, perilune (periselene) and apolune.

3.1.3 The Geometry of the Ellipse

An ellipse is described by its **semi-major axis** a (half the longest diameter) and **eccentricity** e (how elongated it is; 0 for a circle). The orbit equation in polar form is

$$r = \frac{a(1 - e^2)}{1 + e\cos\nu}$$

where ν is the **true anomaly**, the angle from periapsis to the satellite measured at the focus. The periapsis and apoapsis radii are $r_p = a(1 - e)$ and $r_a = a(1 + e)$ (Figure 3.3).

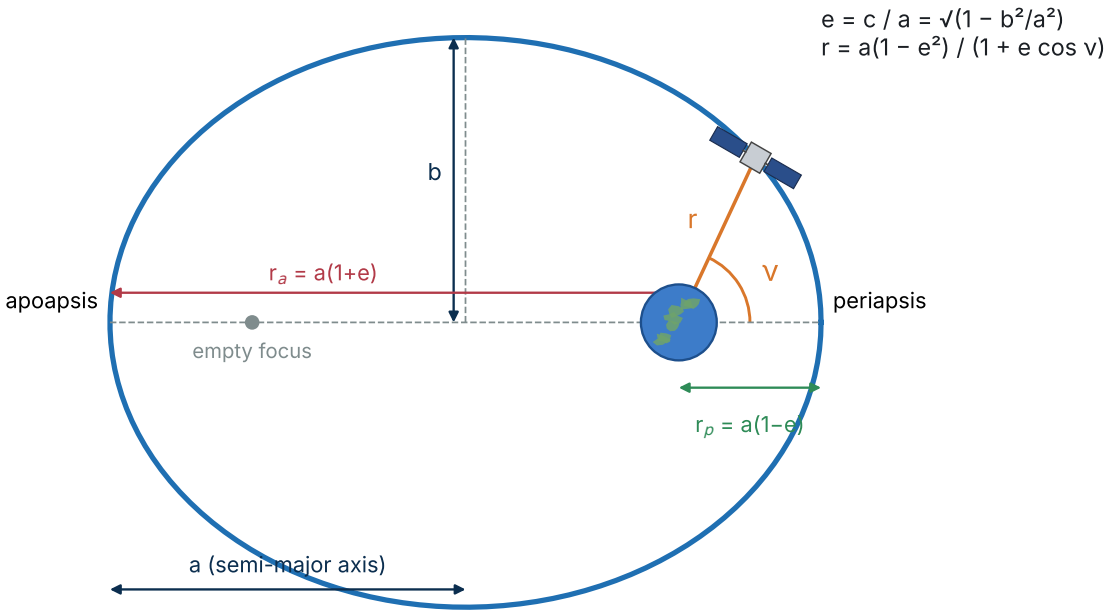


FIGURE 3.3 Geometry of an elliptical orbit. The central body sits at one focus; the other focus is empty. Periapsis and apoapsis lie at the ends of the major axis.

3.1.4 Energy and Speed: the Vis-Viva Equation

Two quantities are conserved in the two-body problem: the **specific mechanical energy** (energy per unit mass) and the **specific angular momentum**:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}, \qquad h = |\mathbf{r} \times \mathbf{v}| = \sqrt{\mu a (1 - e^2)}$$

Rearranging the energy equation gives the most useful single equation in orbital mechanics, the **vis-viva equation**:

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$$

It gives the speed anywhere on an orbit from the current radius and the semi-major axis alone. Special cases:

- **Circular orbit** ($r = a$): $v_c = \sqrt{\mu / r}$.
- **Escape** (parabola, $a \to \infty$): $v_{esc} = \sqrt{2\mu / r} = \sqrt{2}\, v_c$.
- **Hyperbolic excess speed** v_∞ — the speed left over far from the planet: $v^2 = v_{esc}^2 + v_\infty^2$. Mission designers quote launch energy as $C_3 = v_\infty^2$ (km²/s²).

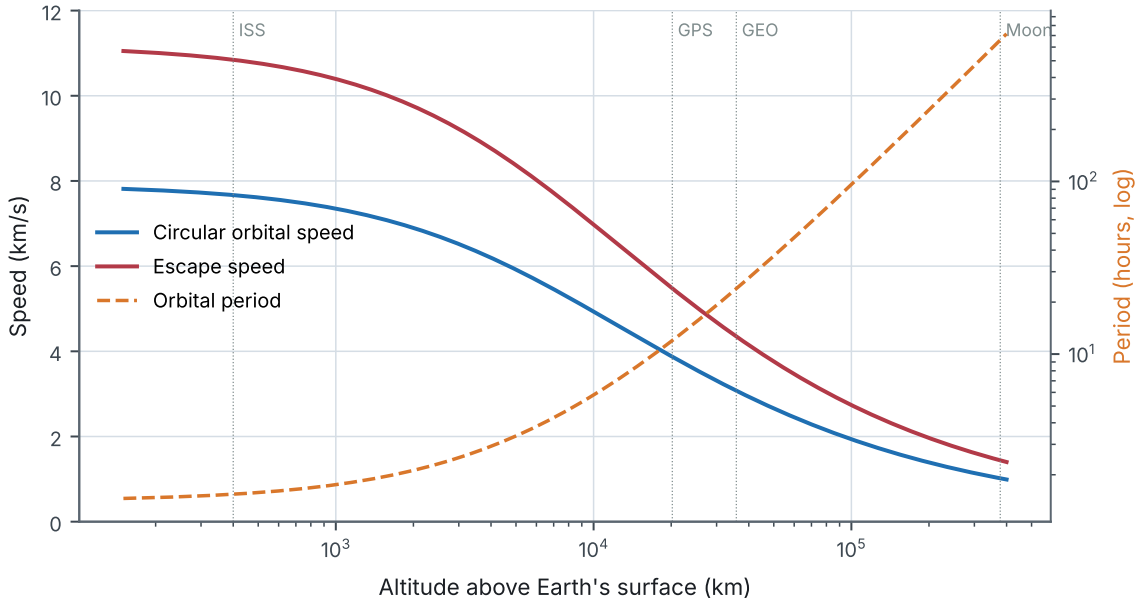


FIGURE 3.4 Circular orbital speed, escape speed and orbital period as functions of altitude above Earth. At the ISS altitude a satellite circles the Earth every 92 minutes at 7.66 km/s; at GEO the period equals one sidereal day (23 h 56 min).

WORKED EXAMPLE — THE INTERNATIONAL SPACE STATION'S ORBIT

The ISS flies at about 420 km, so $r = 6\,378 + 420 = 6\,798$ km.

- Circular speed: $v = \sqrt{398\,600 / 6\,798} = 7.657$ km/s (27 570 km/h).
- Period: $T = 2\pi \sqrt{6\,798^3 / 398\,600} = 5\,578$ s = 93.0 min, so about 15.5 orbits per day.
- Escape speed from this altitude: $\sqrt{2} \times 7.657 = 10.83$ km/s, so leaving Earth from the ISS orbit would need $\Delta v = 3.17$ km/s.

3.2 Describing an Orbit: Orbital Elements and Reference Frames

3.2.1 The Six Classical Orbital Elements

A satellite's position and velocity at an instant (six numbers: $\mathbf{r}$, $\mathbf{v}$) define its two-body orbit completely. An equivalent and more intuitive description uses six **Keplerian (classical) orbital elements** (Figure 3.5):

Element	Symbol	Describes
Semi-major axis	a	Size of the orbit (and the period)
Eccentricity	e	Shape
Inclination	i	Tilt of the orbit plane relative to the equator (0°–180°; > 90° = retrograde)
Right ascension of the ascending node (RAAN)	Ω	Angle in the equatorial plane from the vernal equinox to the point where the satellite crosses the equator northward
Argument of periapsis	ω	Angle in the orbit plane from the ascending node to periapsis
True anomaly (or mean anomaly, or time of periapsis)	ν (M, t_p)	Position of the satellite on the orbit

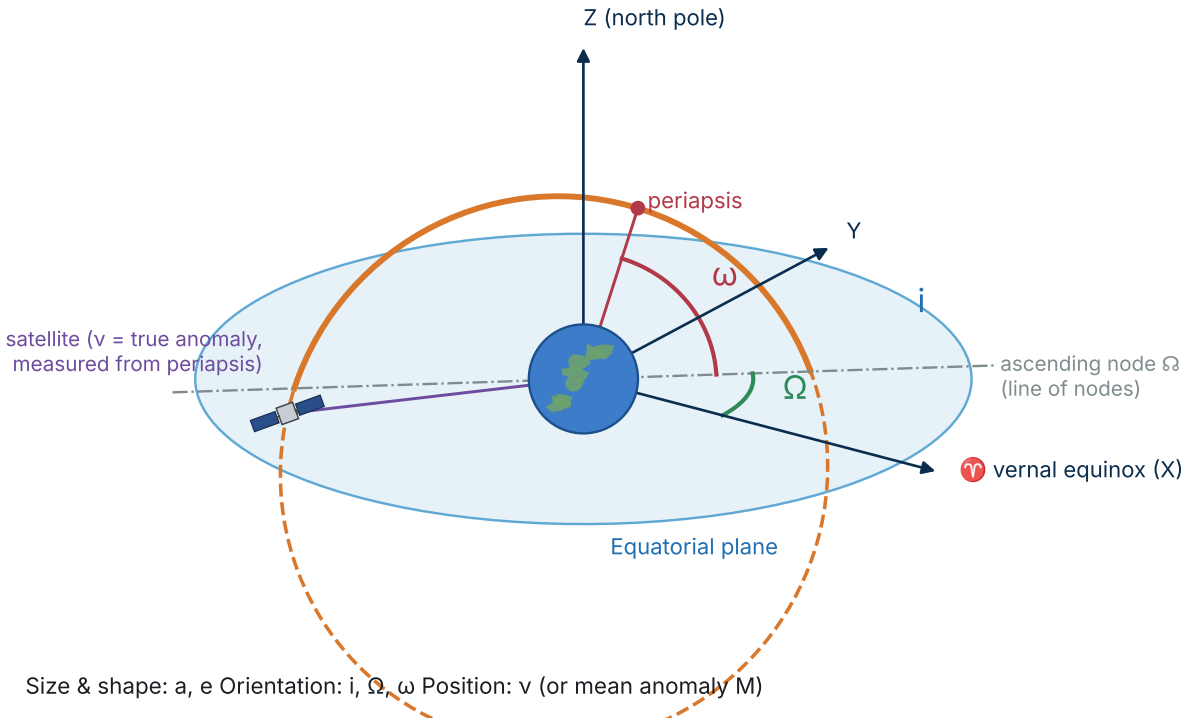


FIGURE 3.5 The classical orbital elements. Two define the size and shape of the orbit (a, e), three its orientation in space (i, Ω, ω), and one the position of the satellite along it (ν).

Some elements are undefined for special orbits: for circular orbits there is no periapsis (ω is replaced by the **argument of latitude** $u = \omega + \nu$), and for equatorial orbits there is no ascending node. For such orbits, engineers use **equinoctial elements**, which have no singularities and are preferred in numerical software and for electric-propulsion trajectory optimisation.

3.2.2 Reference Frames and Time

Orbits are computed in **inertial** (non-rotating) frames, but ground stations, maps and GNSS work in **Earth-fixed** frames, and spacecraft attitude and rendezvous are described in **orbit-fixed** frames (Figure 3.6):

- **Earth-centred inertial (ECI)** frames: today the **Geocentric Celestial Reference Frame (GCRF)**, closely aligned with the older **J2000** (EME2000) frame. The X-axis points to the vernal equinox, Z to the celestial north pole.
- **Earth-centred, Earth-fixed (ECEF)**: the **International Terrestrial Reference Frame (ITRF)** and its GNSS realisations (WGS 84 for GPS). It rotates with the Earth; converting to ECI requires precession, nutation, Earth rotation angle and polar motion, published by the International Earth Rotation and Reference Systems Service (IERS).
- **Orbit frames**: **RTN/RSW** (radial, transverse/along-track, normal) and **LVLH** (local vertical, local horizontal), used for relative motion, attitude pointing and orbit-determination covariance.
- **Body frame**: fixed to the spacecraft structure.

Precise work also needs precise **time scales**. These include International Atomic Time (TAI) and Coordinated Universal Time (UTC), which differs from TAI by an integer number of leap seconds (37 s since 2017). There are also GPS time (TAI − 19 s, no leap seconds), Terrestrial Time (TT = TAI + 32.184 s) and Barycentric Dynamical Time (TDB) for interplanetary ephemerides. Confusing these has caused real errors.

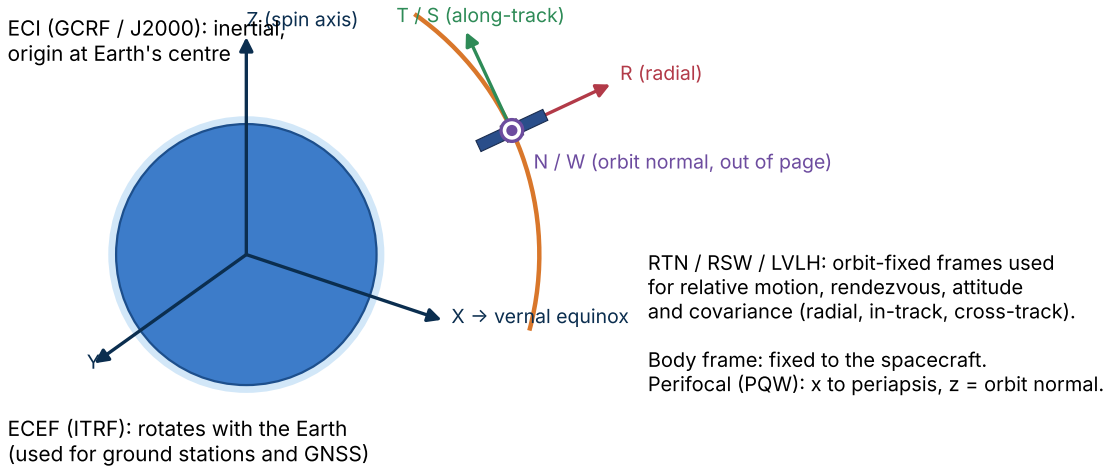
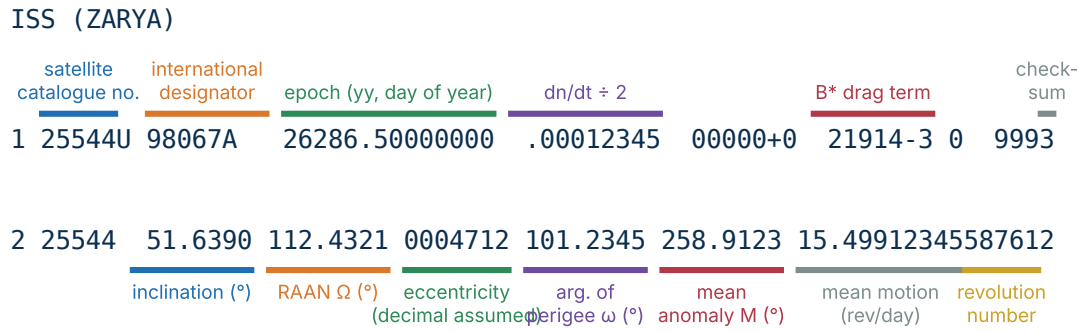


FIGURE 3.6 Common reference frames: Earth-centred inertial and Earth-fixed frames, and the orbit-fixed radial / along-track / cross-track (RTN) frame attached to the satellite.

3.2.3 Two-Line Element Sets

The most widely distributed orbit format is the **two-line element set (TLE)** (Figure 3.7), produced by the US Space Force for the public catalogue. TLEs are *mean* elements fitted for the **SGP4** analytical propagator. They are adequate for antenna pointing and general awareness (errors of about 1 km at epoch, growing by kilometres per day). But they are not adequate for precise conjunction assessment, which uses covariance-carrying ephemerides (Chapter 21). The fixed five-character catalogue-number field is close to exhaustion as tens of thousands of new objects

are catalogued. An interim "Alpha-5" scheme replaces the first digit with a letter. So users are moving to the **CCSDS Orbit Mean-elements Message (OMM)** in XML, KVN or JSON, which has no such limit.



TLEs are mean elements for the SGP4/SDP4 propagator, not osculating Keplerian elements. Illustrative values; current sets are published by CelesTrak and Space-Track.org (also in the newer OMM / CCSDS format to allow catalogue numbers > 99 999).

FIGURE 3.7 Anatomy of a two-line element set (illustrative values). Each field occupies fixed columns; the last digit of each line is a checksum.

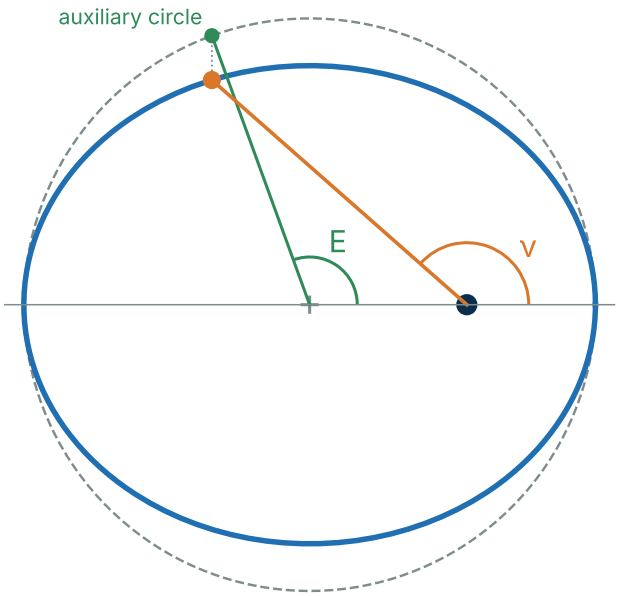
3.3 Where Is the Satellite? Kepler's Equation

Knowing the orbit is not the same as knowing where the satellite will be at a given time. Because the speed varies around an elliptical orbit, the true anomaly does not increase uniformly. Kepler solved this with two auxiliary angles (Figure 3.8): the **mean anomaly** M, which does increase uniformly, $M = n(t - t_p)$ with **mean motion** $n = \sqrt{\mu/a^3}$; and the **eccentric anomaly** E, related to M by **Kepler's equation**:

$$M = E - e \sin E$$

Given M, this transcendental equation is solved for E by iteration (Newton's method converges in a few steps), and then

$$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\,\tan\frac{E}{2}, \qquad r = a(1 - e\cos E)$$



Kepler's equation:
$M = E - e \sin E$
$M = n\,(t - t_p),\ \ n = \sqrt{(\mu / a^3)}$

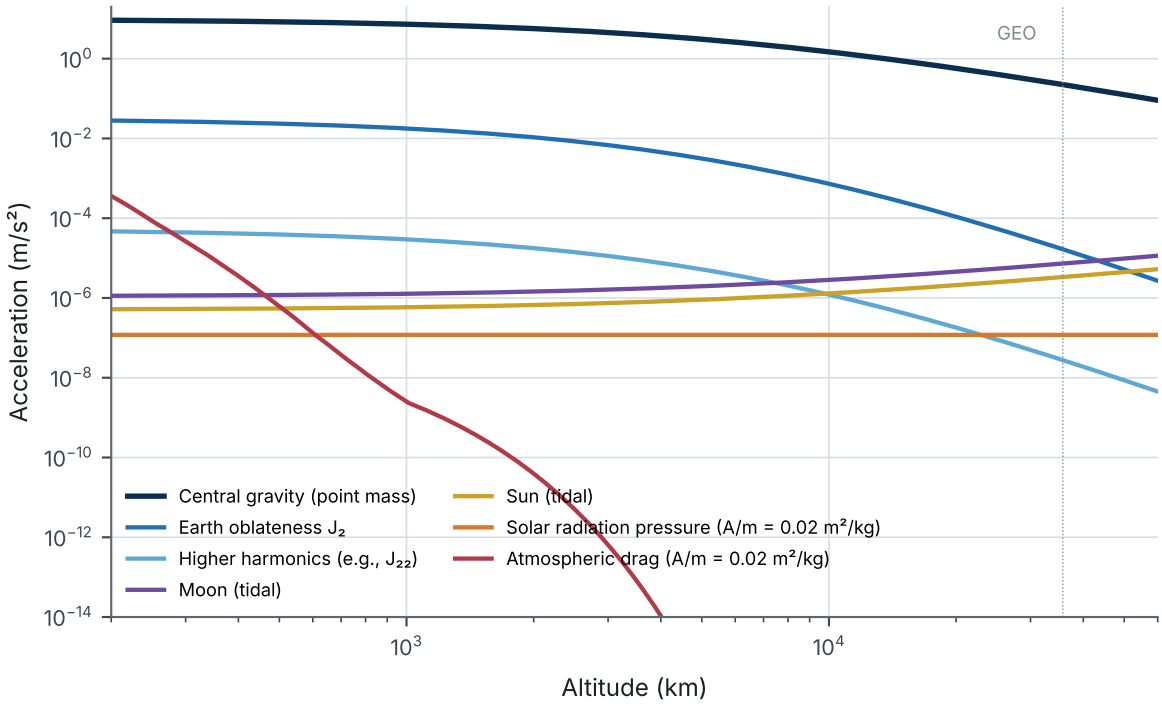
$\tan(\nu/2) = \sqrt{((1+e)/(1-e))}\,\tan(E/2)$

FIGURE 3.8 The eccentric anomaly E is measured at the centre of the ellipse to the point on an auxiliary circle; the true anomaly ν is measured at the focus. Kepler's equation links E to the uniformly increasing mean anomaly M.

Modern software usually avoids the anomalies altogether by integrating the equations of motion numerically (e.g., with Runge–Kutta or Gauss–Jackson integrators), which also allows any perturbing forces to be included. Analytical methods remain important for quick estimates, for onboard software with limited computing power, and for physical insight.

3.4 Orbit Perturbations

Real orbits deviate from Keplerian ellipses because of additional forces. Figure 3.9 compares their sizes.



Orders of magnitude after Montenbruck & Gill (2000). Drag uses moderate solar activity.

FIGURE 3.9 Magnitude of perturbing accelerations as a function of altitude for a typical satellite. Earth's oblateness (J_2) dominates in LEO after central gravity; drag dominates the decay of low orbits; the Sun and Moon dominate high orbits such as GEO.

3.4.1 Earth's Oblateness (J_2)

The Earth is not a sphere: its equatorial radius (6 378.137 km) exceeds its polar radius (6 356.752 km) by 21 km. Its gravity field is expressed as a series of **spherical harmonics**; the largest term after the central field is the **J_2 zonal harmonic** ($J_2 = 1.082\,63 \times 10^{-3}$), about a thousand times larger than any other. The equatorial bulge exerts a torque on an inclined orbit, causing two secular drifts:

Nodal regression — the orbit plane precesses about Earth's axis:

$$\dot{\Omega} = -\frac{3}{2}\, n\, J_2 \left(\frac{R_\oplus}{p}\right)^2 \cos i$$

Apsidal rotation — the line of apsides rotates within the orbit plane:

$$\dot{\omega} = \frac{3}{4}\, n\, J_2 \left(\frac{R_\oplus}{p}\right)^2 \left(5\cos^2 i - 1\right)$$

where $p = a(1 - e^2)$. For the ISS ($i = 51.6^\circ$, 420 km), the node regresses by about 5° per day. This is why its ground track shifts each day, and why launch windows to the ISS move earlier by roughly 22 minutes per day.

These "perturbations" are put to good use in three special orbit families (Figure 3.10):

- **Sun-synchronous orbits (SSO).** Choosing a slightly retrograde inclination (about 97–99° for altitudes of 500–1 000 km) makes $\dot{\Omega} = +0.9856^\circ$/day, exactly matching the Earth's motion around the Sun. The orbit plane then keeps a constant angle to the Sun, so the satellite crosses each latitude at the same **local solar time** every day (e.g., Landsat at 10:00 descending node,

Sentinel-2 at 10:30). This gives consistent lighting for imaging and stable thermal conditions. Dawn–dusk SSOs (06:00/18:00) are almost always in sunlight — ideal for radar satellites and solar-powered missions.

- **Critically inclined orbits.** At $i = 63.4^\circ$ or 116.6°, $5\cos^2 i - 1 = 0$, so the apsides do not rotate. The **Molniya** (12 h period) and **Tundra** (24 h) orbits use this to keep their apogee permanently over high northern latitudes.
- **Frozen orbits.** A suitable choice of e and $\omega = 90^\circ$ (using the J_3 harmonic) keeps the altitude profile over each latitude fixed — useful for altimetry missions such as Sentinel-3 and Jason/Sentinel-6.

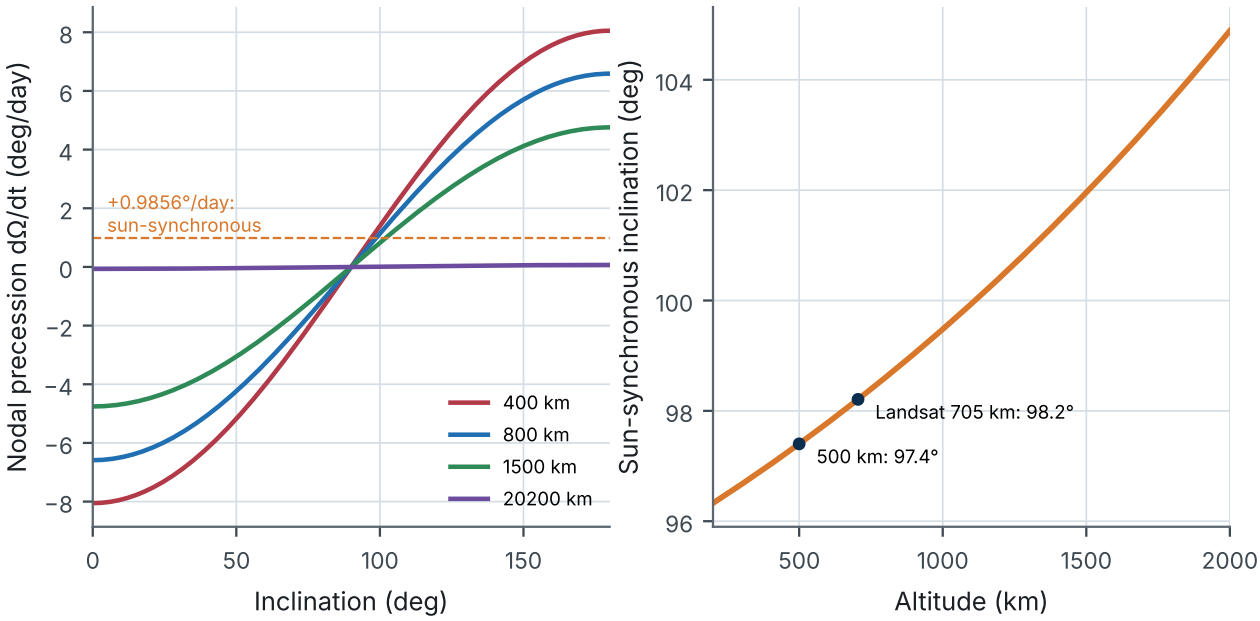


FIGURE 3.10 Left: nodal precession rate versus inclination for several altitudes; the dashed line marks the sun-synchronous rate. Right: inclination required for a sun-synchronous circular orbit as a function of altitude.

3.4.2 Other Perturbations

- **Higher-order gravity.** Tesseral and sectoral harmonics include J_{22}, which is related to the ellipticity of the equator. They cause GEO satellites to drift east or west towards two stable points near 75° E and 105° W. They also cause resonances for orbits with periods that are simple fractions of a day. Precise orbit determination uses models such as EGM2008 or GGM05 to degree and order 100 or more, derived from the GRACE and GOCE missions.
- **Third-body gravity** from the Moon and Sun changes the inclination of GEO satellites by about 0.75–0.95° per year. This must be corrected by north–south station-keeping (about 45–50 m/s per year — the largest item in a GEO propellant budget).
- **Solar radiation pressure (SRP)** — sunlight carries momentum, producing a pressure of about 4.56 μN/m² on an absorbing surface at 1 AU (twice that on a perfect mirror). SRP perturbs eccentricity in GEO and is the main error source in GNSS orbit modelling; it is exploited by solar sails (Chapter 42).
- **Atmospheric drag** — dominant below about 600 km (Chapter 2).
- **Relativity, Earth tides, albedo and thermal re-radiation** — small effects that matter for centimetre-level orbits of geodetic and GNSS satellites.

3.5 Orbital Manoeuvres

A manoeuvre changes the satellite's velocity by firing a thruster. The **velocity change** Δv is the universal currency of mission design, because the rocket equation (Chapter 34) converts it directly into propellant mass:

$$\frac{m_{prop}}{m_0} = 1 - \exp\left(-\frac{\Delta v}{I_{sp}\, g_0}\right)$$

For chemical engines, burns are short compared with the orbital period and are treated as **impulsive** (instantaneous) changes.

3.5.1 The Hohmann Transfer

The minimum-energy two-impulse transfer between two coplanar circular orbits, found by Walter Hohmann in 1925, uses an ellipse tangent to both (Figure 3.11). The first burn at periapsis raises the apoapsis to the target radius; the second burn at apoapsis circularises:

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1\right), \qquad \Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}}\right)$$

The transfer takes half the period of the transfer ellipse: $t = \pi\sqrt{a_t^3/\mu}$ with $a_t = (r_1 + r_2)/2$.

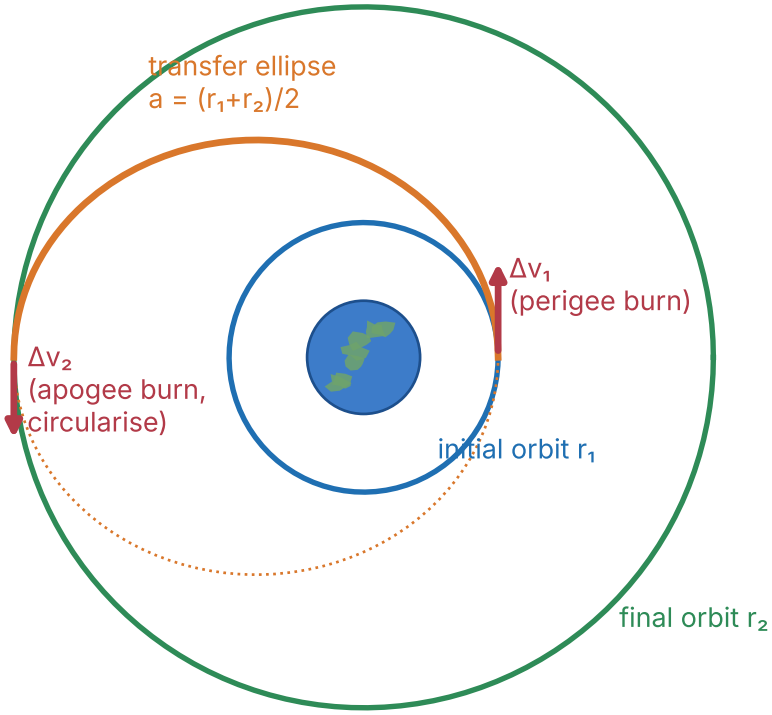


FIGURE 3.11 The Hohmann transfer between circular orbits: a tangential burn at periapsis, a half-revolution coast on the transfer ellipse, and a tangential circularising burn at apoapsis.

WORKED EXAMPLE — FROM A 300 KM PARKING ORBIT TO GEO

$r_1 = 6\,678$ km, $r_2 = 42\,164$ km, so $a_t = 24\,421$ km.

- $v_{c1} = \sqrt{398\,600/6\,678} = 7.726$ km/s; speed at transfer perigee $= \sqrt{398\,600\,(2/6\,678 - 1/24\,421)} = 10.152$ km/s, so $\Delta v_1 = 2.426$ km/s.
- Speed at transfer apogee $= \sqrt{398\,600\,(2/42\,164 - 1/24\,421)} = 1.608$ km/s; $v_{GEO} = 3.075$ km/s, so $\Delta v_2 = 1.467$ km/s.
- Total: **3.89 km/s**; transfer time: $\pi\sqrt{24\,421^3/398\,600} = 18\,990$ s = **5.3 h**.

From Cape Canaveral (28.5°), the satellite must also remove its inclination. Doing this at apogee, where the speed is lowest, and combining it with the circularisation burn: $\Delta v_2 = \sqrt{1.608^2 + 3.075^2 - 2(1.608)(3.075)\cos 28.5^\circ} = 1.83$ km/s. In practice the launcher performs the first burn (injecting into GTO) and the satellite's own engine performs the second, often split into several apogee burns. Launching from Kourou (5.2° N) saves about 0.3 km/s, which is why Kourou is favoured for GEO missions.

3.5.2 Bi-elliptic Transfers

A three-burn **bi-elliptic transfer** first goes beyond the target orbit. It can be cheaper than a Hohmann transfer when the radius ratio exceeds 11.94, and always is when it exceeds 15.58. But it takes much longer (Figure 3.12). It is rarely used for Earth orbits but illustrates that "minimum energy" depends on the constraints.

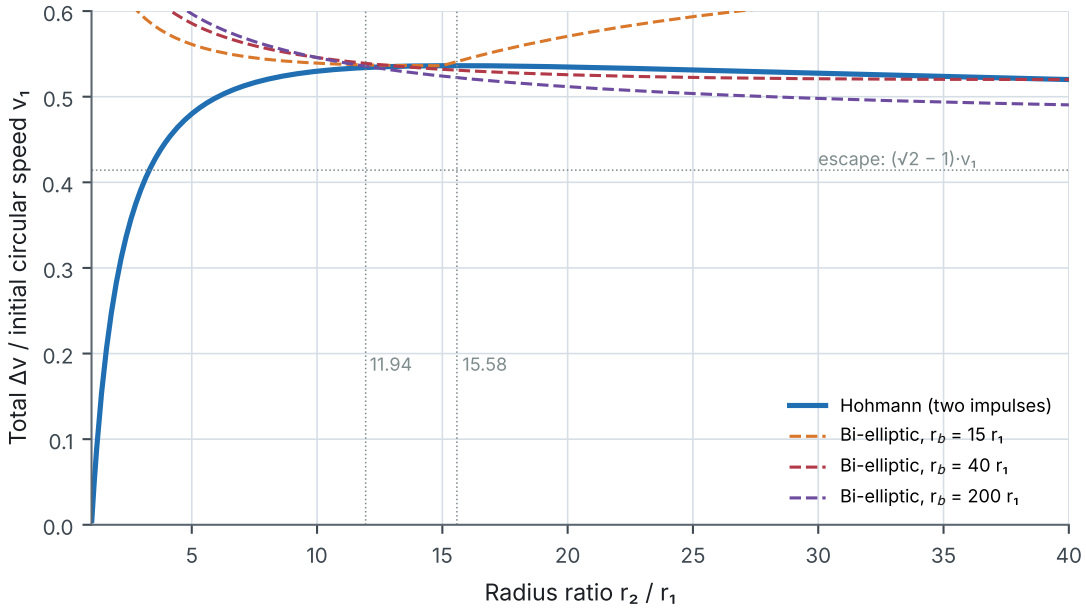


FIGURE 3.12 Total Δv (normalised by initial circular speed) for Hohmann and bi-elliptic transfers. The Hohmann cost peaks at a radius ratio of 15.58; above that, a bi-elliptic transfer with a distant intermediate apoapsis is cheaper.

3.5.3 Plane Changes

Changing the orbit plane is expensive because the whole velocity vector must be rotated. A pure plane change by angle Δi at speed v costs

$$\Delta v = 2v \sin\frac{\Delta i}{2}$$

so it is best performed where the speed is lowest — at apoapsis, or at high altitude — and combined with other burns (Figure 3.13). Changing the plane of a LEO satellite by just 10° costs 1.3 km/s, more than most satellites carry. This is why the launch site latitude and launch azimuth, which set the initial inclination, are so important, and why constellations deploy separate planes from separate launches. Instead of plane changes, some constellations exploit differential nodal precession: satellites placed at slightly different altitudes drift apart in RAAN for free over months (Starlink, Planet).

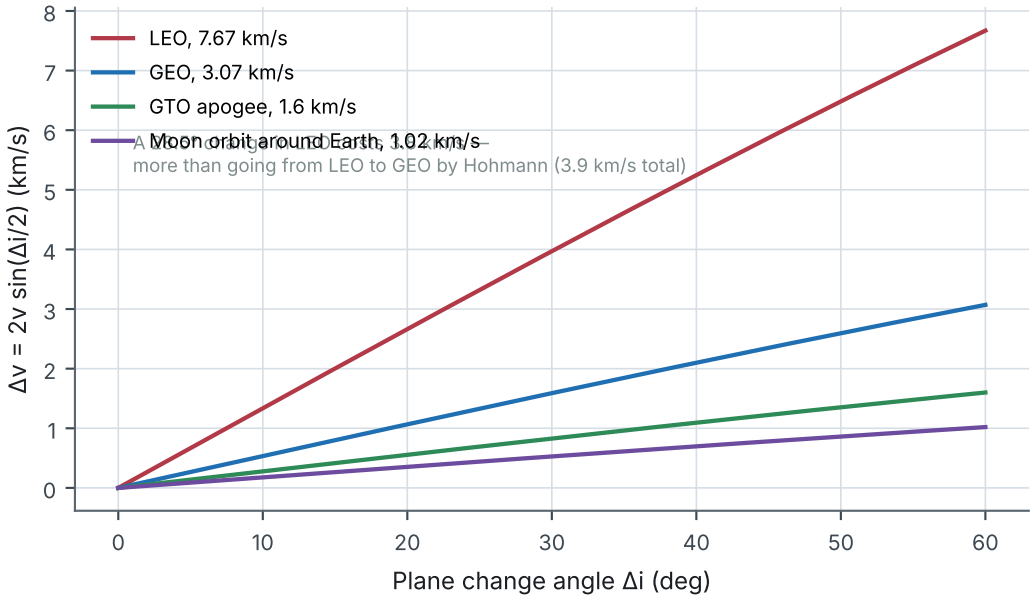


FIGURE 3.13 Δv needed for a plane change as a function of angle, for four orbital speeds. Plane changes are cheapest where the spacecraft moves slowest.

3.5.4 Low-Thrust Transfers

Electric propulsion (Chapter 6) produces tiny thrust (millinewtons to a few newtons) for months. The orbit then spirals gradually; the impulsive approximation no longer applies. For a slow spiral between circular orbits, the Δv approaches the *difference in circular speeds*: $\Delta v \approx v_{c1} - v_{c2}$. This is larger than for a Hohmann transfer (4.6 km/s from LEO to GEO instead of 3.9 km/s). But the propellant efficiency is ten times higher, so far less propellant is used. The **Edelbaum equation** gives the optimal low-thrust transfer with a simultaneous plane change:

$$\Delta v = \sqrt{v_1^2 + v_2^2 - 2 v_1 v_2 \cos\left(\tfrac{\pi}{2}\Delta i\right)}$$

All-electric GEO satellites (Boeing 702SP, Airbus Eurostar Neo, Thales Spacebus Neo) spend four to six months raising their orbits from GTO.

3.6 Rendezvous and Relative Motion

To dock with the ISS, a visiting vehicle must arrive at the same place at the same time with the same velocity. Rendezvous begins with **phasing**: because a lower orbit has a shorter period, a chaser in a slightly lower orbit gains on the target each revolution. When close (tens of kilometres), relative motion is described in the target's RTN frame by the linearised **Hill–Clohessy–Wiltshire (CW) equations**, valid for a circular target orbit with mean motion n:

$$\ddot{x} - 3n^2x - 2n\dot{y} = 0, \qquad \ddot{y} + 2n\dot{x} = 0, \qquad \ddot{z} + n^2z = 0$$

where x is radial, y along-track and z cross-track. Their solutions produce counter-intuitive behaviour (Figure 3.14): a thrust *forward* along the velocity raises the chaser, so it slows down and drops *behind*; a radial burn produces a closed loop with no net drift. Approaches are designed along the **V-bar** (target velocity vector) or **R-bar** (radial line). They include "hold points", corridors and abort trajectories that are passively safe. That is, they never collide with the target if thrusters fail.

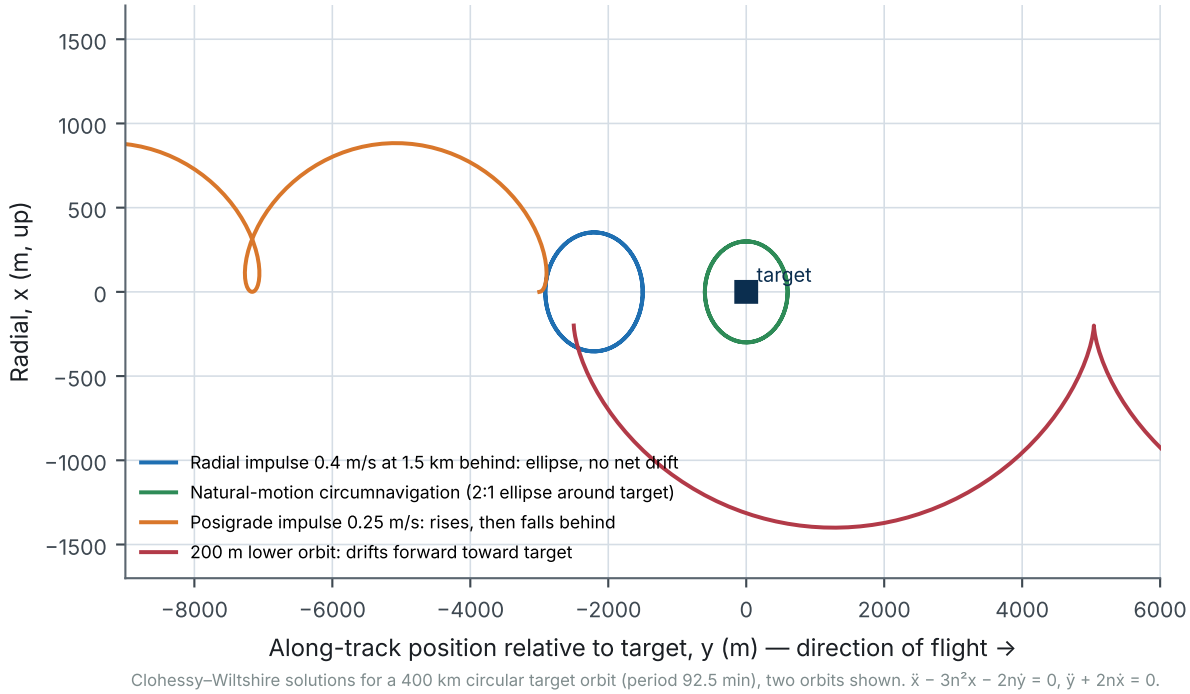


FIGURE 3.14 Relative motion of a chaser near a target in a 400 km orbit, from the Clohessy–Wiltshire solutions. Each trajectory starts from a simple initial condition and is shown for two orbits.

Fast rendezvous profiles now let crews reach the ISS in about 3 hours (Soyuz and Progress since 2020) instead of the traditional 2 days; Crew Dragon typically takes 15–30 hours. Autonomous rendezvous uses relative GNSS, laser rangefinders (lidar) and cameras (Chapter 48).

3.7 Orbit Determination

Orbit determination (OD) estimates a satellite's orbit from measurements. Common measurements are:

- **Range** (distance), from the two-way travel time of a radio signal or laser pulse (satellite laser ranging achieves millimetre precision);
- **Range-rate**, from the Doppler shift of the carrier signal;
- **Angles**, from antennas, optical telescopes or interferometers (e.g., Delta-DOR for deep space, Chapter 24);
- **Onboard GNSS**, which gives LEO satellites their own position to better than 10 m in real time and to centimetres after processing.

Because each measurement is noisy, OD combines many of them with a dynamical model. The **batch least-squares** method adjusts the initial state to minimise the weighted sum of squared residuals over a data arc; the **(extended) Kalman filter** updates the estimate sequentially as each measurement arrives. Both produce a **covariance matrix** — a statistical description of the uncertainty — which is essential for conjunction assessment and for planning manoeuvres. The quality of OD depends as much on the force models (drag, SRP) as on the measurements.

3.8 Ground Tracks and Coverage

The **ground track** is the path traced on the Earth's surface by the point directly beneath the satellite. It combines the satellite's orbital motion with the Earth's rotation beneath it (Figure 3.15). Each successive pass of a LEO satellite crosses the equator about $360^\circ \times T/T_{day}$ further west — about 23° for a 92-minute orbit. The maximum latitude reached equals the inclination (or 180° − i for retrograde orbits).

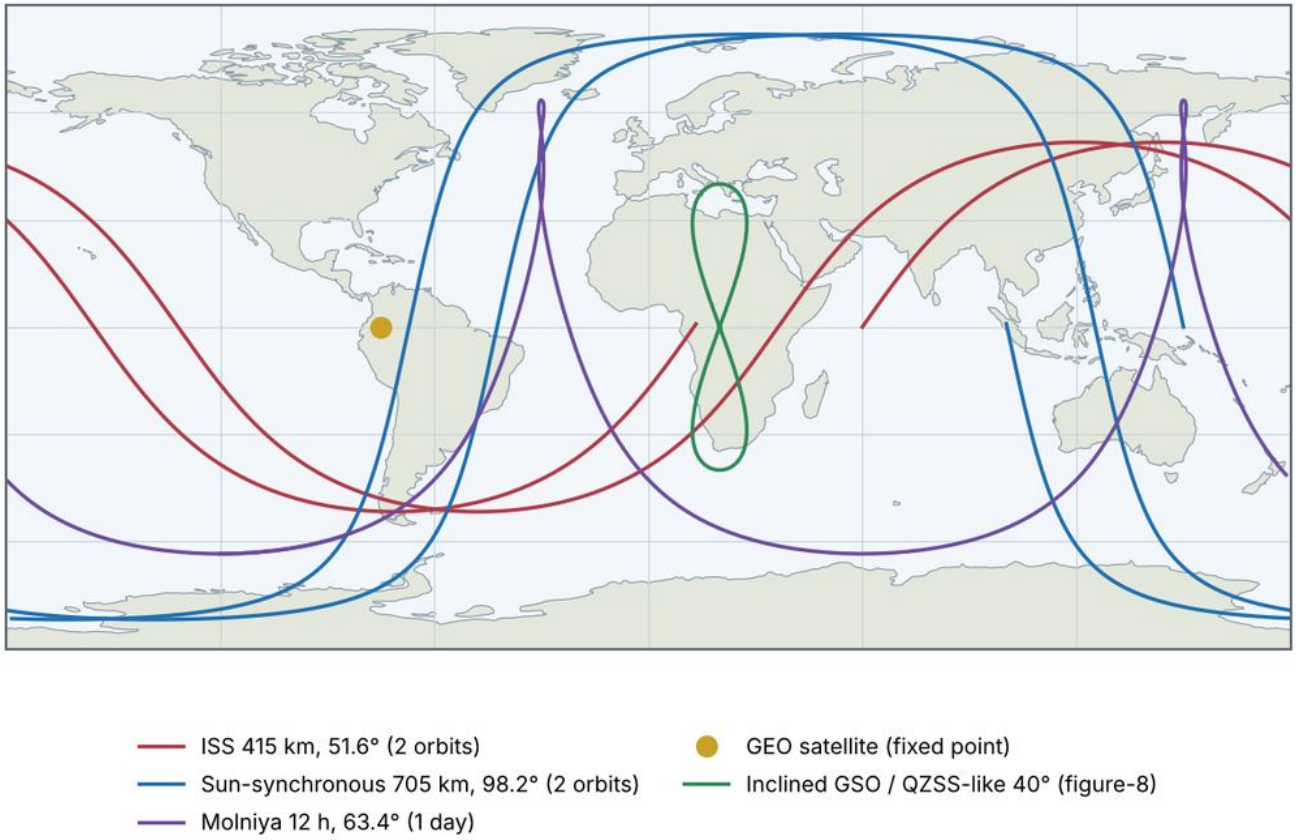


FIGURE 3.15 Ground tracks of representative orbits: the ISS (51.6°), a sun-synchronous imaging orbit (98.2°), a 12-hour Molniya orbit that dwells over the northern hemisphere, a geostationary satellite (a fixed point) and an inclined geosynchronous orbit tracing a figure-eight.

A **repeat ground track** orbit returns over exactly the same points after an integer number of revolutions N in an integer number of days D ($NT = D \times T_{day,nodal}$). Landsat repeats after 233 orbits in 16 days; Sentinel-2 after 143 orbits in 10 days; Jason-3 after 127 orbits in about 9.9 days. Repeat orbits allow consistent comparison of images over time and accurate altimetry along fixed tracks.

The area a satellite can see — its **coverage** — depends on altitude and the minimum elevation angle at which a ground terminal can communicate. Coverage geometry and constellation design are covered in Chapter 11.

Table 3.1 summarises the main Earth-orbit types.

TABLE 3.1 Principal Earth orbit types and their uses.

Orbit	Typical altitude / shape	Inclination	Period	Main uses
Very low Earth orbit	160–350 km, circular	any	87–91 min	High-resolution imaging, technology demos, short-lived missions
Low Earth orbit (LEO)	350–2 000 km	0–90°+	91–127 min	Stations, EO, broadband constellations, science
Sun-synchronous (SSO)	500–1 000 km	97–99.5°	95–105 min	Optical and radar EO, weather (polar), climate
Polar (non-SSO)	700–1 400 km	≈ 85–90°	100–115 min	Iridium (86.4°), global coverage
Medium Earth orbit (MEO)	8 000–23 300 km	0–56°	4.8–14 h	GNSS (≈20 000–23 200 km), O3b mPOWER (8 062 km, equatorial)
Geostationary (GEO)	35 786 km, circular	0°	23 h 56 min	Communications, weather, missile warning, data relay
Inclined geosynchronous (IGSO)	35 786 km	30–60°	23 h 56 min	QZSS, BeiDou IGSO, NavIC (regional navigation)
Molniya	500 × 39 900 km	63.4°	12 h	High-latitude communications, early warning
Tundra	≈ 25 000 × 46 000 km	63.4°	24 h	Sirius XM (formerly), QZSS (variant), Arctic comms
Geosynchronous transfer (GTO)	200 × 35 786 km	launch-site dependent	10.5 h	Transfer to GEO
Graveyard (super-synchronous)	GEO + ≥ 235 km	≈ 0°	> 24 h	Disposal of GEO satellites

3.9 Interplanetary Trajectories

3.9.1 Patched Conics and the Sphere of Influence

A spacecraft travelling to Mars is pulled by the Earth, Mars and the Sun at the same time. The **patched-conic approximation** splits the journey into segments, in each of which only one body matters, and "patches" them together at the boundary of each planet's **sphere of influence (SOI)**:

$$r_{SOI} \approx a_{planet}\left(\frac{m_{planet}}{m_{Sun}}\right)^{2/5}$$

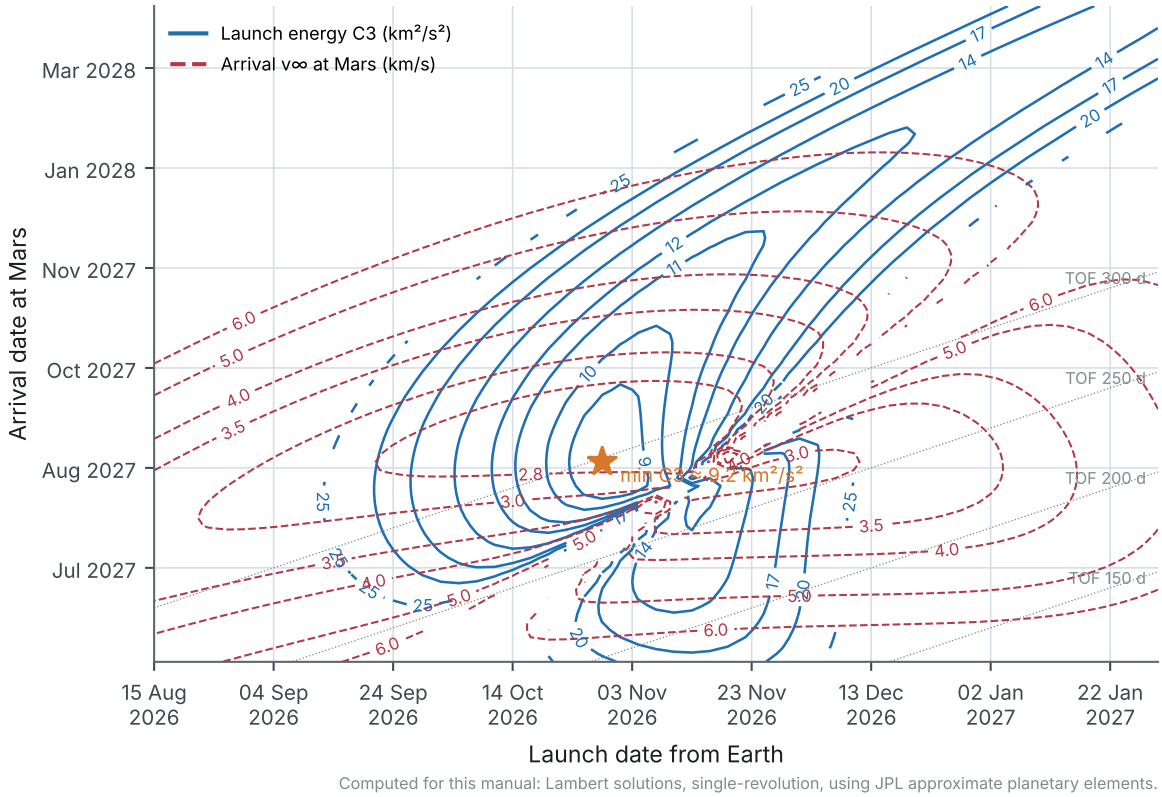
Earth's SOI has a radius of about 925 000 km, Mars' about 577 000 km, and the Moon's (relative to Earth) about 66 000 km. A mission thus consists of a **hyperbolic departure** from Earth, a **heliocentric transfer** ellipse, and a **hyperbolic arrival** at Mars, where the spacecraft either flies by, brakes into orbit, or enters the atmosphere.

WORKED EXAMPLE — A HOHMANN TRANSFER FROM EARTH TO MARS

Treating both orbits as circular and coplanar (1.000 and 1.524 AU): the transfer ellipse has $a = 1.262$ AU, so the flight time is half its period, $0.5 \times 1.262^{1.5}$ years = **259 days**. Earth's orbital speed is 29.78 km/s and the transfer perihelion speed is $29.78\sqrt{2 \times 1.524/2.524} = 32.73$ km/s, so the departure $v_\infty = 2.94$ km/s ($C_3 = 8.7$ km²/s²). From a 300 km parking orbit the injection burn is $\sqrt{8.7 + 2 \times 398\,600/6\,678} - 7.73 = 3.6$ km/s. At Mars the spacecraft arrives with $v_\infty = 2.65$ km/s. Because the planets must be correctly aligned at departure (Mars about 44° ahead of Earth), such opportunities recur only once every **synodic period** — 780 days, or about 26 months.

3.9.2 Lambert's Problem and Porkchop Plots

Real planetary orbits are elliptical and inclined, so mission designers solve **Lambert's problem**: find the orbit that connects position $\mathbf{r}_1$ at time t_1 to position $\mathbf{r}_2$ at time t_2. Solving it for a grid of launch and arrival dates gives a **porkchop plot** (Figure 3.16) — contours of departure energy C_3 and arrival v_∞ — the basic tool for choosing launch periods. The plot for the late-2026 Mars window, computed for this manual, shows a minimum C_3 of about 9.2 km²/s² for launch near 29 October 2026 and arrival in late August 2027. The two lobes are "Type I" (transfer angle less than 180°) and "Type II" (more than 180°) trajectories; the ridge between them is a near-180° transfer that requires a large plane change.



Computed for this manual: Lambert solutions, single-revolution, using JPL approximate planetary elements.

FIGURE 3.16 Porkchop plot for Earth-to-Mars transfers in the 2026 launch opportunity. Solid blue contours: launch energy C3 (km²/s²); dashed red: arrival hyperbolic speed at Mars (km/s); dotted: time of flight.

3.9.3 Gravity Assists

A **gravity assist** (or swing-by) uses a close flyby of a planet to change a spacecraft's heliocentric velocity without propellant (Figure 3.17). In the planet's frame, the hyperbolic flyby only rotates the velocity vector by a turning angle δ, with

$$\sin\frac{\delta}{2} = \frac{1}{1 + r_p v_\infty^2/\mu}$$

but in the Sun's frame the planet's orbital velocity is added before and after, so the spacecraft's speed can increase (passing behind the planet) or decrease (passing in front). Gravity assists made the Voyager "Grand Tour" possible and sent Cassini to Saturn via Venus–Venus–Earth–Jupiter. They slowed MESSENGER and BepiColombo enough to orbit Mercury. Parker Solar Probe uses them (seven Venus flybys) to approach within 6.2 million km of the Sun. Flybys can be "powered" (with a burn at periapsis, exploiting the **Oberth effect**: a burn is more effective where the speed is highest).

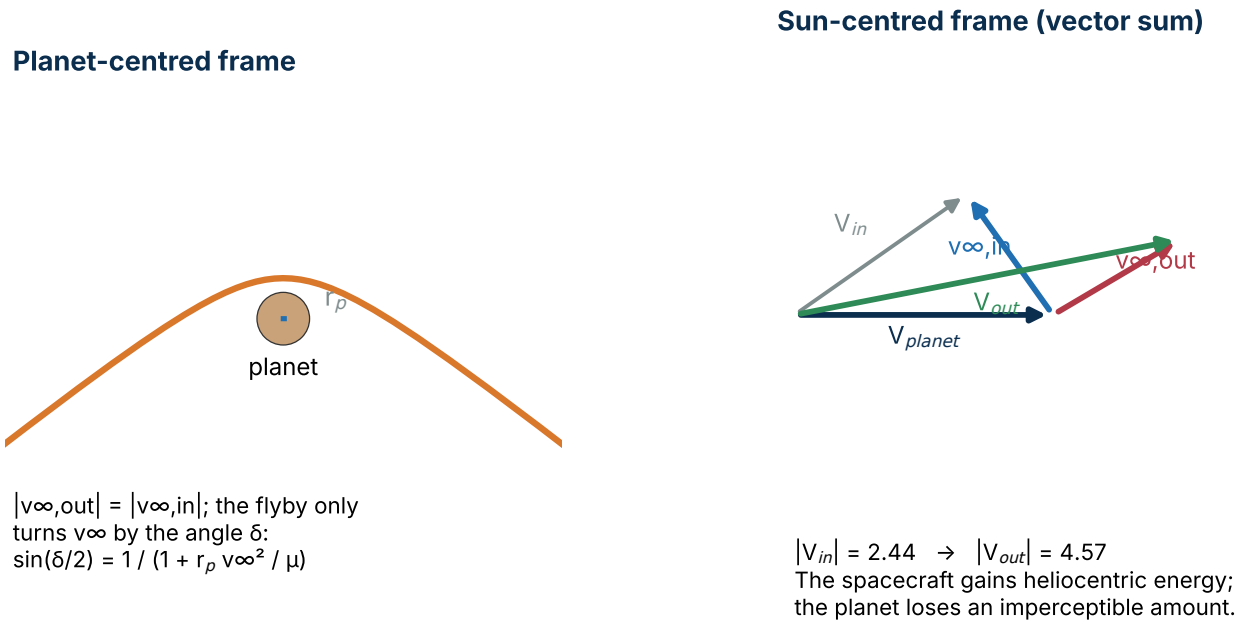
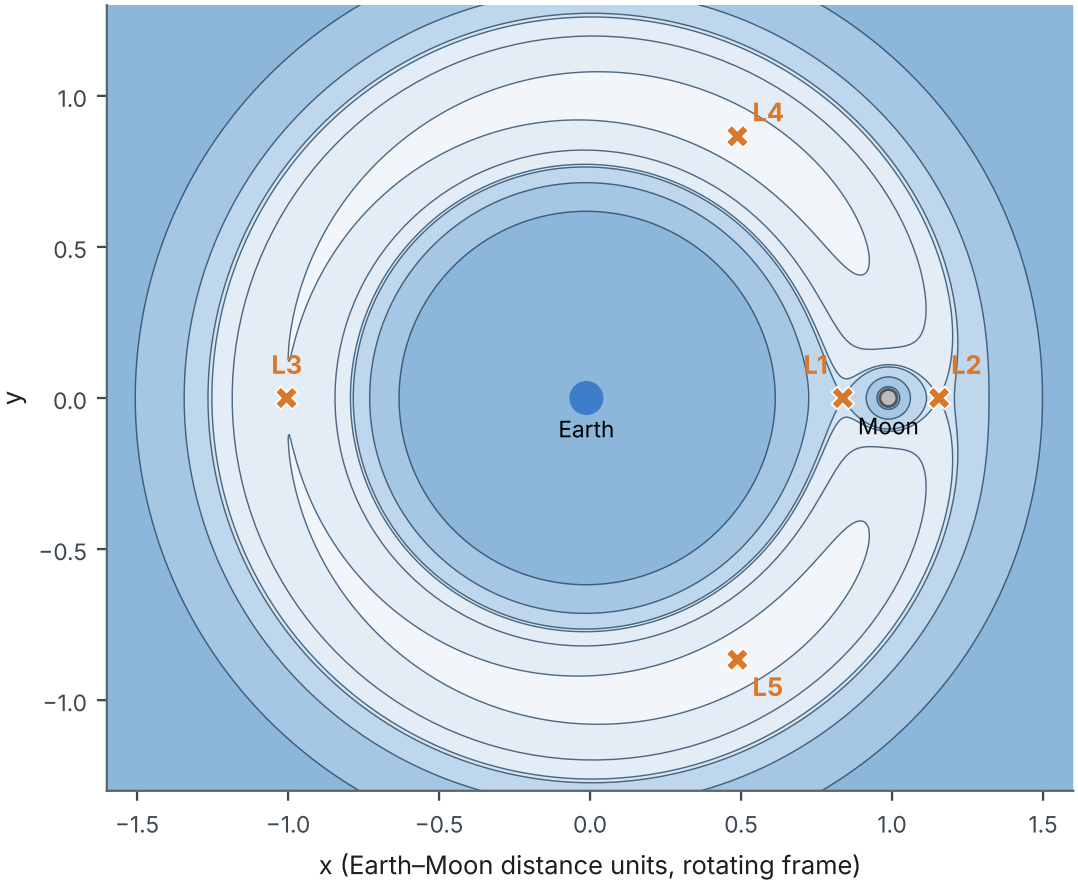


FIGURE 3.17 Geometry of a gravity assist. Left: in the planet's frame the hyperbolic flyby rotates the excess velocity without changing its magnitude. Right: in the Sun's frame, adding the planet's orbital velocity shows a net gain in speed.

3.10 Three-Body Dynamics and the Lagrange Points

When two massive bodies (such as Sun–Earth or Earth–Moon) orbit each other, a third small body moves in the **circular restricted three-body problem (CR3BP)**. In a frame rotating with the two primaries there are five equilibrium points — the **Lagrange points** L1 to L5 (Figure 3.18). L1, L2 and L3 lie on the line through the primaries and are unstable (like a ball on a saddle). L4 and L5 form equilateral triangles with the primaries and are stable when the mass ratio is small. Jupiter's Trojan asteroids gather there.



Contours of the Jacobi constant (zero-velocity curves) for μ = 0.01215. L1 at 326,382 km and L2 at 448,914 km from Earth.

FIGURE 3.18 The Earth–Moon Lagrange points and contours of the Jacobi constant ("zero-velocity curves"), which bound the regions a spacecraft with a given energy can reach in the rotating frame.

Spacecraft cannot sit exactly at an unstable point, but they can fly periodic **halo** or **Lissajous orbits** around it with small station-keeping burns (a few m/s per year):

- **Sun–Earth L1** (1.5 million km sunward) gives an uninterrupted view of the Sun and of the daylit Earth: SOHO, ACE, DSCOVR, Aditya-L1, SWFO-L1 and IMAP.
- **Sun–Earth L2** (1.5 million km anti-sunward) keeps the Sun, Earth and Moon on one side. This is ideal for cold infrared and survey telescopes: WMAP, Planck, Herschel, Gaia, JWST, Euclid and the Nancy Grace Roman Space Telescope.
- **Earth–Moon L1/L2** (about 61 000 km from the Moon) serve as relay and staging locations: China's Queqiao and Queqiao-2 relay satellites orbit near Earth–Moon L2 to support far-side landers.
- **Near-rectilinear halo orbits (NRHO)** are a family of L2 halo orbits that pass close over the lunar poles. They were tested by NASA's CAPSTONE CubeSat (2022) and were chosen for the planned Gateway station. Orion flew **distant retrograde orbits (DROs)**, very stable orbits around the Moon, on Artemis I.

The invariant manifolds of these orbits form low-energy "tubes" that connect them, enabling fuel-efficient transfers known as the **interplanetary superhighway**. The Genesis solar-wind sample-return mission and the GRAIL and CAPSTONE lunar missions used such low-energy trajectories.

3.11 The Δv Budget

Mission designers summarise trajectories as a **Δv budget**: the list of all velocity changes the vehicle must provide, plus margins. Figure 3.19 gives approximate values for the most common legs in the Earth–Moon–Mars system.

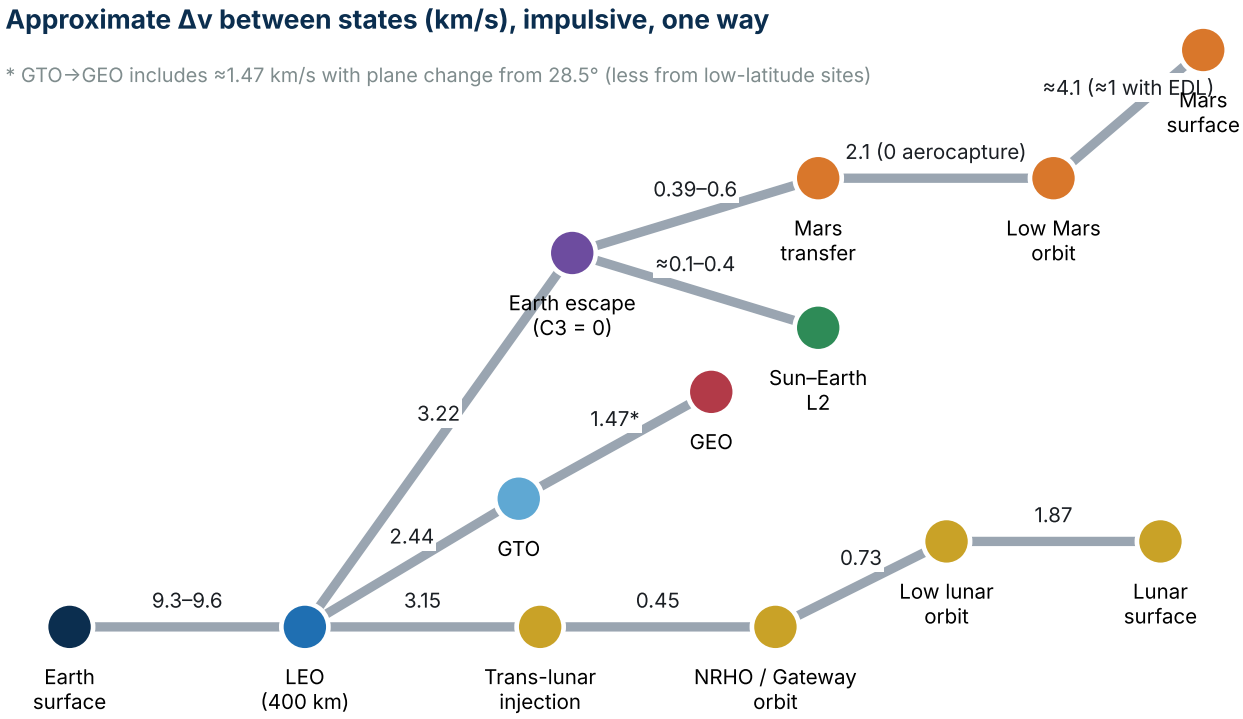


FIGURE 3.19 An approximate "Δv subway map" of the Earth–Moon–Mars system. Values are impulsive, one-way and indicative; real missions add gravity and steering losses, navigation corrections and margins. Aerobraking or aerocapture can replace most of the braking at planets with atmospheres.

TABLE 3.2 Example Δv budget for a geostationary communications satellite (15-year life, launched to standard GTO).

Item	Δv (m/s)	Notes
GTO → GEO (apogee raising, inclination removal from 27°)	1 500–1 800	Several apogee-engine burns; less from supersynchronous or low-inclination GTO
Station acquisition and longitude relocation	10–30	Drift to assigned orbital slot
North–south station-keeping	45–50 per year (≈ 700 total)	Counteracts luni-solar inclination drift; often done with electric thrusters
East–west station-keeping	1–5 per year	Counteracts J_{22} longitude drift and SRP eccentricity
Momentum management	5–20	Depends on ADCS design
End-of-life disposal to graveyard orbit	7–11	≥ 235 km above GEO (IADC guideline)
Margin	3–5 %	Dispersions, contingencies

KEY TAKEAWAYS

- Two-body orbits are conic sections; the vis-viva equation $v^2 = \mu(2/r - 1/a)$ gives the speed anywhere on an orbit, and Kepler's third law gives the period.
- Six orbital elements (a, e, i, Ω, ω, ν) describe an orbit; TLEs are mean elements for SGP4, not precise ephemerides.
- J_2 makes orbit planes precess; this enables sun-synchronous, critically inclined (Molniya/Tundra) and frozen orbits.
- Δv is the currency of mission design: LEO→GEO by Hohmann costs about 3.9 km/s; plane changes are expensive and done where speed is lowest.
- Rendezvous is governed by the counter-intuitive Clohessy–Wiltshire dynamics; orbit determination combines measurements and force models to produce states with covariance.
- Interplanetary missions are designed with patched conics, Lambert solutions and porkchop plots; gravity assists and Lagrange-point orbits extend what is possible with limited propellant.

REVIEW QUESTIONS

1. Use the vis-viva equation to find the speeds at perigee and apogee of a transfer orbit with perigee altitude 250 km and apogee altitude 35 786 km.
2. Calculate the period of a circular orbit at 700 km altitude.
3. List the six classical orbital elements. Which describe the orbit's size and shape, which its orientation, and which the satellite's position?
4. Why are two-line element sets unsuitable for precise collision assessment or precise pointing?
5. Explain how the Earth's J_2 term makes sun-synchronous orbits possible, and why such orbits are slightly retrograde.
6. Calculate the Hohmann-transfer Δv from a 300 km circular orbit to geostationary radius, ignoring the plane change. Why is the plane change usually combined with the apogee burn?
7. In the Clohessy–Wiltshire frame, what happens when a chaser spacecraft makes a small burn forward along its velocity? Why is this counter-intuitive?
8. What is a porkchop plot, and what does a mission designer read from it?
9. Describe the five Lagrange points of the Sun–Earth system and name a mission that uses L1 and one that uses L2.

4 CHAPTER Spacecraft Systems Engineering



Image: NASA/GSFC

A spacecraft is a system of systems that must work first time, in an environment it was never fully tested in, for many years without repair. Systems engineering is the discipline that makes this possible. This chapter explains the project life cycle and its reviews, requirements and architecture, technical budgets and margins, technology readiness, reliability and redundancy, risk management, model-based systems engineering, the verification and test campaign, product assurance, cost and schedule estimation, and the standards that codify decades of hard-won lessons.

IN THIS CHAPTER

4.1 What Systems Engineering Does
4.2 The Project Life Cycle
4.3 Requirements Engineering
4.4 Architecture and Trade Studies
4.5 Technical Budgets and Margins
4.6 Technology Readiness
4.7 Reliability and Redundancy
4.8 Risk Management
4.9 Model-Based Systems Engineering
4.10 Verification and Validation
4.11 Product Assurance and Safety
4.12 Cost and Schedule
4.13 Standards and Bodies of Knowledge

4.1 What Systems Engineering Does

The International Council on Systems Engineering (INCOSE) defines systems engineering as "a transdisciplinary and integrative approach to enable the successful realisation, use and retirement of engineered systems." In plainer words: the systems engineer makes sure that the right system is built, that all its parts work together, and that it is shown to work before it is launched.

In a space project the systems engineering team is responsible for:

- **Defining the problem.** Understanding stakeholder needs and turning them into a concept of operations (ConOps) and verifiable requirements.
- **Designing the architecture.** Choosing the orbit, the segments, the spacecraft configuration and the split of functions between subsystems, hardware and software, space and ground.
- **Managing interfaces.** Every mechanical, electrical, thermal, data and RF interface between subsystems and segments is documented and controlled (interface control documents, ICDs).
- **Allocating and tracking budgets.** Mass, power, propellant, pointing, data, link and cost budgets, each with margins.
- **Managing technical risk.** Identifying what could go wrong and reducing likelihood or consequence.
- **Planning and closing verification.** Proving, by test, analysis, inspection or demonstration, that every requirement is met.
- **Making trade-offs.** Comparing alternatives using clear criteria (performance, cost, risk, schedule).

> **KEY CONCEPT — OPTIMISE THE WHOLE, NOT THE PARTS**
>
> A subsystem engineer naturally wants the best solar array, the lightest structure or the fastest computer. The systems engineer asks a different question: what combination of choices gives the best *mission* within cost, schedule and risk? A slightly heavier structure that simplifies the thermal design and the test campaign may be the better system choice.

4.2 The Project Life Cycle

Large space projects follow a phased life cycle with formal reviews and decision points. NASA's version (NPR 7120.5 and NPR 7123.1) is shown in Figure 4.1; ESA's ECSS-M-ST-10C uses equivalent phases 0, A, B, C, D, E and F.

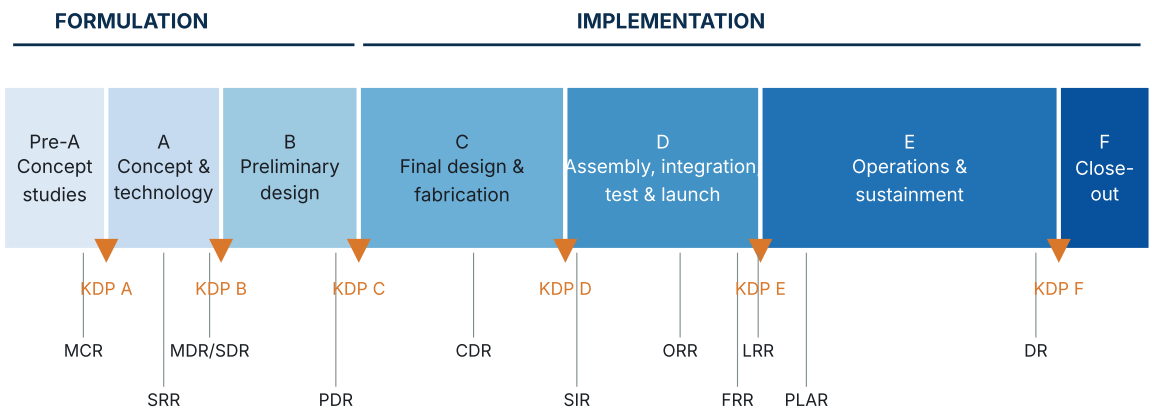


FIGURE 4.1 NASA's project life cycle. Formulation (Pre-Phase A to Phase B) defines and plans the mission; Implementation (Phases C to F) builds, launches, operates and closes it out. Key Decision Points (KDPs) are management gates; technical reviews assess maturity before each gate.

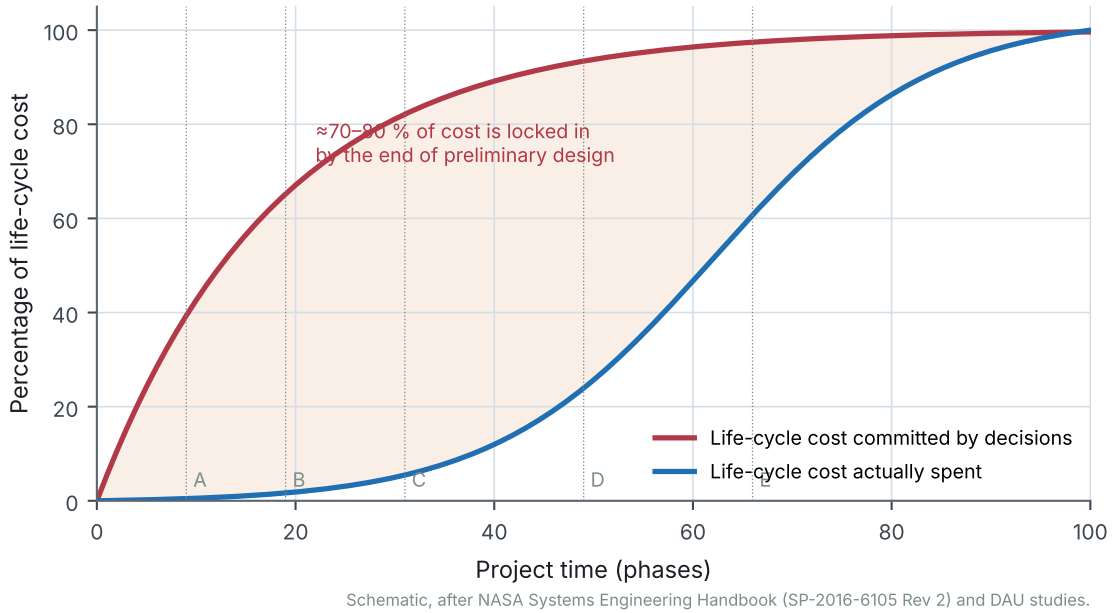
TABLE 4.1 Phases of a space project and their main products.

Phase	Purpose	Typical products	Exit review
Pre-Phase A — Concept studies	Explore the mission idea, identify needs and alternative concepts	Mission concept, feasibility study, rough cost	Mission Concept Review (MCR)
A — Concept and technology development	Define the system baseline, mature critical technologies	ConOps, system requirements, technology development plan, preliminary budgets	System Requirements Review (SRR), Mission/System Definition Review (MDR/SDR)
B — Preliminary design	Complete the preliminary design of all elements; establish cost and schedule baseline	Subsystem specifications, ICDs, preliminary design, verification plan, risk list	Preliminary Design Review (PDR)
C — Final design and fabrication	Complete the detailed design; build and test engineering and qualification models; start flight hardware	Build-to drawings, software code, qualification test results	Critical Design Review (CDR), System Integration Review (SIR)
D — System assembly, integration, test and launch	Integrate and verify the flight system; prepare operations; launch; commission	Verified flight system, operations procedures, trained team	Operational Readiness Review (ORR), Flight Readiness Review (FRR), Launch Readiness Review (LRR), Post-Launch Assessment Review (PLAR)
E — Operations and sustainment	Conduct the mission	Science and service data, operations reports	Critical events reviews, Decommissioning Review (DR)
F — Closeout	Dispose of the spacecraft and close the project	Disposal manoeuvres, data archiving, lessons learned	Disposal readiness

Commercial and “New Space” projects compress these phases dramatically, often merging reviews and replacing paper documents with continuously updated models and agile software practices. A Starlink satellite design iterates in weeks; a flagship science mission such as Europa Clipper took about a decade from Phase A to launch. The principles — know what you need, design to it, prove it works — remain the same.

4.2.1 Why Early Decisions Matter

Most of a project’s life-cycle cost is *committed* by decisions made early, long before most of the money is *spent* (Figure 4.2). Choosing the orbit, launch vehicle, propulsion type or instrument technology in Phase A fixes most of what follows. A change that costs a meeting in Phase A may cost a redesign in Phase C and a mission failure if discovered in orbit. This is why systems engineering effort should be heaviest at the start of a project.



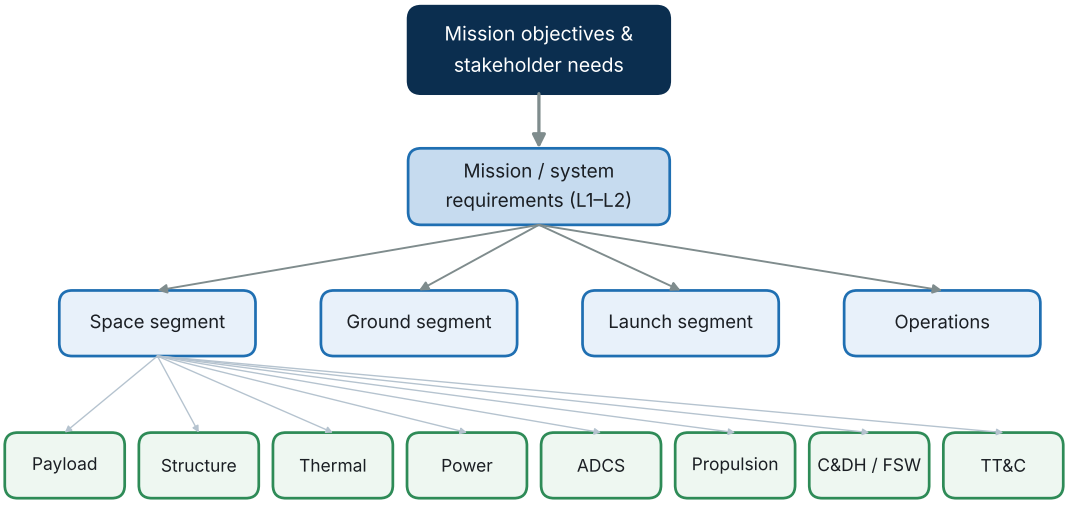
Schematic, after NASA Systems Engineering Handbook (SP-2016-6105 Rev 2) and DAU studies.

FIGURE 4.2 Cumulative life-cycle cost committed by design decisions versus cost actually spent. By the end of the preliminary design phase, roughly three-quarters of the eventual cost is already determined.

4.3 Requirements Engineering

4.3.1 Good Requirements

A requirement states what the system *shall* do or be. Requirements form a hierarchy (Figure 4.3): Level 1 mission requirements (owned by the sponsor), Level 2 system requirements, Level 3 segment and subsystem requirements, and Level 4 component requirements. Every lower-level requirement should trace to a parent; requirements with no parent are "gold plating", and parents with no children are gaps.



Each requirement: unique ID · 'shall' statement · rationale · parent trace · verification method (Test / Analysis / Inspection / Demonstration) · owner

FIGURE 4.3 The requirements hierarchy, from mission objectives through system and segment requirements to subsystem specifications. Each requirement carries a verification method and a trace to its parent.

A good requirement is:

- **Necessary** — traceable to a real need;

- **Unambiguous** — only one interpretation ("the spacecraft shall be lightweight" fails; "the spacecraft wet mass shall not exceed 450 kg" passes);
- **Verifiable** — there is a test, analysis, inspection or demonstration that proves it;
- **Achievable** — technically feasible within the budget;
- **Singular** — one requirement per statement;
- **Solution-independent** where possible — states *what*, not *how*.

> **WORKED EXAMPLE — FROM A MISSION OBJECTIVE TO A SUBSYSTEM REQUIREMENT**
>
> *Mission objective:* "Map crop health across Europe every five days."
>
> *System requirement:* "The system shall acquire multispectral imagery of all land between 35° N and 72° N. The revisit time shall be no more than 5 days, and the ground sampling distance shall be 10 m or better in the red, green, blue and near-infrared bands."
>
> *Derived requirements:* the constellation shall consist of two satellites in a 786 km sun-synchronous orbit phased 180° apart (orbit analysis). The instrument swath shall be ≥ 290 km (coverage analysis), and the instrument focal length and detector pitch shall give 10 m GSD (optics design). The data-handling subsystem shall store ≥ 2.4 Tbit and the X-band downlink shall transmit ≥ 520 Mbit/s (data budget). These numbers are, in fact, those of ESA's Sentinel-2 mission (Chapter 14).

4.3.2 Concept of Operations

A **concept of operations (ConOps)** describes how the system will be used from the user's viewpoint. It covers the mission phases (launch, early orbit, commissioning, nominal operations, contingencies, disposal) and what the spacecraft does on a typical day. It also covers who commands it and how often, how data flows to users, and how anomalies are handled. Writing the ConOps early exposes missing requirements — for example, a need for autonomy because ground contacts are only a few minutes per orbit.

4.4 Architecture and Trade Studies

The **architecture** is the high-level structure of the system: what elements exist and how functions are allocated among them. Typical architectural decisions for an Earth-observation mission include:

- One large satellite or a constellation of small ones?
- Optical or radar payload, or both?
- Onboard processing or raw-data downlink?
- Own ground stations, a commercial ground network or optical relay through GEO?
- Chemical, electric or no propulsion?

Alternatives are compared in **trade studies**. A transparent method is to define weighted criteria (e.g., performance 30 %, cost 25 %, risk 20 %, schedule 15 %, growth potential 10 %). The method then scores each option and tests how sensitive the result is to the weights. A trade study is only as good as its assumptions, so the assumptions must be documented.

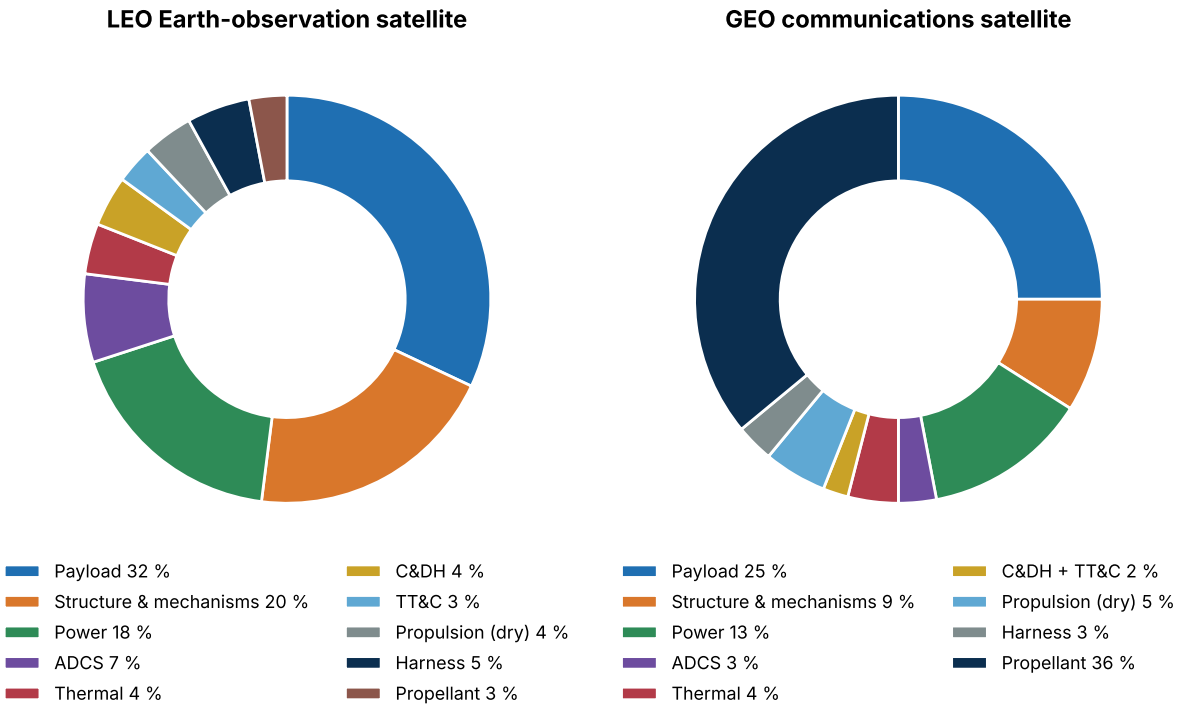
4.5 Technical Budgets and Margins

4.5.1 The Main Budgets

The systems engineer maintains a set of **budgets** — tables that allocate a limited resource among subsystems and track the current best estimate against the allocation:

Budget	Resource tracked	Typical drivers
Mass	kg, including propellant	Launch vehicle capability, structure, propellant for Δv
Power	W (average and peak), Wh per orbit	Payload duty cycle, eclipse length, heaters, end-of-life degradation
Propellant / Δv	m/s and kg	Orbit raising, station-keeping, attitude control, disposal
Data	bits generated, stored, downlinked per day	Payload data rate, compression, ground contact time
Link	dB of margin	Distance, antenna gains, transmitter power, rain fade
Pointing	arcseconds of error and stability	Payload field of view, jitter, alignment, thermal distortion
Thermal	°C ranges for each unit	Dissipation, radiator area, orbit environment
Cost and schedule	money and time	All of the above

Figure 4.4 shows how the launch mass of two typical spacecraft is divided.

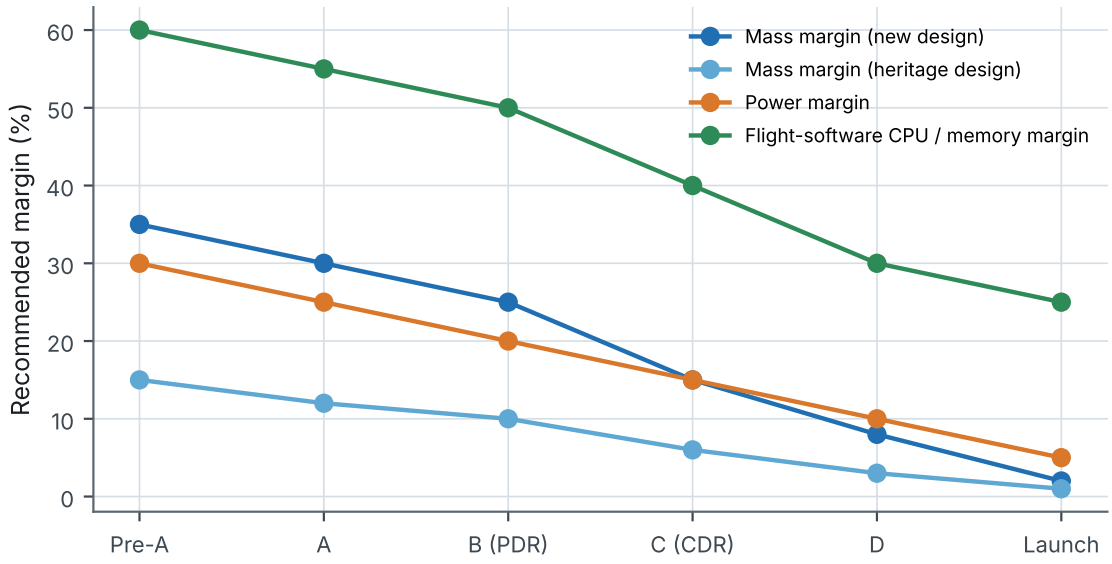


Representative fractions of launch (wet) mass, after SMAD (Wertz et al., 2011) and published spacecraft data.

FIGURE 4.4 Typical distribution of launch mass for a LEO Earth-observation satellite and a GEO communications satellite. The GEO satellite carries a large propellant fraction for orbit raising and 15 years of station-keeping.

4.5.2 Margins and Contingency

Early in a project the estimates are uncertain, and history shows that mass and power almost always *grow* as the design matures. Projects therefore hold **contingency** and **margin**. Contingency is an allowance for growth in the estimate of each item, larger for new designs than for heritage hardware. Margin is a system-level reserve between the total estimate and the hard limit. Figure 4.5 shows typical guidance.



Typical contingency guidance, after AIAA S-120A-2015, ECSS-E-ST-10C and NASA GSFC Gold Rules; organisations differ.

FIGURE 4.5 Typical recommended margins as a function of project phase. Margins are largest when uncertainty is largest and are progressively released as the design matures.

> **FAILURE LESSON — RUNNING OUT OF MARGIN**
>
> The James Webb Space Telescope grew in cost from an initial estimate of about US$1–3.5 billion to about US$10 billion, and its launch slipped from 2007 to 2021. This was in part because the technical complexity of the deployable sunshield and mirror had been underestimated, and early reserves were too small. The telescope ultimately performed better than its requirements. But the experience led NASA to require cost and schedule reserves for major projects to be budgeted at the 70 % confidence level ("joint confidence level", JCL).

4.6 Technology Readiness

Technology Readiness Levels (TRLs) — invented at NASA in the 1970s and now an international standard (ISO 16290) — measure how mature a technology is (Figure 4.6). Most agencies require all critical technologies to reach **TRL 6** (prototype demonstrated in a relevant environment) by PDR. Technologies at TRL 3–5 cross the "valley of death", where research funding ends but projects are not yet willing to adopt them. Related scales include Manufacturing Readiness Levels (MRL) and Software/AI readiness frameworks (Chapter 53).

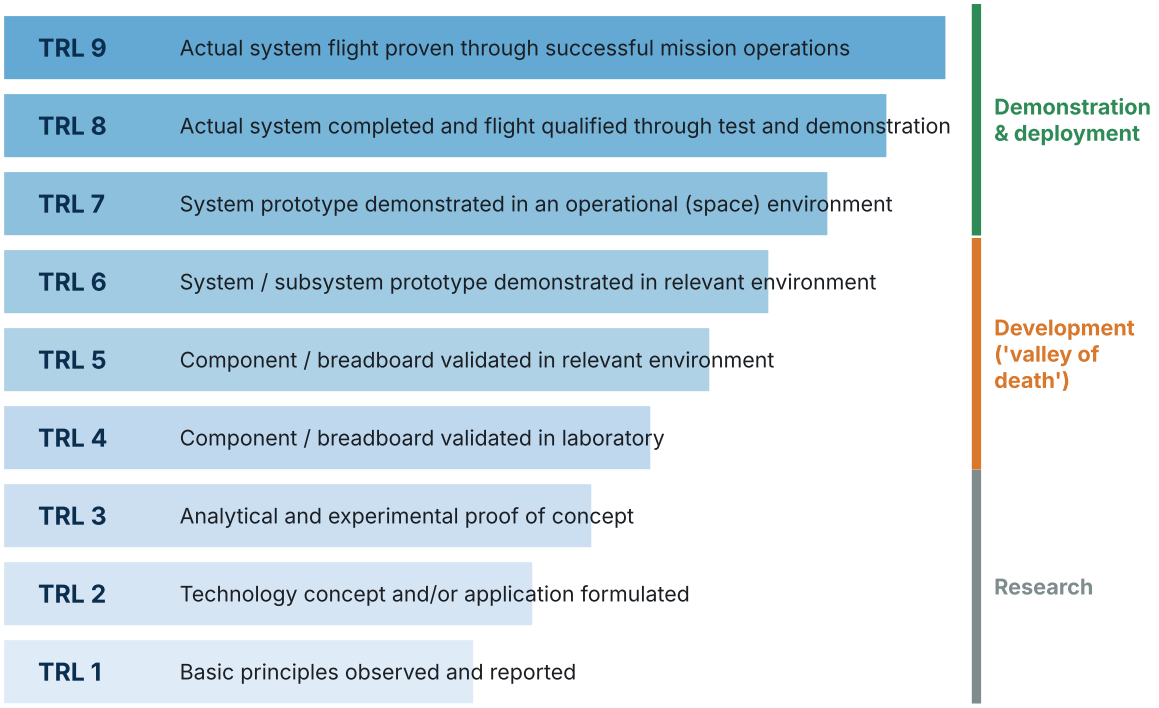


FIGURE 4.6 The nine Technology Readiness Levels. Projects typically require TRL 6 for critical technologies by the Preliminary Design Review.

4.7 Reliability and Redundancy

4.7.1 Reliability Basics

Reliability $R(t)$ is the probability that an item performs its function for time t under stated conditions. During the useful-life period of electronics, failures occur at a roughly constant rate λ, so

$$R(t) = e^{-\lambda t}, \qquad \text{MTTF} = 1/\lambda$$

Failure rates are often given in **FITs** (failures in 10^9 device-hours). Before and after this period, the "bathtub curve" rises (Figure 4.7). Early *infant mortality* failures are weeded out by burn-in, screening and environmental testing. At the end, *wear-out* (battery capacity fade, bearing wear, propellant depletion, solar cell degradation) limits life.

For components in series (all must work), reliabilities multiply: $R_{sys} = \prod R_i$. For parallel redundancy (any one suffices), unreliabilities multiply: $R_{sys} = 1 - \prod (1 - R_i)$.

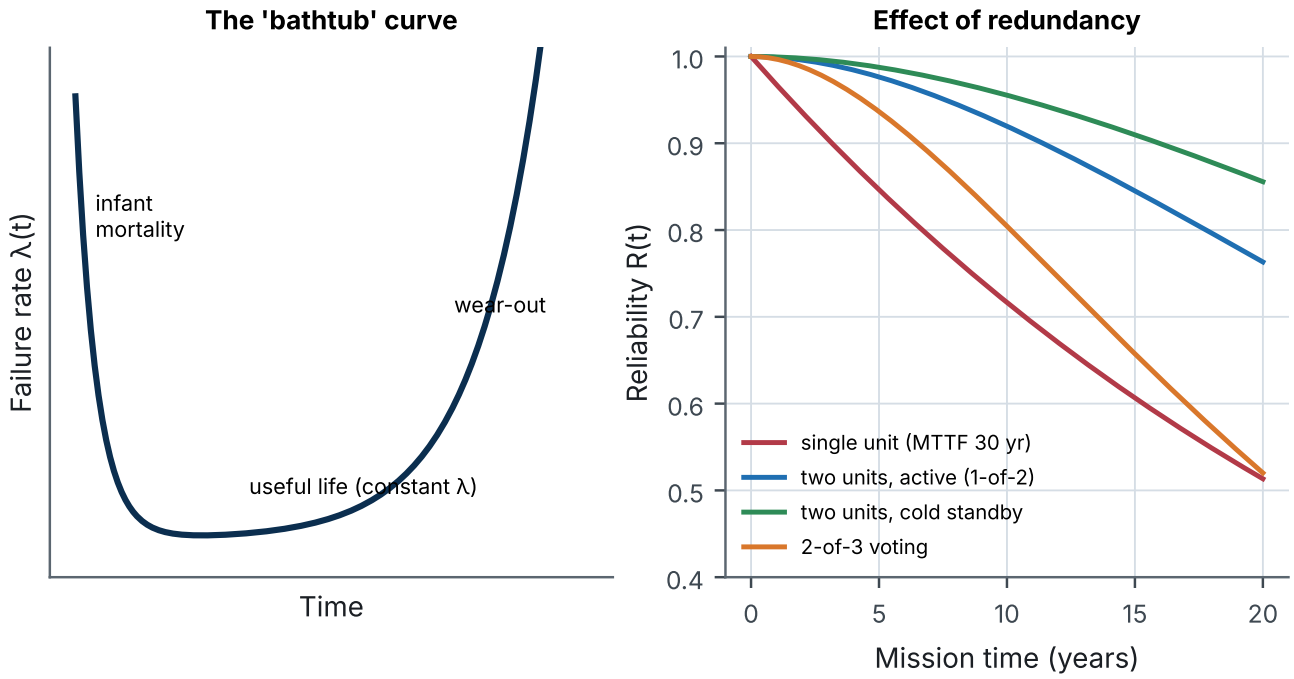


FIGURE 4.7 Left: the bathtub curve of failure rate over life. Right: reliability over a 20-year mission for a single unit and for three redundancy schemes, assuming an MTTF of 30 years per unit.

4.7.2 Redundancy Strategies

- **Hot (active) redundancy** — two units operate together; either can carry the load.
- **Cold (standby) redundancy** — the backup is switched on only when the primary fails; it suffers less wear but needs reliable failure detection and switching.
- **Cross-strapping** — each of two redundant units of subsystem A can work with either unit of subsystem B, avoiding a single failure taking out a whole chain.
- **Voting (k-out-of-n)** — three computers compare outputs and outvote a faulty one (triple modular redundancy); used for launch vehicle avionics and safety-critical systems.
- **Functional redundancy** — a different method achieves the same function (e.g., attitude from star trackers or from Sun sensors and magnetometers).
- **Constellation-level redundancy** — spare satellites in orbit or on the ground, or graceful degradation by design (common in mega-constellations).

Single points of failure (SPFs) are identified by **failure modes, effects and criticality analysis (FMECA)** and either removed or justified. Typical SPFs that are accepted include primary structure, propellant tanks and some deployment mechanisms, which are designed with large margins instead.

TABLE 4.2 Extract of a failure modes, effects and criticality analysis (FMECA) for a small-satellite power subsystem (illustrative).

Item	Failure mode	Cause	Local effect	System effect	Severity	Detection	Mitigation
Solar array string	Open circuit	Interconnect fatigue, MMOD impact	Loss of one string (≈ 4 % power)	Reduced power margin	3 (minor)	String current telemetry	15 % end-of-life array margin; string-level blocking diodes

Item	Failure mode	Cause	Local effect	System effect	Severity	Detection	Mitigation
Battery cell	Internal short	Manufacturing defect, dendrite growth	Cell voltage drop, heating	Loss of battery module if not isolated	2 (major)	Cell voltage and temperature monitoring	Cell screening, cell bypass, n+1 strings
Battery charge regulator	Fails “on”	Transistor short	Overcharge	Battery damage, possible thermal runaway	1 (catastrophic)	Hardware over-voltage detector	Independent hardware cut-off, redundant regulator
Power distribution switch	Fails open	Latch-up, wear	Load unpowered	Loss of unit until switched to redundant	2	Current telemetry, watchdog	Cross-strapped redundant switches
Main bus	Short circuit	Harness chafing, debris	Bus voltage collapse	Loss of mission	1	Undervoltage monitor	Fault-current limiters, harness protection, design reviews

4.8 Risk Management

A **risk** is a potential future event that would harm the project, characterised by its **likelihood** and **consequence**. Risks are recorded in a risk register, scored on a **5 × 5 matrix** (Figure 4.8) and assigned an owner. Each is handled by one of four strategies: *mitigate* (reduce likelihood or consequence), *accept*, *transfer* (e.g., insurance or contract terms), or *avoid* (change the design). Mitigation plans include “burn-down” steps tied to milestones, so that progress is visible.

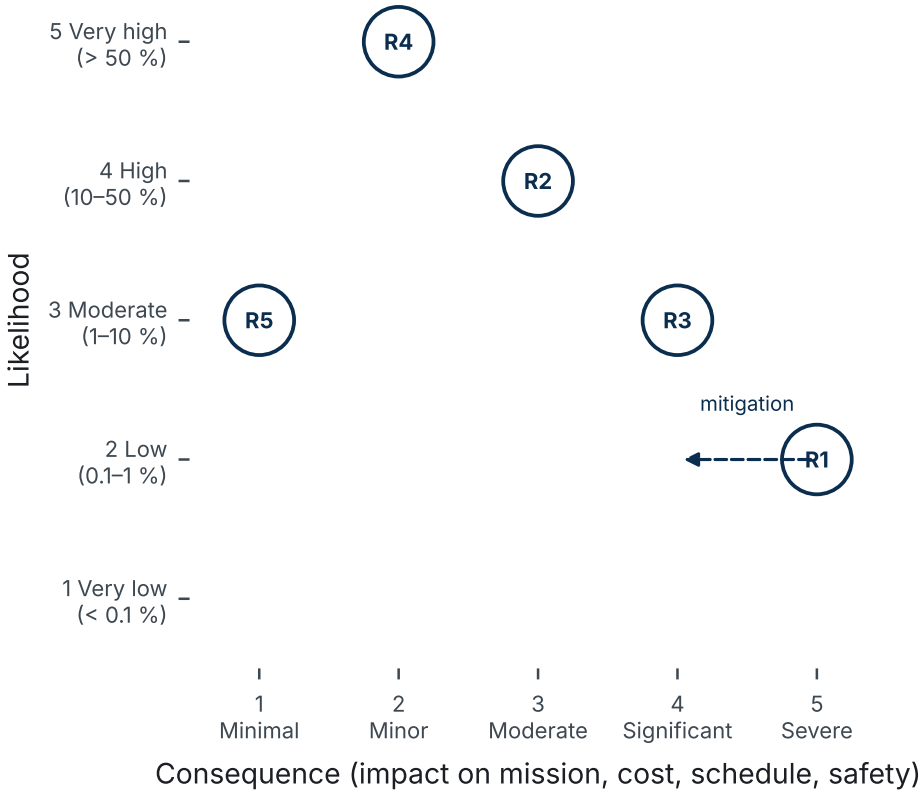


FIGURE 4.8 A 5 × 5 risk matrix. Each risk (R1–R5) is plotted by likelihood and consequence; mitigation actions aim to move risks out of the red zone before key reviews.

Programmes also distinguish **risk classes**. NASA's NPR 8705.4 defines Class A to Class D, plus a "do no harm" category for very low-cost CubeSats. Class A has the highest priority and lowest risk tolerance (e.g., JWST, crewed systems). Class D accepts higher risk for lower cost (many small science missions). The class determines parts quality, redundancy, review depth and test requirements. ESA and other agencies use similar tailoring.

4.9 Model-Based Systems Engineering

Traditionally, the system design lived in hundreds of documents, spreadsheets and drawings, with inconsistencies discovered late. **Model-based systems engineering (MBSE)** replaces documents with a single, connected digital model of requirements, structure, behaviour and parameters (Figure 4.9). The standard language is **SysML** (Systems Modeling Language); **SysML v2**, formally adopted by the Object Management Group in 2025, adds a precise textual notation and an API that lets engineering tools exchange data automatically. MBSE tools link to CAD, finite-element, thermal, orbit and cost models, so that a change in one place propagates everywhere — the foundation of the "digital thread" and **digital twin** concepts (Chapter 52).

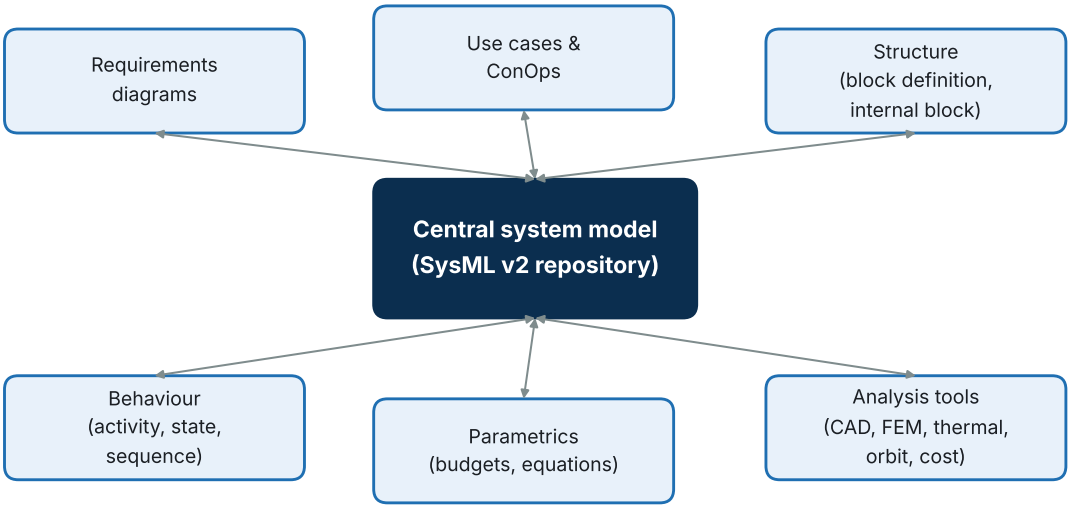


FIGURE 4.9 Model-based systems engineering links requirements, operations, structure, behaviour and parametric analyses in one authoritative system model, connected to specialist engineering tools.

AI IN PRACTICE — ASSISTANTS FOR SYSTEMS ENGINEERS

Large language models are increasingly used to check requirement quality (ambiguity, missing "shall", non-verifiable wording), to trace requirements to design elements, to search lessons-learned databases and to draft SysML models from text. Agencies including NASA and ESA are evaluating such tools under strict data-security rules. Chapter 52 discusses how AI outputs must themselves be verified before they enter a controlled baseline.

4.10 Verification and Validation

Verification asks "did we build the system right?" — does it meet its requirements? **Validation** asks "did we build the right system?" — does it meet the stakeholders' needs in its real environment? The V-model (Figure 4.10) shows how each level of design on the left is matched by a level of verification on the right.

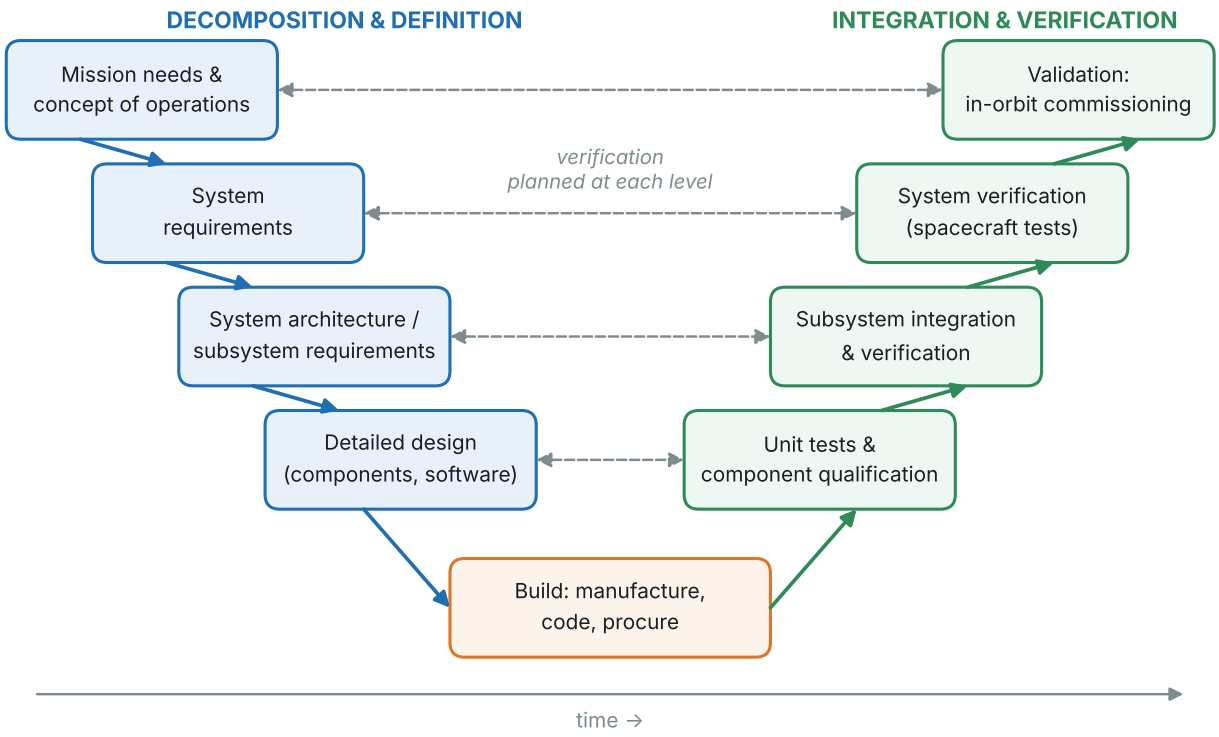


FIGURE 4.10 The systems engineering "V". Requirements are decomposed down the left side, the system is built at the bottom, and verification proceeds up the right side, each level verified against the requirements defined at the same level.

4.10.1 Verification Methods

Every requirement is assigned one or more verification methods:

- **Test** — measurement under controlled conditions (e.g., thermal-vacuum test of the battery).
- **Analysis** — calculation or simulation, used when testing is impossible or too costly (e.g., MMOD survival probability, 15-year radiation dose).
- **Inspection** — visual or dimensional check (e.g., connector labelling, workmanship).
- **Demonstration** — operation showing a function without detailed measurement (e.g., deploying an antenna with gravity offloading).

The **verification matrix** lists every requirement, its method, the verification event and the closure evidence. Closing it is a condition of flight readiness.

4.10.2 Model Philosophy

Projects choose how many physical models to build (Figure 4.11). A classic approach builds a **structural-thermal model (STM)** to verify mechanical and thermal design, and an **engineering model (EM)** to verify functions and interfaces. It also builds a **qualification model (QM)** tested to qualification levels (typically flight levels plus 3–6 dB for vibration and ±10 °C beyond predicted temperatures). Finally, it builds a **flight model (FM)** tested to acceptance levels. The cheaper **protoflight** approach tests the single flight model to qualification levels for acceptance durations, accepting a slightly higher risk of fatigue damage.

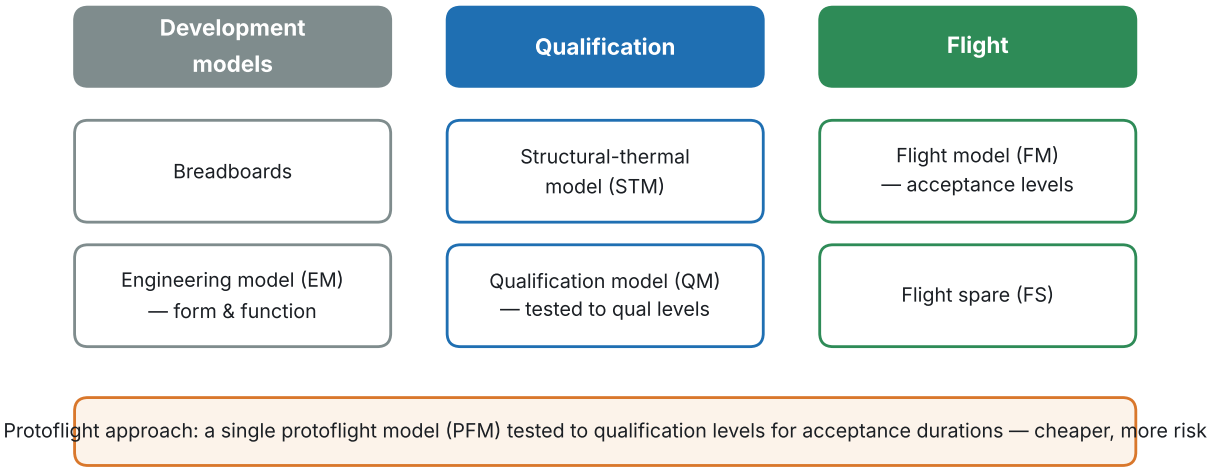
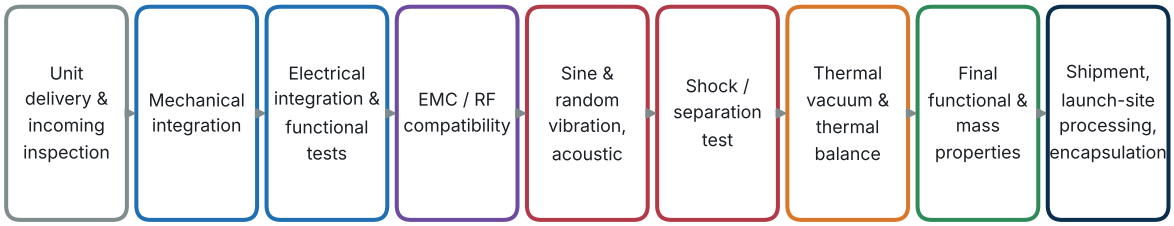


FIGURE 4.11 Model philosophies. Classic approaches qualify a dedicated model before building the flight unit; protoflight approaches qualify and accept the flight unit itself.

4.10.3 The Integration and Test Campaign

Assembly, integration and test (AIT) — called I&T in the US — brings the spacecraft together and exposes it to every environment it will see (Figure 4.12). Key tests include:

- **Functional and performance tests** at unit, subsystem and system levels, repeated after each environmental test (comprehensive performance tests, CPTs, and shorter "aliveness" tests).
- **Mechanical tests**: sine vibration, random vibration (for smaller spacecraft), acoustic (for large spacecraft with light panels), shock and separation tests, modal surveys, and static load tests on structural models.
- **Thermal tests**: thermal balance (to validate the thermal model) and thermal-vacuum cycling (to show the hardware works at the hot and cold extremes; typically 4–8 cycles for units).
- **Electromagnetic compatibility (EMC)**: conducted and radiated emissions and susceptibility (e.g., MIL-STD-461), plus RF compatibility with ground stations and the launcher.
- **Deployment tests** with gravity off-loading (zero-g rigs, balloons, air-bearing tables).
- **End-to-end tests** with the real ground segment and mission operations team, and **mission simulations** that rehearse launch and early operations.
- **Mass properties** (mass, centre of gravity, moments of inertia) and **alignment** measurements.



Functional ("aliveness") tests are repeated after every environmental test to detect damage; a complete spacecraft test campaign typically lasts 6–18 months.

FIGURE 4.12 A typical spacecraft assembly, integration and test (AIT) flow. Environmental tests are separated by functional tests that check for damage; the sequence is arranged so that the most severe mechanical tests precede the final thermal and functional verification.



FIGURE 4.13 The High Bay 1 cleanroom at NASA's Jet Propulsion Laboratory during assembly of the Mars 2020 Perseverance rover and its cruise stage. Cleanrooms control particulate and molecular contamination and, for Mars missions, biological contamination. Image: NASA/JPL (NASA/JPL-Caltech).



FIGURE 4.14 Europa Clipper in JPL's High Bay 1 after installation of its 3 m high-gain antenna in August 2023. The spacecraft was launched in October 2024 and will arrive at Jupiter in 2030. Image: NASA/JPL (NASA/JPL-Caltech).



FIGURE 4.15 A technician inside Sierra Space's Dream Chaser spaceplane preparing for vibration testing at NASA's Neil Armstrong Test Facility in Ohio, home to the world's most powerful spacecraft shaker system and a large vacuum chamber. Image: NASA/GRC (NASA/Jordan Salkin).

FAILURE LESSON — TEST AS YOU FLY

In 1999 NASA's Mars Polar Lander was probably lost because its software interpreted vibrations from the deployment of its landing legs as touchdown, shutting off the descent engines about 40 m above the surface. The fault could have been found by a system-level test of the leg deployment with the flight software — a test that was planned but, after a wiring error was corrected, not repeated. The same year, Mars Climate Orbiter was lost because one team used pound-force seconds and another newton-seconds for thruster impulse data. Both losses led to stricter end-to-end testing and interface verification across NASA.

4.11 Product Assurance and Safety

Product assurance (PA) — called mission assurance in the US — ensures that the hardware and software are built as designed and are reliable. It includes:

- **Quality assurance**: configuration management, nonconformance reporting, material review boards, inspections, workmanship standards (e.g., soldering to ECSS-Q-ST-70-08 or NASA-STD-8739).
- **Electrical, electronic and electromechanical (EEE) parts**: selection from approved lists, screening (burn-in, X-ray), radiation testing, counterfeit avoidance, derating (operating parts well below their maximum ratings).
- **Materials and processes**: outgassing, flammability, stress corrosion, atomic-oxygen resistance.
- **Software product assurance**: coding standards, independent verification and validation (IV&V), software criticality classes (e.g., NASA Class A–E; ECSS criticality A–D).

- **Safety**: hazard analyses for launch-site operations (pressurised systems, pyrotechnics, batteries, propellants), launch-range safety, crew safety for human-rated systems, and re-entry casualty risk. The probability of injuring someone on the ground from re-entering debris must normally be below 1 in 10 000.

4.12 Cost and Schedule

4.12.1 Estimating Cost

Space projects are estimated by three methods: **analogy** (scaling from a similar past project), **parametric** and **engineering build-up** (summing labour and materials for every work-package). The parametric method uses cost-estimating relationships (CERs) that relate cost to mass, power, complexity and heritage. Examples are NASA's Project Cost Estimating Capability (PCEC), the USAF Unmanned Space Vehicle Cost Model and the Aerospace Corporation's Small Satellite Cost Model. Mature projects use all three and reconcile the results.

Typical cost elements for a science mission are the spacecraft bus (25–35 %), instruments (20–35 %) and project management/systems engineering/mission assurance (10–15 %). The others are integration and test (5–10 %), launch (10–25 %), and operations and data analysis (10–20 %). Reserves of 25–50 % are typical before PDR.

4.12.2 Schedule

Space schedules are dominated by long-lead items (detectors, star trackers, reaction wheels, radiation-hardened processors, propellant tanks). They are also dominated by integration and test, which cannot be easily parallelised, and by launch windows. For planetary missions, launch windows occur only every one to two years. A slip of a few weeks in delivering a Mars mission can delay launch by 26 months. This happened to ESA's ExoMars rover, delayed from 2020 to 2022 and then to 2028 after cooperation with Russia ended.

4.13 Standards and Bodies of Knowledge

Systems engineering practice is codified in standards that capture lessons from decades of successes and failures:

TABLE 4.3 Key standards and handbooks for space systems engineering.

Standard / handbook	Publisher	Scope
NASA Systems Engineering Handbook (SP-2016-6105 Rev 2)	NASA	Processes, life cycle, technical management
NPR 7120.5 / NPR 7123.1	NASA	Programme/project management and systems engineering requirements
ECSS-E-ST-10C, ECSS-M-ST-10C, ECSS-Q-ST series	European Cooperation for Space Standardization	Systems engineering, project phasing, product assurance
ISO/IEC/IEEE 15288	ISO / IEEE	System life-cycle processes (all industries)
ISO 24113	ISO	Space debris mitigation requirements

Standard / handbook	Publisher	Scope
GSFC-STD-7000 (GEVS)	NASA Goddard	Environmental verification requirements for payloads and spacecraft
SMC-S-016 / SMC-S-010	US Space Force	Test requirements for launch, upper-stage and space vehicles; systems engineering
AIAA S-120A	AIAA	Mass properties control and growth allowance
CCSDS Blue Books	Consultative Committee for Space Data Systems	Communication and data-system standards (Chapter 7)
Space Mission Engineering: The New SMAD (Wertz, Everett & Puschell, 2011)	Microcosm Press	Classic engineering reference for mission design
INCOSE Systems Engineering Handbook (5th ed., 2023)	INCOSE	General systems engineering body of knowledge

KEY TAKEAWAYS

- Systems engineering optimises the whole mission across segments, subsystems, cost, schedule and risk; most cost is committed in early phases.
- Life cycles move from concept (Pre-A, A) through design (B, C) to integration, launch and operations (D, E) and closeout (F), gated by reviews such as SRR, PDR and CDR.
- Good requirements are necessary, unambiguous, verifiable and traceable; budgets track mass, power, Δv, data, links and pointing with phase-dependent margins.
- Reliability is improved by part quality, derating, redundancy and testing; FMECA identifies single points of failure; risks are managed on a likelihood–consequence matrix.
- MBSE (SysML v2) connects requirements, design and analysis in a single model; verification closes every requirement by test, analysis, inspection or demonstration.
- "Test as you fly": many famous failures (Mars Polar Lander, Mars Climate Orbiter) were interface or end-to-end test gaps, not exotic physics.

REVIEW QUESTIONS

1. Why is most of a project's life-cycle cost committed during the early phases, even though little money has been spent by then?
2. Outline the project phases from Pre-A to F and name the major reviews that close Phase B and Phase C.
3. Rewrite the requirement "The satellite shall have a good camera" so that it is necessary, unambiguous and verifiable.
4. Why do mass and power budgets carry larger margins early in a project? How do the margins change as the design matures?
5. Two identical units are flown in active parallel redundancy, each with a reliability of 0.90 over the mission. What is the reliability of the pair? What assumption does this answer rely on?
6. What is a technology readiness level? Why is TRL 6 normally expected by the preliminary design review?
7. Distinguish verification from validation and list the four verification methods, with an example of each.
8. Explain the principle "test as you fly" using one of the mission failures discussed in this chapter.

CHAPTER

5 Spacecraft Subsystems I: Structures, Thermal Control, Power and Mechanisms



Image: NASA/JPL (NASA/JPL-Caltech/Lockheed Martin)

Every spacecraft, whatever its mission, needs a skeleton to carry it through launch, a way to keep its parts at safe temperatures, a source of electrical power, and mechanisms that release, deploy and point. This chapter explains how structures are designed and built, how thermal control works with insulation, radiators, heat pipes and heaters, how solar arrays, batteries and radioisotope generators produce and store power, and how mechanisms are made reliable in vacuum.

IN THIS CHAPTER

5.1 The Spacecraft Bus
5.2 Structures
5.3 Thermal Control
5.4 Electrical Power
5.5 Mechanisms
5.6 Integrated Design: Example Bus

5.1 The Spacecraft Bus

A spacecraft consists of a **payload**, which performs the mission, and a **bus** (or **platform**), which supports the payload. The bus is divided into **subsystems** (Table 5.1). This chapter covers the "physical" subsystems — structures and mechanisms, thermal control, and electrical power. Chapter 6 covers attitude control, propulsion, data handling and flight software, and Chapter 7 covers communications.

TABLE 5.1 The subsystems of a spacecraft bus and their main functions.

Subsystem	Main functions	Key components
Structure and mechanisms	Carry loads, provide mounting, maintain alignment; release, deploy and point	Primary/secondary structure, adapters, hinges, release devices, drive mechanisms
Thermal control (TCS)	Keep every unit within its allowed temperature range	MLI blankets, coatings, radiators, heat pipes, heaters, louvres, thermostats
Electrical power (EPS)	Generate, store, regulate and distribute power	Solar arrays or RTGs, batteries, regulators, power distribution
Attitude determination and control (ADCS / GNC)	Know and control orientation (and position)	Sensors (star trackers, gyros, Sun sensors), actuators (wheels, thrusters, magnetorquers)
Propulsion	Change orbit, maintain orbit, control attitude, de-orbit	Thrusters, tanks, valves, feed system, electric thrusters
Command and data handling (C&DH / OBDH)	Process commands, run software, collect and store data	Onboard computer, mass memory, data buses, remote interface units
Telemetry, tracking and command (TT&C) and payload data	Link to ground for control and data	Transponders, antennas, high-rate transmitters, optical terminals
Flight software (FSW)	Autonomy, control laws, fault protection, payload operations	Operating system, application software, FDIR

5.2 Structures

5.2.1 Functions and Requirements

The structure carries all the spacecraft's equipment through launch, holds instruments in precise alignment in orbit, and often contributes to thermal control and radiation shielding. The **primary structure** carries the main loads from the launch-vehicle interface; **secondary structures** (brackets, panels, booms) carry individual units; **tertiary structures** include cable trays and small supports.

Structural design is driven by:

- **Strength** — the structure must not yield or break under launch loads multiplied by **factors of safety**. For flight hardware that is tested, these are typically 1.1–1.25 against yield and 1.25–1.4 against ultimate. They are higher for untested designs or pressure vessels.
- **Stiffness** — the fundamental natural frequencies must exceed launcher requirements (e.g., > 20 Hz axial and > 10 Hz lateral for many launchers) to avoid dynamic coupling. Stiffness, not strength, usually sizes the structure.
- **Stability** — dimensional stability under temperature changes and moisture release, which is critical for telescopes and antennas (alignment tolerances of micrometres).
- **Mass** — every kilogram costs launch capacity; structures are typically 15–25 % of dry mass.
- **Interfaces** — standard launch-vehicle adapters (e.g., 937 mm, 1 194 mm, 1 666 mm clamp-band diameters) and CubeSat deployer rails.

5.2.2 Configurations

Figure 5.1 shows the three most common configurations. Large GEO platforms use a **central thrust cylinder** (often carbon-fibre) that carries the propellant tanks and transfers loads to the adapter, with honeycomb **shear panels** and equipment panels. Small and medium satellites use

box structures of bolted honeycomb panels. Stations and very large antennas use **trusses**. Mega-constellation satellites such as Starlink use flat-packed designs that stack dozens of satellites inside a fairing, with the stack itself carrying launch loads.

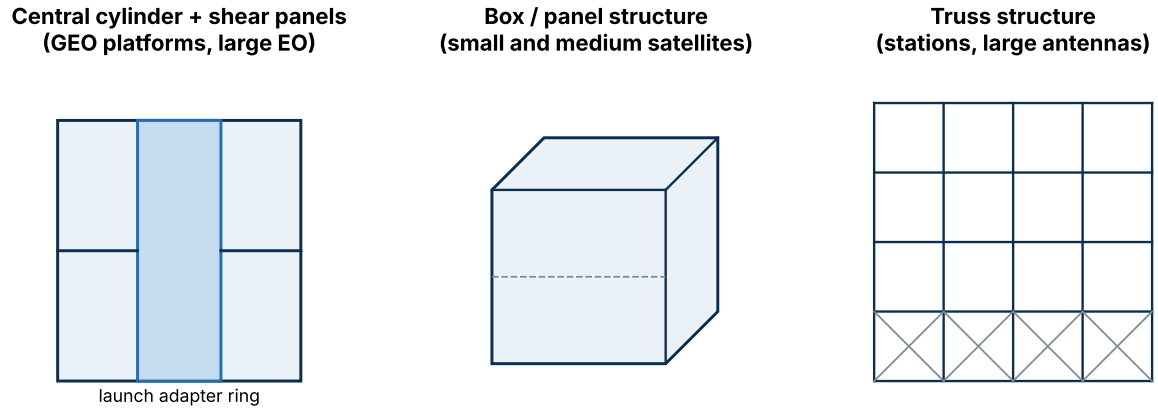


FIGURE 5.1 Common spacecraft structural configurations: central cylinder with shear panels, box/panel structure, and truss.

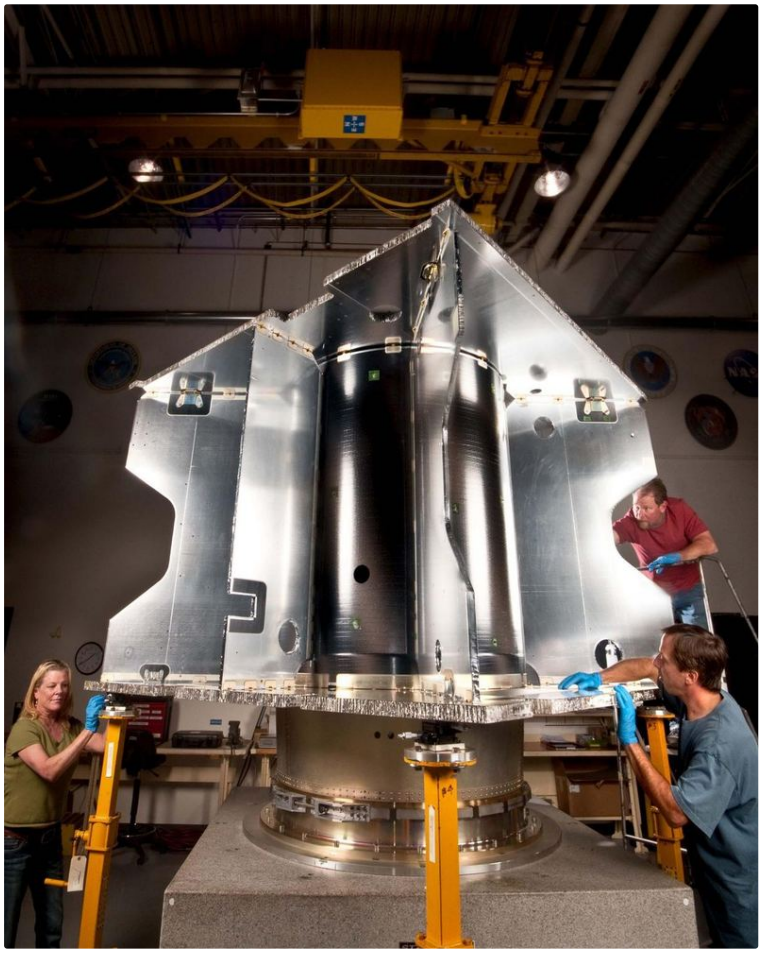
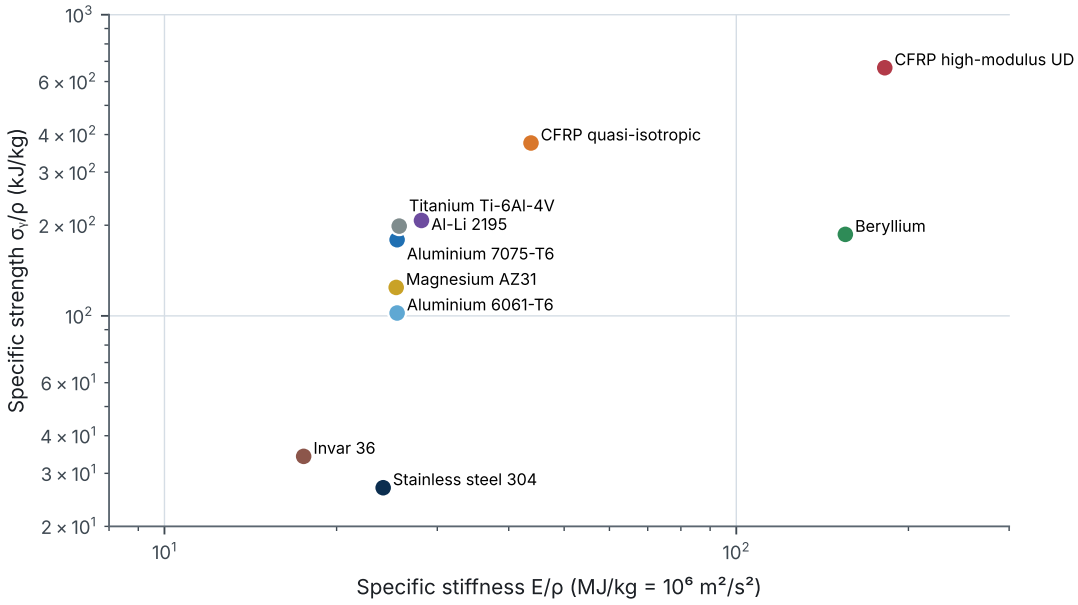


FIGURE 5.2 The primary structure of NASA's MAVEN Mars orbiter during assembly at Lockheed Martin. The central core carries the propellant tanks and transmits launch loads to the launch-vehicle adapter. Image: NASA/GSFC.

5.2.3 Materials

Aluminium alloys remain the workhorse of spacecraft structures — cheap, easy to machine and well characterised. Titanium is used for highly loaded fittings and for its low thermal conductivity. Carbon-fibre reinforced polymers (CFRP) give the highest stiffness per unit mass

and near-zero thermal expansion, ideal for optical benches and antenna reflectors. Beryllium offers extreme stiffness for lightweight mirrors (JWST), and Invar and Super-Invar give dimensional stability (Figure 5.3). Additive manufacturing (3-D printing) of titanium, aluminium and polymer parts is now routine, allowing topology-optimised brackets that are 30–60 % lighter than machined parts (Chapter 52).



Representative room-temperature properties; CFRP values are laminate-dependent. Beryllium: high stiffness for optics (JWST mirrors) but toxic to machine.

FIGURE 5.3 Specific stiffness and specific strength of common spacecraft structural materials. Composites and beryllium dominate where stiffness per kilogram matters most.

5.2.4 Sandwich Panels

The most common structural element is the **honeycomb sandwich panel** (Figure 5.4): two thin face sheets bonded to a lightweight honeycomb core. Like an I-beam, the faces carry bending stresses while the core keeps them apart and carries shear. Equipment is attached with **potted inserts**. Embedded heat pipes can be bonded inside panels to spread heat.

Bending stiffness grows with the square of the core thickness: doubling the core thickness makes the panel ~4× stiffer for a few % more mass.

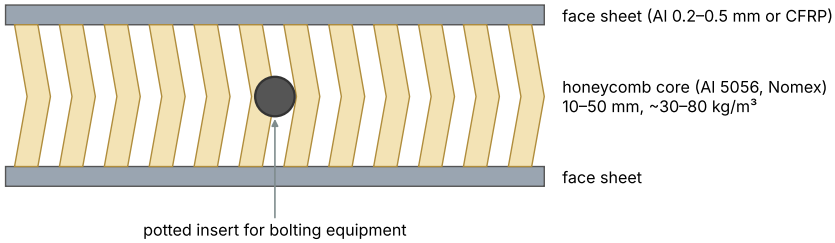


FIGURE 5.4 Construction of a honeycomb sandwich panel. Separating the face sheets with a light core increases bending stiffness roughly with the square of the core thickness.

5.2.5 Structural Analysis

Structures are analysed with the **finite-element method (FEM)**, using codes such as NASTRAN, Abaqus or ANSYS. The main analyses are:

1. **Modal analysis** — natural frequencies and mode shapes.

2. **Coupled loads analysis (CLA)** — the launcher provider combines the spacecraft model with the launcher model and simulates flight events (lift-off, transonic buffet, engine cut-off, stage separation) to predict the real loads at each interface.
3. **Stress analysis** — margins of safety for every part:

$$MS = \frac{\text{allowable stress}}{\text{FS} \times \text{applied stress}} - 1 \geq 0$$

4. **Thermo-elastic analysis** — distortion caused by temperature changes, critical for pointing and optical alignment.
5. **Fracture control** — for pressure vessels and fracture-critical parts, demonstration that a crack of detectable size will not grow to failure within four lifetimes.

5.3 Thermal Control

5.3.1 Why Temperature Matters

Each component has an **operational temperature range** and a wider **survival range** (Table 5.2). The thermal control subsystem (TCS) must keep every unit within these limits during all mission phases, despite the large external variations described in Chapter 2. These phases are launch, eclipse, hot and cold attitudes, safe mode, and end of life.

TABLE 5.2 Typical allowed temperature ranges of spacecraft components.

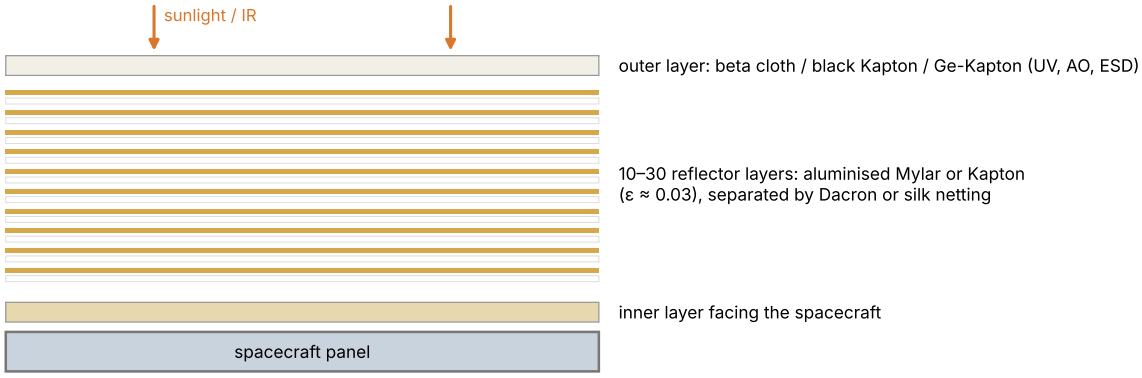
Component	Operating range (°C)	Notes
Li-ion batteries	0 to +30 (charging), −20 to +40 (discharge)	Narrow range; usually the TCS design driver
General electronics	−20 to +50	Reliability falls ≈ 2× per 10 °C rise
Solar arrays	−150 to +110	Efficiency falls as temperature rises
Hydrazine propellant and lines	+7 to +50	Must not freeze (+2 °C)
Reaction wheels / bearings	−10 to +40	Lubricant viscosity
Infrared detectors (HgCdTe)	−233 to −173 (40–100 K)	Cryocoolers or passive radiators
Optical structures	± 0.1–1 °C stability	Thermal distortion budget
CCD/CMOS visible detectors	−40 to +20	Cooler = lower dark current

5.3.2 Passive Thermal Control

Most thermal control is **passive** — it uses no moving parts or power:

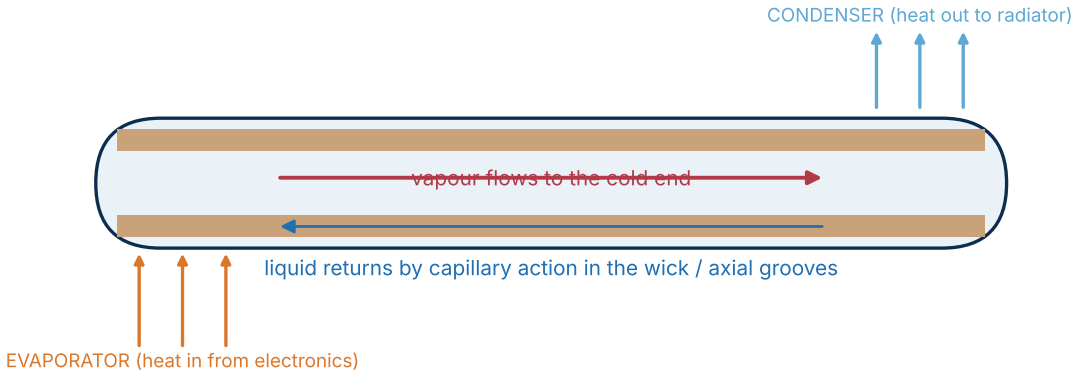
- **Surface finishes** set the balance between absorbed sunlight and emitted infrared (the α/ε ratio, Chapter 2). Radiators use white paints and optical solar reflectors (OSRs — silvered quartz tiles, α ≈ 0.08, ε ≈ 0.8). Low-emittance gold or aluminium surfaces are used where heat must be retained.
- **Multi-layer insulation (MLI)** blankets (Figure 5.5) are the "thermos flask" of spacecraft: many thin layers of aluminised polymer film separated by netting, each layer radiating only to its neighbours. MLI reduces radiative heat flow by a factor of 30–100 compared with a bare surface. The gold-coloured "foil" seen on most spacecraft is the outer Kapton layer of MLI.
- **Radiators** — surfaces with high ε and low α, oriented away from the Sun (and often Earth), that reject waste heat to space (Figure 5.7). The required area is $A = Q/(\varepsilon\sigma(T^4 - T_{sink}^4))$.

- **Conductive paths** — doublers, thermal straps, interface fillers and isolators control where heat flows.
- **Heat pipes** (Figure 5.6) — sealed tubes containing a working fluid (usually ammonia) that evaporates at the hot end and condenses at the cold end, with capillary wicks or grooves returning the liquid. They transport hundreds of watts across a spacecraft with only a few degrees of temperature drop. **Loop heat pipes** extend this over metres with flexible lines; **variable-conductance heat pipes** regulate temperature automatically using a non-condensable gas reservoir.



Effective emittance ε* ≈ 0.01–0.05 in practice (seams, vents and edges dominate); each layer radiates only to its neighbours.

FIGURE 5.5 Construction of a multi-layer insulation blanket. Effective emittances of 0.01–0.05 are typical for real installations, limited by seams, penetrations and compression.



Working fluid: ammonia (−60 to +100 °C), also methanol, water, propylene; conductance ~1000× a solid aluminium bar of the same mass

FIGURE 5.6 Principle of a heat pipe. Vapour carries heat from the evaporator to the condenser; capillary forces in the wick return the liquid without pumps.

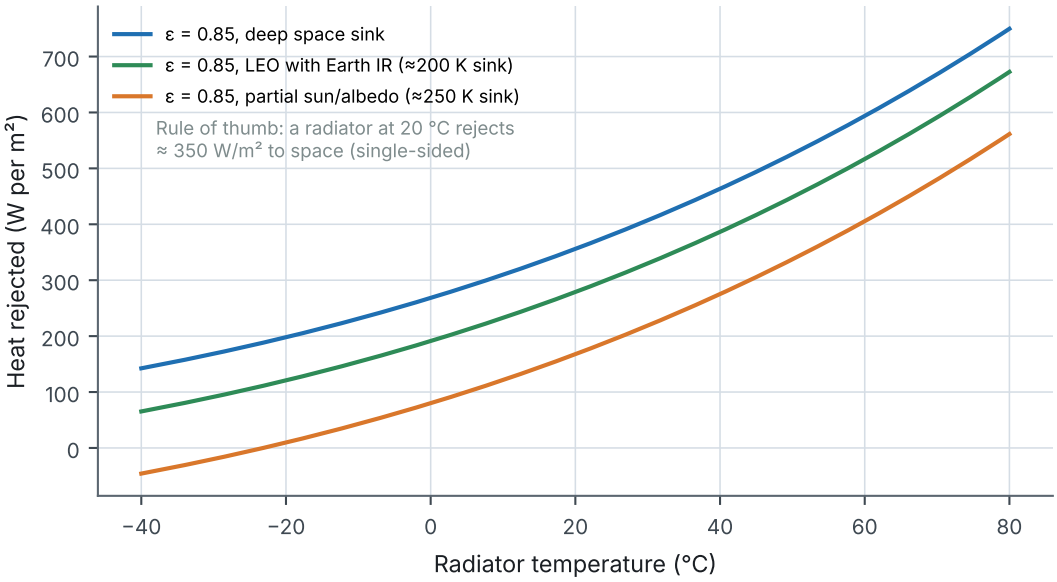


FIGURE 5.7 Heat rejected per square metre by a radiator with ε = 0.85 as a function of its temperature, for three effective sink temperatures. Warmer radiators are smaller, which is why electronics are allowed to run warm when possible.

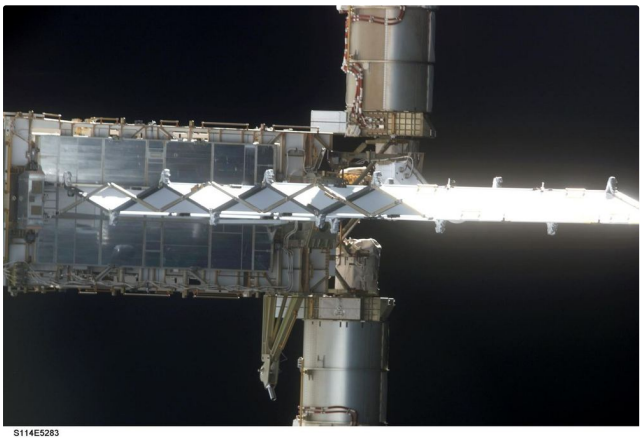


FIGURE 5.8 Radiator panels of the ISS P6 truss photographed during STS-114 in 2005. The station's external active thermal control system pumps ammonia through radiators totalling more than 1 000 m². Image: NASA/JSC.

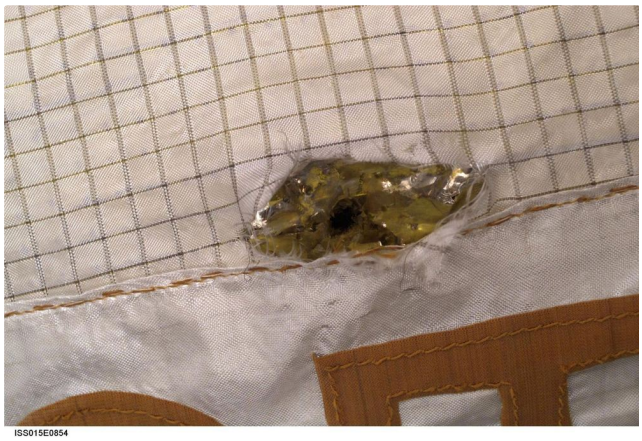


FIGURE 5.9 Damage to a multi-layer insulation blanket on the exterior of the ISS Zarya module, photographed during an Expedition 15 spacewalk in 2007. MLI is lightweight but fragile; outer layers are chosen to resist ultraviolet light, atomic oxygen and handling. Image: NASA/JSC.

5.3.3 Active Thermal Control

Where passive methods are not enough, **active** methods add control:

- **Electric heaters** (Kapton film heaters with thermostats or software control) keep batteries, propellant lines and electronics above minimum temperatures during eclipses and cold attitudes. Heater power is a significant item in the power budget, particularly in safe mode.
- **Louvres** — blinds driven by bimetallic springs that open over radiators when the spacecraft is hot and close when cold (used on Rosetta and many science missions).
- **Mechanically pumped fluid loops** — pumps circulate a coolant (e.g., ammonia, Freon, water in pressurised modules) to carry large heat loads to radiators. The ISS rejects up to 70 kW; Mars rovers use fluid loops to carry heat from their RTG.

- **Cryocoolers** — Stirling, pulse-tube, Joule–Thomson or turbo-Brayton coolers that cool infrared detectors to 4–150 K. JWST's MIRI instrument uses a pulse-tube/Joule–Thomson cryocooler to reach 6.7 K.
- **Thermoelectric coolers** (Peltier devices) for small detectors.

5.3.4 Thermal Analysis

The TCS is designed with a **thermal mathematical model**: the spacecraft is divided into hundreds or thousands of "nodes" connected by conductive and radiative couplings. Tools such as ESATAN-TMS, Thermal Desktop/SINDA and Systema compute the radiative view factors and orbital heat inputs, then solve the energy balance

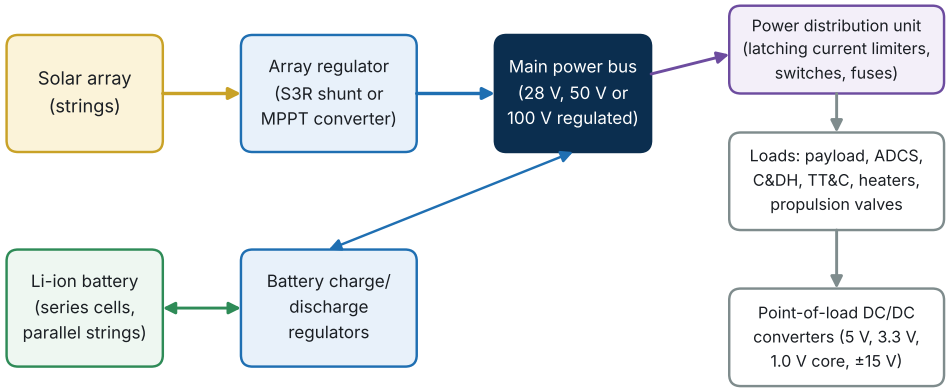
$$m_i c_i \frac{dT_i}{dt} = Q_{ext,i} + Q_{int,i} + \sum_j G_{ij}(T_j - T_i) + \sum_j \sigma R_{ij}(T_j^4 - T_i^4)$$

for each node i, over hot-case and cold-case scenarios (combinations of end-of-life coatings, maximum dissipation, solstice/equinox, attitudes). Predictions carry **uncertainty margins** (typically ±10–15 °C before thermal balance test correlation, ±5 °C after) so that the hardware is qualified beyond the predicted extremes.

5.4 Electrical Power

5.4.1 Power System Architecture

The electrical power subsystem (EPS) generates, stores, conditions and distributes power (Figure 5.10). A typical small satellite needs 50–500 W, and a large Earth-observation satellite 2–5 kW. A modern high-throughput GEO communications satellite needs 15–25 kW. The ISS needs about 120 kW (after the iROSA upgrade, up to 215 kW of array capacity).



Direct-energy-transfer (DET) systems pass array power straight to a regulated bus; MPPT systems extract maximum array power at the cost of converter losses.

FIGURE 5.10 Architecture of a spacecraft electrical power system. Solar array power is regulated onto a main bus that charges the battery and supplies the loads through a protected distribution unit and point-of-load converters.

Two families of regulation exist:

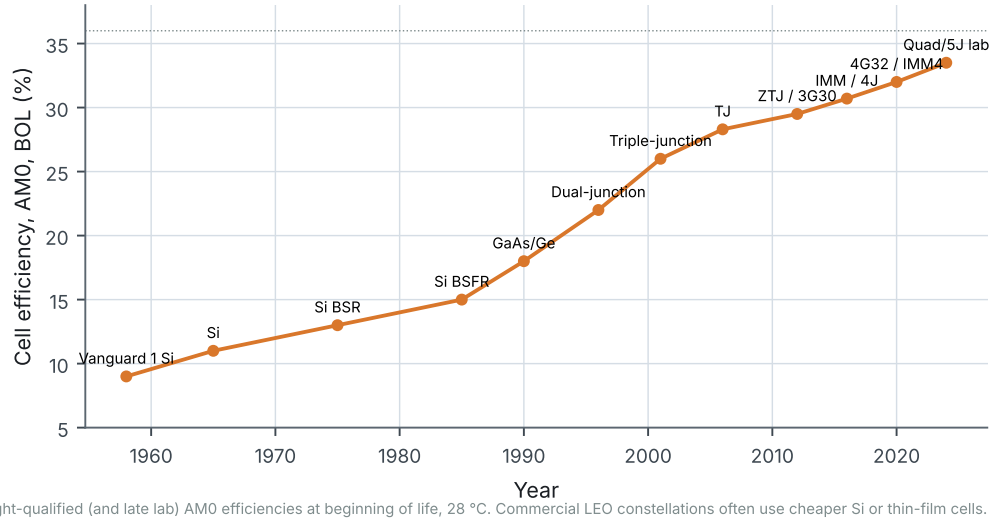
- **Direct energy transfer (DET)** — the array is connected directly (through shunt regulators) to the bus. Simple, efficient and common on GEO platforms (sequential switching shunt regulators, S3R) with regulated 50 V or 100 V buses.

- **Peak power tracking (MPPT)** — a DC/DC converter continuously adjusts the array operating point to its maximum power point (Figure 5.12). This extracts 10–25 % more power when the array is cold or illuminated at varying angles — common in LEO and on CubeSats.

Buses may be **regulated** (fixed voltage) or **unregulated** (voltage follows battery state of charge, typically 22–36 V). Larger spacecraft use higher voltages (100 V on GEO platforms, 120–160 V on the ISS) to reduce current and harness mass. The Lunar Gateway's Power and Propulsion Element was designed for 120 V to feed 50 kW-class Hall thrusters.

5.4.2 Solar Arrays

Photovoltaic **solar cells** convert sunlight into electricity. Early satellites used silicon cells of about 10 % efficiency. Modern space cells are **multi-junction** gallium-arsenide-based devices that stack three or four semiconductor layers, each absorbing a different part of the spectrum. They reach **30–33 %** efficiency at beginning of life under the space (AM0) spectrum (Figure 5.11). Cost-sensitive mega-constellations often use silicon or other lower-cost cells instead.



Approximate flight-qualified (and late lab) AM0 efficiencies at beginning of life, 28 °C. Commercial LEO constellations often use cheaper Si or thin-film cells.

FIGURE 5.11 Evolution of the efficiency of flight-qualified space solar cells. Multi-junction III–V cells now exceed 30 %.

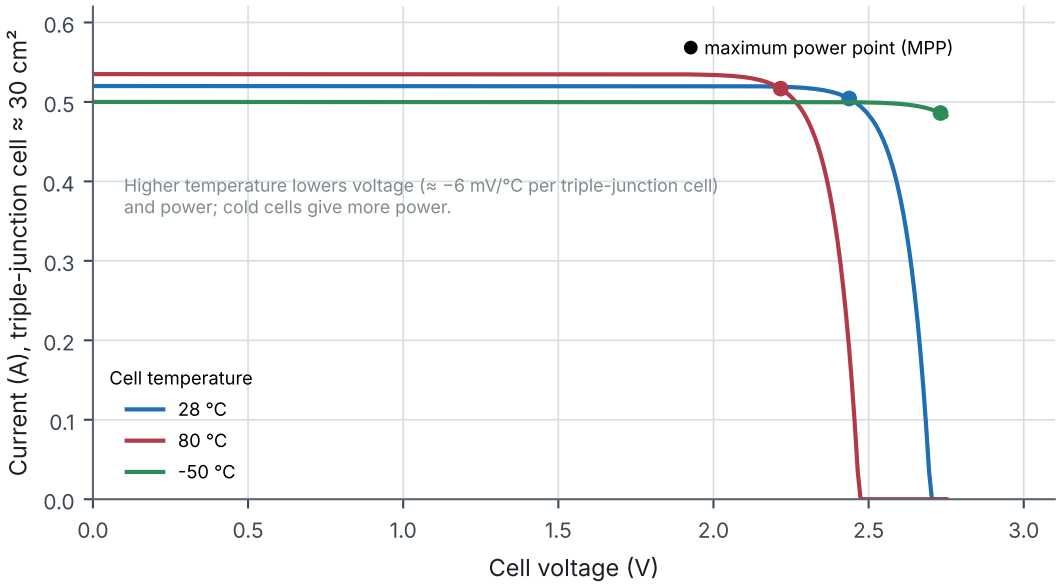


FIGURE 5.12 Current–voltage characteristics of a triple-junction cell at three temperatures. The maximum power point lies at the "knee" of the curve; heating reduces the voltage and the power.

Array output at end of life (EOL) is estimated as

$$P_{EOL} = S\,\eta_{cell}\,A_{cells}\,\cos\theta\;\cdot\;F_{packing}\,F_{temp}\,F_{assembly}\,(1-d)^{L}$$

where θ is the Sun incidence angle, and F terms account for packing factor (≈ 0.85–0.9), temperature and assembly losses. The term d is the annual degradation from radiation, UV and micrometeoroids (typically 1–3 % per year in LEO; more in MEO). Arrays are sized for EOL conditions at the worst season (aphelion in July: 1 316 W/m^2).

Array types:

- **Body-mounted** cells on the spacecraft surface — simple, used on spinning and very small satellites.
- **Rigid deployable panels** — honeycomb panels hinged together, rotated to track the Sun by a **solar array drive assembly (SADA)**.
- **Flexible / roll-out arrays** — thin blankets on booms. The **Roll-Out Solar Array (ROSA)** and its ISS version **iROSA** (≈ 20 kW each, six installed 2021–2023, more delivered later) are lighter and more compact than rigid panels.
- **Concentrator arrays** — lenses or mirrors focus sunlight on fewer cells (e.g., Deep Space 1).

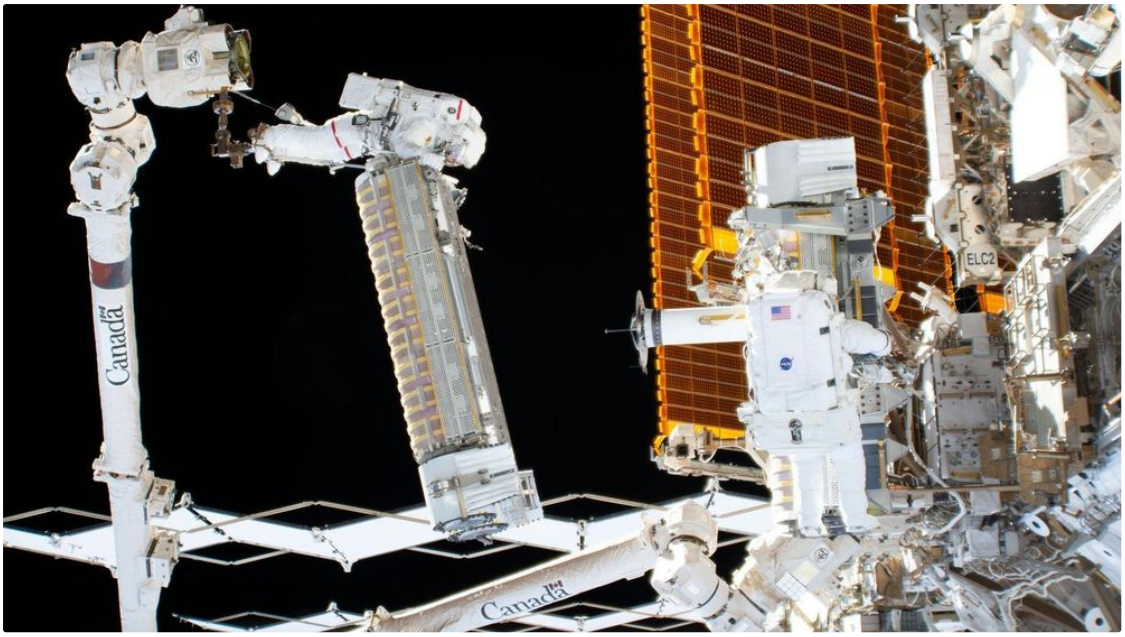


FIGURE 5.13 Spacewalkers install an ISS Roll-Out Solar Array (iROSA) in December 2022. Each iROSA unrolls from a compact canister into a 6 × 19 m flexible wing over the station's original arrays. Image: NASA/JSC.

Solar power falls with the square of the distance from the Sun (Figure 5.14). NASA's Juno is the farthest solar-powered spacecraft (Jupiter, 5.2 AU), using three 9-metre wings that produce about 14 kW at Earth but only about 500 W at Jupiter. ESA's JUICE and NASA's Europa Clipper are also solar-powered at Jupiter, with arrays of 85 m² and about 100 m² respectively. Lucy uses two 7.3 m circular "UltraFlex" arrays to reach Jupiter's Trojan asteroids.

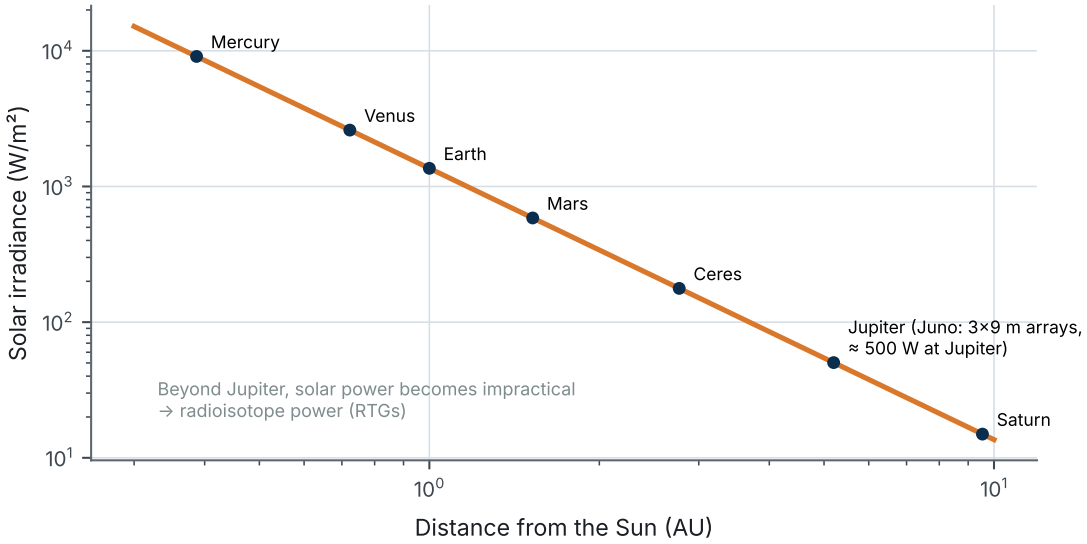


FIGURE 5.14 Solar irradiance versus distance from the Sun. At Jupiter sunlight is 27 times weaker than at Earth; at Saturn about 90 times.

5.4.3 Batteries

Batteries supply power during eclipses, launch, peak loads and emergencies. **Lithium-ion** cells have replaced older nickel-cadmium (Ni-Cd) and nickel-hydrogen (Ni-H_2) technologies since the early 2000s because of their two to four times higher specific energy (Figure 5.15). Battery packs

are built from many small commercial-format cells (e.g., 18650 cylindrical cells) or larger prismatic cells, arranged in series strings (for voltage) and parallel strings (for capacity and redundancy).

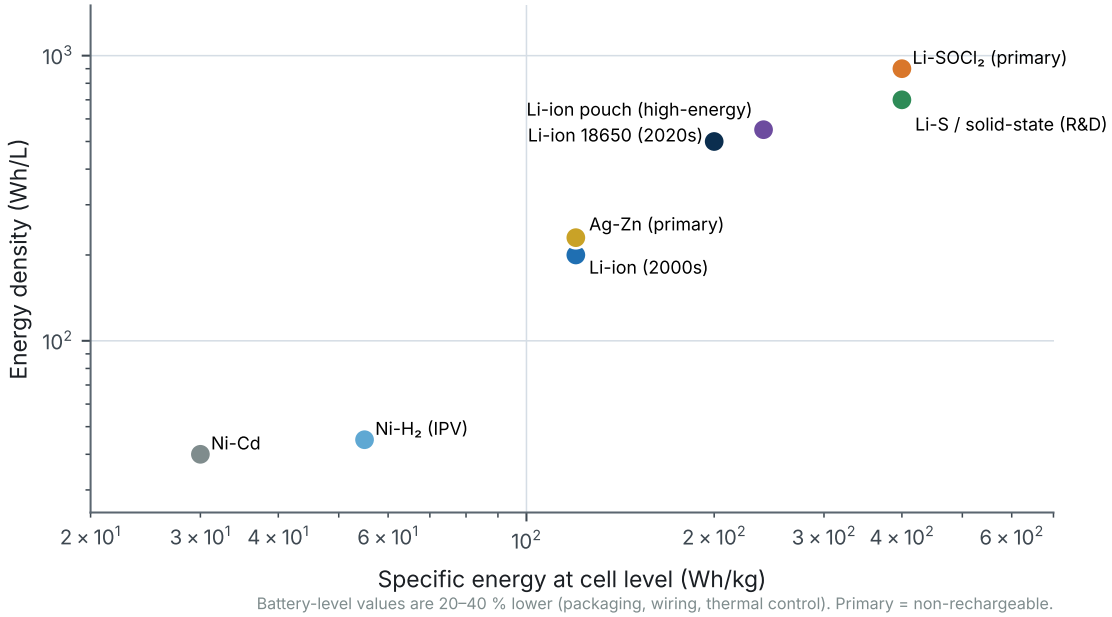


FIGURE 5.15 Specific energy and energy density of space battery chemistries. Li-ion cells made rechargeable storage several times lighter than nickel-based batteries.

Battery life depends strongly on the **depth of discharge (DoD)** in each cycle (Figure 5.16). A LEO satellite cycles about 5 500 times per year, so its batteries are typically limited to 15–30 % DoD. A GEO satellite cycles only 90 times per year (during two eclipse seasons) and can use 60–80 % DoD. Battery capacity is sized as

$$C = \frac{P_{eclipse}\, t_{eclipse}}{\mathrm{DoD}\, \eta_{discharge}\, N_{strings,\,usable}/N_{strings}}$$

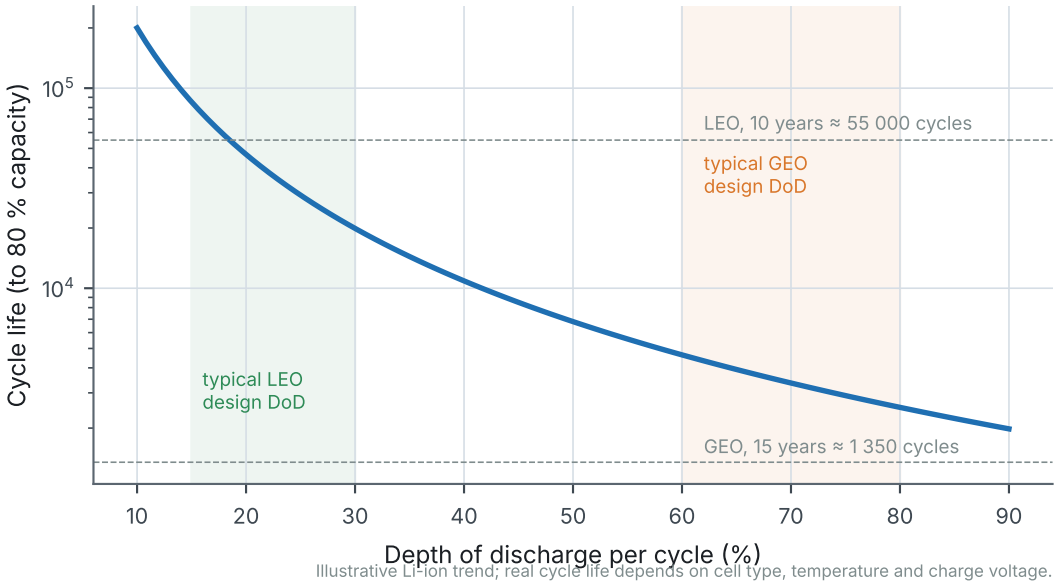


FIGURE 5.16 Illustrative cycle life of Li-ion cells versus depth of discharge. Low-orbit satellites use shallow discharges because they cycle many thousands of times.

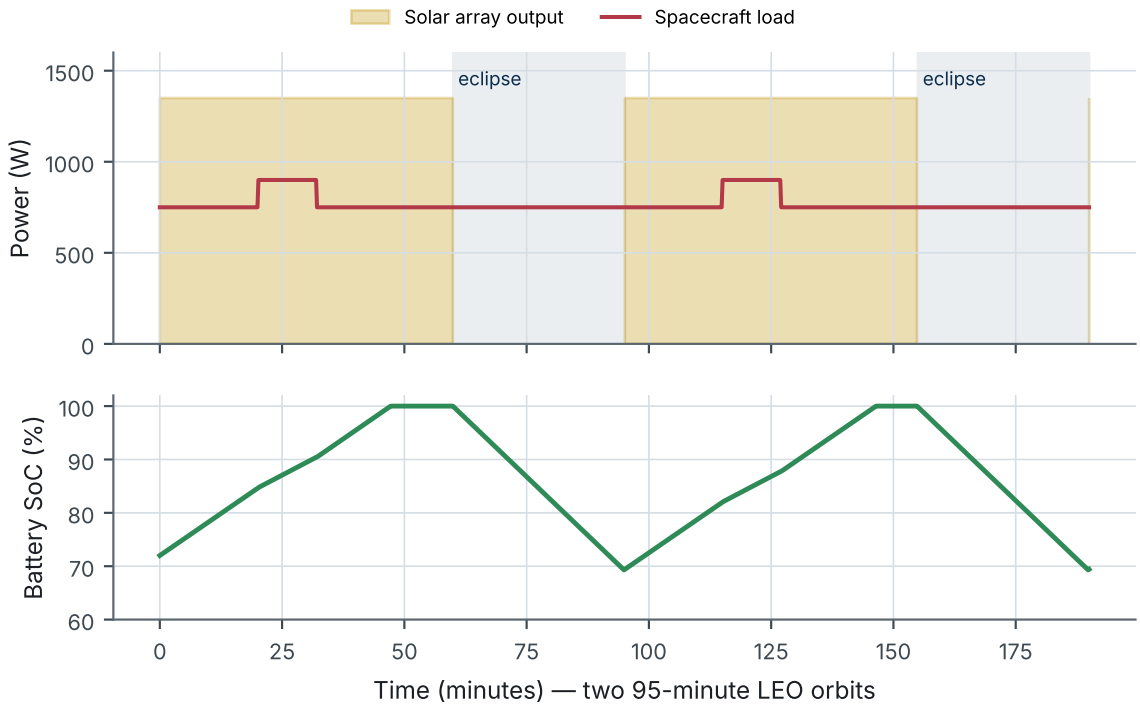


FIGURE 5.17 Power generation, load and battery state of charge over two orbits for a LEO satellite. The battery charges in sunlight and supplies the load in eclipse; the array must be sized to recharge it within the sunlit part of each orbit.

WORKED EXAMPLE — SIZING A LEO POWER SYSTEM

A satellite in a 600 km orbit (period 96.7 min, maximum eclipse 35.5 min) needs 400 W continuously.

Battery: energy in eclipse = 400 W × 35.5/60 h = 237 Wh. With 25 % DoD and 95 % discharge efficiency, plus one failed string out of five: $C = 237/(0.25 \times 0.95 \times 0.8) = 1\,250$ Wh — about 7 kg of Li-ion cells at 180 Wh/kg cell level.

Array: in sunlight (61.2 min) the array must supply the load and recharge 237 Wh / (charge efficiency 0.9) = 263 Wh, so it needs $400 + 263/(61.2/60) = 658$ W at EOL. Assume 30 % cells, a packing factor of 0.88, 0.85 for temperature and other losses, 10 % degradation over 7 years and an average cosine loss of 0.9. Then each square metre gives $1\,361 \times 0.30 \times 0.88 \times 0.85 \times 0.9 \times 0.9 = 247$ W. So about **2.7 m²** of array is needed, plus margin.

5.4.4 Radioisotope Power

Beyond Jupiter, in permanent shadow, or on planetary surfaces with long nights and dust, spacecraft use **radioisotope power systems (RPS)**. A **radioisotope thermoelectric generator (RTG)** converts the heat of plutonium-238 decay into electricity using thermocouples (Figure 5.18). Pu-238 is used because it emits mainly alpha particles (easily shielded) and has an 87.7-year half-life. RTGs have powered Pioneer 10/11, Voyager 1/2 (still operating after 49 years, though at reduced power), Galileo, Cassini, New Horizons, Curiosity and Perseverance; NASA's Dragonfly rotorcraft (launch 2028) will use one on Titan.

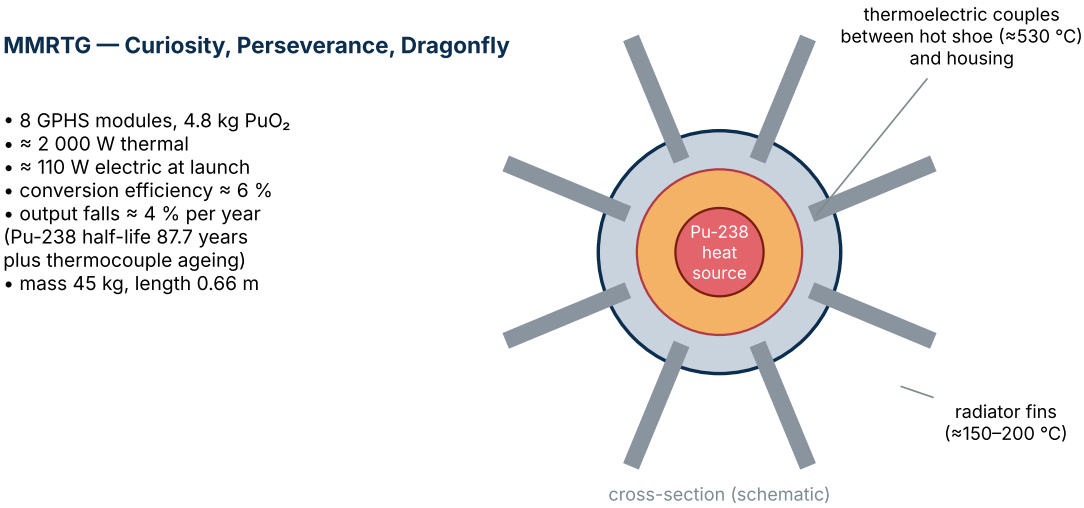


FIGURE 5.18 Schematic of a Multi-Mission RTG. Heat from plutonium dioxide fuel flows through thermoelectric couples to radiator fins; about 6 % of it becomes electricity.

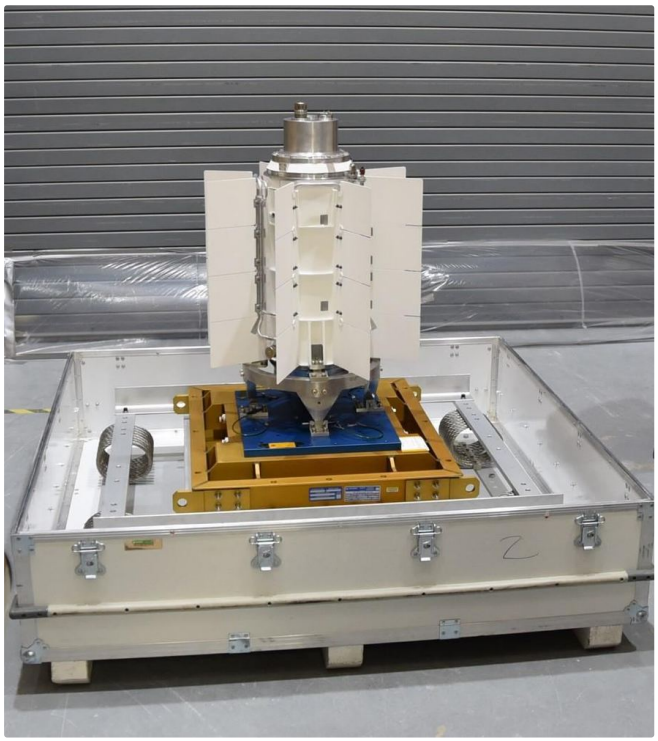


FIGURE 5.19 The Multi-Mission Radioisotope Thermoelectric Generator (MMRTG) for the Mars 2020 Perseverance rover during testing at Idaho National Laboratory. Image: NASA/JPL (NASA/JPL-Caltech).

Smaller **radioisotope heater units (RHUs)**, each producing about 1 W of heat from 2.7 g of PuO_2, keep components warm on many missions (e.g., Mars rovers Sojourner–Opportunity, Chang'e 3/4 landers, which used Chinese-made RHUs). Russia and China use Pu-238 and Po-210 heaters. Future systems aim at higher efficiency: Stirling convertors (≈ 25–30 %), the next-generation RTG with skutterudite couples, and americium-241 RTGs under development in Europe. For high power, **fission reactors** are being developed for lunar bases and nuclear electric propulsion (Chapter 42). Examples are NASA's Kilopower/KRUSTY demonstration (2018) and the Fission Surface Power project. That project targets a lunar reactor of at least 40 kW (and, under 2025 direction, 100 kW-class) by about 2030.

5.4.5 Power Distribution and Protection

The **power distribution unit (PDU)** switches power to each load through **latching current limiters (LCLs)** or fuses, so that a short circuit in one unit cannot collapse the main bus. Harnesses are designed with derated wire gauges, twisted pairs and shielding for electromagnetic compatibility, and separated from RF and pyrotechnic lines. **Point-of-load converters** provide the low voltages (3.3 V, 1.8 V, 1.0 V core) that modern processors need. Fault protection includes undervoltage load shedding: when the battery falls below a threshold, non-essential loads are switched off and the spacecraft enters a power-safe mode.

5.5 Mechanisms

Mechanisms are among the most failure-prone items on spacecraft. They must work after launch vibration, in vacuum and over temperature extremes, often after years of storage. And many can be tested only a few times on the ground. The main families are summarised in Figure 5.20.

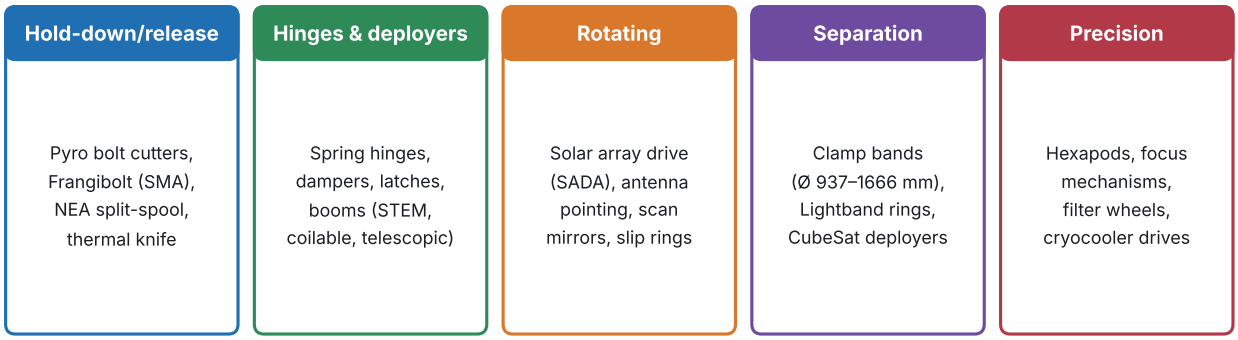


FIGURE 5.20 Families of spacecraft mechanisms and typical examples.

- **Hold-down and release mechanisms (HDRMs)** lock deployables during launch and release them in orbit. Pyrotechnic devices (bolt cutters, separation nuts) are fast and reliable but produce shock; non-explosive actuators (NEA split-spool devices, Frangibolt shape-memory-alloy actuators, thermal knives that cut Dyneema cords on CubeSats) produce little shock.
- **Hinges and deployment devices** — spring-driven hinges with dampers or latches; booms such as coilable lattice masts, STEM/bi-STEM tubes and telescopic masts for magnetometers, antennas and solar sails.
- **Rotating mechanisms** — solar array drives, antenna pointing mechanisms, scan mirrors, reaction wheel bearings; they need space-qualified lubricants and transfer power and signals across rotating joints through slip rings or flexible cables.
- **Separation systems** — clamp bands, motorised Lightband rings and CubeSat deployers (e.g., the standard 3U/6U/12U deployers) separate the spacecraft from the launcher.
- **Precision mechanisms** — hexapods, focus and filter mechanisms, mirror actuators (JWST has 132 actuators to align its 18 primary mirror segments to tens of nanometres).

Design rules for reliable mechanisms include a **torque margin** of at least 2×. ECSS-E-ST-33-01C requires the motorisation torque to exceed twice the sum of resistive torques plus inertia terms, with higher factors early in design. Other rules are redundant springs or motor windings, and

avoidance of cold welding by dissimilar materials and coatings. The rules also call for life testing to several times the required number of cycles, and careful “test as you fly” deployment testing with gravity compensation.

FAILURE LESSON — GALILEO’S HIGH-GAIN ANTENNA (1991)

NASA’s Galileo Jupiter probe carried a 4.8 m umbrella-like high-gain antenna. When commanded to open in April 1991, three of its 18 ribs stuck. This was probably because the lubricant on the rib restraint pins had been lost during repeated truck transports, while the launch was delayed after the Challenger accident. Engineers could never free it. Through heroic software and ground-station upgrades — new data compression and coding, arraying of DSN antennas — the mission still returned about 70 % of its science objectives. It used the low-gain antenna at about 160 bit/s instead of 134 kbit/s. Lessons include lubricant life, the effect of ground handling, and the value of in-flight reprogrammability.

5.6 Integrated Design: Example Bus

To see how these subsystems combine, Table 5.3 lists the main characteristics of a representative 500 kg-class Earth-observation bus.

TABLE 5.3 Representative characteristics of a 500 kg-class LEO Earth-observation satellite bus (illustrative).

Parameter	Value
Launch mass / payload mass	500 kg / 150 kg
Structure	Aluminium honeycomb box, CFRP payload bench, 1 194 mm interface ring
Orbit	600 km sun-synchronous, 10:30 local time
Power	Two deployable GaAs triple-junction wings, 1.2 kW BOL / 0.95 kW EOL; 28 V unregulated bus; 60 Ah Li-ion battery
Thermal	MLI, white-paint and OSR radiators, heat pipes on payload panel, 120 W of thermostatic heaters
ADCS	3-axis stabilised, 4 reaction wheels, 3 magnetorquers, 2 star trackers, fibre-optic gyro, GNSS receiver; pointing accuracy 0.02°
Propulsion	Hydrazine monopropellant, 4 × 1 N thrusters, 40 kg propellant (or Hall thruster, 15 kg xenon/krypton)
Data handling	Radiation-tolerant processor, 4 Tbit mass memory, SpaceWire payload interface
Communications	S-band TT&C (2 kbit/s up, 64 kbit/s down); X-band payload downlink 800 Mbit/s; optional optical terminal
Design life	7 years

KEY TAKEAWAYS

- Structures are usually sized by stiffness, not strength; honeycomb sandwich panels, aluminium, titanium and CFRP dominate; FEM and coupled-loads analysis predict the real launch loads.
- Thermal control is mostly passive (coatings, MLI, radiators, heat pipes), supplemented by heaters, louvres, pumped loops and cryocoolers; models are correlated in thermal-balance tests.
- Multi-junction solar cells reach 30–33 % efficiency; arrays are sized for end-of-life, worst-case conditions; Li-ion batteries are sized by eclipse energy and allowable depth of discharge.
- Beyond Jupiter or on dark, dusty surfaces, RTGs (≈ 6 % efficient) provide power for decades; fission power is being developed for the Moon.
- Mechanisms are high-risk items: use torque margins, redundancy, proper lubrication and realistic deployment testing.

REVIEW QUESTIONS

1. Why are spacecraft structures usually sized by stiffness rather than strength? Why do launch vehicles specify minimum fundamental frequencies?
2. Explain why honeycomb sandwich panels are so widely used in spacecraft structures.
3. A radiator with emittance 0.85 views deep space and must reject 500 W at 290 K. Estimate its area, ignoring absorbed environmental heat.
4. Describe four passive and two active thermal control techniques, and give a typical application of each.
5. A LEO spacecraft needs 800 W during a 35-minute eclipse. Size the battery capacity for a 30 % depth of discharge, ignoring losses.
6. Why are solar arrays sized for end-of-life conditions? List three factors that reduce array output during a mission.
7. Why do missions beyond Jupiter use radioisotope power, and what are its main drawbacks?
8. Why are mechanisms considered high-risk items, and how is that risk reduced?

CHAPTER

6 Spacecraft Subsystems II: Attitude Control, Propulsion, Data Handling and Flight Software



Image: NASA/GRC

A spacecraft must know which way it is pointing and turn when needed, change its orbit, and think for itself between ground contacts. This chapter explains attitude determination and control - sensors, actuators, disturbance torques and control laws; spacecraft propulsion - cold gas, monopropellant, bipropellant and electric thrusters, feed systems and propellant management; command and data handling - onboard computers, memories, buses and radiation-tolerant electronics; and flight software, autonomy and fault protection.

IN THIS CHAPTER

6.1 Attitude Determination and Control
6.2 Spacecraft Propulsion
6.3 Command and Data Handling
6.4 Flight Software

6.1 Attitude Determination and Control

6.1.1 What the ADCS Does

Attitude is the orientation of the spacecraft in space. Almost every mission needs attitude control: cameras must point at their targets, antennas at the Earth, solar arrays at the Sun, radiators away from it, and thrusters along the desired Δv direction. The **attitude determination**

and control subsystem (ADCS) is often combined with orbit control as **guidance, navigation and control (GNC)**. It measures the attitude with sensors and compares it with the desired attitude. It then applies torques with actuators in a feedback loop (Figure 6.1).

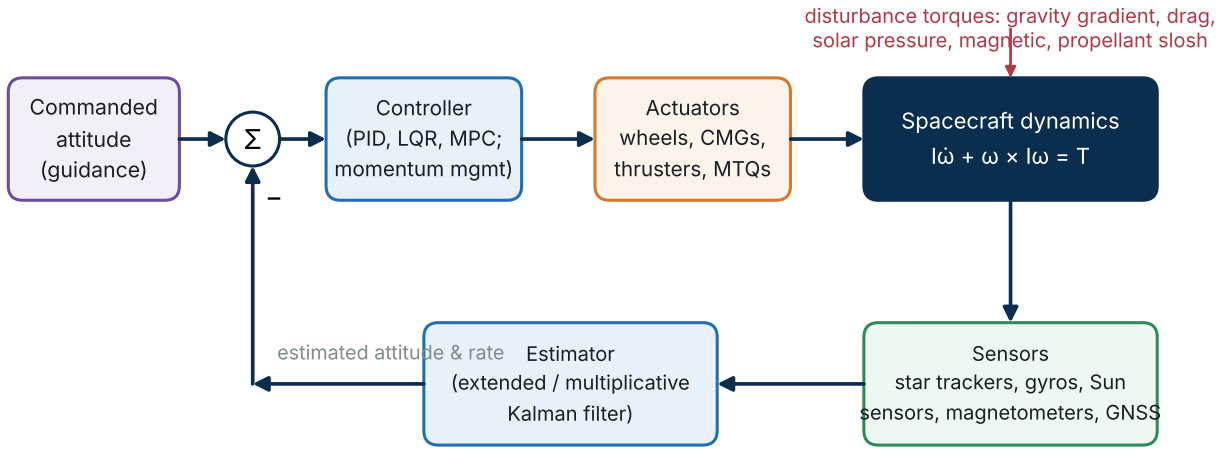


FIGURE 6.1 The attitude control loop. Sensors feed an estimator (usually a Kalman filter); the controller compares estimated and commanded attitude and commands actuators to produce torques that act, together with disturbances, on the spacecraft dynamics.

Pointing requirements range enormously:

- **Coarse**: Sun-pointing for power, ±5–10° (safe modes, simple CubeSats).
- **Moderate**: Earth-pointing communications satellites, ±0.05–0.1°.
- **Fine**: Earth-observation imagers, 0.01–0.02° (tens of arcseconds) with stability of a few µrad/s during imaging.
- **Very fine**: space telescopes — Hubble holds its pointing to 0.007 arcsec (7 milliarcseconds) for hours; JWST about the same with a fine-steering mirror.

Requirements distinguish **accuracy** (how close to the target direction) and **knowledge** (how well the attitude is known after the fact). They also distinguish **stability** or **jitter** (how much it moves over a short time), and **agility** (how fast it can slew from target to target).

6.1.2 Stabilisation Methods

- **Spin stabilisation** — the spacecraft spins (typically 5–60 rpm) and behaves like a gyroscope, keeping its spin axis fixed in inertial space. Simple and robust; used by early satellites, many sounding rockets, the Juno spacecraft (spinning at 2–5 rpm) and some upper stages. **Dual-spin** satellites combine a spinning rotor with a de-spun platform for antennas.
- **Gravity-gradient stabilisation** — an elongated spacecraft tends to align its long axis with the local vertical because the near end feels slightly more gravity than the far end. Accuracy is a few degrees; used on early navigation satellites and some small satellites with deployable booms.
- **Passive magnetic stabilisation** — permanent magnets align a CubeSat with Earth's field, with hysteresis rods for damping.

- **Three-axis stabilisation** — active control of all three axes with wheels or thrusters. This is the standard for modern spacecraft.

6.1.3 Rigid-Body Dynamics

The rotational motion of a rigid spacecraft is governed by **Euler's equation**:

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \mathbf{T}_{control} + \mathbf{T}_{disturbance} - \dot{\mathbf{h}}_{wheels} - \boldsymbol{\omega} \times \mathbf{h}_{wheels}$$

where $\mathbf{I}$ is the inertia tensor, $\boldsymbol{\omega}$ the angular velocity, $\mathbf{T}$ the external torques and $\mathbf{h}$ the angular momentum stored in internal wheels. Attitude itself is represented with **direction-cosine matrices**, **Euler angles** (intuitive but singular at certain angles — "gimbal lock"), or **quaternions** (four parameters, no singularities, the standard in flight software).

6.1.4 Disturbance Torques

Small external torques continually push the spacecraft away from its desired attitude (Figure 6.2):

- **Gravity gradient** — $T = \frac{3\mu}{2r^3}\left|I_z - I_y\right| \sin 2\theta$, important for large, elongated spacecraft in LEO.
- **Aerodynamic** — drag acting at a centre of pressure offset from the centre of mass; dominant below about 500 km.
- **Solar radiation pressure** — the same offset effect for sunlight; dominant in high orbits and deep space.
- **Magnetic** — interaction of residual magnetic dipoles (from currents and magnetised materials) with Earth's field.
- **Internal** — propellant slosh, moving mechanisms, flexible appendages, outgassing and thruster misalignment.

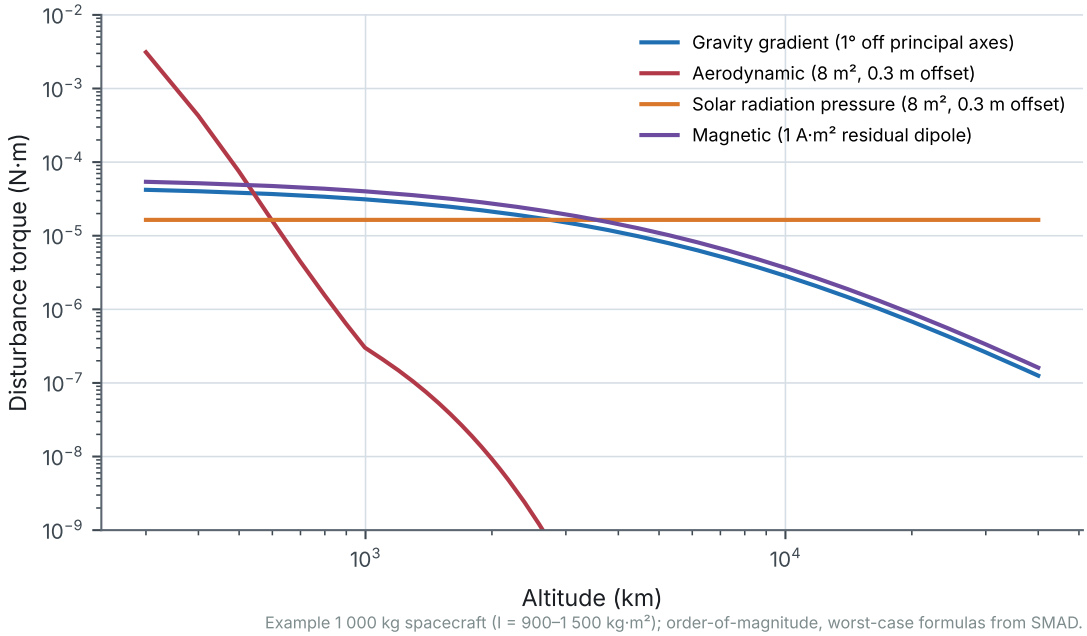


FIGURE 6.2 Order-of-magnitude disturbance torques on a 1 000 kg spacecraft as a function of altitude. Aerodynamic torque dominates in low orbits; gravity-gradient and magnetic torques in mid-LEO; solar pressure in high orbits.

Because these torques act continuously in one direction, momentum accumulates in the reaction wheels until they approach their maximum speed; it must periodically be "dumped" using magnetorquers or thrusters (**momentum management**).

6.1.5 Attitude Sensors

TABLE 6.1 Common attitude sensors.

Sensor	Measures	Typical accuracy	Notes
Coarse Sun sensor	Sun direction (photodiode currents)	1–5°	Cheap, robust; used for safe mode
Fine (digital) Sun sensor	Sun direction (slit or pinhole on detector array)	0.01–0.1°	Field of view up to ±64°
Earth (horizon) sensor	Direction of Earth's IR limb (14–16 µm CO_2 band)	0.03–0.3°	Classic for GEO and LEO Earth-pointing
Magnetometer	Local magnetic field vector	0.5–5° (after comparison with field model)	Also used for magnetorquer control
Star tracker	Full 3-axis attitude by matching stars to a catalogue	1–10 arcsec (cross-boresight); ≈ 5–40 arcsec (roll)	The precision sensor of modern spacecraft; blinded by Sun/Moon/Earth
Gyroscope / IMU	Angular rate (integrates to attitude change)	Drift from 0.001°/h (HRG, FOG) to 10°/h (MEMS)	Bridges star-tracker outages; ring-laser, fibre-optic, hemispherical resonator, MEMS
GNSS receiver	Position, velocity, time (and attitude with several antennas)	Position 1–10 m real-time	Standard in LEO; works up to GEO and beyond with weak-signal receivers
Fine guidance sensor	Offset of a guide star in the science telescope focal plane	Milliarcseconds	Hubble FGS, JWST FGS, Roman

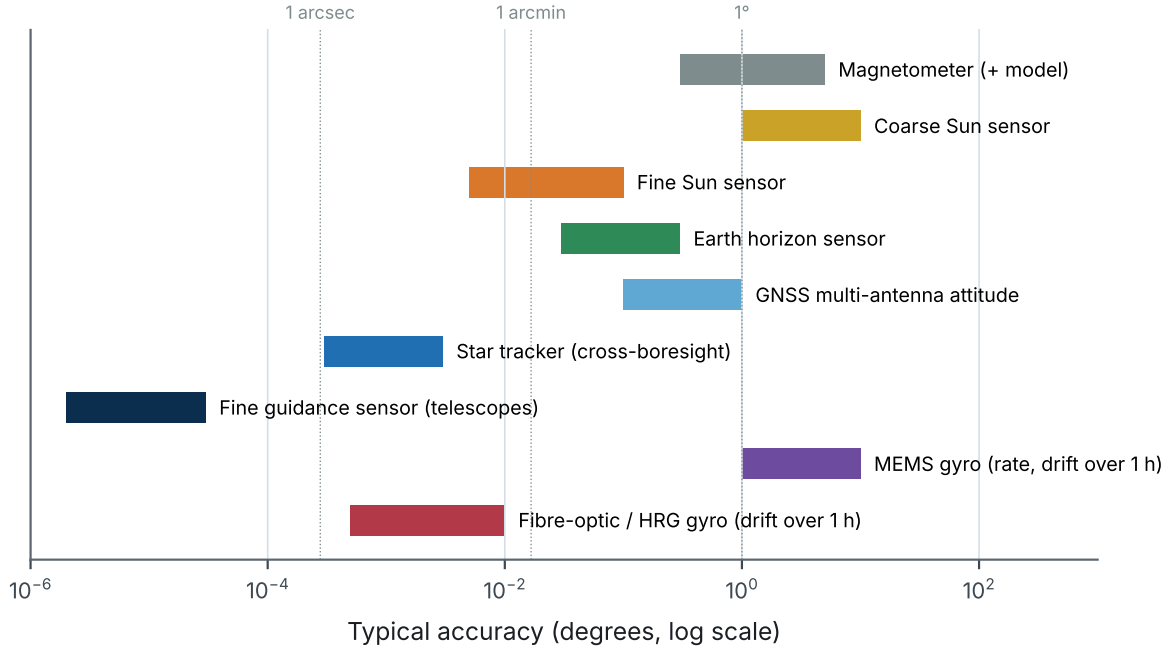


FIGURE 6.3 Typical accuracy ranges of attitude sensors on a logarithmic scale. Star trackers improved attitude knowledge by two to three orders of magnitude over Sun and Earth sensors.

A **star tracker** is a small digital camera with a wide-angle lens (field of view 10–25°) and an onboard catalogue of thousands of stars. A "lost-in-space" algorithm identifies star patterns (triangles or pyramids of stars) without prior knowledge; once locked, it tracks the stars at 1–10 Hz. Modern trackers for small satellites weigh under 300 g; high-end units (e.g., Leonardo's AA-STR, Sodern's Hydra, Ball's CT-2020) serve large missions. Measurements from all sensors are fused by an **extended Kalman filter** (usually a multiplicative quaternion formulation) that also estimates gyro biases.



FIGURE 6.4 Engineers install the star-tracker camera heads on NASA's Europa Clipper before launch (image released by JPL in 2025, when the trackers returned their first in-flight star-field images). Star trackers are the primary attitude sensors of almost every modern three-axis-stabilised spacecraft. Image: NASA/JPL (NASA/JPL-Caltech).

6.1.6 Attitude Actuators

- **Reaction wheels** — flywheels driven by brushless motors; accelerating a wheel in one direction rotates the spacecraft in the other. They give smooth, precise, propellant-free control. Typical wheels provide 0.01–0.3 N·m of torque and store 0.01 N·m·s (CubeSat wheels) to 200 N·m·s. Four wheels in a skewed pyramid (Figure 6.5) provide full control with one redundant wheel.
- **Momentum wheels** — biased wheels spinning at constant high speed for gyroscopic stiffness about one axis (classic GEO "momentum-bias" designs).
- **Control moment gyroscopes (CMGs)** — constantly spinning wheels whose spin axis is tilted by a gimbal motor. The gyroscopic reaction produces torques hundreds of times larger than a reaction wheel of similar mass. This enables very agile spacecraft (WorldView, Pléiades Neo) and the attitude control of the ISS (four 4 760 N·m·s double-gimbal CMGs).
- **Magnetorquers** — electromagnetic coils or rods that interact with Earth's field ($\mathbf{T} = \mathbf{m} \times \mathbf{B}$); used for momentum dumping and for CubeSat attitude control. They can only produce torque perpendicular to the local field.
- **Thrusters** — produce large torques for slews, momentum dumping and control during orbit manoeuvres; necessary beyond Earth's magnetic field.

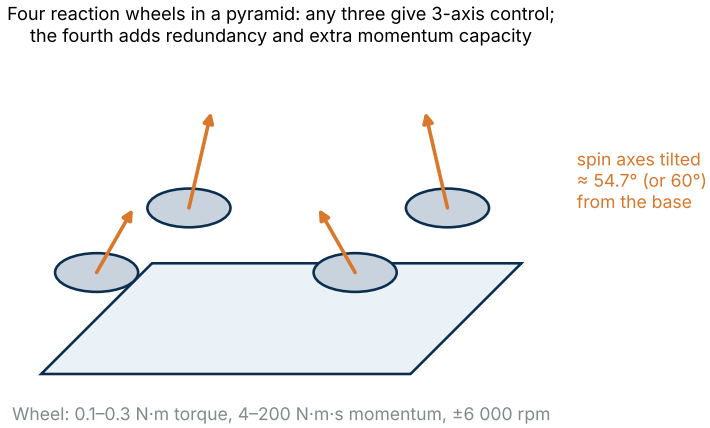


FIGURE 6.5 Four reaction wheels in a pyramid arrangement. The configuration tolerates the failure of any one wheel.

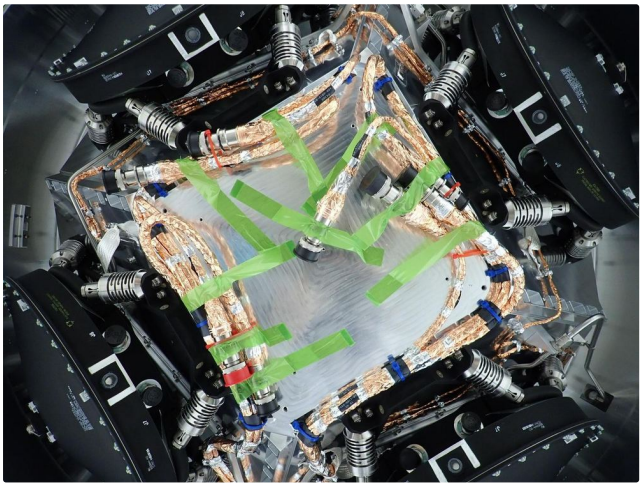


FIGURE 6.6 The four reaction wheels of Europa Clipper installed on the spacecraft's propulsion module in 2022. Image: NASA/JPL (NASA/JPL-Caltech).

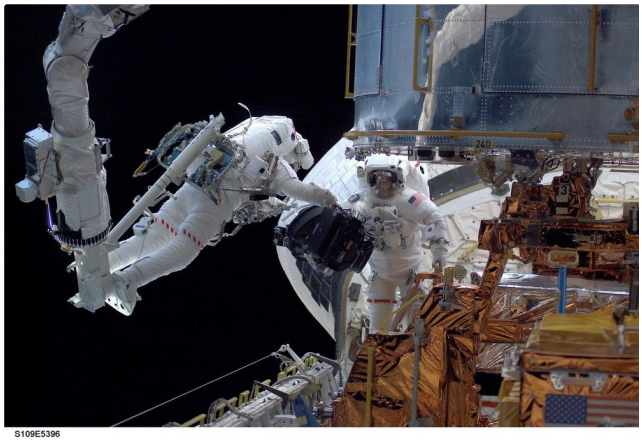


FIGURE 6.7 Astronauts Mike Massimino and James Newman replace a reaction wheel assembly on the Hubble Space Telescope during Servicing Mission 3B in March 2002. Image: NASA/JSC.

FAILURE LESSON — WHEN REACTION WHEELS FAIL

Reaction wheels are a classic life-limiting item. NASA's Kepler planet-hunting telescope lost its second of four wheels in 2013 and could no longer point precisely. Engineers invented the "K2" mission, balancing the spacecraft against solar radiation pressure. This mission continued to discover exoplanets for five more years. The Dawn asteroid mission lost two wheels and completed its mission using hydrazine thrusters and careful propellant management. Hubble has had several gyro failures and since 2024 operates in a one-gyro mode to preserve its remaining gyros. Lessons include bearing lubrication, cage instability, conservative speed limits, and designing control modes that work with fewer actuators.

6.1.7 Control Laws and Modes

Most attitude controllers are variants of **proportional–integral–derivative (PID)** control around each axis. They are designed with classical frequency-domain margins (typically ≥ 6 dB gain margin and ≥ 30° phase margin). They also use filters to avoid exciting flexible modes of solar

arrays and antennas. More advanced methods include **linear-quadratic regulators (LQR)**, **model predictive control (MPC)** for constrained slews, and **eigen-axis slews** that rotate about a single axis for time-optimal reorientation.

A spacecraft has several **ADCS modes**: rate damping after separation (often magnetic "B-dot" control on small satellites), Sun acquisition, safe mode, nominal pointing (nadir, target or inertial), slew, orbit-control (thrusting) mode, and momentum-dumping mode. Mode transitions are managed by the flight software and by the fault-protection system.

6.2 Spacecraft Propulsion

6.2.1 Why Spacecraft Need Propulsion

On-board propulsion provides:

- **Orbit insertion and transfer** — e.g., GTO→GEO, lunar or planetary orbit insertion.
- **Orbit maintenance** — compensating drag in LEO, station-keeping in GEO, constellation phasing.
- **Attitude control and momentum dumping**.
- **Rendezvous, proximity operations and docking**.
- **Collision avoidance**.
- **End-of-life disposal** — de-orbiting within regulatory limits or raising to a graveyard orbit.

Every propulsion system is characterised by two numbers: **thrust** F (newtons) and **specific impulse** I_{sp} (seconds), which measures propellant efficiency:

$$I_{sp} = \frac{F}{\dot{m}\, g_0} = \frac{v_e}{g_0}$$

where v_e is the effective exhaust velocity. Through the rocket equation, I_{sp} determines how much propellant a given Δv requires (Figure 6.8). The landscape of thruster technologies spans eight orders of magnitude in thrust and two in specific impulse (Figure 6.9).

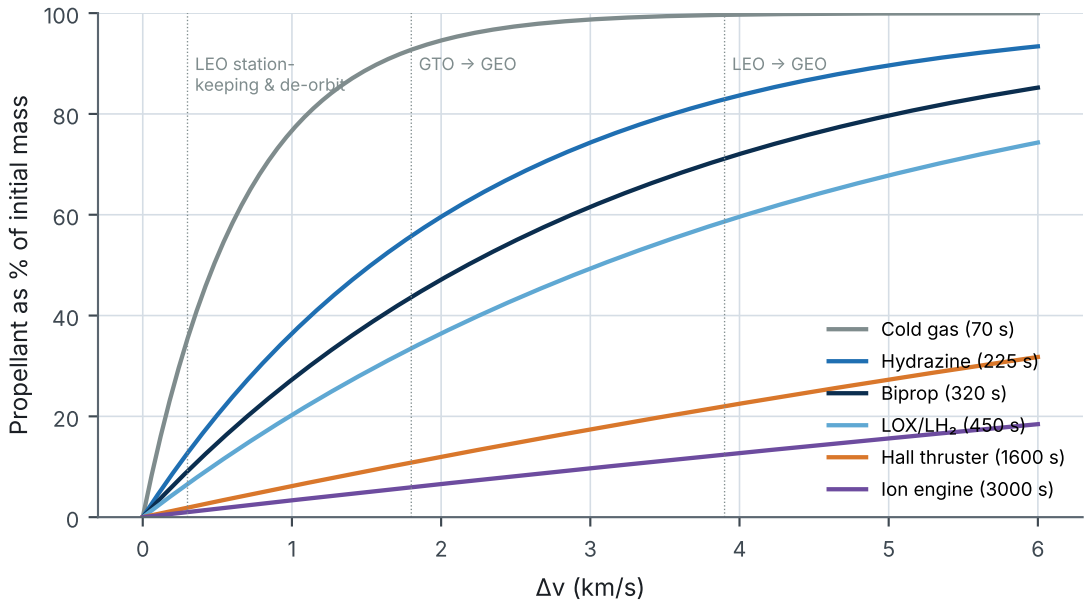


FIGURE 6.8 Propellant required as a fraction of initial mass versus Δv for different propulsion technologies. High-Isp electric propulsion makes large Δv missions possible with modest propellant mass, at the cost of very long thrusting times.

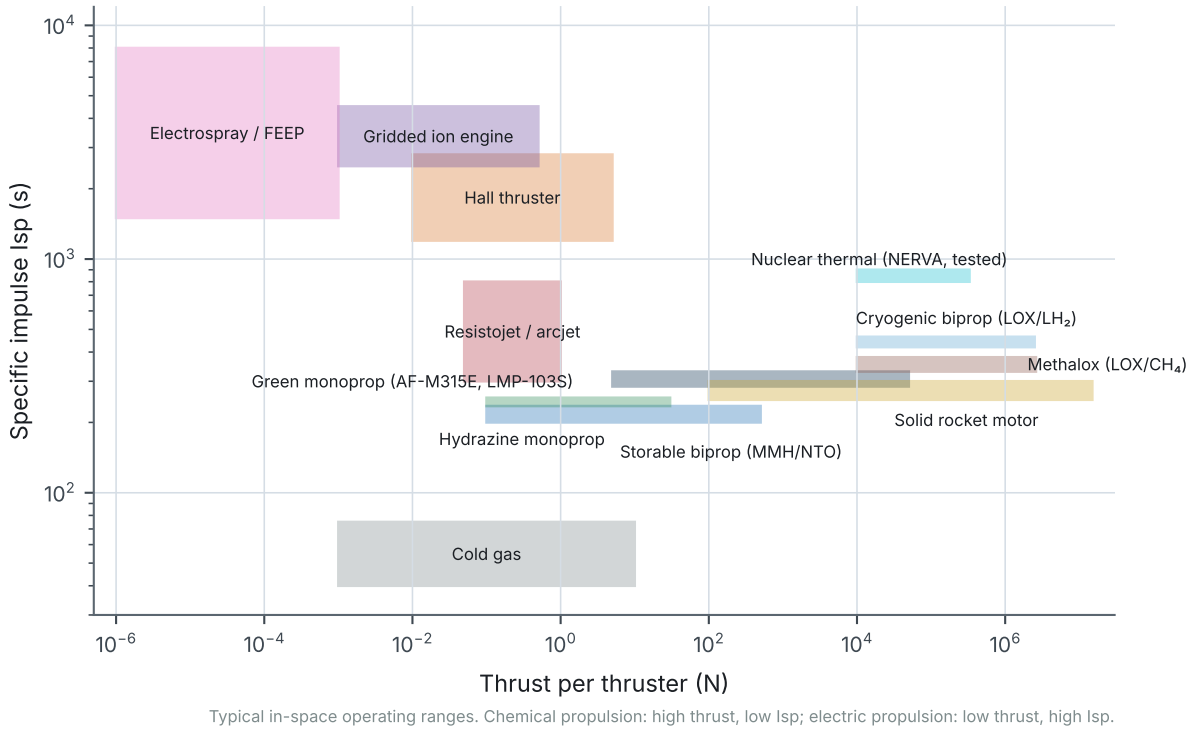


FIGURE 6.9 Operating ranges of in-space propulsion technologies in thrust and specific impulse.

6.2.2 Chemical Propulsion

Cold gas thrusters simply release compressed gas (nitrogen, xenon, butane) through a nozzle. Isp is only 50–75 s, but they are simple, clean and produce precise small impulses — used on CubeSats and for fine control (Gaia used cold gas micro-thrusters).

Monopropellant hydrazine (N_2H_4) has been the standard for satellite attitude and orbit control since the 1970s. Hydrazine decomposes exothermically when it passes over a heated iridium-on-alumina catalyst bed, producing hot ammonia, nitrogen and hydrogen (Figure 6.10). Thrusters range from 0.5 N to several hundred newtons (Isp 220–235 s). Hydrazine is toxic and

carcinogenic, so handling requires protective suits (SCAPE suits) and is expensive. **"Green" monopropellants** — AF-M315E (ASCENT, flown on NASA's Green Propellant Infusion Mission, 2019) and LMP-103S (ammonium dinitramide-based, flown on PRISMA and Skybox/SkySat) — offer higher density-Isp and much safer handling.

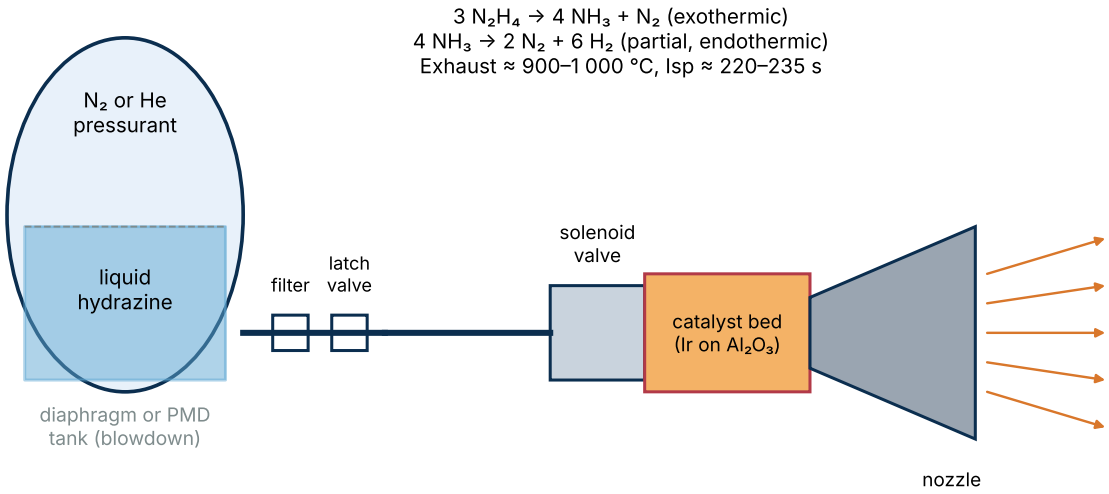


FIGURE 6.10 A hydrazine monopropellant thruster system. Propellant flows from a blowdown tank through filters and valves to a catalyst bed where it decomposes; the hot gases expand through a nozzle.

Bipropellant systems burn a fuel (monomethylhydrazine, MMH, or hydrazine) with an oxidiser (nitrogen tetroxide, NTO, or mixed oxides of nitrogen, MON) that ignite spontaneously on contact (**hypergolic**). Isp is 300–330 s. A typical GEO satellite has a 400–500 N **liquid apogee engine** for orbit raising and 10–22 N thrusters for attitude and station-keeping, fed from regulated tanks (Figure 6.11). Planetary orbiters use biprop engines for orbit insertion (e.g., Juno's 645 N Leros-1b). **Dual-mode** systems use hydrazine both as monopropellant for small thrusters and as fuel for a biprop engine.

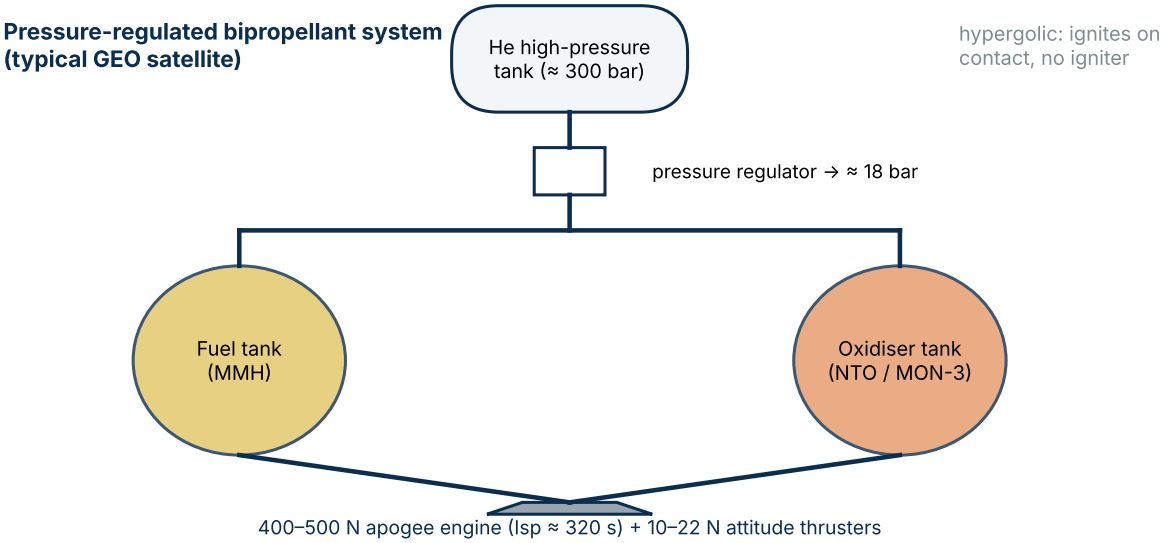


FIGURE 6.11 A pressure-regulated bipropellant system. Helium from a high-pressure tank is regulated to constant pressure and pushes fuel and oxidiser to the engines.

Solid rocket motors provide a single, large impulse — historically used as apogee kick motors and for planetary orbit insertion, and still common in upper stages.

6.2.3 Feed Systems and Propellant Management

Propellant can be fed by **blowdown** or **pressure regulation**. With blowdown, the pressurant is in the same tank, and pressure and thrust fall as the tank empties; this is simple and typical of LEO satellites. Pressure regulation gives constant pressure from a separate high-pressure tank — needed for biprop engines with tight mixture-ratio control. Large launch-vehicle stages use pumps (Chapter 35).

In microgravity, liquid does not settle at the tank outlet. **Propellant management devices (PMDs)** — surface-tension vanes, sponges, galleries and screens — or **elastomeric diaphragms and bladders** keep gas-free liquid at the outlet. Alternatively, a small "ullage" burn settles the propellant before a main burn. **Propellant gauging** is difficult in weightlessness and uses bookkeeping (counting thruster pulses), pressure–volume–temperature methods, or the thermal response of the tank to heating.

6.2.4 Electric Propulsion

Electric propulsion (EP) uses electrical energy to accelerate propellant to much higher exhaust speeds than chemical reactions allow — 15–50 km/s instead of 3–4.5 km/s. Thrust is tiny (millinewtons to a few newtons) because it is limited by available power:

$$F = \frac{2\,\eta\,P}{v_e} = \frac{2\eta P}{g_0 I_{sp}}$$

so a 5 kW Hall thruster at 60 % efficiency and 1 800 s produces about 0.34 N. The main types (Figure 6.12) are:

- **Electrothermal** — resistojets and arcjets heat propellant electrically (Isp 300–800 s); used for GEO station-keeping in the 1990s.
- **Gridded ion engines** — xenon is ionised in a discharge chamber and accelerated through electrically charged grids; a neutraliser emits electrons to keep the spacecraft neutral. Isp 2 500–4 500 s. Flown on Deep Space 1, Hayabusa 1 and 2, Dawn (three NSTAR engines, total Δv ≈ 11 km/s — a record), BepiColombo (four T6 engines), DART (NEXT-C) and Boeing 702 GEO satellites (XIPS).
- **Hall-effect thrusters** — a radial magnetic field traps electrons in an annular channel, creating an axial electric field that accelerates ions. Isp 1 200–2 800 s, higher thrust per kW than ion engines. They are by far the most widely flown EP technology. Starlink uses krypton and (V2 Mini onwards) argon Hall thrusters on every satellite. The 6 kW-class units power Psyche (SPT-140) and all-electric GEO satellites, and 12 kW AEPS thrusters were developed for the Gateway Power and Propulsion Element.
- **Electrospray and field-emission (FEEP)** thrusters accelerate ions or charged droplets from liquid metals or ionic liquids, giving micronewton thrust for precise control (LISA Pathfinder) and compact CubeSat propulsion.
- **Pulsed plasma thrusters** ablate a Teflon block with an electric arc — simple, used on small satellites.

- **Advanced concepts** (VASIMR, magnetoplasmadynamic thrusters, nuclear electric propulsion) are covered in Chapter 42.

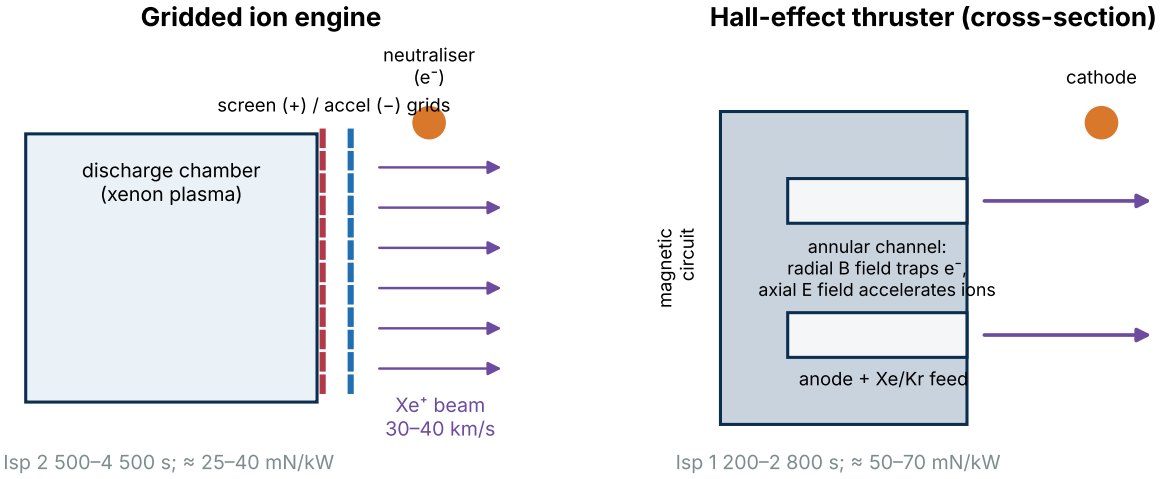


FIGURE 6.12 Principles of gridded ion and Hall-effect thrusters. Both ionise a noble gas and accelerate the ions electrostatically; a cathode neutralises the beam.

DATA BOX — PROPELLANTS FOR ELECTRIC PROPULSION

- **Xenon**: heavy, easily ionised, inert, stores densely at high pressure (≈ 1.5 g/cm^3 at 80–150 bar). Expensive (US$ 3 000–20 000/kg depending on market; prices spiked in 2022).
- **Krypton**: about 10× cheaper than xenon, slightly lower thrust efficiency; used by Starlink V1.
- **Argon**: cheaper still; adopted by Starlink V2 Mini thrusters (≈ 2.4× more thrust and 1.5× more Isp than the earlier krypton thrusters, according to SpaceX).
- **Iodine**: stored as a solid at low pressure — no high-pressure tanks; first flown by ThrustMe on Beihangkongshi-1 in 2020.
- **Water** and other condensables for electrothermal and microwave thrusters on small satellites.

6.2.5 Choosing a Propulsion System

The choice depends on the Δv required, the time available, the power available, cost and safety. Rules of thumb: small LEO satellites needing < 100 m/s use cold gas, green monopropellant or electric propulsion, and GEO satellites increasingly use all-electric or hybrid architectures. Deep-space missions with large Δv use ion or Hall thrusters if solar power is sufficient. Planetary orbit insertion and landing need chemical thrust. Mega-constellations have made Hall thrusters a mass-produced commodity: SpaceX alone has built more electric thrusters than the rest of the world combined.

6.3 Command and Data Handling

6.3.1 Functions and Architecture

The **command and data handling (C&DH)** subsystem — **on-board data handling (OBDH)** in European terminology — is the spacecraft's nervous system. It receives and validates commands, distributes them to subsystems, runs the flight software and keeps time. It collects housekeeping

telemetry and payload data, stores data in mass memory and formats it for downlink. It also supervises the health of the spacecraft (Figure 6.13).

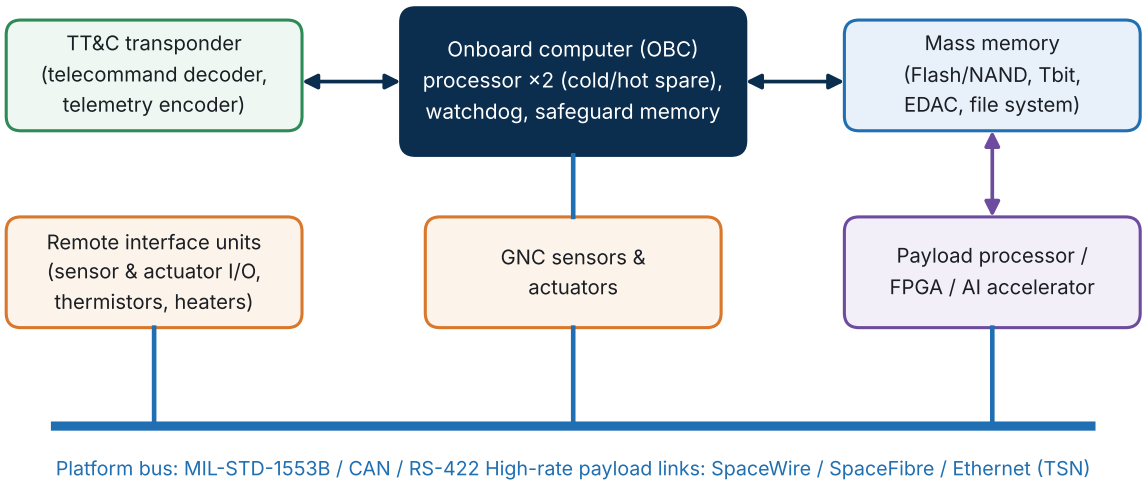


FIGURE 6.13 A typical C&DH architecture: a redundant onboard computer connected through a platform bus to remote interface units and GNC equipment, a high-rate network to the payload and mass memory, and a link to the TT&C transponders.

Architectures may be **centralised** (one computer does everything — common on small satellites) or **federated** (separate computers for GNC, payload and platform). They may also be **distributed/integrated modular avionics**, with partitioned software on shared processors connected by deterministic networks. This is now common on human spacecraft and large platforms.

6.3.2 Data Buses and Networks

TABLE 6.2 Common spacecraft data bus and network standards.

Standard	Data rate	Typical use
MIL-STD-1553B	1 Mbit/s, dual-redundant, command/response	Military and NASA platforms, ISS, Orion
CAN (and CANopen / CubeSat Space Protocol)	up to 1 Mbit/s	Small satellites, ESA platforms (ECSS-E-ST-50-15C)
RS-422 / RS-485, I²C, SPI	kbit/s–tens of Mbit/s	Simple point-to-point and board-level links
SpaceWire (ECSS-E-ST-50-12C)	2–400 Mbit/s, packet-switched	Instrument and mass-memory networks (JWST, Gaia, many ESA/NASA/JAXA missions)
SpaceFibre	2.5–6.25 Gbit/s per lane, multi-lane	High-rate payloads (SAR, hyperspectral)
Time-Triggered Ethernet / TSN	100 Mbit/s–10 Gbit/s, deterministic	Orion, Gateway, launch vehicles, next-generation platforms
SpaceVPX / VITA standards	Backplane	Modular payload processors

6.3.3 Processors and Memory

Spacecraft computers lag their terrestrial counterparts by 10–20 years in performance because they must tolerate radiation (Chapter 2) and are qualified slowly (Figure 6.14). The **BAE Systems RAD750**, a radiation-hardened PowerPC first flown in 2005, has powered more than 150 missions including Curiosity, Perseverance, JWST and Juno, at about 200–260 MIPS. European missions use **LEON** SPARC processors (GR712RC dual-core, GR740 quad-core). NASA's **High-Performance Spaceflight Computing (HPSC)** project with Microchip is producing a radiation-hardened multicore RISC-V processor (PIC64-HPSC) with roughly 100 times the performance of the RAD750, available to missions from about 2025–2026.

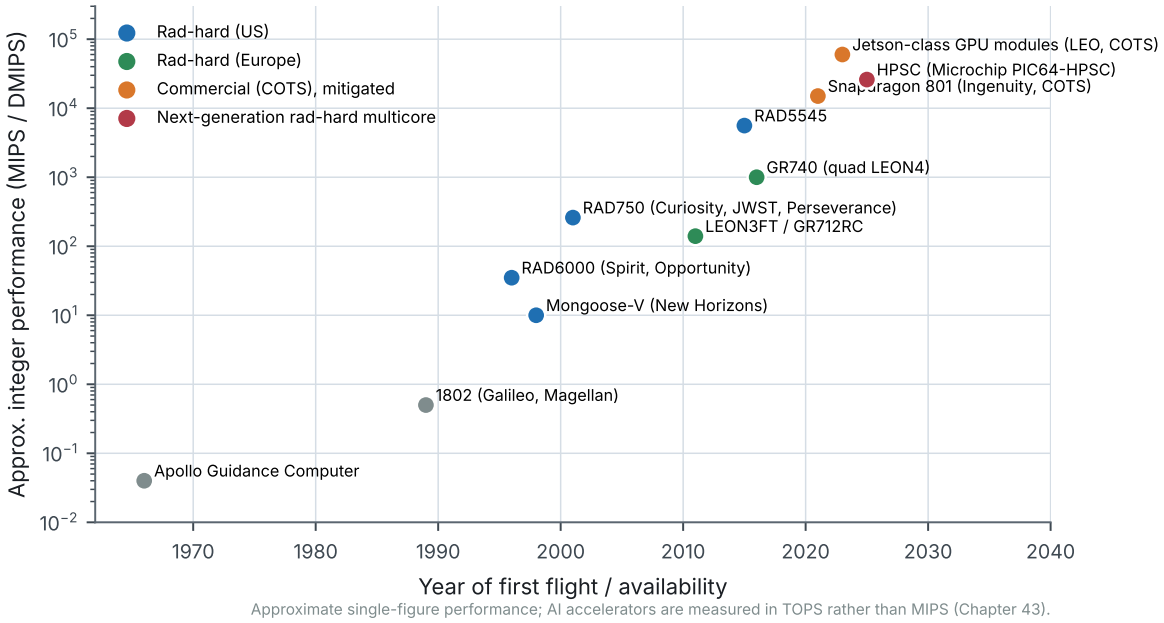


FIGURE 6.14 Approximate integer performance of spaceflight processors over time. Commercial processors with radiation mitigation (e.g., the Snapdragon on the Ingenuity Mars helicopter) offer orders-of-magnitude more performance than rad-hard parts but less proven reliability.

A growing alternative is to fly **commercial (COTS)** processors, FPGAs and GPUs with **mitigation**. This includes screening and radiation testing, triple modular redundancy in FPGA logic, watchdogs and error-correcting memory. It also includes current-limited power with automatic latch-up recovery, and frequent memory scrubbing. Short-duration LEO missions, mega-constellations and technology demonstrations increasingly use this approach — the Ingenuity helicopter on Mars ran on a Qualcomm Snapdragon 801 smartphone processor for nearly three years. Reconfigurable **FPGAs** (e.g., AMD/Xilinx Kintex UltraScale and Versal space-grade parts, Microchip RTG4 and RT PolarFire) process high-rate payload data and run neural networks (Chapter 45).

Memory must cope with single-event upsets. Program memory is often in radiation-tolerant PROM, MRAM or protected Flash; working memory in SDRAM with **error detection and correction (EDAC)** that corrects single-bit errors and detects double-bit errors, continuously **scrubbed** in the background. Mass memories use NAND Flash with file systems (CCSDS File Delivery Protocol for ground transfer) and capacities from gigabits on CubeSats to tens of terabits on Earth-observation satellites.

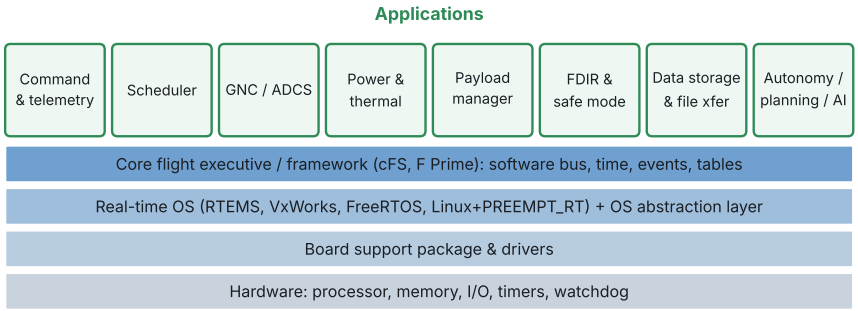
6.3.4 Time

The C&DH maintains **onboard time**, synchronised to UTC or GPS time via GNSS or ground correlation, to time-tag telemetry and to execute **time-tagged commands** stored in an onboard schedule. Earth-observation and science missions need time knowledge to microseconds; navigation satellites carry atomic clocks with nanosecond precision (Chapter 18).

6.4 Flight Software

6.4.1 Architecture

Flight software (FSW) has grown from a few kilobytes on early spacecraft to millions of lines of code (Orion ≈ 2 million; the F-35 fighter about 8 million; Starlink satellites run Linux). Good FSW is layered (Figure 6.15). It has a hardware abstraction layer and drivers, a **real-time operating system** (RTEMS, VxWorks, FreeRTOS, or Linux with real-time extensions), a reusable framework, and mission applications on top. The framework provides a software bus, time, event logging and table services. Examples are NASA's **core Flight System (cFS)**, JPL's **F Prime** (used on Ingenuity and many CubeSats) and ESA's component-based frameworks.



Layering isolates applications from hardware so that they can be reused across missions (NASA cFS has flown on dozens of spacecraft; JPL's F Prime flew on Ingenuity).

FIGURE 6.15 Layered flight-software architecture. A framework such as NASA's cFS or JPL's F Prime separates mission applications from the operating system and hardware.

6.4.2 Development and Assurance

Flight software controls the whole spacecraft and can be the cause of mission loss, so it is developed under strict processes. These include coding standards such as MISRA C and JPL's "Power of Ten" rules (no recursion, no dynamic memory allocation after initialisation, bounded loops, assertions). They also include static analysis, unit and integration tests, hardware-in-the-loop testing on flight-like testbeds ("flatsats"), and **independent verification and validation (IV&V)** for high-criticality software. Most modern spacecraft can be **reprogrammed in flight** — a capability that has rescued many missions.

FAILURE LESSON — SOFTWARE CAN END A MISSION IN SECONDS

On its maiden flight in 1996, Ariane 5 broke up 37 seconds after lift-off. The cause was inertial-reference software reused from Ariane 4, which converted a 64-bit floating-point horizontal velocity into a 16-bit integer. This caused an overflow that shut down both (identical) inertial units. In 2016 the Japanese X-ray observatory Hitomi was destroyed when an incorrect parameter uploaded to its attitude-control software, combined with a star-tracker anomaly, caused it to spin up until it broke apart. In 2023 the ispace HAKUTO-R lander crashed on the Moon after its software rejected valid altimeter data as faulty while passing over a crater rim. Each case shows the importance of verifying software against realistic, end-to-end scenarios — including "off-nominal" ones.

6.4.3 Autonomy and Fault Protection

Spacecraft are out of contact much of the time — a LEO satellite sees a given ground station for only about 10 minutes per pass, and a Mars rover's commands take 5–22 minutes to arrive. They therefore need **autonomy**: the ability to execute plans, react to events and protect themselves.

Fault detection, isolation and recovery (FDIR) — called fault protection at NASA — is organised hierarchically (Figure 6.16). Faults are handled at the lowest level possible. A memory bit-flip is corrected by EDAC, a failed unit is replaced by its redundant twin, and a subsystem anomaly triggers a mode change. If all else fails, the spacecraft enters **safe mode** — pointing its solar arrays at the Sun, switching off non-essential loads, and waiting for instructions from the ground. Safe modes must be simple, robust, and independent of the failed hardware (many spacecraft have a separate "survival" computer or hardware-implemented safe mode).

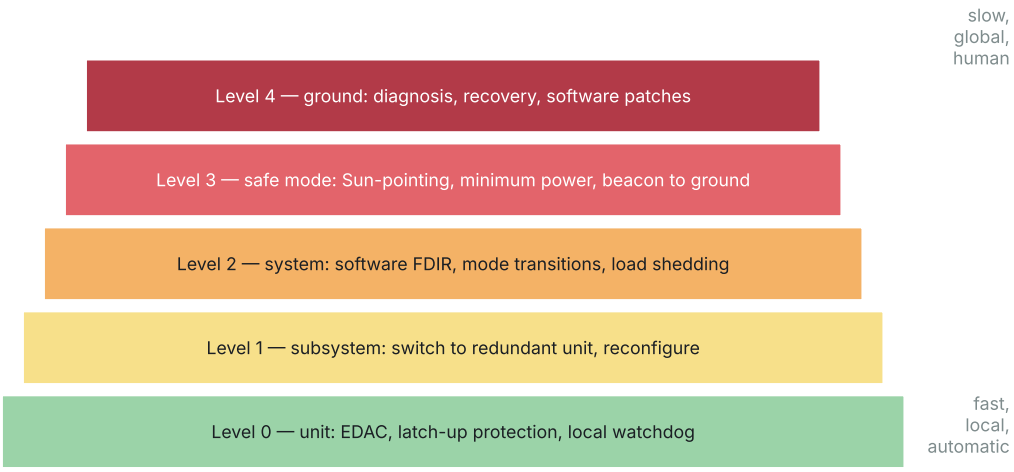


FIGURE 6.16 The hierarchy of fault detection, isolation and recovery. Fast local responses handle simple faults; system-level safe mode and ground intervention handle complex ones.

Higher levels of autonomy — onboard planning and scheduling, autonomous navigation, opportunistic science and model-based diagnosis — are increasingly implemented with artificial intelligence techniques, discussed in depth in Chapters 48 and 49.

KEY TAKEAWAYS

- The ADCS determines and controls attitude with sensors (star trackers, gyros, Sun sensors, magnetometers, GNSS) and actuators (reaction wheels, CMGs, magnetorquers, thrusters) in a Kalman-filter-based feedback loop. Momentum must be managed against persistent disturbance torques.
- Propulsion is characterised by thrust and Isp. Chemical systems (cold gas 70 s, hydrazine 225 s, biprop 320 s) give high thrust. Electric propulsion (Hall 1 200–2 800 s, ion 2 500–4 500 s) gives high efficiency at low thrust and is now standard on constellations and many GEO and deep-space missions.
- Microgravity requires propellant management devices; feed systems are blowdown or pressure-regulated.
- C&DH uses radiation-hardened or mitigated commercial processors, EDAC-protected memory and deterministic buses (1553, CAN, SpaceWire, TSN); rad-hard processors lag commercial ones by a decade or more.
- Flight software is layered (RTOS + framework + applications), rigorously verified and reprogrammable; hierarchical FDIR and safe modes protect the spacecraft between ground contacts.

REVIEW QUESTIONS

1. Compare spin stabilisation with three-axis stabilisation and give one advantage of each.
2. List four attitude sensors and rank them by typical accuracy.
3. Why do reaction wheels need momentum management, and which actuators are used to unload them?
4. A reaction wheel can store 2 N m s. A constant disturbance torque of 10 μN m acts on the spacecraft. How long does the wheel take to saturate if it starts at zero speed?
5. Compare cold gas, hydrazine monopropellant, bipropellant and Hall-effect propulsion in specific impulse and thrust, and give a typical application of each.
6. A 200 kg satellite (initial mass) needs a Δv of 300 m/s. Compare the propellant mass needed with hydrazine (I_{sp} = 225 s) and with a Hall thruster (I_{sp} = 1 500 s).
7. Why do space-qualified processors lag commercial processors by a decade or more, and how can commercial parts be used safely?
8. Describe the layers of a typical flight-software architecture and the role of safe mode in fault protection.

CHAPTER

7 Space Communications and the Ground Segment



Image: NASA/JPL (NASA/JPL-Caltech)

> *Without communications a spacecraft is useless - it cannot be controlled, and its data cannot be used. This chapter explains how radio and laser links work, from frequency bands and antennas to the link budget that decides whether a signal can be received; how modulation and error-correcting codes approach the theoretical limits of communication; the ground stations, networks and mission control centres that make up the ground segment; deep-space communications; and the international standards that let spacecraft and ground systems from different organisations work together.*

IN THIS CHAPTER

7.1 The Role of Communications
7.2 Frequency Bands and Spectrum
7.3 Antennas
7.4 The Link Budget
7.5 Modulation and Coding
7.6 Optical (Laser) Communications
7.7 The Ground Segment
7.8 Space Data Standards: CCSDS

7.1 The Role of Communications

Every spacecraft has at least one radio link. Communications serve four purposes:

1. **Telecommand (TC)** — instructions from the ground to the spacecraft (uplink), typically at a few kbit/s with strong error protection and authentication.
2. **Telemetry (TM)** — health and status data from the spacecraft (housekeeping), typically kbit/s to tens of kbit/s.
3. **Tracking** — measuring range, range-rate and angles to determine the orbit (Chapter 3).

4. **Payload (mission) data** — the product of the mission: images, radar data, science measurements, or, for communications satellites, the user traffic itself. Rates range from bits per second (deep space) to tens of gigabits per second (laser crosslinks).

The first three are grouped as **TT&C** (telemetry, tracking and command). Payload data usually uses a separate, higher-frequency, higher-rate link.

7.2 Frequency Bands and Spectrum

7.2.1 The Radio Spectrum for Space

Radio frequencies are a limited, shared natural resource. The **International Telecommunication Union (ITU)** allocates frequency bands to services (e.g., Earth exploration-satellite, space research, fixed-satellite, mobile-satellite, radionavigation-satellite) in its **Radio Regulations**, revised at World Radiocommunication Conferences every three to four years. Satellite operators must file their networks with the ITU through national administrations and coordinate with others to avoid interference. This process can take years and is a central part of any commercial constellation business plan.

Space engineers use the IEEE radar letter designations for bands (Figure 7.1, Table 7.1).

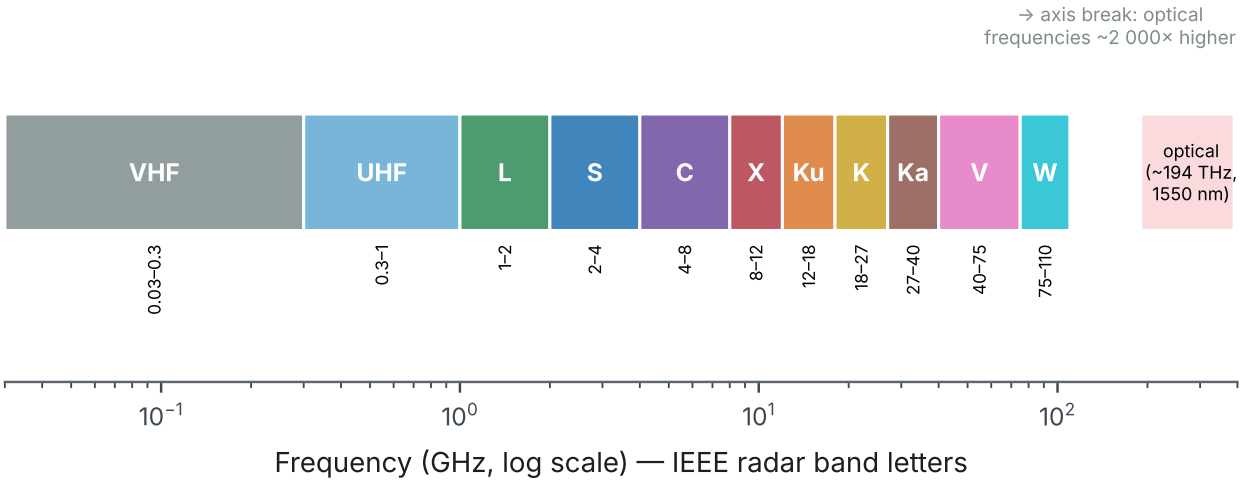


FIGURE 7.1 Microwave frequency bands used for space communications, with their approximate frequency ranges in GHz. Optical communication uses frequencies about 2 000 times higher than Ka-band.

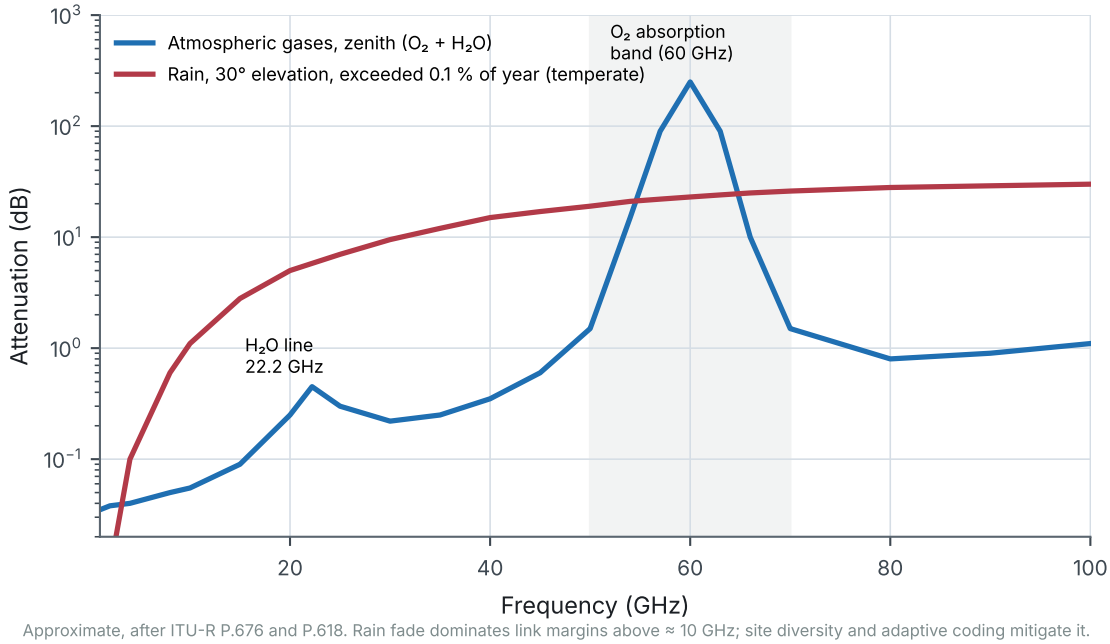
TABLE 7.1 Principal frequency bands for space communications and their uses.

Band	Frequency range	Typical space uses	Notes
VHF / UHF	30–300 MHz / 300 MHz–1 GHz	Amateur and CubeSat TT&C, military UHF SATCOM (MUOS), AIS ship tracking, IoT	Low gain, crowded, ionospheric effects
L-band	1–2 GHz	GNSS (1 164–1 610 MHz), Iridium, Inmarsat, Globalstar uplink, direct-to-device	Penetrates foliage and rain; small antennas
S-band	2–4 GHz	TT&C (2 025–2 110 MHz up, 2 200–2 290 MHz down), Globalstar, some EO, radar	The classic TT&C band

Band	Frequency range	Typical space uses	Notes
C-band	4–8 GHz	GEO TV distribution, rain-robust trunking, SAR (Sentinel-1, RADARSAT)	Little rain fade; large ground dishes
X-band	8–12 GHz	EO payload downlinks (8 025–8 400 MHz), deep space (7.1–7.2 GHz up, 8.4–8.45 GHz down), military SATCOM, SAR	Workhorse for science downlinks
Ku-band	12–18 GHz	Direct-to-home TV, VSAT, Starlink/OneWeb user links, data relay	Rain fade becomes significant
K / Ka-band	18–40 GHz	High-throughput satellites, gateways, deep-space Ka (31.8–32.3 GHz), EO high-rate (25.5–27 GHz)	Wide bandwidths, strong rain fade
Q / V-band	33–75 GHz	Feeder links for next-generation HTS and constellations	Experimental-to-early operational
W-band	75–110 GHz	Experimental links and radiometry	Research
Optical	≈ 200 THz (1 064 / 1 550 nm)	Laser crosslinks, data relay, deep-space optical	Unregulated by ITU (so far); blocked by cloud

7.2.2 Why Higher Frequencies?

Higher frequencies offer more **bandwidth** (and therefore higher data rates) and allow small antennas to form narrow, high-gain beams. Their disadvantages are greater attenuation by rain and atmospheric gases (Figure 7.2), the need for more precise antenna pointing, and more expensive components. The trend over 60 years has been steadily upward: from VHF and S-band to X and Ka for downlinks, and now to optical.



Approximate, after ITU-R P.676 and P.618. Rain fade dominates link margins above ≈ 10 GHz; site diversity and adaptive coding mitigate it.

FIGURE 7.2 Approximate atmospheric attenuation through the whole atmosphere as a function of frequency: gaseous absorption at zenith and rain attenuation exceeded 0.1 % of the time at 30° elevation in a temperate climate. Above 10 GHz, rain fade dominates link design.

7.3 Antennas

An antenna converts electrical signals into electromagnetic waves and focuses them in a direction. Its key property is **gain** G — how much it concentrates power compared with an isotropic radiator, expressed in dBi. For a parabolic dish of diameter D at wavelength λ and aperture efficiency η (typically 0.55–0.70):

$$G = \eta\left(\frac{\pi D}{\lambda}\right)^2, \qquad \theta_{3dB} \approx 70\,\frac{\lambda}{D}\ \text{degrees}$$

Gain rises with the square of frequency for a fixed size, while beamwidth shrinks (Figure 7.3). A 34 m Deep Space Network antenna has a gain of about 68 dBi at X-band and 80 dBi at Ka-band. Its Ka beam is only 0.017° wide, so pointing must be accurate to a few millidegrees.

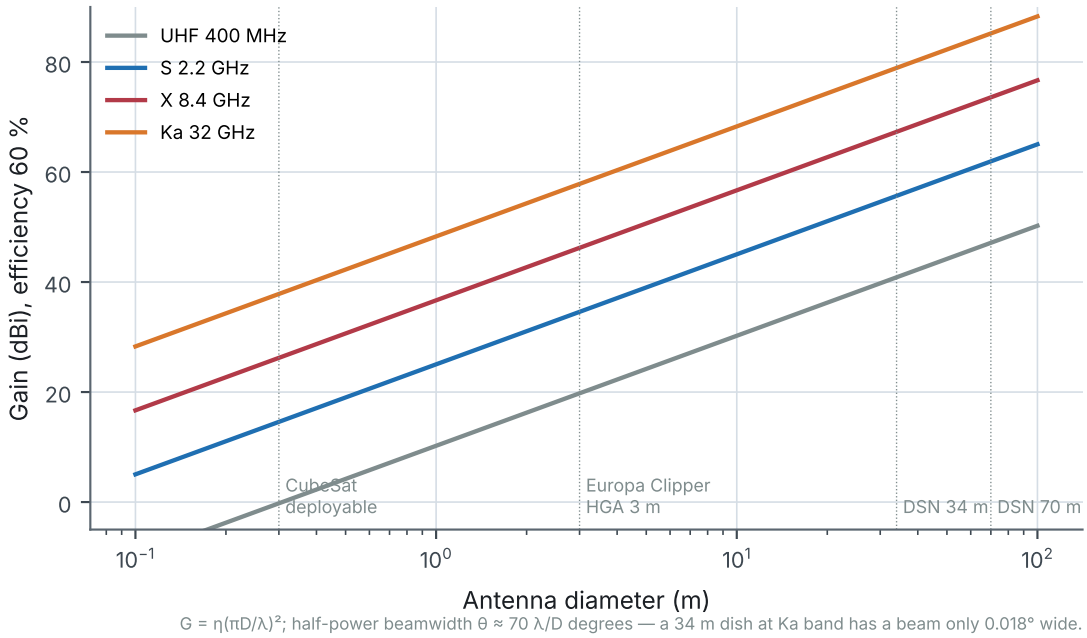


FIGURE 7.3 Gain of parabolic antennas as a function of diameter for four frequency bands.

Spacecraft use several antenna types (Figure 7.4):

- **Low-gain antennas (LGAs)** — monopoles, patches, helices with almost hemispherical coverage, used for TT&C in any attitude and in safe mode.
- **Medium-gain antennas** — horns, helices and small arrays with tens of degrees of beamwidth.
- **High-gain antennas (HGAs)** — fixed or steerable parabolic reflectors; deployable mesh reflectors up to 22 m (e.g., on mobile-satellite GEO spacecraft such as TerreStar and AST SpaceMobile's large phased arrays in LEO).
- **Phased arrays** — many small elements whose signals are shifted in phase to steer the beam electronically, with no moving parts. They are now standard on LEO broadband satellites and user terminals (the Starlink user terminal is a phased array of more than a thousand elements) and on SAR satellites.

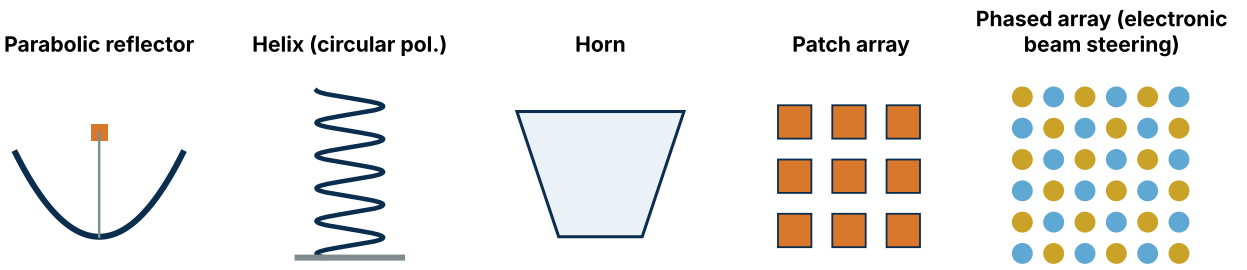


FIGURE 7.4 Common antenna types used on spacecraft and ground terminals.

7.4 The Link Budget

7.4.1 The Link Equation

The **link budget** is the accounting that shows whether a link closes — whether the received signal is strong enough compared with noise to deliver the required data rate at the required error rate. In decibels, the received carrier power is:

$$C = P_t + G_t - L_t - L_{fs} - L_a - L_{pol} - L_{point} + G_r - L_r \quad [\text{dBW}]$$

where P_t is transmitter power, G_t and G_r are antenna gains, L_t and L_r line losses, and L_a atmospheric losses. The term L_{fs} is the **free-space path loss**, which describes how power spreads over the surface of an ever-larger sphere:

$$L_{fs} = \left(\frac{4\pi d f}{c}\right)^2 \quad \Rightarrow \quad L_{fs}[\text{dB}] = 92.45 + 20\log_{10} d[\text{km}] + 20\log_{10} f[\text{GHz}]$$

The product $P_t + G_t - L_t$ is the **effective isotropic radiated power (EIRP)**. Path loss grows enormously with distance (Figure 7.5): about 177 dB for a 2 000 km LEO link at X-band, 205 dB to GEO at Ku-band, and about 280 dB to Mars at X-band.

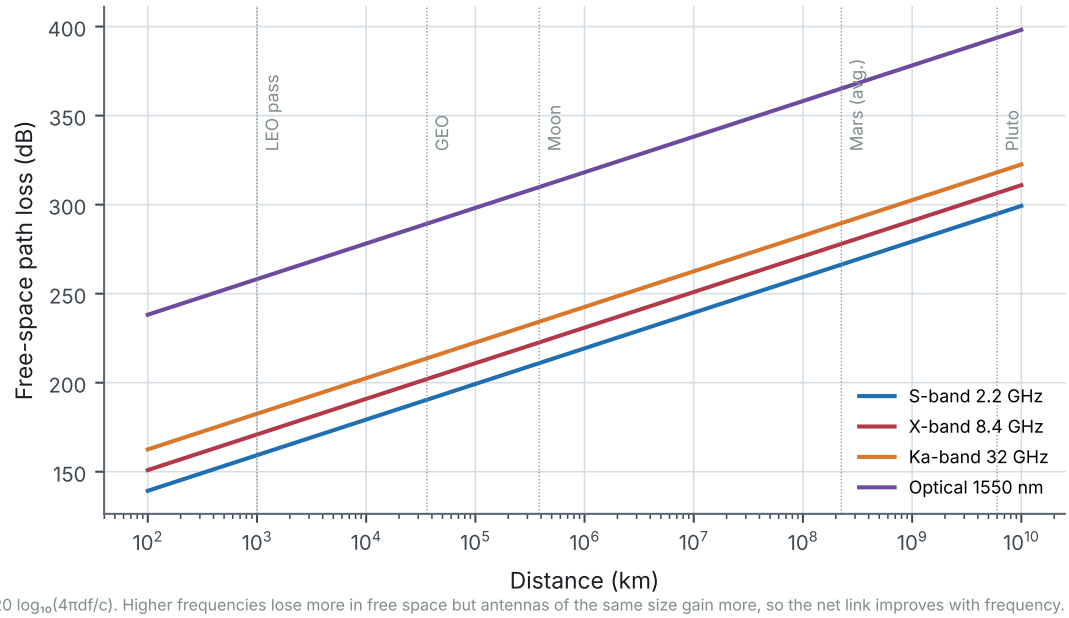


FIGURE 7.5 Free-space path loss versus distance for four frequency bands. Each tenfold increase in distance costs 20 dB.

The noise that competes with the signal is characterised by the **noise power spectral density** $N_0 = kT_{sys}$, where $k = 1.38 \times 10^{-23}$ J/K (−228.6 dBW/K/Hz) is Boltzmann's constant and T_{sys} is the system noise temperature (antenna, sky, receiver). A receiving system's quality is summarised by its **figure of merit** G/T in dB/K. The ratio that matters is

$$\frac{C}{N_0} = \text{EIRP} - L + \frac{G}{T} + 228.6 \quad \text{[dB-Hz]}, \qquad \frac{E_b}{N_0} = \frac{C}{N_0} - 10\log_{10} R_b$$

where R_b is the bit rate and E_b/N_0 the energy per bit relative to noise density. The link **margin** is the difference between the achieved and the required E_b/N_0; designs typically require 3 dB of margin, more for rain-affected links.

WORKED EXAMPLE — AN X-BAND EARTH-OBSERVATION DOWNLINK

A LEO satellite transmits 10 W (10 dBW) through 1.5 dB of losses into a 6 dBi patch antenna (EIRP = 14.5 dBW) at 8.2 GHz. At 5° elevation from a 600 km orbit, the slant range is about 2 000 km, so L_{fs} = 176.7 dB; atmosphere and rain cost 1.5 dB, pointing and polarisation 1 dB. The ground station has a 5.4 m dish (51 dBi) and T_{sys} = 150 K, so $G/T = 51 - 21.8 = 29.2$ dB/K. Then (Figure 7.6)

- $C = 14.5 - 176.7 - 2.5 + 51 = -113.7$ dBW
- $C/N_0 = -113.7 - 10\log(1.38 \times 10^{-23} \times 150) = -113.7 + 206.8 = 93.1$ dB-Hz
- For R_b = 500 Mbit/s (87.0 dB-Hz): $E_b/N_0 = 6.1$ dB.

With LDPC-coded QPSK requiring about 3 dB, the margin is about 3 dB — adequate. At higher elevations the range shrinks to 600 km and the margin rises by 10 dB, which modern systems exploit with **variable coding and modulation (VCM)**, increasing the data rate during the pass.

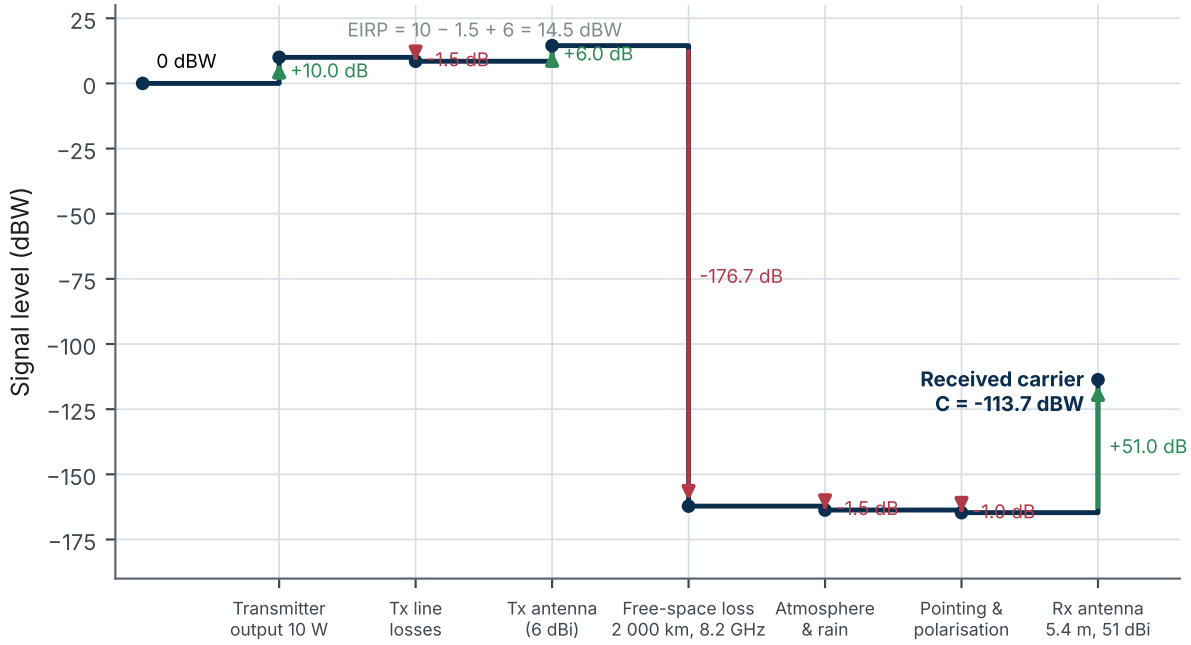


FIGURE 7.6 Signal level through the example X-band downlink: free-space loss dominates, and is recovered mainly by the gain of the ground antenna.

7.5 Modulation and Coding

7.5.1 Modulation

Modulation imprints digital data on the carrier wave by changing its phase, amplitude or frequency. Phase-shift keying is dominant in space (Figure 7.7):

- **BPSK** — two phases, 1 bit per symbol, very robust; used for TT&C and deep space.
- **QPSK / OQPSK** — four phases, 2 bits per symbol, same power efficiency as BPSK with twice the spectral efficiency; the workhorse of EO downlinks.
- **8PSK, 16APSK, 32APSK** — more bits per symbol, at the cost of needing more E_b/N_0. Amplitude-phase shift keying (APSK) arranges points on concentric rings. This lets it tolerate the non-linear power amplifiers (travelling-wave tube amplifiers, TWTAs) used in space. DVB-S2/S2X and CCSDS 131.2 support them with adaptive switching.
- **GMSK and filtered modulations** — constant-envelope, spectrally compact schemes.

Ranging signals (tones or pseudo-noise codes) can share the carrier with data to measure distance.

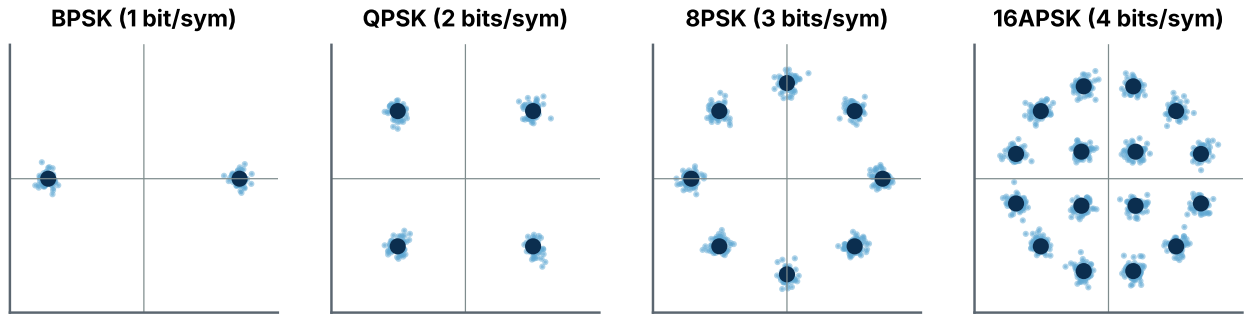


FIGURE 7.7 Constellation diagrams of common modulation schemes with noise. Denser constellations carry more bits per symbol but need a stronger signal to separate the points.

7.5.2 Error-Correcting Codes

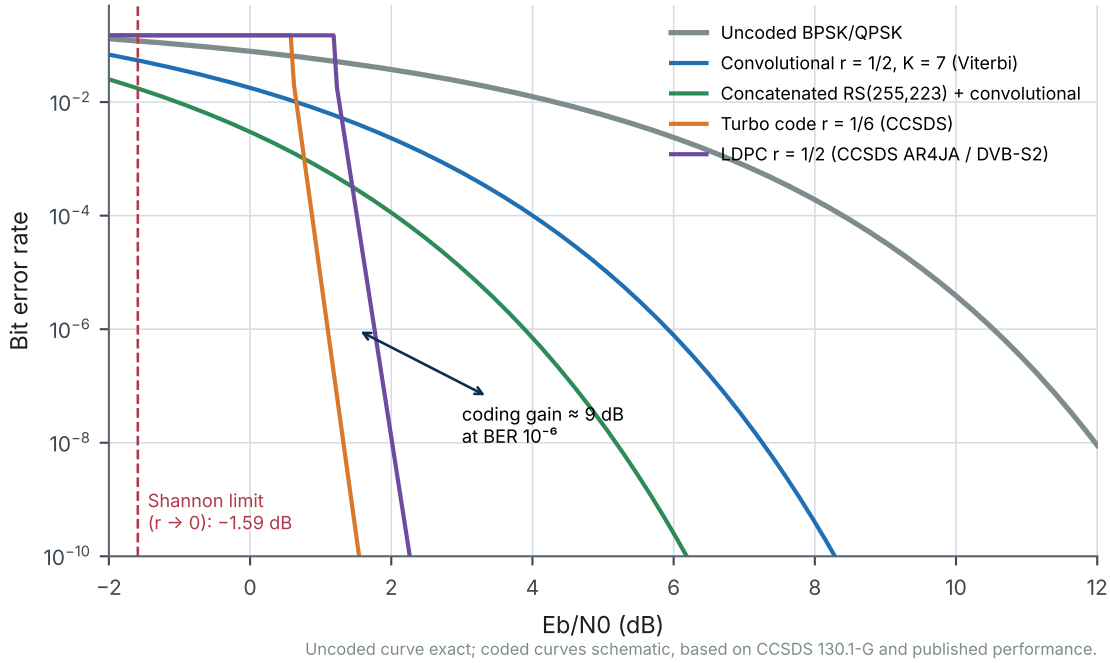
Claude Shannon showed in 1948 that the maximum error-free data rate (capacity) of a channel of bandwidth B is

$$C = B \log_2 \left(1 + \frac{S}{N}\right)$$

and that, for any rate below capacity, a code exists that makes errors arbitrarily rare. The ultimate limit is $E_b/N_0 = -1.59$ dB. **Forward error correction (FEC)** codes add redundant bits that let the receiver correct errors; the improvement is called the **coding gain** (Figure 7.8). Space communications have driven coding theory for 60 years:

- **Reed–Muller codes** on Mariner 6/7/9 (1969–1971).
- **Convolutional codes with Viterbi decoding** (rate 1/2, constraint length 7) — the standard from Voyager onwards.
- **Concatenated Reed–Solomon + convolutional codes** — Voyager at Uranus and Neptune, Galileo, and many EO missions.
- **Turbo codes** (invented 1993; CCSDS standard 1999) — within 1 dB of the Shannon limit; used on MRO, New Horizons, MESSENGER and many deep-space missions.

- **Low-density parity-check (LDPC) codes** — near-Shannon performance with efficient parallel decoding; standardised in CCSDS 131.0 (AR4JA codes for deep space, rate 7/8 for near-Earth) and DVB-S2 for commercial satellite broadband.



Uncoded curve exact; coded curves schematic, based on CCSDS 130.1-G and published performance.

FIGURE 7.8 Bit error rate versus Eb/N0 for uncoded and coded transmission. Modern turbo and LDPC codes operate within about 1–2 dB of the Shannon limit, a gain of roughly 9 dB over uncoded transmission — equivalent to an eightfold increase in transmitter power or antenna area.

Data is also **compressed** before transmission. CCSDS standards include lossless compression (CCSDS 121.0, based on Rice coding), image compression (CCSDS 122.0, wavelet-based) and multispectral/hyperspectral compression (CCSDS 123.0). Onboard AI that discards cloudy images or extracts only events of interest is an increasingly powerful form of "compression" (Chapter 45).

7.6 Optical (Laser) Communications

Free-space optical communications use lasers at near-infrared wavelengths (1 064 or 1 550 nm). The wavelength is about 10 000 times shorter than X-band. So a 10 cm telescope produces a beam about 10 µrad wide — far narrower than any radio beam — which concentrates the power on the receiver. Optical links offer data rates of gigabits to hundreds of gigabits per second, use spectrum that is not (yet) regulated, and are very hard to intercept or jam. Their challenges are clouds (which block the beam entirely), precise pointing, acquisition and tracking (to microradians while both terminals move), and atmospheric turbulence.

Milestones include:

- **LLCD** (NASA, 2013) — 622 Mbit/s from lunar orbit on LADEE.
- **EDRS** (ESA/Airbus "SpaceDataHighway", 2016 onwards) — GEO relay nodes receiving 1.8 Gbit/s laser links from Sentinel-1 and Sentinel-2 satellites and relaying the data to Europe in near-real time.
- **LCRD** (NASA, 2021) — a GEO laser relay, linked with the **ILLUMA-T** terminal on the ISS in 2023.

- **TBIRD** (NASA/MIT Lincoln Laboratory, 2022) — a CubeSat terminal that downlinked at 200 Gbit/s.
- **DSOC** (NASA/JPL, on the Psyche mission, 2023–2025) — the first deep-space laser link. It achieved 267 Mbit/s from 33 million km and continued to link from beyond Mars' distance. It used a superconducting nanowire photon-counting detector at Palomar Observatory.
- **Optical inter-satellite links in mega-constellations** — Starlink satellites carry three or more laser terminals, forming a mesh network in space that carries many petabytes per day. The US Space Development Agency's Transport Layer also uses optical crosslinks, and China's constellations are adopting them.
- **Artemis II's O2O terminal** (2026) — a laser link returning high-definition video from Orion during the crewed lunar flight.

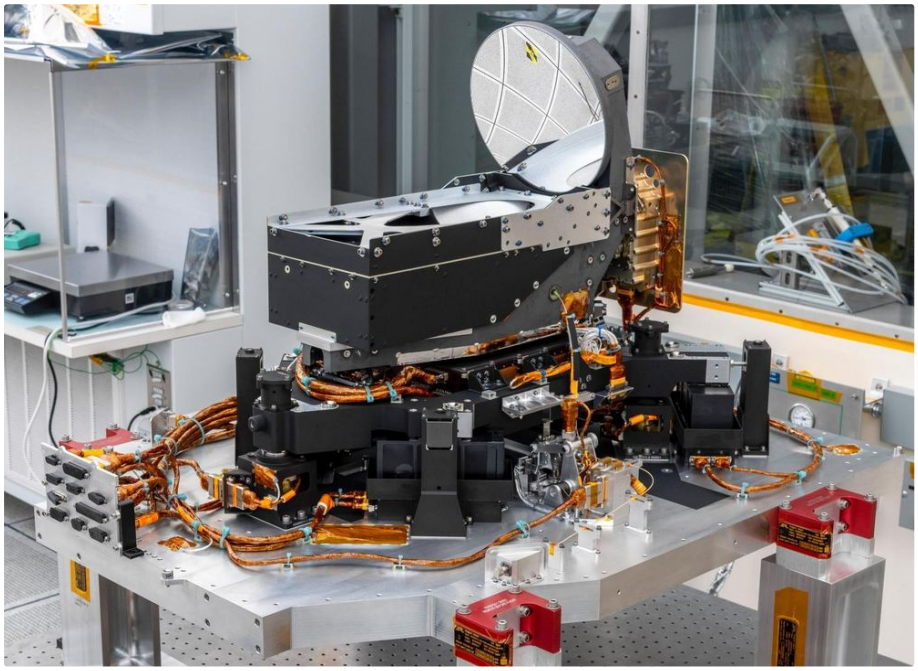


FIGURE 7.9 The Deep Space Optical Communications (DSOC) flight laser transceiver before integration with NASA's Psyche spacecraft. Its 22 cm telescope sent the first high-rate laser signals from deep space. Image: NASA/JPL (NASA/JPL-Caltech).



FIGURE 7.10 Artist's illustration of the ILLUMA-T terminal on the International Space Station relaying data by laser through the LCRD relay satellite in geostationary orbit. Image: NASA/JSC.

Figure 7.11 shows how the best achievable data rates have grown by more than ten orders of magnitude since the first planetary missions.

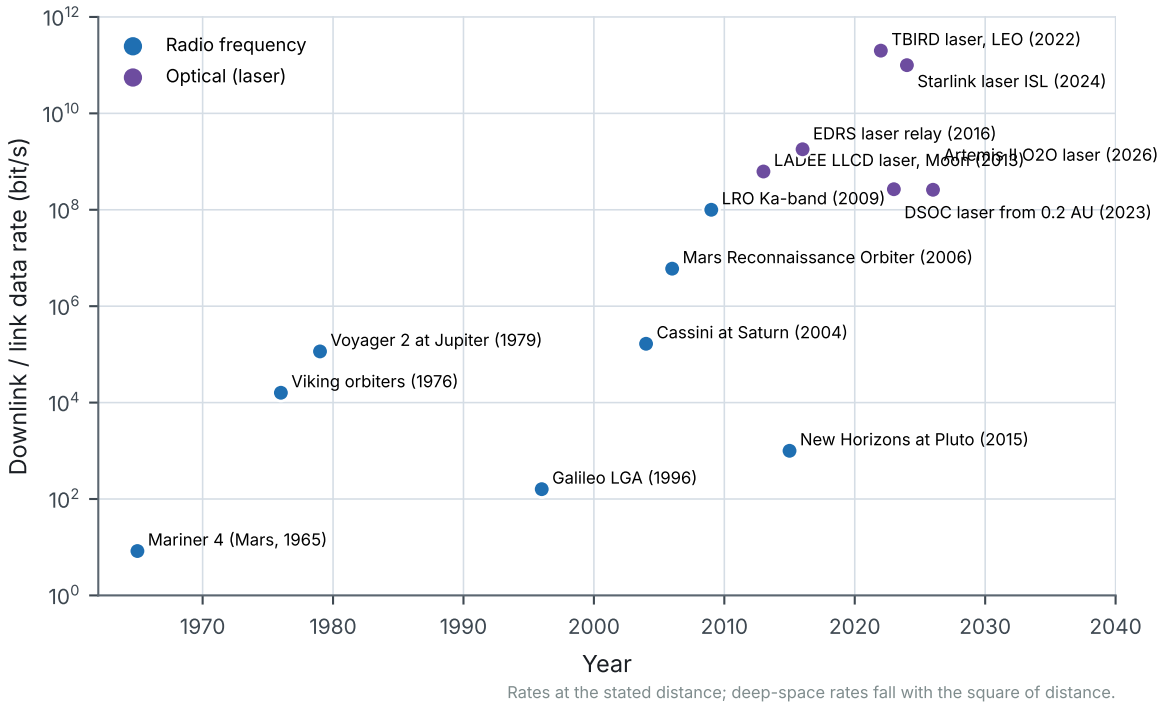


FIGURE 7.11 Representative space communication data rates since 1965. Laser links have pushed rates into the tens and hundreds of gigabits per second.

7.7 The Ground Segment

7.7.1 Architecture

The **ground segment** comprises all the ground facilities needed to operate the spacecraft and exploit its data (Figure 7.12):

- **Ground stations** with antennas, RF equipment (low-noise amplifiers, high-power amplifiers, up/down-converters), modems/baseband units, timing, and network connections.
- **Mission operations centre (MOC)** — the control room where flight controllers monitor telemetry, send commands, and respond to anomalies. They use a **mission control system** (e.g., ESA's SCOS-2000 and its successor EGS-CC, NASA's ITOS and AMMOS, or open-source frameworks such as Yamcs and OpenMCT).
- **Flight dynamics** — orbit determination, manoeuvre planning, attitude reconstruction, collision avoidance, and product generation (ephemerides, pointing predictions).
- **Mission planning and scheduling** — turning user requests into conflict-free activity timelines that respect power, thermal, data-volume and visibility constraints (Chapter 49 discusses AI planners).
- **Payload data ground segment (PDGS)** — receiving, processing, calibrating, archiving and distributing payload data. Processing levels follow a standard ladder: Level 0 (raw, time-ordered), Level 1 (calibrated, geolocated), Level 2 (geophysical variables), Level 3 (gridded, composited), Level 4 (model output) (Chapter 13).
- **Security** — encryption and authentication of telecommands (e.g., CCSDS Space Data Link Security), network segmentation, intrusion detection, and protection of ground systems, which are the most frequently attacked part of space systems.

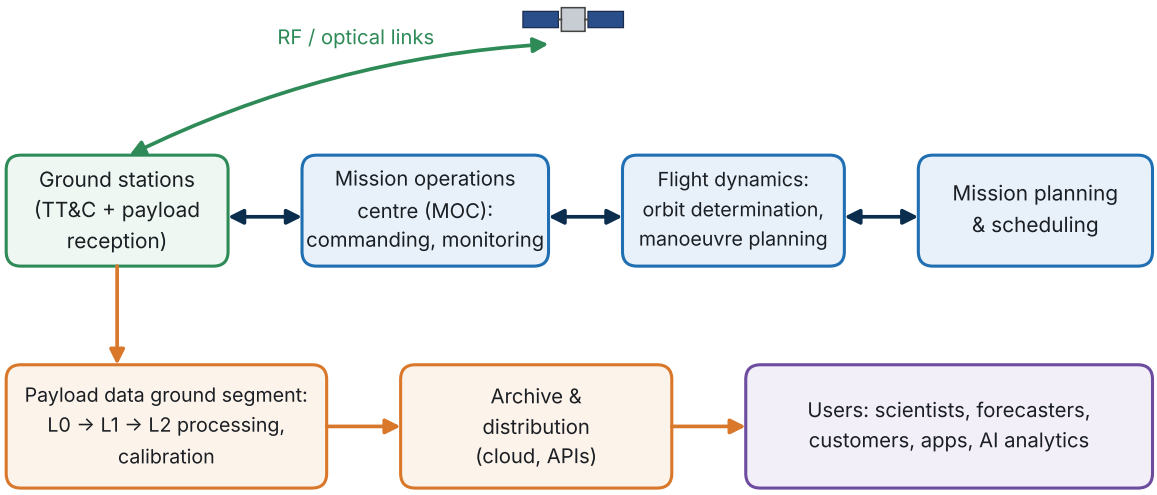


FIGURE 7.12 Elements of a typical ground segment and their data flows.

7.7.2 Ground Station Networks

A LEO satellite is visible from a given station for only 5–15 minutes per pass, and from a mid-latitude station for only a few passes a day. Polar-orbiting satellites, however, pass over the poles on every orbit, so stations at high latitudes see every orbit (Figure 7.13). This is why **Svalbard** (78° N, KSAT's SvalSat with more than 100 antennas), **Inuvik**, **Kiruna**, **Fairbanks** and Antarctic stations such as **Troll** (72° S) and **McMurdo** host so many antennas.

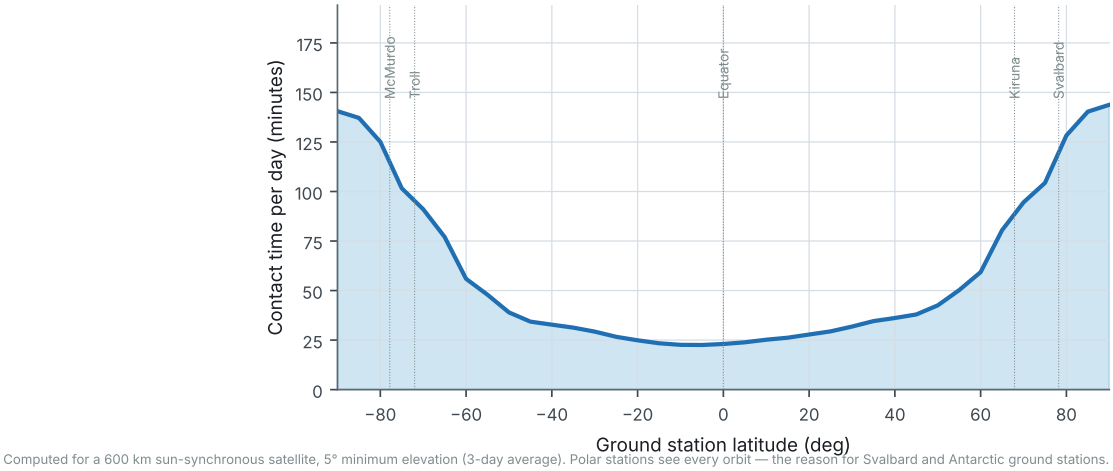


FIGURE 7.13 Daily contact time between a 600 km sun-synchronous satellite and a ground station, as a function of station latitude, computed for this manual. A polar station sees the satellite for several times longer than an equatorial one.

Operators increasingly buy **ground-station-as-a-service**, paying per pass or per minute on shared networks: KSAT, SSC (Swedish Space Corporation), AWS Ground Station, Microsoft Azure Orbital (partnered networks), Leaf Space, Atlas Space Operations, RBC Signals and others. Electronically steered **phased-array ground stations** and software-defined radios allow one site to track several satellites at once. Data relay satellites in GEO — NASA's **TDRS** constellation, ESA's **EDRS**, China's **Tianlian**, Russia's **Luch** — provide nearly continuous contact for the ISS, Hubble, Tiangong and Earth-observation satellites.

7.7.3 Deep Space Communications

Deep-space missions require the largest antennas, the most sensitive receivers and the most powerful transmitters on Earth. NASA's **Deep Space Network (DSN)** has three complexes spaced about 120° apart in longitude: **Goldstone** (California), **Madrid** (Spain) and **Canberra** (Australia). This spacing means that any spacecraft beyond a few thousand kilometres is always visible from at least one of them (Figure 7.14). Each complex has a 70 m antenna and several 34 m beam-waveguide antennas; new 34 m antennas were added under the DSN Aperture Enhancement Project (DSS-53 in Madrid in 2022, DSS-56 and others). Antennas can be **arrayed** to combine their signals, as at Madrid in 2024, where six antennas were arrayed together for the first time to receive Voyager-class signals.

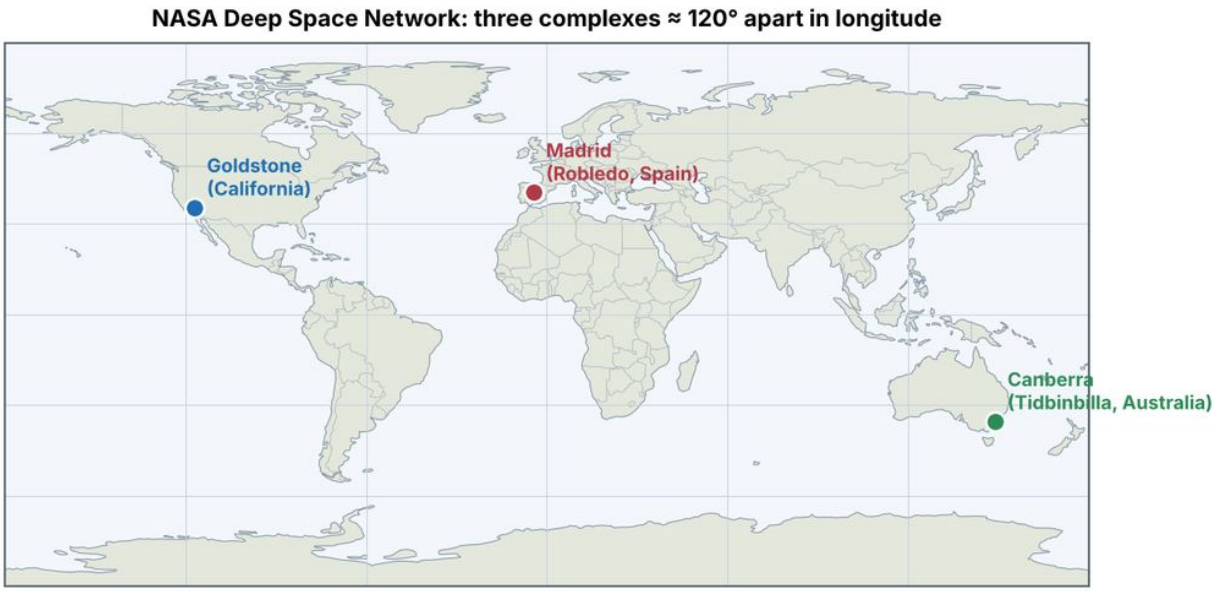


FIGURE 7.14 Locations of the three Deep Space Network complexes.



FIGURE 7.15 Six DSN antennas at the Madrid complex arrayed for the first time in April 2024 to receive a single spacecraft's signal. Image: NASA/JPL (MDSCC/INTA, Francisco "Paco" Moreno).

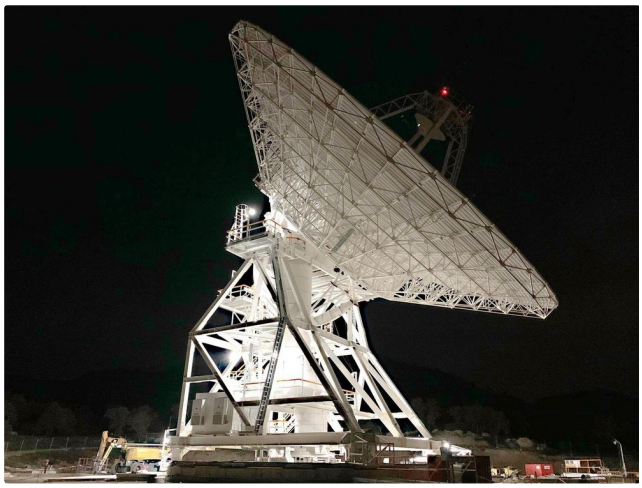


FIGURE 7.16 The 34 m beam-waveguide antenna DSS-53, added to the Madrid complex in 2022. Image: NASA/JPL (NASA/JPL-Caltech).

Other agencies operate complementary networks, such as ESA's **ESTRACK** with 35 m deep-space antennas at New Norcia (Australia), Cebreros (Spain) and Malargüe (Argentina). China's deep-space network has 35 m and 66 m antennas at Kashgar, Jiamusi and Zapala (Argentina). There are also India's IDSN at Byalalu, JAXA's 54 m Misasa antenna and Russia's network. **Cross-support** agreements using CCSDS standards let agencies track each other's missions.

The DSN is heavily oversubscribed: in the late 2020s it supports more than 40 missions, and demand from lunar missions (Artemis, CLPS landers, international orbiters) is growing. NASA is complementing it with commercial lunar communication services and with optical ground stations.

DATA BOX — TALKING TO VOYAGER 1

- Distance (October 2026): about 171 AU, or 25.6 billion km; one-way light time ≈ 23.7 hours (it reaches one light-day from Earth in late 2026).
- Transmitter: about 22 W at 8.4 GHz through a 3.7 m dish (gain ≈ 48 dBi, EIRP ≈ 61 dBW).
- Free-space loss ≈ 319 dB; received power at a 70 m DSN antenna ≈ 4×10^{-19} W (−184 dBW).
- Downlink rate: 160 bit/s (science), reduced over the years as power and distance changed.
- Commands are sent with transmitters of 20 kW or more; a round-trip command-and-response takes almost two days.

7.7.4 Delay- and Disruption-Tolerant Networking

The Internet assumes continuous end-to-end connectivity and round-trip times of milliseconds. In space, links are intermittent and delays can be minutes or hours. **Delay/Disruption-Tolerant Networking (DTN)** uses the **Bundle Protocol** to store data at each node until the next link is available ("store-and-forward"), forming an interplanetary internet. NASA has used DTN on the ISS since 2016 and in the PACE Earth-observation mission (2024). It is a core element of **LunaNet**, the planned interoperable network of lunar relays and services, and of ESA's **Moonlight** initiative.

7.8 Space Data Standards: CCSDS

The **Consultative Committee for Space Data Systems (CCSDS)**, founded in 1982 by the major space agencies, develops open standards ("Blue Books") for space communications and data handling. Using CCSDS standards lets spacecraft use any compliant ground station and lets agencies share hardware and software. Figure 7.17 shows the main layers.

Application	File delivery (CFDP), image/data compression (CCSDS 121/122/123), Mission Operations services
Transport / network	Space Packet Protocol, Encapsulation, IP over CCSDS, Delay/Disruption-Tolerant Networking (Bundle Protocol)
Data link — protocol	TM / TC / AOS / Proximity-1 / USLP transfer frames; Space Data Link Security (SDLS)
Data link — coding & sync	Reed–Solomon, convolutional, turbo, LDPC codes; randomisation; frame sync
Physical	RF modulation (BPSK, QPSK, GMSK, APSK), ranging; optical (CCSDS 141/142 O3K, HPE)

FIGURE 7.17 The CCSDS protocol stack used by most civil spacecraft, from physical-layer modulation up to file delivery and mission operations services.

Key CCSDS standards include Space Packets (133.0), TM/TC/AOS frames (132.0/232.0/732.0), the Unified Space Data Link Protocol (USLP, 732.1) and channel coding (131.0, 231.0). Others cover the CCSDS File Delivery Protocol (CFDP, 727.0), Space Data Link Security (355.0), the Bundle Protocol (734.2) and optical communications (141.0, 142.0). Commercial communications

satellites use the DVB-S2/S2X family; LEO broadband constellations use proprietary protocols; and **3GPP non-terrestrial network (NTN)** standards (from Release 17 onwards) integrate satellites into 5G mobile networks (Chapter 17).

KEY TAKEAWAYS

- TT&C links control the spacecraft; payload links deliver its product. Frequencies are allocated by the ITU; higher bands give more bandwidth but suffer rain fade.
- The link budget — EIRP minus losses plus G/T — determines C/N_0 and E_b/N_0; free-space loss grows 20 dB per decade of distance.
- Modern LDPC and turbo codes operate within 1–2 dB of the Shannon limit; adaptive coding and modulation use every available dB.
- Laser links deliver gigabit-to-hundreds-of-gigabit rates and now connect satellites in constellations, relays in GEO and even deep-space probes.
- The ground segment includes stations, mission control, flight dynamics, planning and payload data processing; polar stations and relay satellites maximise contact time; the DSN's three complexes provide continuous deep-space coverage.
- CCSDS standards enable interoperability; DTN extends the Internet to intermittent space links.

REVIEW QUESTIONS

1. Distinguish TT&C links from payload data links and give typical frequency bands for each.
2. Calculate the gain of a 3.7 m ground antenna with 60 % efficiency at 8.2 GHz.
3. Calculate the free-space path loss at 2.2 GHz over a slant range of 2 000 km.
4. Write the link equation for C/N_0 and explain each term. Why does −228.6 dBW/(K Hz) appear?
5. Why do modern links use LDPC or turbo codes? What does "within 1–2 dB of the Shannon limit" mean?
6. Give two advantages and two disadvantages of laser communication compared with radio.
7. Why are many ground stations for polar-orbiting satellites located at high latitudes?
8. Why does deep-space communication need very large antennas and three stations spaced around the world?
9. What problem does delay- and disruption-tolerant networking solve, and how?